Extreme Series

CHEMISTRY

GRADE 11-12

Based on Common Currently Used
National and International Curriculum

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2016 E.C

Mengesha Moges

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UNIT 1

Atomic Structure and Periodic Classification

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1.1 Historical Development of the Atomic Nature of Substances

- ☑ The idea of small, invisible particles being the building blocks of matter can be traced back more than 2000 years to the Greek philosophers **Democritus and Leucippus**
- ☑ These particles considered to be small and indestructible that they could not be divided in to small particles, were called **atoms.**
- ☑ The English name **atom** comes from the Greek word, **atomos** meaning indivisible.
- ☑ This early concept of atoms was not based on experimental evidence but was simply a result of thinking and reasoning on the part of philosophers.
- ☑ **Democritus** believed that matter is discrete meaning division has limit.
- ☑ **Arstotle and Plato** believed that matter is continuous meaning division has no limit.

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1.2 Dalton's Atomic Theory and the Modern Atomic Theory

Atomic theory is a theory that describes the structure and behavior of substances.

John Dalton proposed some basic assumptions about atoms based on scientific experimentation and observations at that time.

- ☑ **Dalton's theory** was based on two chemical laws, law of conservation of mass and law of definite proportion . Later, he formulated the law of multiple proportion.
- ☑ **Dalton** is often referred to as the father of modern atomic theory.

Some of the Dalton's Postulates are :

1. All matter is made of very small, discrete , particles called atoms.
2. Atoms cannot be subdivided, created or destroyed.
3. All atoms of an element are alike in mass, and other properties.
4. Atoms of different elements combine in simple whole number ratio to form chemical compounds.
5. In chemical reactions, atoms are combined, separated or rearranged.

Note : I. Postulates 2 and 3 were found to be wrong by later works.

II. Postulates 1, 4 and 5 are retained in modern atomic theory.

Modern atomic theory is simply modified Dalton's postulates plus the rest. This implies :

1. All matter is made up of atoms
2. Atoms are divisible in to sub - atomic particles such as electron, proton and neutron
3. Atoms of the same elements have different mass and physical properties due to isotopes.
4. Atoms of different elements combine in simple whole number ratio to form compounds.

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5. In chemical reactions, atoms are combined, separated or rearranged.

Illustrative Examples

1. The Greek philosopher that proposed the first atomic theory was

   A. Plato
   B. Parmenides
   C. Democritus
   D. Aristotle

Solution

Plato - believed that the earth, fire and water are comprised of unchanging geometric solids.

Parmenides - believed the all changes are illusion

Democritus - proposed the first atomic theory and believed that the world is made up of atom and empty space.

Arstotle - was the opponent of Democritus .

Answer : C

2. Which of the following is not true about Dalton's atomic theory ?

   A. Elements are made up of atoms
   B. Atoms of different elements are different
   C. Atoms of different elements combine simple whole number ratio to form compound
   D. Atoms are divisible in to sub atomic particles

Solution

Atoms are indivisible according to Dalton. **Answer : D**

3. Which of the following statements is accepted both in Dalton's and modern atomic theories ?

   A. Atoms of the same elements may have different masses
   B. Atoms cannot be sub-divided, or destroyed
   C. Atoms of different elements combine in simple whole number ratio
   D. Atoms of a given element are identical in mass and other properties.

Answer : C

4. Explain the difference between Democritus and Arstotle ideas about atomic theory.

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Solution

Democritus	Aristotle
➥ Believes in atom	➥ Does not believe in atom
➥ Discrete theory (division has limit)	➥ Continuous theory (division has no limit)

5. What are the basis of Dalton's atomic theory ?

Solution

The two chemical laws : law of conservation of mass and law of definite proportion.

6. What mass of hydrogen and chlorine can be produced from.
   a. 36.5 g of HCl b. 1 g of HCl

Solution

To solve the answer for such type of question, first determine the percentage composition of each element.

$$\% H = \frac{M_H \times 100}{M_{HCl}} = \frac{1\,g}{36.5\,g} \times 100 = 2.74\ \%$$

a. mass of hydrogen $= \%H \times m_{HCl} = \frac{2.74}{100} \times 36.5g = 1.00\,g$

mass of chlorine = mass HCl - mass of H = 36. 5 -1 = 35. 5g

b. mass of $H = \frac{2.74}{100} \times 1g = 0.0274\,g$

mass of Cl = 1g - 0.0274 g = 0.9726g

7. A 17.52 g sample of element M reacts with 46.52 g a sample of element N, to form the compound M_3N_2. What is the mass of reactants and product ? what can you conclude from the result ?

Solution

Based on law of conservation of mass :

Mass of reactants = mass of product = 17.52 g + 46.52 g = 64.04 g.

Thus, mass is conserved in chemical reaction

8. Show how the five oxides of nitrogen illustrate the law of multiple proportion.

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Solution

Oxide	Mass ratio of N to O	Ratio of O dividing each by the smallest
N_2O	$28:16 \Rightarrow \frac{28}{28}:\frac{16}{28}=1:\frac{4}{7}$	$\frac{4}{7}\div\frac{4}{7}=1$
N O	$14:16 \Rightarrow \frac{14}{14}:\frac{16}{14}=1:\frac{8}{7}$	$\frac{8}{7}\div\frac{4}{7}=2$
N_2O_3	$28:48 \Rightarrow \frac{28}{28}:\frac{48}{28}=1:\frac{12}{7}$	$\frac{12}{7}\div\frac{4}{7}=3$
NO_2	$14:32 \Rightarrow \frac{14}{14}:\frac{32}{14}=1:\frac{16}{7}$	$\frac{16}{7}\div\frac{4}{7}=4$
N_2O_5	$28:80 \Rightarrow \frac{28}{28}:\frac{80}{28}=1:\frac{20}{7}$	$\frac{20}{7}\div\frac{4}{7}=5$

Thus, the ratio of oxygen in the five oxides of nitrogen by fixing the mass of nitrogen is **1 : 2 : 3 : 5.**

1.3 Early Experiments to Characterize the Atom

I. The Discovery of Electron

- The discovery of the electron as the first sub - atomic particle is credited to J . J Thomson.
- He used an evacuated tube connected to a spark coil.
- As the voltage across the tube was increased, a beam become visible. This was referred to as **cathode ray.**
- Thomson found that the beam was deflected by both electric and magnetic fields.
- He concluded that cathode rays are made up of very small, negatively charged particles, called **electrons.**

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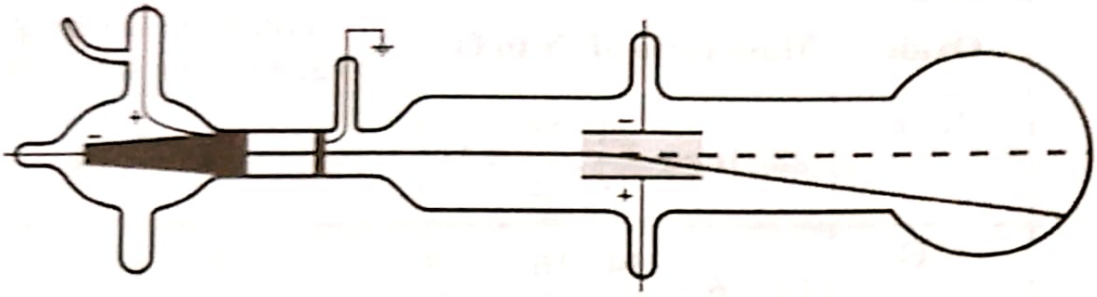

Fig 1.1 Cathode Ray Used by Thomson

Properties of Cathode Rays

- They travel in straight line
- They are deflected by both magnetic and electric fields
- They are deflected towards positive plate (anode).
- They are found in all elements

Note : i The charge to mass ratio (e^-/m) of an electron was determined By J . J Thomson

$$e^-/m = -1.76 \times 10^8\ C/g$$

ii The charge of an electron (e^-) was determined by **Robert Millikan using oil drop experiment**

$$e^- = -1.602 \times 10^{-19}\ C$$

iii **Robert millikan** used Thomson's e^-/m and e^- to calculate mass of an electron (me^-)

$$M_{e^-} = \frac{e^-}{e/m} = \frac{-1.602 \times 10^{-19} C}{-1.76 \times 10^8\ C/g}$$

$$M_{e^-} = 9.1 \times 10^{-28} g = 9.1 \times 10^{-31}\ kg$$

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Illustrative Examples

9. Which of the following is not true about an electron ?

A. It has the smallest mass

B. It has negative charge

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C. It has the highest charge to mass ratio

D. It was discovered by J . Dalton.

Solution

An electron was discovered by J . J Thomson.

Answer : D

10. Which of the following was determined by the Millikan's oil drop experiment ?

A. Charge on the electron
C. Mass of an electron
B. e/m ratio for an electron
D. Velocity of an electron

Answer : A

11. Calculate the mass of 1 mole of electrons.

Solution

1mole of electrons = 6.02×10^{23} electrons

Mass of 1mole of electrons = $9.10 \times 10^{-28} \frac{g}{electron} \times 6.02 \times 10^{23} \frac{electrons}{mol}$

$= 4.48 \times 10^{-4}$ g/mol

12. Determine the charge of 1 mole of electrons.

Solution

Charge of 1mole of elecrons = $-1.602 \times 10^{-19} \frac{C}{electron} \times 6.02 \times 10^{23} \frac{electrons}{mol}$

$= -9.64 \times 10^{4}$ C/mol

13. Why an electron has the highest charge to mass ratio ?

Explanation

Because it has the smallest mass.

2. Radio Activity and Discovery of the Nucleus

Radio activity - is the spontaneous emission of particles or radiation from unstable nuclei. It is also known as **radioactive decay.**

➥ **Elements with unstable nuclei are termed as radioactive elements.**

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Examples

Uranium (U - 238), Carbon (C- 14), chromium (C r - 55), Radium (Ra - 226) etc.

In genera, elements whose atomic number is greater than or equal to 83 are radio active

There are three types of particles and /or rays emitted by radio active elements. These are

1. Alpha particles 2. Beta particles 3. Gamma rays

Table 1.1 Properties of Alpha, Beta and Gamma Rays

Particle or Ray	Symbol	Mass in Gram	Charge	Deflection	Penetration
Alpha particle	${}^{4}_{2}\alpha$ or ${}^{4}_{2}He^{2+}$	6.65×10^{-24}	+2	Deflected toward negative plate	Stopped by paper
Beta particle	${}^{0}_{-1}\beta$ or ${}^{0}_{-1}e^{-}$	9.11×10^{-28}	- 1	Deflacted toward positive plate	Stopped by aluminum
Gamma ray	${}^{0}_{0}\gamma$	0	0	undeflected	Stopped by thick lead sheet.

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Discovery of Nucleus

Ernest Rutherford performed a gold foil experiment that had tremendous implications for atomic structure.

He alloted alpha particles to pass through the gold foil. As a result, three main observations were made.

1. Most of the alpha particles went through the thin gold foil with out any change in path. This indicated that most of the space in an atom was empty
2. A few alpha particles were deflected through small angles. This showed that the nucleus has **positive charge.**
3. Few alpha particles deflected with large angles because of **the heavy nucleus.**

Note : This experimental finding of E. Rutherford discarded the **plum - pudding model of Thomson.**

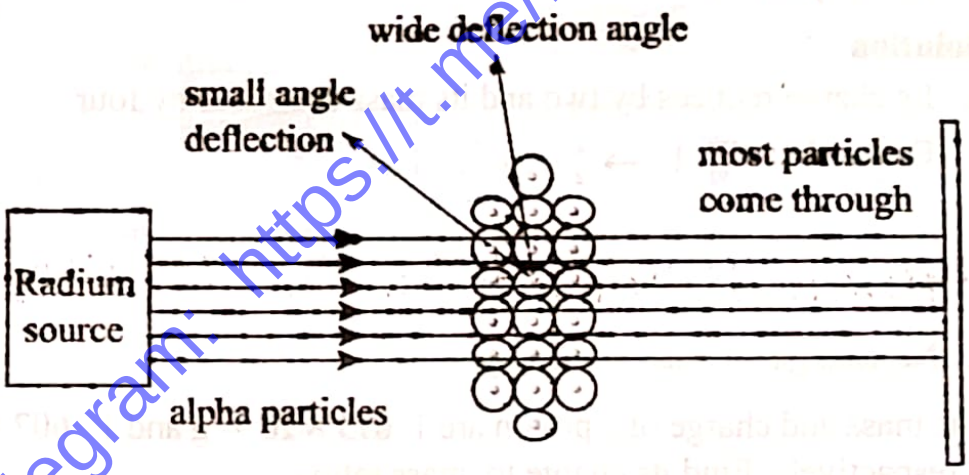

Fig 1.2 Rutheford's Experiment

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E. Rutherford predicted the existence of neutral particles in the nucleus. Later, his student **James chad wick** , discovered this particles, **the neutrons.**

14. Which of the following types of rays combine to form atoms of helium ?

A. Gamma rays
B. Beta rays
C. Alpha rays
D. X - rays

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Solution

Experiments showed that alpha rays were nuclei of helium atoms

Answer : C

15. What did Rutheford's alpha particle experiment show ?
    A. Electrons have a negative charge
    B. A proton is a hydrogen atom with out electron
    C. Electrons circle the nucleus of an atom in orbits.
    D. Most of the mass and all of the positive charge of an atom is found in the tiny nucleus.

Answer : D

16. Who discarded the Thomson model by using alpha scattering experiment ?

Answer: E. Rutheford

17. What happens to the charge and mass of a radio active element if it emits.
    a. An alpha particle
    b. A beta particle
    c. A gamma ray

Solution

a. Its charge reduces by two and its mass decreases by four

Example : $^{228}_{92}U \rightarrow {}^{4}_{2}\alpha + {}^{234}_{90}Th$

b. Its charge increases by one and its mass remains same.

Example : $^{16}_{6}C \rightarrow {}^{0}_{-1}\beta + {}^{14}_{7}N$

c. No change on mass and charge

18. The mass and charge of a proton are 1.673×10^{-24} g and 1.602×10^{-19} C respactively. Find its charge to mass ratio.

Solution

$$\frac{\text{charge}}{\text{mass}} = \frac{e}{m} = \frac{1.602 \times 10^{-19} C}{1.673 \times 10^{-24} g} = 9.58 \times 10^{4}\ C/g$$

19. Sulfur - 35 emits an alpha particle. Find the mass and charge of the new isotope formed. Identify the isotope.

Solution

Let the new isotope be A

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$^{35}_{16}S \longrightarrow {}^{4}_{2}He + {}^{x}_{y}A$ since mass and charge are conserved

$x + 4 = 35 \Rightarrow x = 31$

$y + 2 = 16 \Rightarrow y = 14$

$\Rightarrow {}^{x}_{y}A = {}^{31}_{14}A$, mass = 31, charge = 14 Thus, the isotope is Silicon ($^{31}_{14}Si$).

20. If thorium ($_{90}{}^{234}Th$)emits a beta particle, what will be the charge and mass of the new isotope ?

Solution

$^{234}_{90}Th \rightarrow {}^{0}_{-1}\beta + {}^{a}_{b}X \Rightarrow a + 0 = 234 \Rightarrow a = 234$

$b - 1 = 90 \Rightarrow b = 91$ $\quad {}^{a}_{b}X = {}^{234}_{91}X\ (^{234}_{91}Pa)$

Make up of the Nucleus

An atom has two regions, nucleus and atomic shell.

1. **Nucleus** is the center of an atom.

- It has proton and neutron
- The mass and positive charge of an atom are concentrated in the nucleus.
- Protons and neutrons are collectively known **nucleons**

The only atom without neutron is ordinary hydrogen (protium)

- Neutron is important to reduce the proton - proton repulsion.

2. **Atomic shell / energy level /** is region around the nucleus that contains electron.

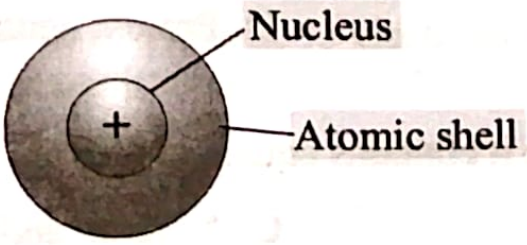

Fig 1.3 Regions of an Atom

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Table 1.2 The Three Fundamental Sub Atomic Particles

Particle	Symbol	Charge in Coloumb	Relative Charge	Mass in Gram	Mass in amu
Electron	e	-1.602×10^{-19}	-1	9.1×10^{-28}	5.486×10^{-4}
Proton	p	$+1.602 \times 10^{-19}$	+1	1.673×10^{-24}	1.007276
Neutron	n	0	0	1.675×10^{-24}	1.008665

21. Which of the following sub atomic particles are found in the nucleus ?

A. Proton and neutron
C. Neutron and electron
B. Proton and electron
D. Proton, neutron and electron

Answer : A

22. Atom is neutral because it contains equal number of

A. Proton and neutron
C. Neutron and electron
B. Proton and electron
D. Proton, neutron and electron

Solution

An atom has equal number of electron and proton that is why it is neutral

Answer : B

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23. Write the importance of neutron.

Solution

- It reduces the strong electrostatic repulsion forces between protons.
- It contributes half of the mass of an atom.

24. Compare the mass of a proton (m_p) with mass of an electron (m_e).

Solution

$$\frac{m_p}{m_e} = \frac{1.673 \times 10^{-24} g}{9.1 \times 10^{-28} g} = 1838 \Rightarrow m_p = 1838\, m_e$$

25. What are the two regions of an atom ?

Answer: Nucleus and atomic shell

Atomic Number and Mass Number

Atomic number (Z) is the number of protons in the nucleus of an atom.

$Z = p$

- for neutral atom: $Z = p = e$
- Atomic number was determined by **Henry Mosely** through the use of X - ray.

Mass number (A) is the sum of atomic number and neutron number.

$A = z + n = p + n$

Mass number is the same as number of **nucleons**

A → mass number

Represertion of an atom: X → symbol

z → atomic number

Example

Aluminum has 13 protons, 13 electrons and 14 neutrons. Represent aluminium in terms of atomic number and mass number.

Solution

$A = Z + n = p + n = 13 + 14 = 27$

$Z = p = 13$

Representation: $^{27}_{13}Al$

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Note : For charged atom (ion) : e = p - charge

Example

$^{24}_{12}Mg^{2+}$: e = 12 − (+ 2)=10

$^{52}_{16}S^{2-}$: e = 16 − (−2) = 18

Isotopes, Isotones, Isobars and Isoelectronics

Isotopes are atoms of the same element with the same atomic numbe protons number, electron number and chemical property but differe neutron number, mass number and physical property.

Example

Carbon has three isotopes : $^{12}_{6}C$, $^{13}_{6}C$, $^{14}_{6}C$.

Isotones are atoms of different elements with the same number of neutrons

Example

$^{23}_{11}Na$ and $^{24}_{12}Mg$

Isobars are atoms of different elements with the same mass number.

Example

$^{14}_{6}C$ and $^{14}_{7}N$

Iso electronics are different species with the same electron number.

Example

$_{10}Ne$, $_{11}Na^{+}$ and $_{9}F^{-}$

Atomic Mass

☑ The actual mass of an atom is very small and determined by using **mass spectrometer**

Example

The actual mass of carbon - 12 is 2×10^{-23} g

☑ **Atomic mass unit (amu)** is the unit of a scale of relative atomic masses of the elements.

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$$1 \text{ amu} = \frac{1}{12} \times \text{mass of C - 12} = \frac{1}{12} \times 2 \times 10^{-23} \text{g}$$

$$1 \text{ amu} = 1.67 \times 10^{-24} \text{ g}$$

$$\text{Atomic mass of an atom} = \frac{\text{Absolute mass of an atom}}{1.67 \times 10^{-24} \text{ g /amu}}$$

Example

The absolute mass of magnesium atom is 4×10^{-23} g. Calculate its atomic mass.

Solution

$$\text{Atomic mass of Mg} = \frac{\text{Absolute mass of Mg}}{1.67 \times 10^{-24} \text{ g / amu}}$$

$$= \frac{4 \times 10^{-23} \text{g}}{1.67 \times 10^{-24} \text{ g /amu}} = 24 \text{ amu}$$

Average or Relative atomic Mass (A V. Mass) is the average mass of an atom in a representative sample atoms of an element.

$$\text{Average mass} = A_1 f_1 + A_2 f_2 + \ldots A_n f_n$$

- $A_1, A_2, \ldots$ An are atomic masses of first, second --- and n^{th} isotopes respectively.
- f_1, f_2, f_n are percentage abundances of $1^{st}, 2^{nd}$ --- n^{th} isotopes respectively.
- $f_1 + f_2 + f_n = 100$.

$$\text{Average atomic mass} = A_1 X_1 + A_2 X_2 + \text{---} A_n X_n$$

- X_1, X_2 --- X_n fraction abundance of $1^{st}, 2^{nd}$, ---- n^{th} isotopes respectively.

$$X_1 + X_2 + \text{------} X_n = 1$$

Example

Copper has two naturally occurring isotopes with the given mass and percentage abundances.

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Isotope	Mass (amu)	Relative Abundance
Cu - 63	62.93	69. 08 %
Cu - 65	64. 93	30. 91 %

Calculate the average atomic mass of copper.

Solution

$A_1 = 62.93,\ f_1 = 69.09\%,\ A_2 = 64.93,\ f_2 = 30.91\%$

$AV.\ mass = A_1 f_1 + A_2 f_2$

$= 62.93 \times \frac{69.09}{100} + 64.93 \times \frac{30.91}{100} = 63.55\,amu$

Illustrative Examples

26. What is the difference between chlorine - 35 and chlorine - 37 ?

A. Chlorine - 37 has two more protons than chlorine - 35

B. Chlorine - 37 has two more neutrons than chlorine - 35

C. Chlorine - 37 has two more electrons than chlorine - 35

D. Chlorine - 37 has two more protons and one more neutron than chlorine - 35

Solution

C1- 35 , p = 17 , n = 35 - 17 = 18

C1 - 37 , p = 17, n = 37 - 17 = 20

C1 - 37 has two more neutrons than C1 - 35

Answer : B

27. The number of electrons, protons and neutrons in $^{31}_{15}P^{3-}$ respectively are:

A. 15 , 15 , 16 **C. 16 , 15 , 18**

B. 18 , 15 , 16 **D. 16 , 15 , 15**

Solution

p = Z = 15 , e = p - charge = 15 - (- 3) = 18

n = A - p = 31 - 15 = 16 . **Answer : B**

28. A mono atomic ion has 20 protons and a + 2 charge .

A. Has 16 protons C. Has 18 neutrons

B. Has the symbol Ar^{2+} D. Is iso electronic with Ar

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Solution

➥ e = p - charge = 20 - (+ 2) = 18

➥ Argon has 18 electrons

Answer : D

29. Which of the following has 45 neutrons, 35 protons and 36 electrons ?

A. $^{80}_{35}Hg$ B. $^{103}_{45}Rh^-$ C. $^{80}_{35}Br^-$ D. $^{71}_{35}Lu^-$

Solution

A = p + n = 35 + 45 = 80

Charge = p - e = 35 - 36 = - 1

An element with 35 protons is bromine (Br) $\Rightarrow$ $^{80}_{35}Br^-$

Answer : C

30. Natural copper is a mixture of only two isotopes one of which has 29 protons, 34 neutrons and a relative abundance of 69. 17 % . The relatively atomic mass is 63.6 amu. How many neutrons must be there in the second isotope ?

A. 29 B. 36 C. 32 D. 35

Solution

$A_1 = p + n_1 = 29 + 34 = 63, f_1 = 69.17\ \%$

$A_2 = p + n_2 = 29 + n_2, f_2 = 100 - f_1 = 30.83\ \%$

$63.6 = \dfrac{63 \times 69.17 + (29 + n_2)\ 30.83}{100}$ solving for n_2 , $n_2 = 36$

Answer : B

31. In which pair of the elements do the nuclei of the atoms contain the same number of neutrons ?

A. $^{7}_{3}Li$ and $^{9}_{4}Be$ C. $^{14}_{7}N$ and $^{16}_{8}O$

B. $^{23}_{11}Na$ and $^{24}_{12}Mg$ D. $^{32}_{16}S$ and $^{35}_{17}Cl$

Answer : B

32. Determine the number of protons, neutrons and electrons in

a. $^{59}_{27}Co^{2+}$ b. $^{79}_{34}Se^{2-}$

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Solution

a. $p = 27$, $e = 27 - (+2) = 25$, $n = 59 - 27 = 32$

b. $p = 34$, $e = 34 - (-2) = 36$, $n = 79 - 34 = 45$

33. Why do isotopes have similar chemical properties ?

Explanation

Because they have the same electron number

34. Lithium has two naturally occurring isotopes, Li - 6 and Li - 7. The average atomic mass of lithium is 6.94 amu. Which isotope is more abundant in nature ?

Solution

Li - 7 is more abundant isotope because the average atomic mass is nearer to it .

35. The element gallium has two stable naturally occurring isotopes, ^{69}Ga and ^{71}Ga . Given that

Isotope	Mass in amu
Ga - 69	68.9
Ga - 71	70. 9

The average atomic mass of gallium is 69.7 amu. Calculate the relative percentage abundance of each isotope.

Solution

$A_1 = 68.9$, $f_1 = 100 - f_2$, $A_2 = 70.9$, $f_2 = 100 - f_1$

Average mass $= A_1 f_1 + A_2 f_2$

$$69.7 = \frac{68.9 \times f_1}{100} + 70.9 \times \frac{(100 - f_1)}{100}$$ solving for f_1 ,

$f_1 = 60\%$ and $f_2 = 40\%$

36. The atomic mass of the two stable isotopes of boron $^{10}_{5}B$ (0.1978) and $^{10}_{5}B$ (0.8022) are 10.0129 amu and 11.0093 amu, respectively. Calculate the relative atomic mass of boron ?

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Solution

$A_1 = 10.0129$, $x_1 = 0.1978$, $A_2 = 11.0093$, $x_2 = 0.8022$

Average mass = $A_1 x_1 + A_2 x_2$

$= 10.0129 \times 0.18 + 8 + 11.0093 \times 0.8022 = 10.81$ amu

37. The absolute mass of sodium atom is 3.81×10^{-23} g. What is its atomic mass ?

Solution

$$\text{Atomic mass of Na} = \frac{\text{Absolute mass of Na}}{1.67 \times 10^{-24} \text{ g/amu}} = \frac{3.81 \times 10^{-23} \text{g}}{1.67 \times 10^{-24} \text{ g/amu}}$$

$= 22.8$ amu

38. The atomic mass of aluminium is 27 amu . What is the absolute mass of aluminium atom ?

Solution

Absolute mass = atomic mass $\times 1.67 \times 10^{-24}$ g / amu

$= 27$ amu $\times 1.67 \times 10^{-24}$ g / amu $= 4.5 \times 10^{-23}$ g

1.4 Electromagnetic Radiation and Atomic Spectra

☑ **Electro Magnetic Radiation (E M R)** is radiation that can be described in terms of oscillating or wave like electric and magnetic fields in space.

☑ It is mainly characterized by wave length (λ), frequency (v) and speed (c).

☑ **Wave length (λ)** is the distance between two adjacent crests or troughs in a particular wave.

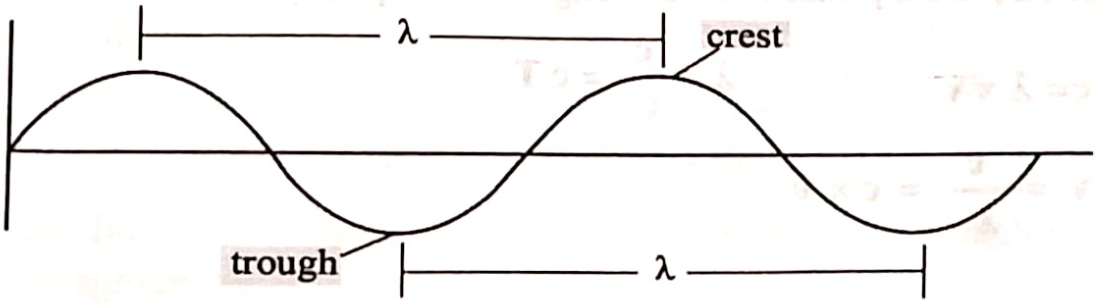

Fig 1.4 Electromagnetic Wave

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Wave number ($\bar{\upsilon}$) is the reciprocal of wave length expressed in centimater inverse (cm^{-1})

$$\bar{\upsilon} = \frac{1}{\lambda} = \lambda^{-1}$$

Example

The wave length of a wave is 100nm. Determine its wave number.

Solution

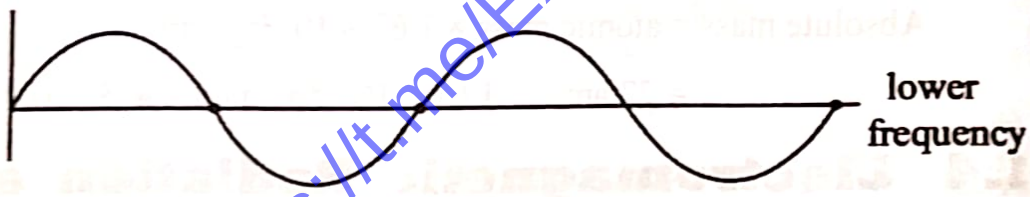
$$\lambda = 100\,nm = 100 \times 10^{-9}\ m = 100 \times 10^{-9} \times 100\,cm = 1 \times 10^{-5}\ cm$$

$$\bar{\upsilon} = \frac{1}{\lambda} = \frac{1}{1 \times 10^{-5}\,cm} = 1 \times 10^{5}\ cm^{-1}$$

Frequency (ν) is the number of waves which can pass through a point in one second.

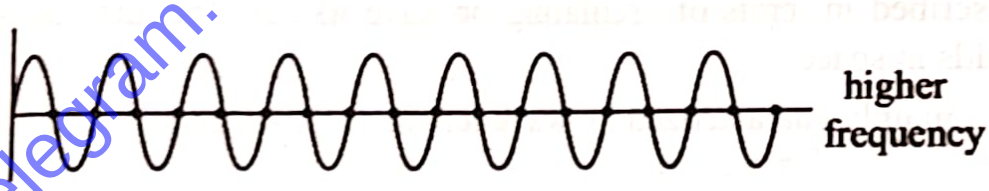
☑ The SI unit frequency is second inverse (s^{-1}) or Hertz (Hz)

☑ Frequency is the reciprocal of period (T), $\nu = \frac{1}{T}$



ν = 2 cycles per second = 2Hz

one cycle = 1 crest + 1 trough



ν = 9 cycles per second = 9Hz

Speed (c) is the product of wavelength and frequency .

$$c = \lambda \times \nu \qquad \lambda = \frac{c}{\nu} = c\,T$$

$$\nu = \frac{c}{\lambda} = c \times \bar{\upsilon}$$

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All electromagnetic radiations travel through vacuum with the same speed, $c = 3 \times 10^{8}\ m/s = 3 \times 10^{10}\ cm/s$

Example

The frequency of a radio wave is 100 MHz, find its wave length.

Solution

$$v = 100\,MHZ = 100 \times 10^{6}\ H_z = 1 \times 10^{8}\ Hz$$

$$\lambda = \frac{c}{v} = \frac{3 \times 10^{8}\ m/s}{1 \times 10^{8}\ s^{-1}} = 3m$$

Electro Magnetic Spectrum

It is region that contains electromagnetic radiations in order of increasing wave lengths or decreasing frequencies and enegies

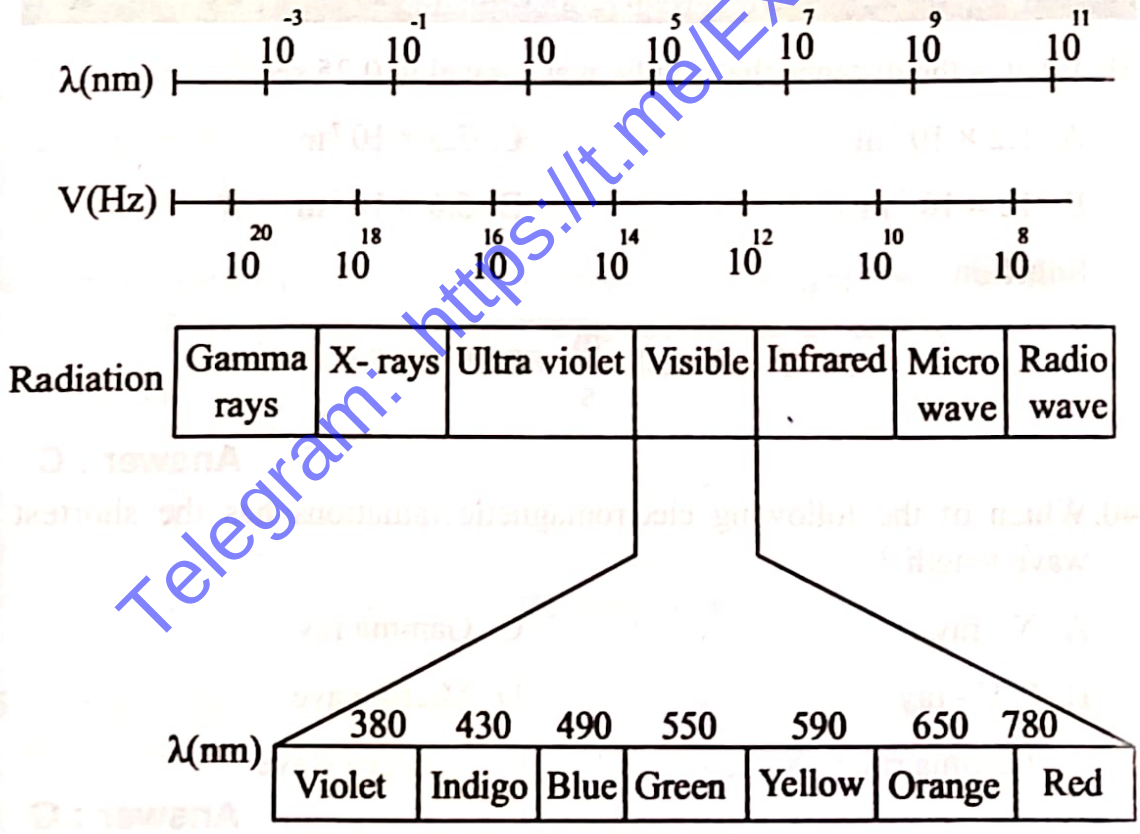

Fig 1.5 Electromagnetic Spectrum

Amplitude (A) is the height of the crest or the depth of the trough of the electromagnetic wave

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It is related to the intensity of radiation in the visible region of the electromagnetic spectrum. The higher the amplitude, the more intense is the light.

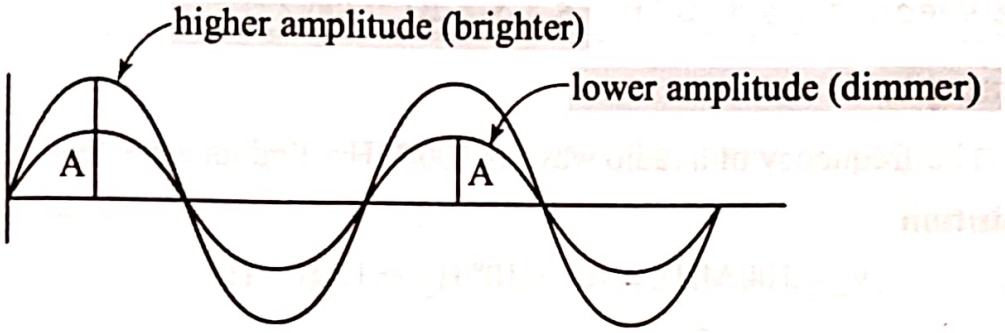

Fig 1.6 Different Amplitudes of a Wave

According to, Max Well theory, electro magnetic wave has two components, electric and magnetic field components. These Two components have the same wave length, frequency and speed, but they travel in mutually perpendical planes.

Illustrative Examples

39. What is the distance that a radio wave travel in 0.25 sec ?

A. 1.2×10^7 m
C. 7.5×10^7 m
B. 12×10^7 m
D. 5.6×10^7 m

Solution

$$\lambda = \frac{c}{v} = c \times T = 3 \times 10^8 \ \frac{m}{s} \times 0.25 \ s = 7.5 \times 10^7 \ m$$

Answer : C

40. Which of the following electromagnetic radiations has the shortest wave length ?

A. X - ray
C. Gamma ray
B. U V - ray
D. Micro wave

λ : Gamma ray < X - ray < U V - ray < micro wave

Answer : C

41. Which of the following are not electro magnetic waves ?

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A. Infrared waves
C. Radio waves
B. Gamma wave
D. Sound waves

Answer : D

42. What is the frequency of a radiation with a wave length of 9×10^{-9} nm ?

Solution

$$\lambda = 1 \times 10^{-9}\ \text{nm} = 1 \times 10^{-18}\text{m},\ v = \frac{c}{\lambda} = \frac{3 \times 10^{8}\ \text{m/s}}{1 \times 10^{-18}\ \text{m}} = 3 \times 10^{26}\ \text{s}^{-1}$$

43. Some diamonds appear yellow because they contain nitrogeneous compounds that absorb purple light of frequency 7×10^{14} Hz. Calculate the wave length in

a. nm b. pm c. A° d. μm

Solution

a. $1\text{m} = 1 \times 10^{9}$ nm $\lambda = \frac{c}{v} = \frac{3 \times 10^{8}\ \text{m/s}}{7 \times 10^{14}\ \text{s}^{-1}} = 4.29 \times 10^{-7}\ \text{m} = 429\ \text{nm}$

b. $1\text{m} = 1 \times 10^{12}\ \text{pm} \Rightarrow 4.29 \times 10^{-7}\ \text{m} = 4.29 \times 10^{5}\ \text{pm}$

c. $1\text{m} = 1 \times 10^{10}\ \text{A}^0 \Rightarrow 4.29 \times 10^{-7}\ \text{m} = 4.29 \times 10^{3}\ \text{A}^0$

d. $1\text{m} = 1 \times 10^{6}\ \mu\text{m} \Rightarrow 4.29 \times 10^{-7}\ \text{m} = 4.29 \times 10^{-1}\ \mu\text{m}$

44. The frequency of the green light from a traffic signal is centered of 5.75×10^{14} Hz. Calculate the wave number of the green light.

Solution

$$\lambda = \frac{c}{\lambda} = \frac{3 \times 10^{10}\ \text{cm/s}}{5.75 \times 10^{14}\ \text{s}^{-1}} = 5.22 \times 10^{-5}\ \text{cm}$$

$$\bar{\upsilon} = \frac{1}{\lambda} = \frac{1}{5.22 \times 10^{-5}\ \text{cm}} = 1.92 \times 10^{4}\ \text{cm}^{-1}$$

45. Explain the difference and similarity between electro magnetic radiations.

Solution

Similarity : All electromagnetic radiations travel through vacuum with the same speed.

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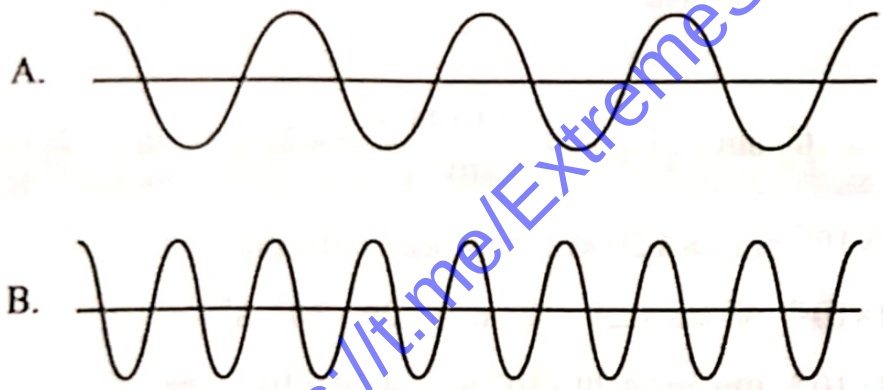
Difference : Different electro magnetic radiations have different wave length, frequency and energy.

46. Give some examples of electro magnetic radiations.

Solution

- The light from the sun.
- The energy used to cook food in micro wave oven.
- The X - ray used by dentists
- The radiant heat from a fire place.

47. Compare the wavelength and frequency of the following waves.



Solution

Wave length : A > B

Frequency : B > A

The Quantum Theory and Photon

☑ Classical physics assumed that radiant energy was **continuous** ; that is it could be emitted or absorbed in any amount. This idea was disproved by **Max Planck.**

☑ Planck proposed that radiant energy could only be emitted or absorbed in **discrete quantities**, like small **packages or bundles.**

☑ Planck gave **quantum or photon** to the smallest quantity of energy that can be emitted or absorbed in the form of electro magnetic radiation.

$$E_{photon} = h\nu = \frac{hc}{\lambda} = hc\,\bar{\upsilon}$$

E_{photon} = Energy of photon

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h = Planck's constant = 6.626×10^{-34} J . s

According to Planck's quantum theory, energy is always emitted or absorbed in an integral multiple of h v . For example, h v , 2h v , 3h v, - - - Quantization

Note : Energy is directly proportional frequency but inversely proportional to wave length: $E \propto \frac{v}{\lambda}$

Example

Calculate the energy of a photon with wave length of 5×10^{-2} nm.

Solution

$\lambda = 5 \times 10^{-2}\,nm = 5 \times 10^{-11}$ m, $c = 3 \times 10^{8}$ m/s

$h = 6.626 \times 10^{-34}$ J.s, E = ?

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\,J.s \times 3 \times 10^{8}\,m/s}{5 \times 10^{-11}\,m} = 3.98 \times 10^{-15}\,J$$

Photo Electric Effect

- ☑ It is a phenomenon in which electrons are ejected from the surface of a metal exposed to light of at least a certain minimum frequency. This frequency is known as **threshold frequency** (v_o)
- ☑ The minimum amount of energy required to eject electrons is known as **work function** (ϕ).

$$\boxed{\phi = h v_o = \frac{hc}{\lambda_{max}}}, \quad \lambda_{max} - \text{maximum wave length}$$

- ☑ If there is excess energy of light , it will be converted into kinetic energy (K E) .

$$\boxed{E_{photon} = hv = \phi + KE = hv_o + KE}, \quad \boxed{KE = hv - hv_o = h(v - v_o)}$$

$$KE = \frac{1}{2} mV^2 = h(v - v_o) = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_{max}}\right)$$

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m - mass of an electron ejected

V - velocity of an ejected electron.

$$V = \sqrt{\frac{2h\,(v - v_o)}{m}}$$

☑ Electro magnetic radiations exhibit wave and particle properties.

$E = mc^2$, Einstein's famous equation

$E = h\nu = \frac{hc}{\lambda}$, Max Planck's equation

☑ Relating these equations result :

$$mc^2 = \frac{hc}{\lambda} \Rightarrow \boxed{\lambda = \frac{h}{mc}},$$

This equation is used if the wave nature dominates.

☑ For a particle like electron with velocity, V the equation becomes :

$\boxed{\lambda = \frac{h}{mV}}$, This equation is known as **de Broglie's equation**

Illustrative Example

48. What is the magnitude of a quantum energy and the frequency for object whose wave length is 0.6×10^{-6} m ?

Solution

$$E = h\nu = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\,J.s \times 3 \times 10^{8}\,m/s}{0.6 \times 10^{-6}\,m}$$

$$E = 3.31 \times 10^{-19}\,J,\ \nu = \frac{c}{\lambda} = \frac{3 \times 10^{8}\,m/s}{0.6 \times 10^{-6}\,m} = 5 \times 10^{14}\,Hz$$

49. The energy required to remove an electron from an iron atom amou to 7.21×10^{-19} J. What is the maximum wave length of light that c perform this ?

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Solution

$$\phi = h\,v_o = \frac{hc}{\lambda_{max}} \qquad \lambda_{max} = \frac{hc}{\phi} = \frac{6.626\times10^{-34}\ J.s \times 3\times10^{8}}{7.21\times10^{-19}\ J}\ m/s$$

$$\lambda_{max} = 2.76\times10^{-7}\ m = 276\ nm$$

50. Explain the difference between Planck's theory and classical perceptions of the nature of energy.

Explanation

Planck's theory suggested that energy is not continuous, rather expressed in discrete packets. However, prior this, it was believed that energy is continuous without limit.

51. The threshold frequency for metallic potassium is 5.46×10^{14} Hz. **Calculate**

a. Work function
b. Kinetic energy
c. Velocity of the emitted electron, if the wave length of shining on the potassium surface is 350 nm and mass of an electron is 9.1×10^{-31} kg.

Solution

a. $\phi = h\,v_o = 6.626\times10^{-34}\ J.s \times 5.46\times10^{14}\ s^{-1} = 3.62\times10^{-19}\ J$

b. $v = \frac{c}{\lambda} = \frac{3\times10^{8}\ m/s}{3.5\times10^{-7}\ m} = 8.57\times10^{14}$ Hz. $KE = h\,(v - v_o)$

$= 6.626\times10^{-34}\ J.s\ (8.57\times10^{14} - 5.46\times10^{14})s^{-1}$, $KE = 2.06\times10^{-19}\ J$

c. $V = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2\times 2.06\times10^{19}\ kg.m^2/s^2}{9.1\times10^{-31}\ kg}} = 6.7\times10^{5}\ m/s$

52. What is the wave length of a proton having a velocity of 1.38×10^{7} cm/s ?

Solution

Mass of a proton = 1.673×10^{-27} kg

$V = 1.38 \times 10^{7}$ cm/ s = 1.38×10^{5} m /s

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$$\lambda = \frac{h}{mV} = \frac{6.626 \times 10^{-34}\ \text{kg. m}^2/\text{s}}{1.673 \times 10^{27}\ \text{kg} \times 1.38 \times 10^{5}\ \text{m/s}} = 2.88 \times 10^{-12}\text{m}$$

$\lambda = 2.88$ pm

53. Calculate the wave length associated with an electron travelling at 40 % of the velocity of light.

Solution

$$\text{V of electron} = 40\% \times c = \frac{40}{100} \times 3 \times 10^{8}\text{m/s} = 1.2 \times 10^{8}\text{m/s}$$

$$\lambda = \frac{h}{mV} = \frac{6.626 \times 10^{-34}\text{kg. m}^2/\text{s}}{9.1 \times 10^{31}\text{kg} \times 1.2 \times 10^{8}\text{m/s}} = 6.07 \times 10^{-12}\text{m} = 6.07\,\text{pm}$$

54. A microwave oven operates with a radiation having a wave length of 1.20 cm. Calculate the energy of this radiation in

a. J / photon b. J / mol c. kJ / mol

Solution

$$\text{a. } E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J.s} \times 3 \times 10^{10}\ \text{cm/s}}{1.20\,\text{cm}} = 1.66 \times 10^{-23}\text{J / photon}$$

$$\text{b. } E = 1.66 \times 10^{-23}\ \text{J/photon} \times 6.02 \times 10^{23}\ \text{photon/mol} = 9.99\ \text{J/mol}$$

$$\text{c. } E = 9.99 \times 10^{-3}\ \text{kJ/mol}$$

55. Calculate the energy of a photon of light with a wave length of 600 nm. Determine the number of photons of energy 1 J.

Solution

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J.s} \times 3 \times 10^{8}\ \text{m/s}}{6 \times 10^{-7}\ \text{m}} = 3.313 \times 10^{-19}\ \text{J/photon}$$

1 photon $= 3.313 \times 10^{-19}$ J

? = 1 J

$$\text{Number of photons} = \frac{1\ \text{J} \times 1\ \text{photon}}{3.313 \times 10^{-19}\ \text{J}} = 3.02 \times 10^{18}\ \text{photons.}$$

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56. Explain the dual nature of light using diagram.

Explanation

Light has wave nature and particle nature

Light as a stream of photons

Light as a wave phenomenon

Atomic or Line Spectra

Unlike those of the sun or a white - hot iron bar, the emission spectra of atoms in the gas phase do not show a **continuous** spread of wave lengths from red to violet ; rather, the atoms produce bright lines in distinct parts of the visible spectrum. These line spectra are the emission of light of only at specific wave lengths.

- Every element has a unique emission spectrum, so the characterstic lines in atomic spectra can be used in chemical analysis **to identify** elements.
- Atomic spectra are also important to study composition of sun and stars.
- Atomic spectra are fine lines separated by colorless or blank spaces.
- The intensity of the lines in an atomic spectrum vary with temperature and pressure affects their width.
- The spectrum produced by sun light is called **continuous spectrum** because one color merges into the next without any gaps or missing colors.

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Illustrative Examples

57. Which of the following produces atomic or line spectra ?

A. Sun light
B. Red - hot iron bar
C. Heated solid
D. Excited atoms

Solution

When an atom is heated , it excites and wants to return to the ground state by releasing its energy. When the released energy is passed

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through a prism,line spectra are produced.

Answer : D

58. Which of following is true about atomic spectra ?

A. Atomic spectra are also called dot spectra

B. Atomic spectra are a series of fine lines of individual colors separated by colored spaces.

C. There are fewer energy states than electrons.

D. The intensity of the lines in an atomic spectrum very with temperature.

Answer : D

59. Write the applications of atomic spectra.

Solution

- For identifying elements
- To study composition of stars and sun
- To estimate the temperature of the surface of stars and planets.
- To estimate the gas pressure and magnetic field strength on and around planets and stars.

60. Describe how rainbow is formed in the sky.

Solution

After it rains, the sky is filled with moisture in the form of water droplets, when the sun comes out, the white light it produces reflects off the water droplets, the water droplets act like a prism and separate the white light in to spectrum colors. All of the colored light or light frequencies bend in different ways to form a rainbow in the sky.

1.5 The Bohr Model of the Hydrogen Atom

Bohr applied to the hydrogen atom the concept that the electron can exist only in certain energy levels with out an energy change.

Bohr used the quantum theory of Max Planck and photo electric theory of Einstein to formulate his model of the hydrogen atom.

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1. The electron in a hydrogen atom travels around the nucleus in a circular path called **orbits.**
2. The orbits are quantized
3. The only orbits that are allowed are those for which the angular momentum of an electron is an integral multiple of $h / 2\pi$
4. The energy of an electron is directly proportional to the value of the principle quantum number n.

$$E_n = \frac{-R_H.Z^2}{n^2} \qquad R_H = \text{Rydberg constant} = 2.18 \times 10^{-18}\ J$$

Z = atomic number (for H = 1), n = 1,2,3,---

$$\text{For H - atom: } E_n = \frac{-2.18 \times 10^{-18}\ J}{n^2}$$

Note : The negative sign (-) implies as n increases, energy also increases.

5. As long as an electron stays in a given orbit, it neither gains nor loses energy.
6. A definite amount of energy is absorbed when an electron, jumps from lower to higher orbit and emits when it falls from higher to lower orbit.

$$\Delta E = E_f - E_i = h\nu \quad E_f = \frac{-2.18 \times 10^{-18}\ J}{n_f^2},\ E_i = \frac{-2.18 \times 10^{-18}\ J}{n_i^2}$$

$$\Rightarrow \Delta E = \left(\frac{-2.18 \times 10^{-18}\ J}{n_f^2}\right) - \left(\frac{-2.18 \times 10^{-18}\ J}{n_i^2}\right)$$

$$\Delta E = 2.18 \times 10^{-18}\ J \left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

Note : The radius (r) of an orbit can be determined by the formula :

$$r = \frac{n^2 a_0}{Z} = \frac{0.053\,nm.n^2}{Z}, \text{for H, } Z = 1$$

$$r = 0.053\,nm.n^2 = 0.53\ A^0 n^2$$

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Example: Calculate the radius of the first and second orbits.

Solution

First orbit (n = 1) : $r = 0.053\ nm \times 1^2 = 0.053\ nm$

Second orbit (n = 2) : $r = 0.053\ nm \times 2^2 = 0.212nm$

Bohr's model can be represented by the diagram below.

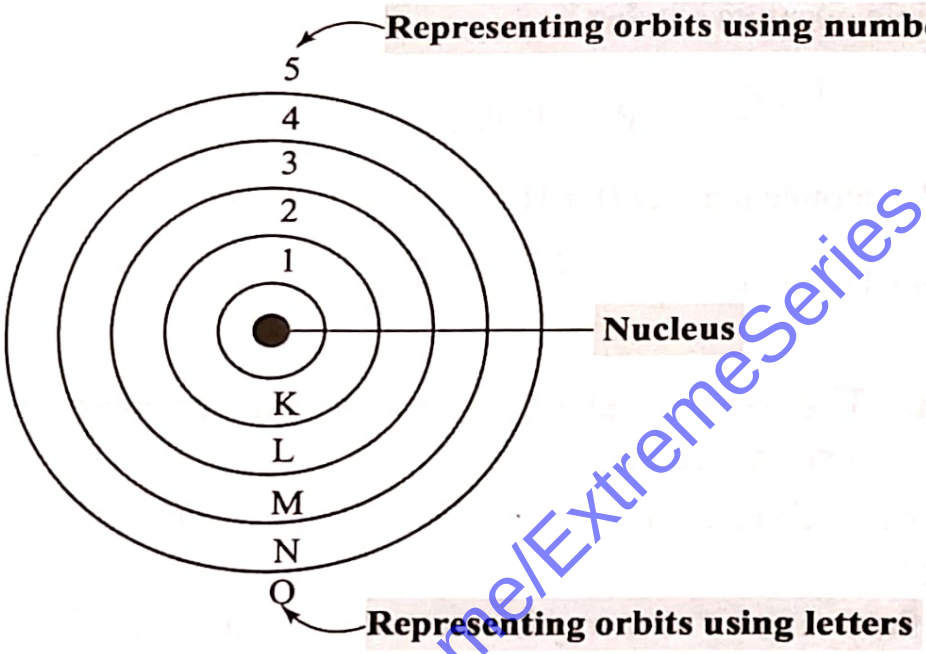

Fig 1.7 Bohr's Electron Orbits of the Hydrogen Atom

Note : The emission spectrum of hydrogen covers a wide range of wave lengths from the infrared to the ultraviolet. They are named after their discoverers .

Table 1.3 The Various Series in Atomic Hydrogen Emission Spectrum.

Series	n_f	n_i	Spectrum region
Lyman	1	2, 3, 4, ---- ∞	Ultra vielet
Balmer	2	3, 4, 5, --- ∞	Visible and ultra violet
Paschen	3	4, 5, 6, ---- ∞	Infrared
Brackett	4	5, 6, 7, ---- ∞	Infrared
Pfund	5	6, 7, 8, ---- ∞	Infrared

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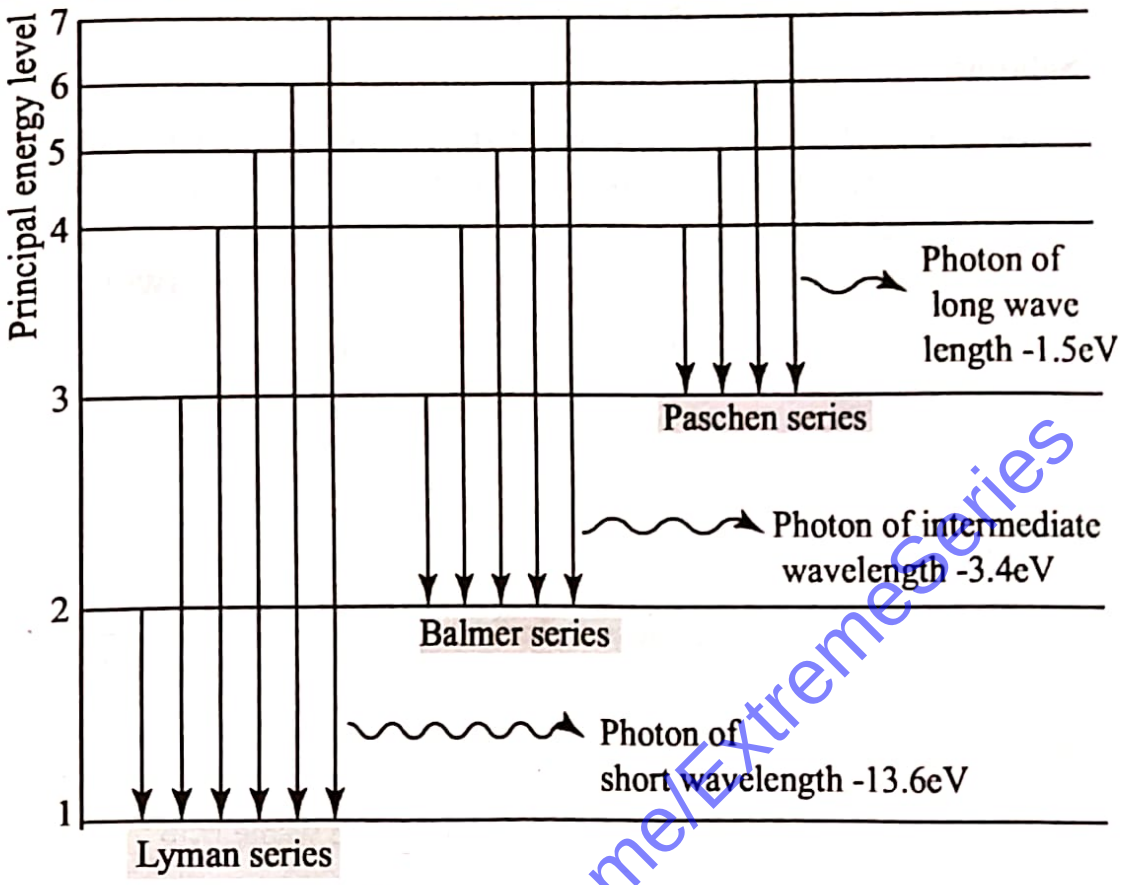

e V = electron Volt

one eV = one ev = 1.6×10^{-19} J

Limitations of the Bohr Theory

- ☑ He failed to explain the atomic spectra of species that contain more than one electrons.
- ☑ He failed to explain the further splitting of spectral lines on the application of magnetic field in hydrogen atom.

Illustrative Examples

61. What new concept did Bohr adapt and use to formulate his model of the hydrogen atom ?

A. Electromagnetic theory developed by Max Well

B. The quantum concept developed by Planck

C. The photo electric effect developed by Thomson

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D. Neutron theory developed by Chad wick.

Solution

Bohr used the quantum concept developed by Planck and the photoelectric effect developed by Einstein to formulate his model of the hydrogen atom.

Answer : B

62. The ratio of the radii of first orbit of H, He^+ and Li^{2+} is

A. 1 : 2 : 3 C. 1 : 4 : 9

B. 6 : 3 : 2 D. 9 : 4 : 1

Solution

$r = \frac{a_0 . n^2}{Z}$, for H: Z = 1, $He^+ = 2$, $Li^{2+} = 3$ $r_H = a_0 . n^2$,

$r_{He^+} = \frac{a_0 . n^2}{2}$, $r_{Li^{2+}} = \frac{a_0 . n^2}{3}$ $r_H : r_{He^+} : r_{Li^{2+}} =$

$a_0 . n^2 : \frac{a_0 . n^2}{2} : \frac{a_0 n^2}{3} = 1 : \frac{1}{2} : \frac{1}{3}$, multiplying the whole by 6 = 6 : 3 : 2

Answer : B

63. According to Bohr's atomic model

A. Electron in an atom of hydrogen atom can have only certain values of angular momentum.

B. Electron has particle as well as wave character

C. Atomic spectrum of an element should contain only five lines.

D. An electron emits energy when it jumps from lower to higher orbit.

Answer : A

64. The wave length of a spectral line for an electron transition in an atom is inversely proportional to

A. Velocity of electron

B. The difference of energy between energy levels involved in the transition.

C. Number of electrons undergoing transition

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D. The value of the Planck's constant.

Solution

$$\Delta E = E_f - E_i = h\nu = \frac{hc}{\lambda} \Rightarrow \lambda \propto \frac{1}{\Delta E}$$

Answer : B

65. Bohr's Model could explain successfully

A. The spectrum of helium

B. The spectrum of species containing only one electron

C. The spectrum of multi electrons atoms

D. The spectrum of the hydrogen molecule

Answer : B

66. Which of the following electron transitions requires the smallest energy to be absorbed by the hydrogen atom ?

A. From $n = 1$ to $n = 2$

B. From $n = 2$ to $n = 3$

C. From $n = 3$ to $n = 4$

D. From $n = 4$ to $n = 5$

Solution

The further the transition from the nucleus , the smaller is the energy absorbed .

Answer : D

67. The wave lengths of which series lie toward the ultra violet region ?

A. Lyman series

B. Balmer series

C. Paschen series

D. Brackett series

Answer : A

68. Calculate the frequency and wave length of a photon emitted during an electron transition from $n = 5$ state to $n = 2$ state in the hydrogen atom.

Solution

$$\Delta E = 2.18 \times 10^{-18}\ J \left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right) = 2.18 \times 10^{-18}\ J \left(\frac{1}{5^2} - \frac{1}{2^2}\right)$$

$$= -4.578 \times 10^{-19}\ J$$

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The negative sign implies emision of energy, omit it to solve frequency and wave length.

$$\Delta E = h\,v \qquad \Rightarrow v = \frac{\Delta E}{h} = \frac{4.578 \times 10^{-19} J}{6.626 \times 10^{-34} J.s} = 6.91 \times 10^{14}\, s^{-1}$$

$$\lambda = \frac{c}{v} = \frac{3 \times 10^{8}\, m/s}{691 \times 10^{14}\, s^{-1}} = 4.34 \times 10^{-7}\, m = 434\, nm$$

69. The first line of the Balmer series occurs at a wave length of 656 .3 nm. What is the energy difference between the two energy levels involved in the emission that result in this spectral line ?

Solution

Method I

$$\Delta E = h\,v = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} J.s \times 3 \times 10^{8}\, m/s}{6.563 \times 10^{-7}\, m} = 3.03 \times 10^{-19} J$$

Method II

In Balmer series, $n_f = 2$ and for the 1st line, $n_i = 1+2 = 3$

$$\Delta E = 2.18 \times 10^{18} J \left(\frac{1}{n_i^{\,2}} - \frac{1}{n_f^{\,2}} \right) = 2.18 \times 10^{-18} J \left(\frac{1}{3^2} - \frac{1}{2^2} \right)$$

$$\Delta E = 3.03 \times 10^{-19} J$$

70. An electron in an orbit of principal quantum number n_i in a hydrogen atom makes a transition to an orbit of principal quantum number 2. The photon emitted has a wave length of 434 nm. Calculate n_i.

Solution

$$\Delta E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\, J.s \times 3 \times 10^{8}\, m/s}{4.34 \times 10^{-7}\, m} = 4.58 \times 10^{-19} J$$

$$\Delta E = \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \times 2.18 \times 10^{-18} J \text{ solving for } n_i,\ n_i = 5$$

71. How much energy,in kilo joules per mole is released when an electron makes a transition from n = 4 to n = 3 in a hydrogen atom ? Is this energy sufficient to break the H - H bond (436 kJ /mol is needed to break this bonds) ?

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Solution

$$\Delta E = 2.18 \times 10^{-18}\ J \left(\frac{1}{4^2} - \frac{1}{3^2} \right) = -1.06 \times 10^{-19}\ J/\text{photon}$$

$$\Delta E = 1.06 \times 10^{-19}\ \frac{J}{\text{photon}} \times 6.02\ 10^{23}\ \frac{\text{photon}}{\text{mol}}$$

$$= 6.38 \times 10^{4}\ J/\text{mol} = 63.8\ kJ / mol$$

This energy is not sufficient to break the H-H bond.

72. Calculate the energy required to remove an electron from the lowest level of the hydrogen atom to produce hydrogen ion.

Solution

$$n_i = 1 \text{ and } n_f = \infty,\ \Delta E = 2.18 \times 10^{-18}\ J \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$= 2.18 \times 10^{-18} J \left(\frac{1}{12} - \frac{1}{\infty^2} \right) = 2.18 \times 10^{-18}\ J$$

73. Calculate the radius of the forth orbit of the hydrogen atom.

Solution

$$r = 0.53\ A^0 \times n^2 = 0.53 \times (4)^2\ A^0 = 8.48\ A^0$$

1.6 The Quantum Mechanical Model (Wave Mechanical Model or Quantum Mechanics)

Macroscopic objects do exhibit a wave like motion but the wave length is too small for humans to perceive.

A wave is associated with any object, even a moving elephant but is dominated by the particle nature.

Electron is a microscopic particle which moves with a very high speed. As a result, it is very difficult to trace the exact location and speed of an electron at the same time around the nucleus.

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Heisenberg uncertainity principle states that it is impossible to know simultaneously both the momentum (mass times velocity) and the position of electron with certainty.

Mathematically

$$\Delta X \cdot \Delta P \geq \frac{h}{4\pi}$$

ΔX – change in position

ΔP – change in momentum

h – Planck's constant.

The concept of electron density gives the probability that an electron will be found in a particular region of an atom. This region is called **orbital**.

Orbital is a region in space around the nucleus where there is high probability of getting an electron.

Quantum Numbers

In quantum mechanics, three quantum numbers are required to describe the distribution of electrons in atoms. They are called the principal quantum number, angular quantum number and magnetic quantum number.

A fourth quantum number, the spin quantum number describes the behavior of a specific electron and completes the description.

1. The principal quantum number (n)

➥ It has positive integral values. n = 1, 2, 3, -----

Uses of n

i. To identify main energy levels or atomic shells

Example

n = 1, First energy level (K - shell)

n = 2, Second energy level (L- shell)

n = 3, Third energy level (M - shell)

n = 4, Fourth energy level (N - shell)

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ii. To determine energy of orbitals

➥ As the value of n increases, the energy of orbitals also increases.

iii. To determine size of orbitals.

➥ As the value of n increases, size of orbitals also increases.

iv. To find number of orbitals in main energy levels.

$$\text{Number of orbitals} = n^2$$

Example

Calculate the number of orbitals in the third energy level (n = 3)

Solution

Number of orbitals = $n^2 = 3^2 = 9$

Note : The maximum capacity of an orbital is 2 electrons

v. To calculate the maximum number of electrons in main energy levels.

$$\text{Number of electrons} = 2n^2$$

Example

Determine the maximum number of electrons in the fourth atomic shell (n = 4).

Solution

Number of electrons = $2n^2 = 2 \times 4^4 = 32$

vi. To determine the number of sub - levels in main energy levels.

$$\text{Number of sub - levels} = n$$

Example

Find the number of sub - shells in the fourth atomic shall (n = 4)

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Solution

Number of sub - shells = n = 4 (s, p, d, f)

vii. To determine the average distance of the electron from the nucleus in a particular orbital.

➥ The larger the values of n, the greater is the average distance of an electron in the orbital from the nucleus.

2. Angular momentum quantum number (ℓ)

➥ It is also called **azimutal quantum number or orbital shape quantum number or subsidiary quantum number.**

☑ It has values from **0** to **n - 1**

☑ Its minimum value is **0** and its maximum value is **n - 1.**

Example

➥ n = 1, ℓ = 0

➥ n = 2, ℓ = 0, 1

➥ n = 3, ℓ = 0, 1, 2

➥ n = 4, ℓ = 0, 1, 2, 3

➥ n = 5, ℓ = 0, 1, 2, 3, 4 e t c

Uses of ℓ

i. To identify sub - shells

Value of ℓ	0	1	2	3	4
Sub - shell	s	p	d	f	g

ii. To find number of orbitals in sub - shells

$$\text{Number of orbitals} = 2\ell + 1$$

Example

Find number of orbitals in s, p and d sub - shells.

Solution

➥ s - sub - shell (ℓ = 0) , Number of orbital = 2 × 0 + 1 = 1

➥ p - sub shell (ℓ = 1) , Number of orbitals = 2 × 1 + 1 = 3

➥ d - Sub shell (ℓ = 2), Number of orbitals = 2 × 1 + 1 = 5

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iii. To determine maximum number of electrons in sub shells

Number of orbitals = $4\ell + 2$

Example

Calculate the maximum number of electrons in f - sub shell

Solution

f - sub - shell ($\ell = 3$)

Number of electrons = $4\ell + 2 = 4 \times 3 + 2 = 14$

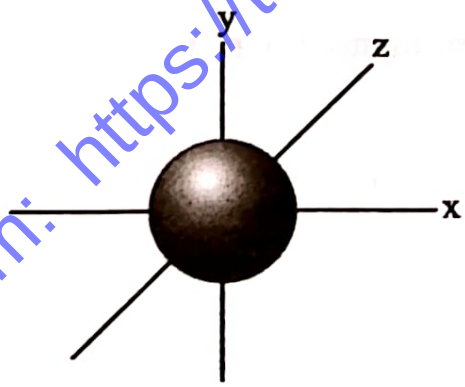
iv. Energy of sub - shells with same principal quantum number.

Example

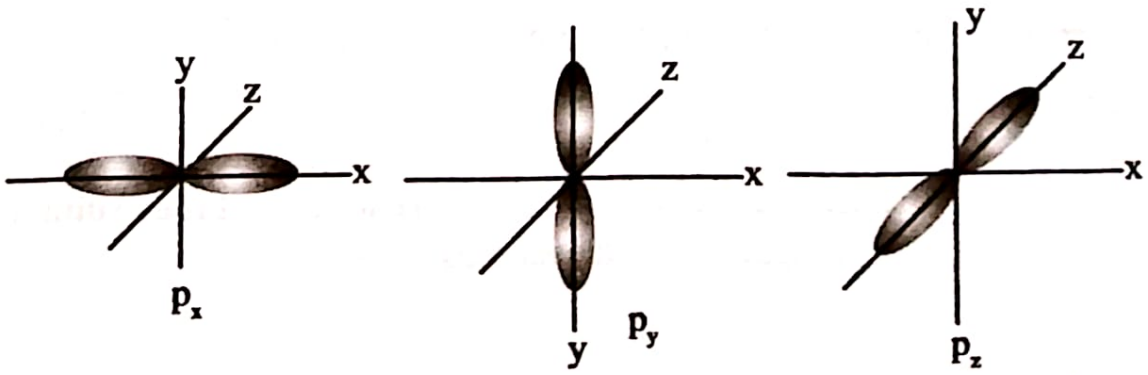
- Energy of 2s < 2p
- Energy of 3s < 3p < 3d

v. To describe shape of atomic orbitals

- **s - Orbital** has spherical shape



- **p - Orbitals have dump - bell shapes.**



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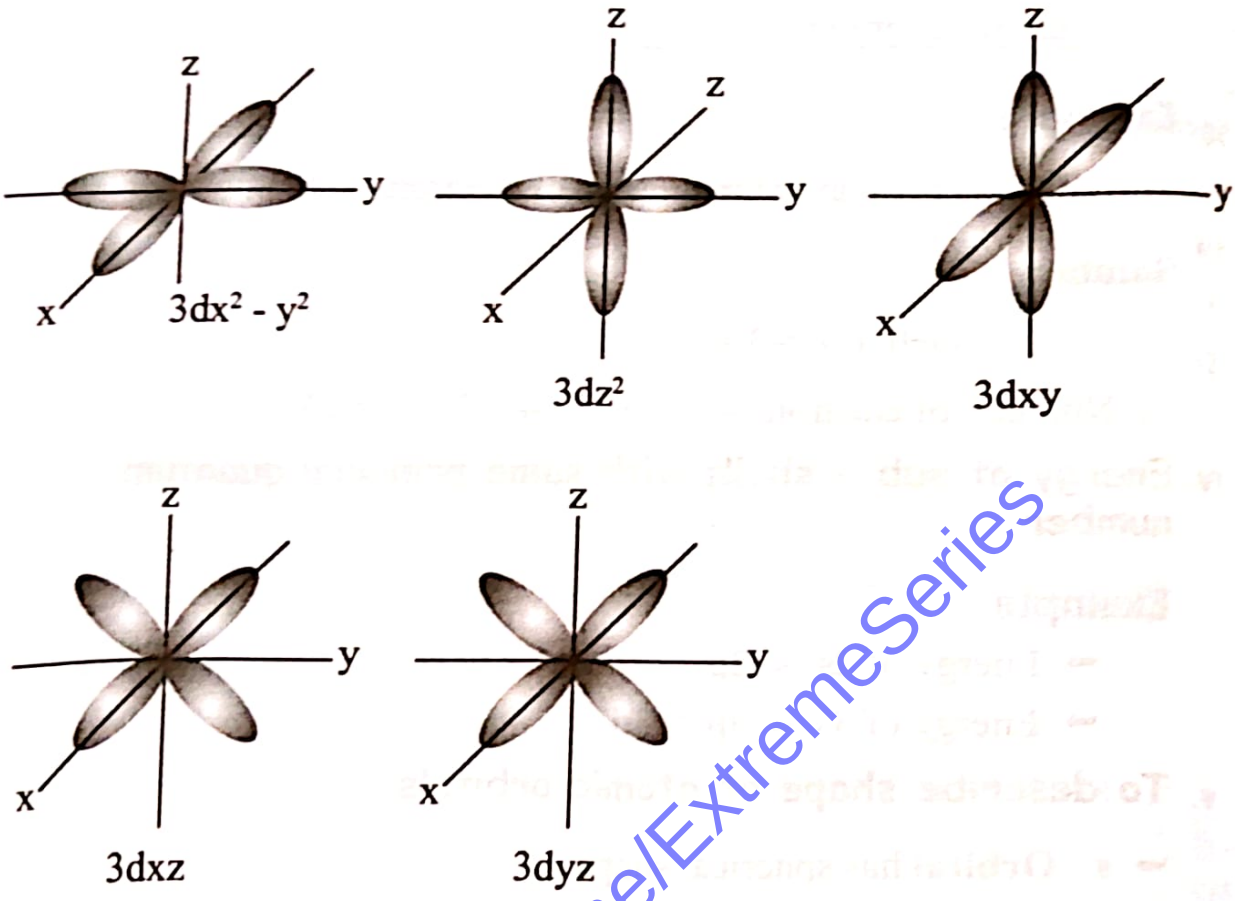
➥ d - Orbitals



3. Magnetic quantum number (m_ℓ)

☑ It is also called orbital orientation quantum number

☑ It is an integer having values from - ℓ to + ℓ

Example

➥ $\ell = 0$, $m_\ell = 0$

➥ $\ell = 1$, $m_\ell = -1, 0, 1$ or $0, \pm 1$

➥ $\ell = 2$, $m_\ell = -2, -1, 0, 1, 2$ or $0, \pm 1, \pm 2$

➥ $\ell = 3$, $m_\ell = -3, -2, -1, 0, 1, 2, 3$ or $0, \pm 1, \pm 2, \pm 3$

The possible values of m_ℓ is $2\ell + 1$

☑ Magnetic quantum number prescribes the **orientation of the orbital** in three dimensional space about the nucleus.

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Orbital	s	p	d	f
Number of oreintation in space (x,y, z axis)	1	3	5	7

4. Spin quantum number (m_s)

☑ To specify the electrons 's spin, we use the **electron spin quantum number (m_s)** since there are two possible directions of spin, opposite each other, m_s, has two possible values; **+ 1/2** and **- 1/2**

☑ It describes the direction of spin of an electron.

(a) clock wise (b) counter clockwise

Fig 1.8 (a) clockwise and (b) counter clock wise spins of an electron. The magnetic fields generated by these two spinning motions are analogous to those from the two magnets The upward and downward arrows are used to denote the direction of spin.

Illustrative Examples

4. What are the four quantum numbers for an electron in a 2s orbital ? (ESSLCE)

A. 2, 0, 1, $+\frac{1}{2}$

B. 2, 0, 0, $+\frac{1}{2}$

C. 2, 1, 1, $+\frac{1}{2}$

D. 2, 1, 0, $+\frac{1}{2}$

Solution

In 2s, $n = 2$, $\ell = 0$ (s - orbital), $m_s = +1/2$ or $-1/2$

since it is one electron.

Answer : B

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75. What values of angular momentum quantum number ℓ are permitted for an electron with $n = 4$?

A. 1, 2, 3 C. 0, 1, 2, 3, 4

B. 1, 2, 3, 4 D. 0, 1, 2, 3

Solution

Value ℓ is from 0 to n - 1: $\ell = 0, 1, 2, 3$

Answer : D

76. Which of the following electrons, identified only by their n and ℓ quantum numbers have the highest energy ?

A. $n = 2, \ell = 0$ C. $n = 3, \ell = 2$

B. $n = 4, \ell = 1$ D. $n = 4, \ell = 2$

Solution

The higher the value of n, the higher is the energy. With same value of n, the higher the value of ℓ , the greater is the energy.

Answer : D

77. What is /are the possible value /s/of azimutal quantum number for an electron in the 4f sub - shell ?

A. 0, 1, 2, 3 B. 1, 2, 3 C. 3 D. 4

Solution

If the value of n only is given, ℓ will be from 0 to n -1. However, if both n and sub shell are given, ℓ will have single value that represent the sub - shell. Thus, in 4f $\ell = 3$.

Answer : C

78. For an electron that has quantum numbers $n = 4$ and $m\ell = 0$, which of the following is true ?

A. It must have quantum number $n = 0$

B. It must have quantum number $\ell = 0$

C. It must have quantum number $m_s = \pm 1/2$

D. It may have quantum number $\ell = 0, 1, 2, 3$

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Solution

For n = 4, ℓ may be 0, 1, 2, or 3.

Answer : D

79. Which of the following quantum numbers is not allowed in the hydrogen atom ?

A. n = 2, ℓ = 1, m_ℓ = 1 C. n = 8, ℓ = 7, m_ℓ = -6

B. n = 1, ℓ = 0, m_ℓ = 0 D. n = 2, ℓ = 0, m_ℓ = 2

Solution

The value of m_ℓ is from $-\ell$ to $+\ell$.

Answer : D

80. What is the maximum number of electrons in an atom that can have the quantum numbers n = 5 and m_ℓ = + 1 ?

A. 8 B. 16 C. 32 D. 64

Solution

For n = 5 : ℓ = 0, m_ℓ = 0

ℓ = 1, m_ℓ = -1, 0, 1

ℓ = 2, m_ℓ - 2, -1, 0, 1, 2

ℓ = 3, m_ℓ = - 3, -2, - 1, 0, 1, 2, 3

ℓ = 4; m_ℓ = - 4, -3, - 2, -1, 0, 1, 2, 3, 4

Here, there are four orbitals with m_ℓ = + 1 orientations. Thus,the maximum number of electrons is 4 × 2 = 8.

Answer : A

1. How many orbitals are there in an atom with

a. n = 4 b. ℓ = 3 c. n = , ℓ = 2, m_ℓ = 0

Solution

a. Number of orbitals = $n^2 = 4^2 = 16$

b. Number of orbitals = $2\ell + 1 = 2 \times 3 + 1 = 7$

c. Number of orbitals = 1 because only one value of m_ℓ is given.

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82. Determine the maximum number of electrons in an atom with

a. $n = 4$

b. $\ell = 2$

c. $n = 4, \ell = 3, m_\ell = +2$

d. $n = 3, \ell = 2, m_\ell = 1, m_s = +1/2$

Solution

a. 32 b. 10 c. 2 d. 1

83. Identify the sub shells represented by the quantum numbers ?

a. $n = 3, \ell = 2$ c. $n = 5, \ell = 1$

b. $n = 4, \ell = 3$ d. $n = 2, \ell = 0$

Solution

a. 3 d b. 4 f c. 5 p d. 2s

84. Give the values of the quantum numbers associated with the following orbitals.

a. 3s b. 4 P c. 6 d d. 5 f

Solution

a. $n = 3, \ell = 0, m_\ell = 0, m_s = \pm 1/2$

b. $n = 4, \ell = 1, m_\ell = 0, \pm 1, m_s = \pm \frac{1}{2}$

c. $n = 6, \ell = 2, m_\ell = 0, \pm 1, \pm 2, m_s = \pm 1/2$

d. $n = 5, \ell = 3, m_\ell = 0, \pm 1, \pm 2, \pm 3, m_s = \pm 1/2$

85. Arrange the following orbitals in increasing order of energy.

a. 3s, 3p, 3d c. 2s, 3s, 4s

b. 4 s, 4f, 4d, 4 p

Solution

a. $3s < 3p < 3d$
b. $4s < 4p < 4d < 4f$
} due to the value of ℓ

c. $2s < 3s < 4s$: due to the value of n.

86. List the possible sub - shells and orbitals associated with the principal quantum numbers, n = 5.

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Solution

5 s = one orbital

5 p = three orbitals

5 d = five orbitals

5 f = seven orbitals

5 g = nine orbitals

87 How many sub shells are there in the shell designated by n = 6 ?

Solution

Number of sub shells = n = 6 (6 s, 6 d, 6 f, 6 g, 6 h)

88. What are degenerate orbitals ? Give examples.

Solution

Degenerate orbitals are orbitals with the same energy but different orientations.

Example

$2p_x$, $2p_y$, $2p_z$; $3d_{x^2-y^2}$, $3dz^2$, $3d_{xy}$, $3d_{xz}$, $3dyz$

89. An electron in a hydrogen atom is known to have a velocity of 5×10^6 m /s ± 1 percent using the uncertainity principle, calculate the minimum uncertainity in the position of the electron and give that the diameter of the hydrogen atom is less than 1 angstrom (A^0), comment on the magnitude of this uncertainity compared to the size of the atom.

Solution

$M_{e^-} = 9.1 \times 10^{-31}$ kg , $h = 6.626 \times 10^{-34}$ kg . m^2/s

ΔV = velocity = 1 % × 5 × 10^6 m /s = 5 × 10^4 m /s

$\Delta x . m\Delta V \geq \frac{h}{4\pi}$, $\Delta P = m\Delta V$

$$\Delta x = \frac{h}{4\pi m \Delta V} = \frac{6.626 \times 10^{-34} \text{ kg.m}^2/\text{s}}{4 \times 3.14 \times 9.1 \times 10^{-31} \text{ kg} \times 5 \times 10^4 \text{m/s}}$$

$\Delta x = 1.0 \times 10^{-9}$ m

$\Delta x \geq 1.0 \times 10^{-9}$ m or $\Delta x \geq 10A^0$. The minimum uncertainity in position Δx is $10A^0$ .The uncertainity in the electron's position is10 times larger than theatom.

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90. List some orbitals that do not exist based on the values of quantum numbers.

Solution

1 p, 2 d, 3 f, 4 g etc

1.7 Electronic Configuration and Orbital Diagram

Electron Configuration is distribution of electrons in atomic shells, sub shells or orbitals based on different principles. There are different types of electron configurations. These are :

1. Main shell electrons configuration
2. Bohr's diagram configuration
3. Sub - shell electron configuration
4. Orbital diagram electron configuration
5. Molecular orbital electron configuration

The electron configuration of atoms or ions or molecules is based on the following three principles

1. The Aufbau (building) principle: states that atomic shells or sub shells with lowest energy filled first.

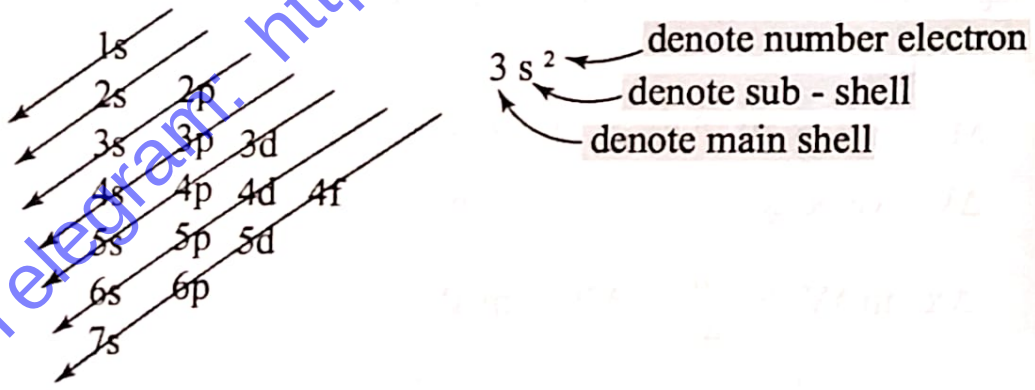

Fig 1.8 Diagonal rule of Filling Atomic Orbitals

Example

a. Write electron configuration for $_{11}Na$.

Solution

Electrons = 11

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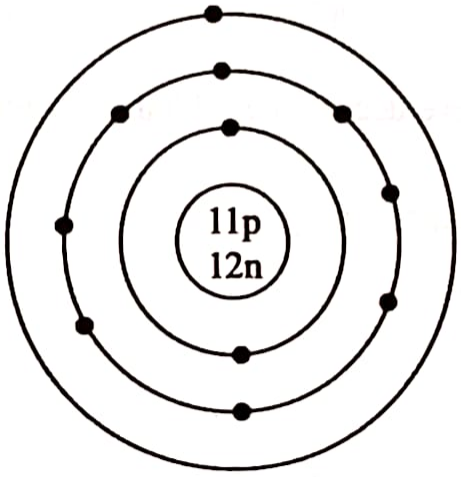
- $1s^2\ 2s^2\ 3s^1$: Sub - shell electron configuration
- 2 8 1 : Main shell configuration
- [Ne] $3s^1$: Noble gas core configuration (condensed configuration)
- n = A - Z = 23 - 11 = 12



Bohr's diagram configuration

b. Write electron configuration for

a. $_{15}P^{3-}$ b. $_{26}Fe^{2+}$

Solution

a. Electrons = 15 - (-3) = 18

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ = [Ar]

2, 8, 8

b. electrons = 26 - (+2) = 24

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$, the two electrons are lost from 4s.

[Ar] $3d^6$

i. Note : Exception to building up principle due to stability of orbitals

Copper (Z = 29) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^9$: Expected

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$: Observed.

Chromium (Z = 24) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^4$: expected

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$: observed

For these two elements write their electron configuration based on the observed results.

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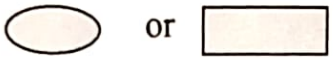
ii. The electron configuration of all elements except hydrogen and helium can be represented using **a noble gas core.** Here , the inner electrons are represented by the nearest noble gas using brackets.

2. Hund's rule states that the most stable arrangment of electrons in orbitals of equal energy is the one in which the number of electrons with same spin is maximized.

➥ It is applied for elements that have degenerate orbitals which are not fulfilled.

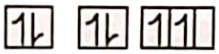
Note : An orbital can be represented by ____,

 or

Example

Consider carbon (Z = 6)

$1s^2$ $2s^2$ $2px^1$ $2py^1$ $2pz^0$ or using orbital diagram



Note : Orbital diagram electron configuration is very important to determine the number of unpaired electrons.

➥ Carbon has two unpaired electrons

Sub - shell	s	p	d	f	g
Maximum number of unpaired electron	1	3	5	7	9

3. The Pauli Exclusion principle : states that no two electrons in an atom can have the same four quantum numbers. At least they differ in spin quantum.

Example

$_2$He : $1s^2$ First electron : $n = 1, \ell = 0, m_\ell = 0, m_s = +1/2$

Second electron : $n = 1, \ell = 0, m_\ell = 0, m_s = -1/2$

Illustrative Example

91. Which one of the following is the diagramatic representation of $^{35}_{17}Cl$? (ESSLCE)

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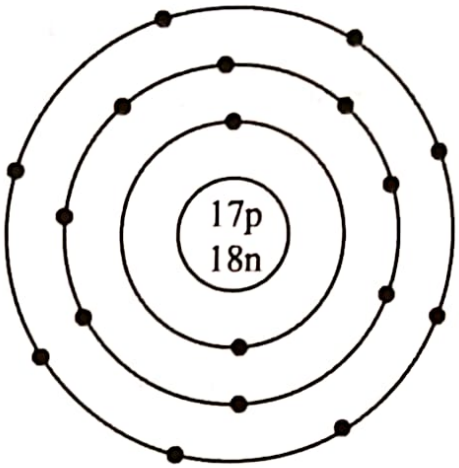
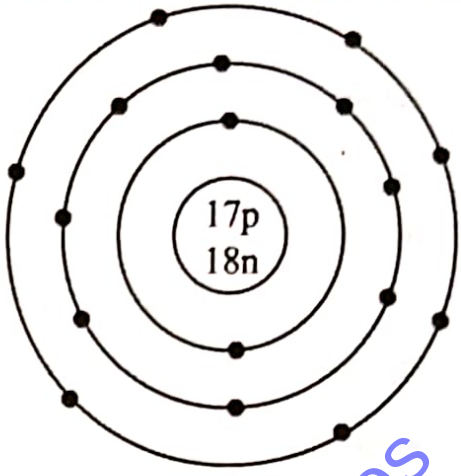
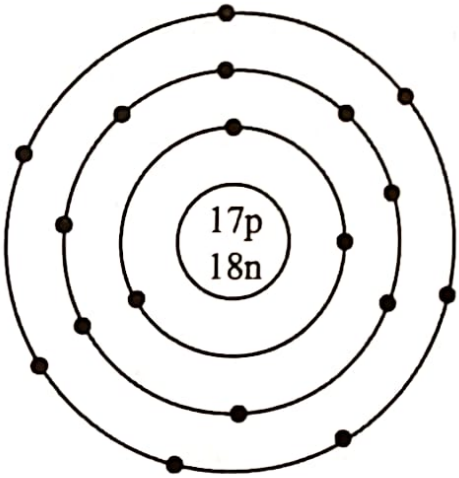
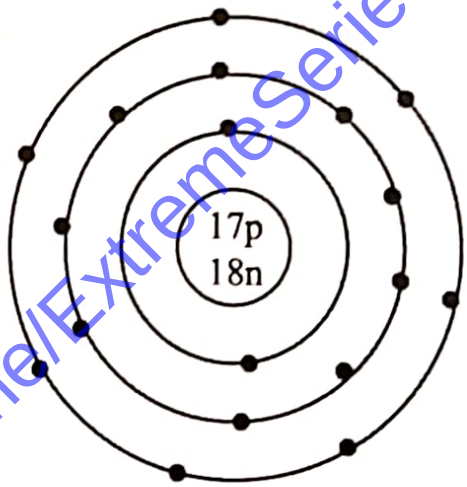
A. 

C. 

B. 

D. 

Answer : C

92. The correct electron configuration of N^{3-} (Z = 7) is

A. $1s^2\ 2s^2$

B. $1s^2\ 2s^2 2p^3$

C. $1s^2\ 2s^2\ 2p^1$

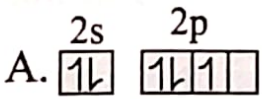
D. $1s^2\ 2s^2\ 2p^6$

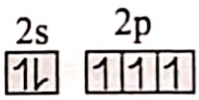
Solution

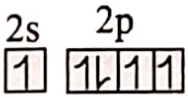
N^{3-} e = 7- (- 3) = 10 : $1s^2\ 2s^2\ 2p^6$

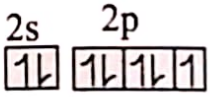
Answer : D

93. The orbital diagram in which the Aufbau principle violated is

A. 

C. 

B. 

D. 

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Answer : B

94. Which of the following electronic configurations is correct

A. $1s^2\ 2s^2\ 2px^2\ 2py^2\ 2pz^2\ 3s^2 3px^2$

B. $1s^2\ 2s^1\ 2px^1\ 2py^1\ 2pz^1$

C. $1s^2\ 2s^2\ 2p^6 3s^2\ 3p^6\ 4s^2\ 3d^4$

D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3px^1\ 3py^1\ 3pz^1$

Solution

Degenerate orbitals are singly filled with electrons before pairing sta rts.

Answer : D

95. What is the maximum number of unpaired electrons in a d - sub shell ?

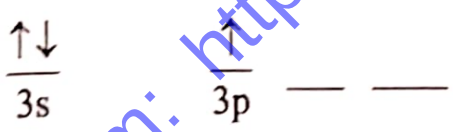
A. 2 B. 5 C. 3 D. 4

Solution

d - sub shell has five orbitals and contains a maximum of five unpaired electrons.

Answer : B

96. The following energy level diagram represents outer most shell of what ground state element ?



A. B B. He C. Al D. Be

Solution

Write the complete electron configuration and find the atomic number of the element .

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$: $Z = 2 + 2 + 6 + 2 + 1 = 13$. Thus, the element with atomic number 13 is Al.

Answer : C

97. What can you conclude from the figure below ?

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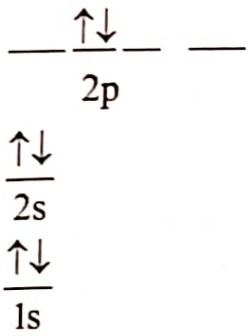

A) Hund's rule has been violated

B) Pauli exclusion principle has been violated

C) The Aufbau principle has been violated

D) This is valid diagram.

Answer : A

98. The Pauli exclusion principle _____

A. Tells the energy of an electron

B. Tells the shape of an electron

C. Describes the position in space of an electron

D. Limits the number of electrons in an orbital to be two.

Solution

Pauli limited the number of electrons in an orbital to be two.

Answer : D

99. Which one of the Following is the correct electron configuration for Cu (Z = 29) ?

A. [A r] $4s^1\ 3d^9$

B. [A r] $4s^1 3d^{10}$

C. [A r] $4s^2\ 3d^9$

D. [A r] $3d^{10}$

Solution

Because of stability of orbital, the configuration is [A r] $4s^1\ 3d^{10}$.

Answer : B

100. A neutral atom has a ground state electron configuration that ends with 4f half - filled orbital. Determine the atomic number of the element.

Solution

For neutral atom : Z = e

[X e] $6s^2\ 4f^7$, Z for X e is 54. Therefore, Z for the element is :
Z = 54 + 2 + 7 = 63

101. Describe the valence electrons of potassium (Z = 19) using the four

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quantum numbers.

Solution

Electron configuration : [A r] $4s^1$

Valence shell configuration : $4s^1$: n = 4 , $\ell = 0$, $m_\ell = 0$, $m_s = +\frac{1}{2}$

102. Write the electron configuration of the following atoms and ions.

a. Ni (Z = 28)
b. Cr^{3+} (Z = 24)
c. Br^- (Z = 35)
d. Co^{2+} (Z = 27)

Solution

a. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^8$ = [A r] $4s^2\ 3d^8$

b. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3$ = [A r] $3d^3$

c. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6$ = [K r]

d. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^7$ = [A r] $3d^7$

103. Two electrons in an orbital always have opposite spin. Why ?

Solution

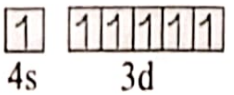
The spin of an electron is magnetic property not an actual rotation like a magnet with its south and north poles. Electrons which are paired have opposite magnetic properties which lead to opposing spins.

104. Determine the number of unpaired electrons in

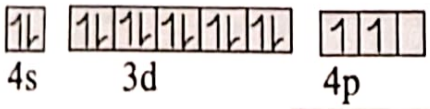
a. Cr
b. Ge
c. Zr^{3+}
d. Se^{2-}

Solution

a. Number of unpaired electrons are found in the valance shell of an atom.

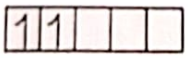
[Ar] $4s^1\ 3d^5$:  **6 unpaired electrons**

4s 3d

b. [Ar] $4s^2\ 3d^{10} 4p^2$: 

4s 3d 4p

2 unpaired electrons

c. [Kr] $4d^2$:  **2 unpaired electrons**

d. [Kr] : **No unpaired electrons**

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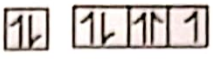
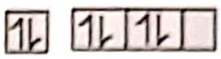
105. Portions of orbital diagrams representing the ground state electron configurations of certain elements are shown here. Which of them violate Hund's rule ? Which of them violate Pauli exclusion principle

I.

III.

II. 

IV. 

Solution

I and III violate the Pauli exclusion principle

II and IV violate the Hund's rule

106. Which of the following species has the greatest number of unpaired electrons = S^+, S or S^- ? (Z = 16)

Solution

S^+: [Ne] $3s^2 3p^3$ = 3 unpaired electrons

S : [Ne] $3s^2 3p^4$ = 2 unpaired electrons

S^-: [Ne] $3s^2 3p^5$ = 1 unpaired electrons

107. An atom may absorb a quantum of energy and excites. When it happens, we say that the atom is in excited state. Identify these atoms and write their ground state configurations.

a. $1s^1\ 2s^1$

b. $1s^2\ 2s^2\ 2p^2\ 3d^1$

c. [A r] $4s^1\ 3d^{10}\ 4p^4$

d. $1s^2\ 2s^2\ 2p^6\ 4s^1$

e. [Ne] $3s^2\ 3p^4\ 3d^1$

Solution

	Atom	Ground state configuration
a	He (Z = 2)	$1s^2$
b	N (Z = 7)	$1s^2 2s^2 2p^3$
c	A s (Z = 33)	[A r] $4s^2\ 3d^{10}\ 4p^3$
d	Na (Z = 11)	$1s^2\ 2s^2\ 2p^6\ 3s^1$
e	Cl (Z = 17)	[Ne] $3s^2\ 3p^5$

108. Write orbital diagram configuration for

a. O b. P c. Co

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Solution

a. ↿⇂ ↿⇂ ↿⇂ ↿ ↿
1s 2s 2p

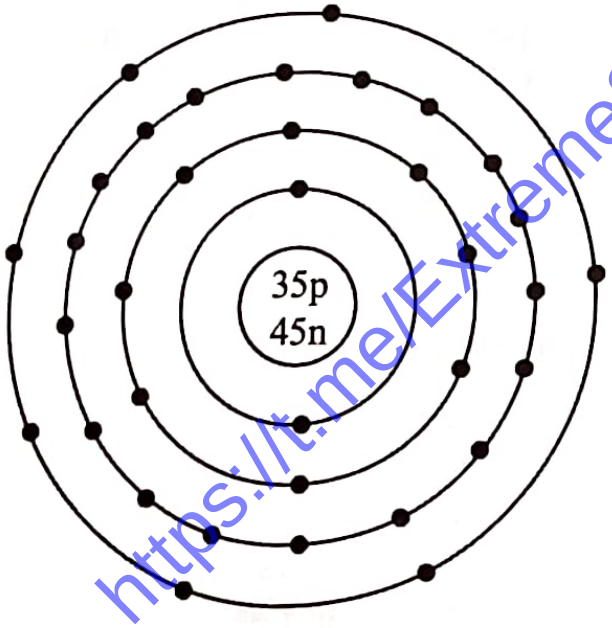
b. ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿ ↿ ↿
1s 2s 2p 3s 3p

c. ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿ ↿ ↿
1s 2s 2p 3s 3p 4s 3d

109. Draw Bohr diagram for bromine ($_{35}^{80}$ B r)

Solution

P = 35, n = 80 - 35 = 45, configuration : 2, 8, 18, 7



110. The electrons identified by quantum numbers n and ℓ have increa order of energy .

i. n = 4, ℓ = 1
ii. n = 4, ℓ = 0
iii. n = 3, ℓ = 2
iv. n = 3, ℓ = 1

Solution

Identify the sub shells first.

i. 4 p ii. 4 s iii. 3 d iv. 3 p

iv < ii < iii < i

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1.8 Historical Development of the Periodic Table

1. Dobernier's triad

☑ It is called **the law of triad**

☑ He arranged three elements in increasing order of atomic masses and said the **atomic mass** of the middle element is the average of the other two.

Example

^{7}Li, ^{23}Na, ^{39}K is one triad

$$\text{Atomic mass of Na} = \frac{\text{Atomic mass Li} + \text{Atomic mass K}}{2}$$

$$23 = \frac{7+39}{2} = 23$$

☑ Dobernier's triad was the first attempt for classification of elements.

Limitation

☑ The law of triad was not applied to all elements even discovered at that time.

2. Newland's law of octaves

☑ He likened this to music and called it the **law of octaves**

☑ He states that every eighth element has similar property with the first element if they are arranged in increasing order of **their atomic masses.** He arranged 56 known elements in order of increasing atomic weight.

Li	Be	B	C	N	O	F
Na	Mg	Al	Si	P	S	Cl
K	Ca					

Example

Be and Mg
N and P
F and Cl
} have similar properties

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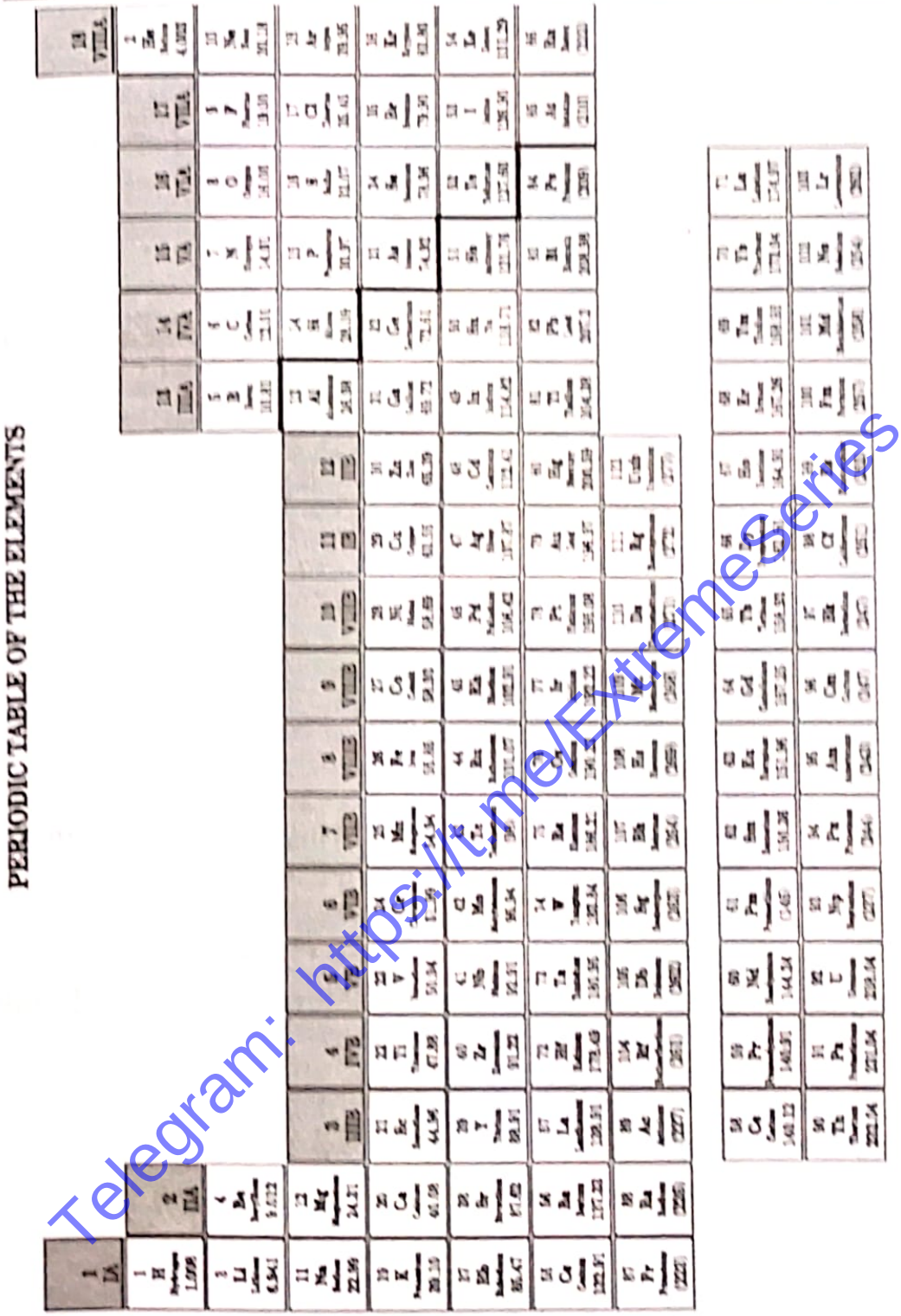

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Classification of Elements

1. Representative Elements

- ☑ They are 'A' group elements (found in IA to VIIIA)
- ☑ They are s - block and p - block elements

I V A	Carbon family	$ns^2 np^2$
V A	Nitrogen family	$ns^2 np^3$
VIA	Oxygen family	$ns^2 np^4$
V II A	Halogens (salt makers)	$ns^2 np^5$
V III A	Noble gases, inert gases or 'o' group	$ns^2 np^6$ except He ($1s^2$)

☑ It has seven periods (horizontal rows).

Table 1.5 The Number of Elements in Each Period

Period	1	2	3	4	5	6	7
Number of element	2 (H and He)	8 (Li - Ne)	8 (Na - Ar)	18 (K- Kr)	18 (Rb - Xe)	32(Cs - Rn)	Incomplete Fr = ?

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PERIODIC TABLE OF THE ELEMENTS

1 IA												13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1 H Hydrogen 1.008	2 IIA		3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8 VIIIB	9 VIIIB	10 VIIIB	11 IB	12 IIB	5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	2 He Helium 4.003
3 Li Lithium 6.941	4 Be Beryllium 9.012																10 Ne Neon 20.18
11 Na Sodium 22.99	12 Mg Magnesium 24.31											13 Al Aluminium 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	15 Ar Argon 39.95
19 K Potassium 39.10	20 Ca Calcium 40.08		21 Sc Scandium 44.96	22 Ti Titanium 47.88	23 V Vanadium 50.94	24 Cr Chromium 51.99	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.39	31 Ga Gallium 69.72	32 Ge Germanium 72.61	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80
37 Rb Rubidium 85.47	38 Sr Strontium 87.62		39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
55 Cs Cesium 132.91	56 Ba Barium 137.33		57 La Lanthanum 138.91	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
87 Fr Francium (223)	88 Ra Radium (226)		89 Ac Actinium (227)	104 Rf Rutherfordium (261)	105 Db Dubnium (262)	106 Sg Seaborgium (263)	107 Bh Bohrium (264)	108 Hs Hassium (269)	109 Mt Meitnerium (268)	110 Ds Darmstadtium (271)	111 Rg Roentgenium (272)	112 Uub Ununbium (277)						

58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.04	71 Lu Lutetium 174.97
90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.04	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (254)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (254)	103 Lr Lawrencium (262)

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Classification of Elements

1. Representative Elements

- ☑ They are 'A' group elements (found in IA to VIIIA)
- ☑ They are s - block and p - block elements

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☑ They include metals, non - metals and metalloids

☑ For representative elements.

Group = valence electrons (electrons in outer most shell)

Period = Number of shells or valance shells

Example

Determine the group and period of an element with atomic number 14

Solution

☑ Using main shells : 2, 8, 4

⇒ Valence electrons = 4 ∴ group = 4 A or I V A

⇒ Number of shells = 3 ∴ period = 3

☑ Using sub - shells : $1s^2 2s^2 2p^6 3s^2 3p^2$

Valance shell configuration : $3s^2 3p^2$

☑ Valance electrons = 2 + 2 = 4 ∴ group = 4 A or I V A

☑ Valance shell = 3 ∴ Period = 3

Thus an element with atomic number 14 is located in group I V A and period 3

2. Transition Elements

☑ They are ' B' group elements (found in I B - VIII B)

☑ They are d - block elements (their last electron enters in the d - su shell)

☑ They are found between II A and III A elements

☑ They form colored and complex compounds

☑ They have variable valance number

☑ They are found in four series by filling **3d, 4d, 5d, and 6d.**

Period = Number of shells

☑ To determine the group of a transition element use the simple formul given below.

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Valence Shell Configuration	Group
$ns^1 (n-1) d^{10}$	I B (only s - electron)
$ns^2 (n-1) d^{10}$	II B (only s - electrons)
$ns^2 (n-1) d^1$	III B (s + d electrons)
$ns^2 (n-1) d^2$	I V B (s + d electrons)
$ns^2 (n-1) d^3$	V B (s + d electrons)
$ns^2 (n-1) d^5$	V I B (s + d electrons)
$ns^2 (n-1) d^5$	V II B (s + d electrons)
$ns^2 (n-1) d^6$	V III B
$ns^2 (n-1) d^7$	
$ns^2 (n-1) d^8$	

Example

Find the group and period of an element with atomic number 25.

Solution

[A r] $4s^2 3d^5$

Valence shell = 4 $\therefore$ period = 4

Group = s - electrons + d - electrons = 2 + 5 = 7 B (V I I B)

3. Inner Transition Elements

- ☑ They are f - block elements
- ☑ They are found in two series lanthanides ($4f^1 - 4f^{14}$) actinides ($5f^1 - 5f^{14}$)

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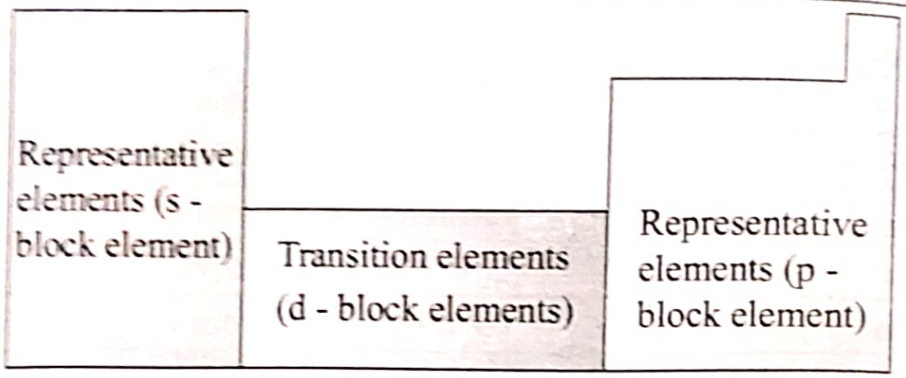

Fig 1.10 Classification of Elements

Illustrative Examples

116. Which of the following types of elements are good oxidizing agents ?

A. Alkali metals
C. Lanthanides
B. Halogens
D. Transition elements

Solution

Metals are generally considered as reducing agent and non - metals are oxidizing agent.

Answer : B

117. Which group of elements is characterized by ns^2 np^2 outer electron configuration ?

A. Group 2 A
C. Group 4 B
B. Group 4 A
D. Group 3 B

Solution

Valence electrons = 2 + 2 = 4 $\Rightarrow$ group = 4 A

Answer: B

118. In which of the following period of the periodic table is an element with atomic number 20 is placed ?

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A. 1 B. 2 C. 3 D. 4

Solution

$Z = 20$, [A r] $4s^2$: valance shell = 4, period = 4 **Answer : D**

119. All of the following are representative elements, **EXCEPT**

A. $_{11}Na$ B. $_{15}P$ C. $_{26}Fe$ D. $_{35}Br$

Solution

Na : [Ne] $3s^1$ = s - block
P : [Ne] $3s^2$ = p - block
B r : [A r] $4s^2 = 3d^{10}\ 4p^5$ = p - block
} are representative element

F e : [A r] $4s^2\ 3d^6$ = d - block = transition element.

Answer : C

120. Determine the group, period and block of the following elements.

a. Rb (Z = 37 b. Mn (Z = 25) c. Se (Z = 34)

Solution

a. [K r] $5s^1$: group = I A, period = 5, block = s

b. [A r] $4s^2\ 3d^5$: group = V ll B, period = 4, block = d

c. [A r] $3d^{10}\ 4s^2\ 4p^2$: group = I V A, period = 4, block = p

121. Two elements X and Y are found in group I IA and V A respectively. Write the formula of the compound formed between X and Y.

Solution

Group I I A elements are metals. Their valance electrons are the same as their valance number. Group V A elements are non - metals. Their valance number is eight minus their valence electrons.

$\Rightarrow$ Valence number of X = 2

$\Rightarrow$ Valance number of Y = 8 - 5 = 3

$$X^{2}\ Y^{3} = X_3Y_2$$

122. What is the most important relationship among elements in the same

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group in the periodic table ?

Solution

Elements in the same group have the same valence electrons. As a result, they have similar chemical properties.

123. Write the flame colors of

a. Lithium
b. Sodium
c. Potassium
d. Rubidium
e. Cesium
f. Calcium
g. Strontium
h. Barium
i. Chlorine
j. Bromine
k. Iodine

Solution

Element	a	b	c	d	e	f
Color	Crimson	Yellow	Lilac	Red	Blue	Brick - red

g	h	i	j	k
Crimson	Apple green	Pale green	Red - brown	Purple black

Periodic properties of elements down a group and across a period

1. Number of Valence Electrons

☑ Increases across a period from left to right but remains same down a group.

2. Number of Shells

☑ Increases down a group but remains same across a period from left to right.

Li 2,1							
Na 2,8,1	Mg 2,8,2	Al 2,8,3	Si 2,8,4	P 2,8,5	S 2,8,6	Cl 2,8,7	Ar 2,8,8
K 2, 8, 8, 1							

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Fig 1.11 Valence Electrons and Number of Shells of Group IA and Period 3 Elements

3. Nuclear Charge is the total positive charge in the nucleus of an atom.

Example

Mg (Z = 12) : Nuclear charge = + 12

☑ It increases down a group and across a period.

4. Effective nuclear charge (Z_{eff}) is the difference between atomic number and inner or core electrons (all electrons out of valence electrons)

$\boxed{Z_{eff} = Z - S}$ Z - atomic number

S - number of innerelectrons

Example

Determine Z_{eff} of Ca (Z = 20)

Solution

First, write electron configuration and determine number of inner electrons.

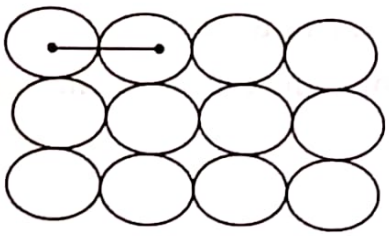
$\Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$

(S) = 2 + 2 + 6 + 2 + 6 = 18

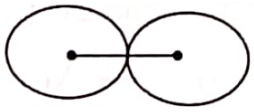
$Z_{eff} = Z - S = 20 - 18 = +2$

Z_{eff} remains same down a group but increases across a period

5. Atomic size (radius) is the distance between the nuclei of any two closest atoms in a substance



(a)



(b)

a. Atomic radius in metals is defined as half the distance between

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adjacent metal atoms

b. Atomic radius in non - metals is defined as half the distance between bonded identical atoms in a molecule.

☑ Atomic size increases down a group but decreases across a period from left to right.

H 0.037							He 0.08
Li 0.134	Be 0.125	B 0.090	C 0.077	N 0.075	O 0.073	F 0.071	Ne 0.160
Na 0.154	Mg 0.145	Al 0.130	Si 0.118	P 0.102	S 0.102	Cl 0.099	Ar 0.190

Fig 1.12 Atomic radii in nm of the First Three Period Elements

☑ The more positive the cation, the smaller is the size. For elements in the same period.

Example : $Al < Al^{+} < Al^{3+}$; $Al^{3+} < Mg^{2+} < Na^{+}$

☑ The more negative the anion, the larger is its ionic radius.

Example : $Cl < Cl^{-}$; $F^{-} < O^{2-} < N^{3-} < C^{4-}$

6. Ionization Energy (IE)

☑ It is the minimum energy required to remove an electron from an atom in the gas phase.

$$Na(g) + IE \rightarrow Na^{+}(g) + e^{-}$$

☑ First ionization energy (IE) removes first valence electron.

☑ Second ionization energy (IE_2) removes second valence electron.

☑ Third ionization energy (IE_3) removes third valence electron.

☑ Fourth ionization energy (IE_4) removes the fourth valence electron

$$M(g) + IE_1 \rightarrow M^{+}(g) + e^{-}$$

$$M^{+}(g) + IE_2 \rightarrow M^{2+}(g) + e^{-}$$

$$M^{2+}(g) + IE_3 \rightarrow M^{3+}(g) + e^{-}$$

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$M^{3+}(g) + IE_4 \rightarrow M^{4+}(g) + e^-$

$I E_1 < I E_2 < IE_3 < IE_4$

☑ Ionization energy decreases down a group and increase across a period

Exception in periods 2 and 3 due to stability of orbitals

Period 2 : IE of Be > B ; N > O

Period 3 : IE of Mg > Al ; P > S

Li	Be	B	C	N	O	F	Ne
520	900	801	1086	1402	1314	1681	2081
Na	Mg	Al	Si	P	S	Cl	Ar
496	738	578	789	102	1000	1251	1521

Fig 1.13 First ionization energy is KJ/mol of period 2 and 3 elements

7. **Electron affinity (E A)** is the energy released when one mole of electrons added to gaseous atom.

Example

$Cl(g) + e \rightarrow Cl^-(g) + EA,\ EA = -348\ kJ/mol$

☑ Electron affinity decreases down a group and increases across a period.

8. **Electro negativity (E N)** is the power of an atom to pull shared electrons to itself when it is in a molecule.

☑ Electro negativity decreases down a group and increases across a period.

The three most electro negative elements are fluorine, oxygen and nitrogen respectively

9. **Metallic character (electro positivity)** is the property of metals.

☑ Metals are malleable, ductile, luster and good conductors of heat and electricity.

☑ All metals are solid except mercury which is liquid at room temperature

☑ They have high meting and boiling points

☑ Metallic character increases down a group and decreases across a period from left to right.

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Example

Consider period 3 and group V A

Period 3

Na	Mg	Al	Si	P	S	Cl	Ar

- Na, Mg and Al are metals
- Si is metalloid
- P, S, Cl and Ar are non - metals.

VA

N
P
As
Sb
Bi

* N and P are non - metals
* As and Sb are metalloids
* Bi is metal

Importance of Periodic Table

1. It is useful in predicting properties of elements
2. It is also used to predict properties of compounds
3. It is useful in predicting formula of compounds
4. It is used to refer symbols, atomic masses and atomic numbers.
5. It is simple and easy way to remember properties of elements by writing electron configuration.

Illustrative Examples

124. Which of the following elements has similar chemical property with calcium (Ca) ?

A. Na B. Mg C. Al D. Si

Solution

Elements with the same valence electrons or in the same group have similar chemical property.

Answer : B

125. Arrange the elements Li, Be and B in order of increasing ionization

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energy :

A. Li < Be < B

B. Li < B < Be

C. Be < B < Li

D. B < Be < Li

Solution

I E : Li < Be and B < Be due to stability of orbital

⇒ Li < B < Be

Answer : B

126. With in a specific period, an increase in atomic number is usually accompanied by

A. An increase in atomic radius and an increases in electro negativity.

B. A decrease in atomic radius and an increase in electro negativity.

C. An increase in atomic radius and a decrease in electro negativity

D. A decrease in atomic radius and a decrease in electro negativity

Solution

➥ As atomic size increases, electro negativity decrease and vice - versal.

➥ As atomic number increases, atomic size usually decreases.

Answer : B

127. In which case the metals are arranged in order of increasing metallic character ?

A. Mg, Ca, K, Cs

B. Mg, Ca, Cs, K

C. Ca, Mg, Cs , K

D. K, Mg, Ca, Cs

Solution

➥ Mg and Ca are found in l l A and K and Cs in l A.

➥ Metallic character increases down a group but decreases across a period.

Answer : A

128. A periodic property

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A. Changes according to periods

B. Repeats itself in the same manner with increase in atomic number.

C. Changes according to ionization energy

D. Changes with temperature and mass number.

Answer : B

129. One of the following is the property of metals. They are

A. Solid at room temperature

B. Soft

C. Good conductors

D. Insulators

Answer : C

130. Arrange the following in decreasing order of ionization energy.

a. F e, Fe^{3+}, Fe^{2+}, Fe^{+}

b. B, Be, N, O

c. Al, Mg, S, P

d. I, Cl, B r, F

Solution

a. $Fe^{3+} > Fe^{2+} > Fe^{+} > Fe$

b. N > O > Be > B

c. P > S > Mg > Al

d. F > Cl > B r > I

131. Arrange the following in increasing order of atomic radius.

a. Na^{+}, F^{-}, Al^{3+}, Mg^{2+}

b. Li , Be, Na, Mg

c. C, B, F, Na

d. As^{3-}, Se^{2-}, P^{3-}, S^{2-}

Solution

a. $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$

b. Be < Li < Mg < Na

c. F < B < C < Na

d. $S^{2-} < P^{3-} < Se^{2-} < As^{3-}$

132. Write the four most electro negative elements in descending order.

Solution

F > O > N > Cl

133. Arrange the following elements in each of the following groups in order of increasing electron affinity.

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a. Li, Na, K, Rb b. F, Br, Cl, I

Solution

a. Rb < K < Na < Li

b. I < B r < F < Cl, Chlorine exceptionally has the highest negative electron affinity.

Solved Problems

134. Bromine has two stable isotopes. For bromine the isotopic masses are 78.9183 amu (relative abundance, 50.69 %) and 80.9183 amu (relative abundance, 49.31 %) . Calculate the average atomic weight of bromine.

Solution

Average atomic mass = $A_1\ f_1 + A_2\ f_2$

= 78 . 9183 × 50.69 % + 80.9183 × 49.31 %

= 79. 90 amu

135. The metallic element chromium has four stable isotopes.

Exact	Fraction
49.9461	0.0435
51.9405	0.8379
52.9407	0.0950
53.9389	0.0236

Calculate the average atomic mass of chromium

Solution

A V. atomic mass = $A_1\ X_1 + A_2\ X_2 + A_3 X_3 + A_4 X_4$

= 49. 9461 × 0.0435 + 51.9405 × 0.8379 + 52.9407 × 0.950 + 53.9389 × 0.0236 = 51.9 amu

136. A retina of a human eye can defect light when radiant energy incident on it is at least 4.0×10^{-17} J. For light of 585 nm wavelength, how many photons does this energy correspond to ?

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Solution

$$E_{Photon} = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ J.S \times 3 \times 10^{8}\ m/s}{5.85 \times 10^{-7}\ m}$$

$$\Rightarrow 1 photon = 3.4 \times 10^{-19}\ J$$

$$? = 4.0 \times 10^{-17}\ J$$

$$\text{Number of photons} = \frac{1 photon \times 4.0 \times 10^{-17}\ J}{3.4 \times 10^{-19}\ J} = 118 \text{ photons}$$

137. What is the wavelength in nanometer of radiation that has an energy content of 1.0×10^{3} k J /mol ?

Solution

$$1 \times 10^{3}\ kJ/mol = 1 \times 10^{6}\ J/mol$$

$$1\ mol = 6.02 \times 10^{23}\ photons = 1 \times 10^{6}\ J, 1\ photon = ?$$

$$\Rightarrow E_{photon} = \frac{1 photon \times 1 \times 10^{6}\ J}{6.02 \times 10^{23}\ photons} = 1.66 \times 10^{-18}\ J$$

$$\lambda = \frac{hc}{E_{photon}} = \frac{6.626 \times 10^{-34}\ J.s \times 3 \times 10^{8}\ m/s}{1.66 \times 10^{-18}\ J} = 1.20 \times 10^{-7} m$$

$$= 120nm$$

138. Light with a wavelength of 400nm strikes the surfaces of metallic cesium in a photocell and the maximum kinetic energy of the ejected is 1.54×10^{-19} J. Calculate the work function and the longest wavelength of light that is capable of ejecting electrons from that metal. Determine the velocity of an electron. (Mass of an electron = 9.1×10^{-31} kg)

Solution

$$v = \frac{c}{\lambda} = \frac{3 \times 10^{8}\ m/s}{4 \times 10^{-7} m} = 7.5 \times 10^{14}\ H_z$$

$$KE = h(v - v_o)$$

$$v_o = v - \frac{kE}{h} = 7.5 \times 10^{14}\ Hz - \frac{1.54 \times 10^{-19} J}{6.626 \times 10^{-34}\ J.s} = 5.18 \times 10^{14}\ Hz$$

$$\phi = h v_o = 6.626 \times 10^{-34}\ J.s \times 5.18 \times 10^{14} s^{-1} = 3.43 \times 10^{-19}\ J$$

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$$\lambda_{max} = \frac{c}{v_o} = \frac{3 \times 10^8 \text{ m/s}}{5.18 \times 10^{14} \text{ s}^{-1}} = 5.79 \times 10^{-7} \text{ m} = 579 \text{ nm}$$

139. Compare the wavelength for an electron (mass = 9.1×10^{-31} kg) traveling at a speed of 1×10^7 m /s with that for a ball (mass = 0.10kg) traveling at 35 m /s.

Solution

$$\lambda_{electron} = \frac{h}{mV} = \frac{6.626 \times 10^{-34} \text{ kg. m}^2/\text{s}}{9.1 \times 10^{-31} \text{ kg} \times 1 \times 10^7 \text{ m/s}} = 7.3 \times 10^{-11} \text{ m}$$

$$\lambda_{ball} = \frac{h}{mV} = \frac{6.626 \times 10^{-34} \text{ kg. m}^2/\text{s}}{0.10 \text{ kg} \times 35 \text{ m/s}} = 1.89 \times 10^{-34} \text{ m}$$

As mass increase, wavelength decreases.

140. Calculate the wavelength of the third line in the Paschen series for hydrogen atom.

Solution

In Paschen series : $n_f = 3$ and $n_i = 3 + 3 = 6$

$$\Delta E = 2.18 \times 10^{-18} \text{ J} \left(\frac{1}{n_i^4} - \frac{1}{n_{f^2}} \right) =$$

$$2.18 \times 10^{-18} \text{ J} \left(\frac{1}{6^2} - \frac{1}{3^2} \right) = -1.82 \times 10^{19} \text{ J}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{6.626 \times 10^{-34} \text{ J.s} \times 3 \times 10^8 \text{ m/s}}{1.82 \times 10^{-19} \text{ J}} = 1.09 \times 10^{-6} \text{ m}$$

$$= 1.09 \mu\text{m}$$

141. How many photons of light of 7×10^3 A^0 are equivalent to 1.00J of energy ?

Solution

Solve by your own !

Answer : 3.52×10^{18} photons

142. When a certain metal was irradiated with light of frequency

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147. Arrange the following in decreasing order of ionic radius Co, Co^{+}, Co^{3+} Co^{2+}

Solution

$Co > Co^{+} > Co^{2+} > Co^{3+}$

148. Arrange the following in increasing order of electro - negativity.
Be, Mg, Ca, Sr

Solution

Sr < Ca < Mg < Be

149. What sub - shell (s) is / are being filled in each of the following regions of the periodic table.

a. Group I A and I I A
b. Group I I I A - V I I I A
c. Transition elements
d. The lanthanides and actinides

Solution

a. n s
b. n p
c. (n - 1) d
d. (n - 2) f

150. Arrange the atoms (ions) in the following groups in order of increasing size.

a. Be , C , B , F , Li
b. Al^{3+}, F^{-}, Na^{+}, Mg^{2+}, O^{2-}
c. C^{4}, F^{-}, Ne , N^{3-}, Na^{+}
d. Ca^{2+}, Cl^{-}, As^{3-}, K^{+}, S^{2-}

Solution

a. F < C < B < Be < Li
b. $Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-}$
c. $Na^{+} < Ne < F^{-} < N^{3-} < C^{4-}$
d. $Ca^{2+} < K^{+} < Cl^{-} < S^{2-} < As^{3-}$

Supplementary Exercise

I. Multiple Choice Items

1. The lowest principal quantum number that an electron can have is

A. 0 B. 1 C. 2 D. 3

2. The sub - shell that is found in all energy levels is :

A. s B. d C. p D. f

3. An orbital may never be occupied by

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A. 1 electron
C. 3 electrons
B. 2 electrons
D. 0 electron

4. An atom of aluminum contains 13 electrons, 13 protons and 14 neutrons. The mass number of this atom is

A. 26 B. 40 C. 27 D. 14

5. Radio active changes differ from ordinary chemical changes because radio active changes

A. involves protons
C. absorb energy
B. are explosive
D. release energy

6. Atoms of ^{235}U and ^{238}U differ in structure by three

A. Electrons
C. Protons
B. Isotopes
D. Neutrons

7. How do the elements of group I I A form ions ?

A. By losing 2 electrons
C. By gaining one electron
B. By gaining 2 electrons
D. By losing one electron

8. Which one of the following describes the quantum mechanical model ?

A. Electrons moved around the nucleus of an atom in circular orbits.

B. An atom is a sphere with a positively charged matter, in which electrons are embedded

C. An atom is represented by a solid indestructible sphere.

D. The locations of electrons in an atom are described in terms of probability.

9. Which elements has an outer orbital configuration of $2s^2\ 2p^2$?

A. C B. B C. O D. F

10. If E is the kinetic energy of the particle then which of the following expressions is correct for de Broglie wave length of the particle ?

A. $\lambda = \frac{h}{2mE}$
C. $\lambda = \frac{h}{\sqrt{2mE}}$
B. $\lambda = \sqrt{\frac{h}{2mE}}$
D. $\lambda = \sqrt{\frac{2mE}{h}}$

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Solution

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c. C^{4}, F^{-}, Ne , N^{3-}, Na^{+}
d. Ca^{2+}, Cl^{-}, As^{3-}, K^{+}, S^{2-}

Solution

a. F < C < B < Be < Li
b. $Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-}$
c. $Na^{+} < Ne < F^{-} < N^{3-} < C^{4-}$
d. $Ca^{2+} < K^{+} < Cl^{-} < S^{2-} < As^{3-}$

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C. $\lambda = \frac{h}{\sqrt{2mE}}$
B. $\lambda = \sqrt{\frac{h}{2mE}}$
D. $\lambda = \sqrt{\frac{2mE}{h}}$

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11. An element , M has an atomic mass of 19 and atomic number 9, its ion is represented by

A. M^{+} B. M^{2+} C. M^{-} D. M^{2-}

12. The two particles A and B have de Broglie wave lengths 1 nm and 5 nm respectively. If mass of A is four times the mass of B, the ratio kinetic energies of A and B would be :

A. 4 : 5 B. 25 : 4 C. 5 : 4 D. 4 : 25

13. In an atom, which has 2K, 8 L, 18 M and 2 N electrons having magnetic quantum number, $m_\ell = 0$ is

A. 6 B. 10 C. 7 D. 14

14. Which of the following expressions imposes the condition of quantization of energy of an atom electron in an atom ?

A. $E = mc^2$ C. $E = h\nu$

B. $\lambda = \frac{h}{mV}$ D. $mVr = \frac{nh}{2\pi}$

15. How many electrons in an atom can have quantum numbers $n = 3, \ell = 2, m_\ell = +2, m_s = +1/2$?

A. 5 B. 1 C. 3 D. 9

16. The ratio of radius of second orbit of hydrogen to the radius of first orbit is

A. 4 : 1 B. 2 : 1 C. 1 : 2 D. 6 : 1

17. Particles in cathode rays have same charge to mass ratio as

A. x - particles C. Protons

B. $\propto$ - rays D. β - rays

18. Azimuthal quantum number of last electron of sodium ($_{11}Na$) is

A. 1 B. 2 C. 0 D. 3

19. An electron spin quantum number, $m_s = +1/2$ and magnetic quantum number, $m_\ell = +1$. It can not be present in

A. s - orbital B. p - orbital C. d - orbital D. f - orbital

20. Non - metals when compared to metals :

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A. are less electronegative and have smaller radii

B. have greater ionization energies and larger radii

C. are more electro negative and have smaller radii

D. are less electro positive and have smaller ionization energy

21. An atom containing two electrons with n = 3, ℓ = 1, m_ℓ = 1, m_s = -1/2 and n = 3, ℓ = 1, m_ℓ = 1, m_s = - 1/2 may not exist based on

A. Hund's rule

B. Pauli's exclusion principle

C. Heisenberg's uncertainty principle

D. Building up principle

22. All of the following increases down a group. **EXCEPT**

A. Metallic character

C. Electro positivity

B. Atomic size

D. Ionization energy

23. The classification of elements in the modern periodic table is mainly based on :

A. Mass number

C. Physical property

B. Natural abundance

D. Electron arrangment

24. A particular shaped volume of space in an atom where the probability of finding an electron is maximum is known as ____

A. An orbital

C. A sub shell

B. An orbit

D. A shell

25. Thomson, Dalton and Bohr, all support which of the following statements ?

A. All matter is composed of atoms

B. Atoms are mostly composed of empty space

C. Atoms are composed of an electron and a proton

D. Electron orbits the nucleus of an atom at distinct energy levels.

26. The atomic number of an element with valence electron configuration $3s^2\ 3p^5$ is

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A. 7 B. 17 C. 15 D. 18

27. The elements ^{14}C and ^{14}N have the same

A. Chemical property
C. Neutron number
B. Mass number
D. Proton number

28. Which group in the modern periodic table has elements in the three states of matter ?

A. Alkali metals
C. Boron family
B. Chalcogens
D. Halogens

29. The quantum numbers for a given orbital are n = 2, $\ell = 1$, $m_\ell = 0$. We would usually represent this as

A. 1s B. 2s C. 2px D. 3dxy

30. How many different sub - shells exist with principal quantum number, n = 3 ?

A. 3 B. 6 C. 9 D. 18

31. Which of the following is the most electronegative element ?

A. Li B. Na C. Be D. K

32. Which electronic sub shell do the lanthanides have incompletely filled ?

A. 4f B. 5f C. 6f D. 5g

33. In the reaction $^{223}_{87}Fr \rightarrow {}^{223}_{88}Ra + X$,particle X is :

A. A neutron
C. A beta particle
B. An alpha particle
D. A proton

34. How many different values can m_ℓ assume in the electron sub shell designated by quantum numbers n = 5, $\ell = 4$?

A. 9 B. 4 C. 8 D. 3

35. The energy of a photon having a frequency of 1×10^9 MH_z is

A. 6.626×10^{-19} J
C. 6.626×10^{-17} J
B. 6.626×10^{-25} J
D. 6.626×10^{-15} J

36. All of the following are isoelectronic. EXCEPT

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A. K^+ B. S^{2-} C. Ar D. Al^{3+}

37. The total number of s - electrons in Mg^{2+} are :

A. 6 B. 4 C. 10 D. 12

38. Which of the following is invalid electronic configuration ?

A. 2, 8, 8, 2 B. 2,8, 6 C. 2, 8, 9, 1 D. 2, 8, 18, 2

39. How many 3d electrons are present in the ground state of copper atom

A. 10 B. 9 C. 8 D. 5

40. Which of the following is correct decreasing order of size ?

A. $O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

B. $Al^{3+} > Mg^{2+} > Na^+ > F^- > O^{2-}$

C. $Na^+ > Mg^{2+} > Al^{3+} > F^- > O^{2-}$

D. $Mg^{2+} > Al^{3+} > Na^+ > F^- > O^{2-}$

41. The element whose atoms have the electron configuration of $(Kr)\,4d^{10}\,5s^2\,5p^2$:

A. is in group I I A C. is a transition element

B. is in period 4 D. has 50 protons.

42. The ratio of energy of photon $\lambda = 150$ nm to that of $\lambda = 300$ nm is

A. 1 : 2 B. 2 : 1 C. 4 : 1 D. 1 : 4

43. In an atom, the maximum number of electrons having the quantum numbers $n = 6$, $\ell = 3$, $m_s = +1/2$ is

A. 14 B. 7 C. 6 D. 1

44. The threshold frequency of a metal is $1 \times 10^{15}\ s^{-1}$. The ratio of maximum kinetic energies of the photo electrons when the metal is irradiated with radiations of frequencies $1.5 \times 10^{15}\ s^{-1}$ and $2.0 \times 10^{15}\ s^{-1}$ respectively, would be

A. 3 : 4 B. 2 : 1 C. 1 : 2 D. 4 : 3

45. Which of the following radiations has the highest wave number ?

A. X - ray B. UV C. IR D. Visible

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II Short answer question

46. Which one has larger size ?

a. Al or Al^{3+} c. Cu^{+} or Cu^{2+}

b. S or S^{2-} d. Be or B

47. Give the quantum number associated with each of the following .

a. Orbital shape c. Electron motion

b. Orbital energy d. Orbital orientation

48. When $\ell = 3$, the sub shell is _____

49. When n = 3, the value of ℓ can be _____, _____ and _____

50. When $\ell = 1$, the values of m_ℓ can be ___, _____, and ___

Answer to supplementary exercise

Part I

1. B	10. C	19. A	28. D	37. B
2. A	11. C	20. C	29. C	38. C
3. C	12. B	21. B	30. A	39. A
4. C	13. D	22. D	31. C	40. A
5. A	14. C	23. D	32. A	41. D
6. D	15. B	24. A	33. C	42. B
7. A	16. A	25. A	34. A	43. B
8. D	17. D	26. B	35. A	44. C
9. A	18. C	27. B	36. D	45. A

Part I I

46. a. Al b. S^{2-} c. Cu^{+} d. Be

47. a. ℓ b. n c. m_s d. m_l

48. f

49. 0, 1, and 3

50. - 1, 0 and 1

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UNIT 2

Chemical Bonding

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2.1 Introduction

Most of the substances around us are **compounds** and **molecules**.

Noble gases are stable or unreactive because they have full electron energy level. It would seem that when elements react to form compounds they do so to achive full electron energy levels. This idea forms the basis of the electronic theory of chemical bonding.

WHAT IS CHEMICAL BONDING?

- ☑ **Chemical bonding** is the attraction force that holds atoms or ions together.
- ☑ Species that have ns^2np^6 in their valance shell obey **the octet rule.**
- ☑ All nobel gases except **helium** obey the octet rule. The rest of the elements obey the octet rule either by losing,gaining or sharing electron
- ☑ The type of electrons that participate in bonds formation are **valance electrons.**

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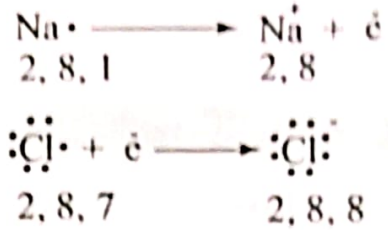
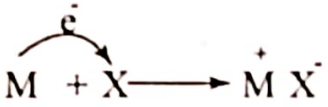
Example



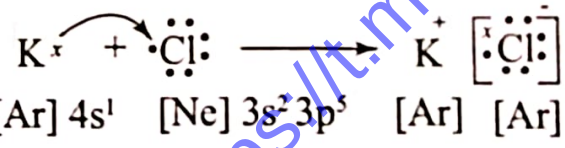
There are three type of bonds .These are ionic, covalent and metallic bonding.

2.2 Ionic Bonding

- ☑ It is also known as **electrovalent bond.**
- ☑ It is an attraction force between positive ion (cation) and negative ion (anion).It is usually formed between metal and non-metals.



It is formed by complete transfer of electrons from metal to non-metal.



Lewis electron dot symbols is an elemental symbol surrounded by dots,where each dot represent a valence electron.

Example

Lewis dot symbols of period 3 elements

Lewis structure: reprsentation of ionic compound cation without dot and anion with dots and their charges.

Example

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Illustrative Examples

1. Which of the following is not a form of chemical bonding?

A. Covalent bonding
C. metallic bonding
B. Ionic bonding
D. hydrogen bonding

Solution

Hydrogen bonding is an intermolecular force.It is not a chemical bond.

Answer: D

2. Which of the following compound is unlikely to contain ionc bonds?

A. CO
B. NaF
C. LiCl
D. $MgBr_2$

Solution

Ionic bond is usually formed between metals and non metals or ammonium ion and non metals.

Answer :A

3. Which of the statements below best explains when atoms react, chemically with each other ?

A. When atoms react,they gain proton and are more stable.
B. When atoms react,they lose all electrons and become more stable.
C. When atoms react,they lose ,gain or share electrons and are less stable.
D. When atoms react,they lose ,gain or share electrons to attain full outer energy level and are then more stable

Solution

All atoms except nobel gases react chemically to lose ,gain or share electrons to be stable by lowering their energy.

Answer D

4. Four elements A,B,C and D have atomic number ,Z-1, Z, Z+1 and Z

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+2 respectively. If Z is 9,then the bond between which pair of elements will be ionic?

A. A and C B. C and D C. B and D D. B and C

Solution

Metals have usually 1 to 3 valence electrons and non-metals from 4 to 8. Ionic bond is formed between metal and non-metals.

Element	Atomic number	Electron configration	Valence electron
A	Z - 1 = 9 + 1 = 8	2,6	6
B	Z = 9	2,7	7
C	Z + 1 = 9 + 1 = 10	2,8	8
D	Z + 2 = 9 + 2 = 11	2,8,1	1

Thus, an ionic bond is formed between D and A or D and B.

Answer: C

5. What is the difference between cation and anion?

Solution

Cations are positive ions formed by losing of electrons.

Anions are negative ions formed by gaining of electrons.

6. Which of the following forms ionic bond with bromine? Why?

Boron, Carbon, Oxygen, Potassium

Explanation

Potassium forms ionic bond with bromine because it is metal.

7. Draw the Lewis electron dot symbol for period 2

Solution

Li· ·Be· ·B· ·C· ·N· ·O· :F· :Ne:

8. Draw the Lewis structure for the following ionic compounnds.

a. NaCl b. Mg_3N_2 c. Al_2O_3 d. CaO

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Solution

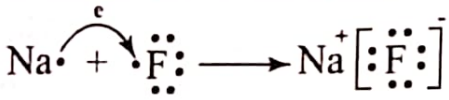
a. $Na^{+}[:\ddot{Cl}:]^{-}$ b. $Mg^{2+}_{3}[:\ddot{N}:]^{3-}_{2}$ c. $Al^{3+}_{2}[:\ddot{O}:]^{2-}_{3}$ d. $Ca^{2+}[:\ddot{O}:]^{2-}$

9. How many ionic bonds result from the formation of:

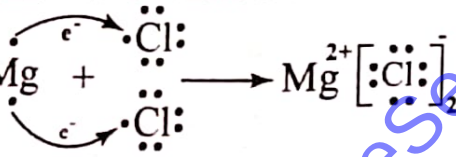
a. NaF b. $MgCl_2$ c. Li_3N

Solution

a. One ionic bond because a valence electron is transfered from Na to F.



b. Two ionic bonds since two valence electrons are transfered from Mg to Cl atoms.



c. Three ionic bonds because three electrons are transfered from three Li to N.

10. Ionic bond is also known as electrovalent bond. Why?

Explanation

Reason: An ionic bond is electrostatic force of attraction betw cations and anions.

11. Use sub-shell notation and Lewis dot symbol to reprsent the electro configuration of the following ions

a. Li^{+} b. P^{3-} c. Al^{3+} d. S^{2-}

Solution

Sub - shell Notation	Lewis Symbol
a. $1s^2$ or [He]	$[Li]^{+}$
b. $1s^2 2p^2 2p^6 3s^2 3p^6$ or [Ar]	$[:\ddot{P}:]^{3-}$

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c. $1s^2 2s^2 2p^6$ or [Ne] $[Al]^{3+}$

d. $1s^2 2s^2 2p^6 3s^2 3p^6$ or [Ar] $[:\ddot{S}:]^{2-}$

12. How elements obey the octet rule?

Solution

Nobel gases except helium obey the octet rule in their atomic form.

Most elements obey the octet rule by losing, gaining or sharing electrons.

13. Write the three type of bonds.

Solution

- Ionic bonding
- Covalent bonding
- Metallic bonding

Formation of Ionic Bonding

An ionic bond is formed by transfer of electrons.

Factors Affecting Formation of Ionic Bonding

i. Ionization Energy (IE)

Low ionization energy favors formation of ionic bonding.

ii. Electron Affinity(EA)

High electron affinity favors formation ionic bonding.

iii. Lattice energy(U): is the amount of energy required to convert a mole of ionic solid to its component ions in the gas phase .In the reverse ,it is the energy released when a mole of ionic compound is formed from isolated gaseous cation and anion.

☑ The magnitude of lattice energy is the measure of ionic compound's **stability.** The greater the lattice energy ,the more stable the is compound.

☑ Lattice energy is affected by **the size** and **magnitude** of the charge of ions.

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$U \alpha \frac{\text{Magnitude of charge}}{\text{size}}$ $U \alpha$ melting point

As lattice energy increases ,melting point also increasees.

Table 2.1 Lattice Energies of Selected ionic Compound

Compound	Lattice energy(KJ/ mol	Melting point(^{0}C)
LiF	1017	845
LiCl	860	610
LiBr	787	550
LiI	732	450
NaCl	788	801
NaBr	736	750
NaI	686	662
KCl	699	772
KBr	689	735
KI	632	680

Example

The lattice energy of: Na F > NaCl > NaBr > NaI

Reason: size of $F^- < Cl^- < Br < I^-$

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Lattice energy is calculated by using Hess's law through the Born-Haber cycle

☑ **Hess's law** states that enthalpy change, ΔH is independent on the paths the reactants proceed to form products.

Mathematically: $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 + ...$

Consider the formation of lithium fluoride,LiF

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Step 1: Conversion of lithium solid to lithium gas

$Li(s) \rightarrow Li(g)$, $\Delta H_{sub} = 155kJ$

Step 2: Decomposition of $F_2(g)$

$F_2(g) \rightarrow 2F(g)$, $\Delta H_{diss} = 158kJ/mol$

Step 3: Ionization of lithium atoms.

$Li^+(g) \rightarrow Li^+(g) + e^-$, $\Delta H_{IE} = 520kJ$

Step 4: Ionization of fluorine atoms

$F(g) + e^- \rightarrow F^-(g)$, $\Delta H_{EA} = -328kJ/mol$

Step 5: Bringing togther one mole of $Li^+(g)$, and $F^-(g)$ to form one mole of LiF(s).

$Li^-(g) + F^-(g) \rightarrow LiF(s)$, $U = -1349kJ$

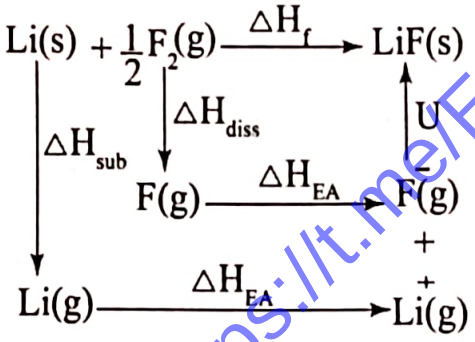

Fig 2.1 The Born - Haber Cycle for the Formation of Lithium Fluoride, LiF

Based on **Hess's law**:

$$\Delta H_f = \Delta H_{sub} + \Delta H_{diss} + \Delta H_{IE} + H_{EA} + U$$

Enthalpy (ΔH) is an extensive property ,changing its cofficent changes its quantity .

$1/2F_2(g) \rightarrow F(g)$, $\Delta H_{diss} = 1/2 \times (+158kJ) = +79kJ$

$\Rightarrow \Delta H_f = 155 + 79 + 520 - 328 - 1349 = -923kJ$

Illustrative Examples

14. The energy required to completely separate one mole solid ionic compound into gaseous ions is a/an (ESSLCE)

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A. Bond energy C. Activation energy

B. Lattice energy D. Ionization energy

Answer :B

15. Which of the following compounds would be expected to have the highest melting point?

A. BaF_2 B. $BaCl_2$ C. $BaBr_2$ D. BaI_2

Solution

As the size of the ions decreases both lattice energy and melting point increase.

Answer :A

16. Which of the following ionic compounds has the smallest lattice energy?

A. $Mg(OH)_2$ B. $Ba(OH)_2$ C. $Sr(OH)_2$ D. $Ca(OH)_2$

Answer: B

17. Formation of ionic bond is favored between elements that have

A. Low ionization energy and low electron affinity

B. High ionization energy and low electron affinity

C. High ionization energy and hight electron affinity

D. Low ionization energy and hight electron affinity

Solution

Elements with low ionization energy (metals) and high electron affinity (non - matals) favor formation of ionic bonding.

Answer: D

18. Which of the following is the most stable ionic compound?

A. NaCl B. NaBr C. NaF D. NaI

Solution

The higher the lattice energy, the more stable is the compound.

Answer: C

19. Arrange the following ionic compounds in decreasing order of solubility in water. NaCl, RbCl, KCl, CsCl

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Solution

As lattice energy increases, solubility of ionic compound in water decreases.

CsCl > RbCl > KCl > NaCl > LiCl

20. Calculate the lattice energy of cesium chloride, CsCl and draw the Born - Haber cycle.

ΔH_{sub} = 44kJ /mol

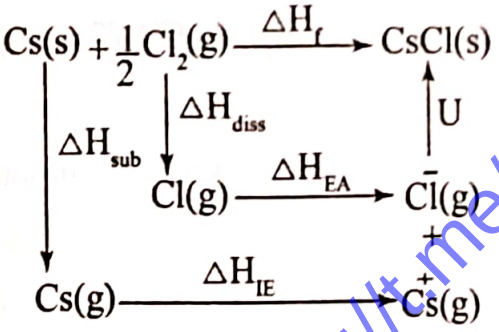
$\Delta H_{diss}(Cl_2)$ = 243kJ /mol

ΔH_{EA} (Cl) = -349 kJ/mol

ΔH_f (Cs - Cl (s)) = -423 kJ/ mol

ΔH_{IE} (Cs) = 376 kJ/ mol

Solution



$$\Delta H_{diss} = \frac{1}{2} \times 243 = 121.5 \text{ kJ/mol}$$

$$\Delta H_f = \Delta H_{sub} + \Delta H_{diss} + \Delta H_{IE} + \Delta H_{EA} + U$$

$$U = \Delta H_f - (\Delta H \text{ sub} + \Delta H\text{diss} + \Delta H\text{IE} + \Delta H_{EA} + U)$$

$$= -423 - (44 + 121.5 + 376 - 349) = -615.5 \text{ kJ/mol}$$

Exception to Octet Rule in Ionic Compounds

☑ If there is no eight electrons in the valence shell of cation, such type of ionic compound violates the octet rule. This occur in two ways;

i. Ionic Compounds Below Octet

Ionic compounds of lithium and berylium have less than eight electrons in the valence shell of the cations.

Example

LiF, Li_2O, BeF_2, BeO

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$\Rightarrow Li^{+} : 1s^2$ and $Be^{2+} : 1s^2$

ii. Ionic Compounds Above Octet

☑ The cations of such type of ionic compounds possess above eigh electrons in their valence shell.

Example

$FeSO_4$, ZnO, GaF_3, CuO etc.

$\Rightarrow Fe^{2+} : 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$

Valence shell configuration : $3s^2\ 3p^6\ 3d^6$

Number of valence electrons: 2 + 6 + 6 = 14.

The Eighteen Electrons Rules ($ns^2\ np^6\ nd^{10}$) (Pseudo Noble Gas Cofiguration)

☑ Some cations have 18 electrons in their valence shells. Compounds tha contain such cations obey the 18 electrons rule.

Example

Cu^{+}, Zn^{2+}, Ga^{3+}, Ag^{+}, Sn^{4+} etc

Note: Heavier post transition metals lose electrons from outer most and s orbitals. Transition metals lost electrons from the outr most s - orbital and then from d- orbital.

Properties of Ionic Compounds

1. Ionic compounds are aggregates of positive and negative ions.

Example

$Na^{+}\ Cl^{-}\ Na^{+}\ Cl^{-}\ Na^{+}\ Cl^{-}$

$Cl^{-}\ Na^{+}\ Cl^{-}\ Na^{+}\ Cl^{-}\ Na^{+}$

$Na^{+}\ Cl^{-}\ Na^{+}\ Cl^{-}\ Na^{+}\ Cl^{-}$

$Cl^{-}\ Na^{+}\ Cl^{-}\ Na^{+}\ Cl^{-}\ Na^{+}$

2. Ionic compounds are crystalline solids. They are not molecules.
3. They have high melting and boiling points

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4. They are hard and brittle.
5. They have high density
6. They are resistant to heat but many will be broken by heat.
7. Ionic compounds are usually soluble in polar solvents (water) but insoluble in non-polar solvents because they are polar compounds.
8. They are non - volatile
9. They are electrolytes (transinit elecricity in molten and aqueous states but not in soild state)

Example

NaCl(ℓ) - molten
NaCl(ℓ) - aqueous solution } Transmit electricity because ions are free to carry charge

NaCl(ℓ) - solid } Does not transmit electricity since the ions are not free to move and carry charge

Illustrative Example

21. All of the following are charactestics of most ionic compounds in their solid phases. **EXCEPT**

A. High melting point
B. Soluble in water
C. High electrical conductivity
D. Insoluble in organic solvents

Solution

Ionic compounds do not transimit electricity in solid state because the ions are not mobile.

Answer: C

22. Which of the following ionic compounds does not obey the octet rule?

A. NaI B. MgO C. Li_3N D. Al_2O_3

Solution

Li_3N does not obey the octet rule because there are only 2 electrons in the Li^+.

Answer: C

23. Which of the following cations obeys the 18 - electrons rule?

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A. V^{3+} B. Mn^{4+} C. Cd^{2+} D. Ca^{2+}

Solution

V^{3+} (Z = 23): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2$: 2 + 6 + 2 = 10

Mn^{4+} (Z = 25): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3$: 2 + 6 3 = 11

Cd^{2+} (Z = 48): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}$ = 18

Ca^{2+} (Z = 20): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$: 2 + 6 = 8

Answer: C

24. Check whether the following ionic compounds obey the octet rule or not.

a. SnO b. AlF_3 c. NiF_2 d. KCl

Solution

a. Sn^{2+} (Z = 50): (Kr) $4d^2\ 5p^2$, thus SnO does not obey the octet rule.

b. Al^{3+} (Z = 13): $1s^2\ 2s^2\ 2p^6$, thus AlF_3 obeys the octet rule.

c. Ni^{2+} (Z = 28): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_6\ 3d^8$, thus NiF_2 does not obey the octet rule.

d. K^+ (Z = 19): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$, thus KCl obeys the octet rule.

25. Write the formula and name of ionic compounds formed between the following elements.

a. Calcium and nitrogen
b. Aluminum and fluorine
c. Magnesium and oxygen
d. Barium and sulfur

Solution

a. Ca_3N_2 - Calcium nitride
b. AlF_3 - Aluminum fluoride
c. MgO - Magnesium oxide
d. BaS - Barium sulphide

2.3 Covalent Bonding

Covalent dond is the inter - atomic attraction resulting from sharing of electrons between the atoms.

☑ The shared electrons between the nuclei of the bonded atoms are called **boning paires (bp).**

☑ The unshared electrons are known as **non-bonding pairs or lone pairs (LP)**

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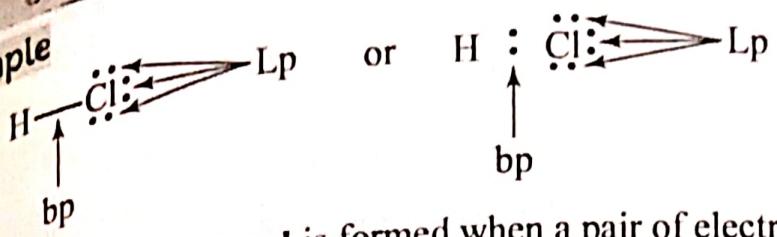
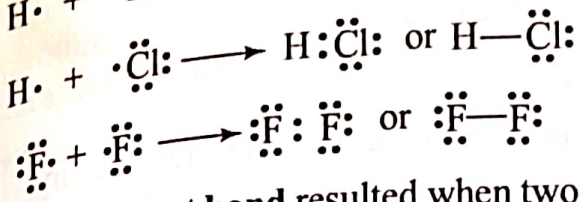
Example



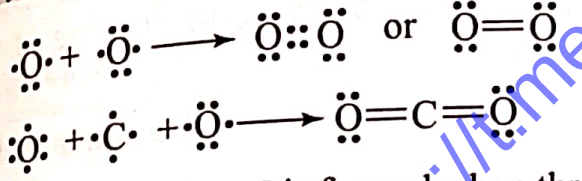
☑ **Single convalent bond** is formed when a pair of electrons shared.

Example



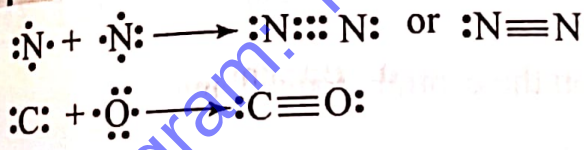
☑ **Double covalent bond** resulted when two pairs of electrons are shared.

Example



☑ **Triple covalent bond** is formed when three pairs electrons are shared.

Example



Bond strength: Triple bond > double bond > single bond

Bond length: Single bond > double bond > triple bond

Lewis structure for Covalent Compounds

It is a representation of covalent bonding in which shared electrons pairs are shown either as dashes or as pairs of dots between two atoms and lone pair are shown as pairs of dots on individual atoms.

The basic steps to draw the correct Lewis structure of a covalent compound are as follows .

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1. Determine the total valence electrons and pair them.

Example

Consider SF_4 (Z: S = 16 , F = 9)

S : 2 , 8 , 6 : valence electrons = 6

F: 2 , 7 : valence electrons = 7

S: $1 \times 6 = 6$

F: $4 \times 7 = 28$

Total valenc electrons = 6 + 28 = 34

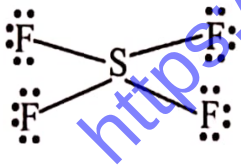
pairs = 34/2 = 17

2. Identify the **central atom** (more positive or larger in size atom)

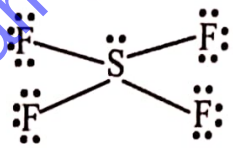
⇒ S is the central atoms in SF_4.

Hydrogen and fluorine cannot be used as central atoms because hydrogen has the smallest size and fluorine is the most electronegative element.

3. Draw the skeletal structure and fulfill the octet of the terminal atoms except for hygrogen.



4. Place the raining electrons on the central atoms if any.



bp = 4
Lp = 13

5. Make multiples bonds (double or triple) by using lone pair of electrons on the terminal atoms to fulfill the octet of the central atoms if necessary.

Example:

Consider SO_3

S : 2, 8 , 6 O : 2 , 6

Total valence electrons = $1 \times 6 + 3 \times 6 = 24$

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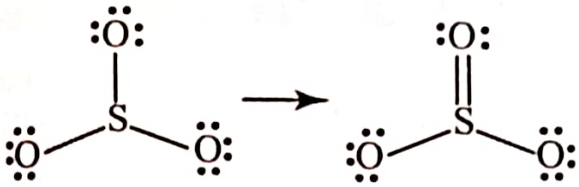
pairs = 24\2 = 12

centeral atoms : S



Formal charge = valecne electrons - associated electrons

To determine the number of electrons associated with an atoms in a Lewis structure, keep in mind the following .

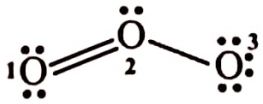
- All the atom's non bonding electrons are associated with atoms.
- Half of the atom's bonding electrons are associated with the atoms.

Example

Determine the formula Charge of each oxygen atom in O_3.

Solution

First draw the Lewis structure of O_3.



For O - 1 : Formal charge

= 6 - (4unshared electrons +4 shared electron/2)

= 6 - (4 + 2) = 0

For O - 2: Formal charge

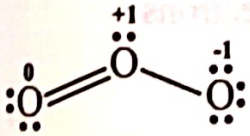
= 6 - (2unshared electrons + 6 shared electrons/2

= 6 - (2 + 3) = +1

For O - 3: formal charge

= 6 - (6unshared electrons + 2shared electrons/2)

= 6 - (6 + 1) = -1



sum of formal charges must be equal to the charge of the molecule.

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$\Rightarrow$ Charge $O_3 = 0 + 1 - 1 = 0$

$\Rightarrow$ Formal charges do not represent actual charges on atoms in molecule.

Illustrative Examples

26. How many non-bonding pairs of electrons are present in the valence shell of the central atom in ClF_3? (ESSLCE)

A. 2 B. 3 C. 0 D. 1

Solution

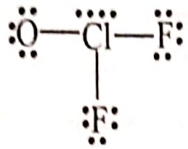
Cl : $1 \times 7 = 7$

F: $3 \times 7 = 21$

Sum 28

Pairs $= \frac{28}{2} = 14$

Central atom Cl



Lp on Cl = 2

Answer :A

27. How many bonding pairs and lone pairs receptively does the ion ICl_4^- have?

A. 3, 2 B. 4, 2 C. 5, 1 D. 4, 1

Solution

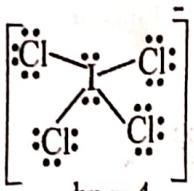
I : $1 \times 7 = 7$

Cl: $4 \times 7 = 28$

Charge $= \frac{+1}{36}$

Sum = 36

Pairs = 18



bp = 4

Lp on I = 2

Answer :B

28. What does the correct Lewis structure for CCl_4 molecules indicate?

A. 5 bonds C. No unshared electrons

B. 24 unshared electrons D. 10 shared electrons

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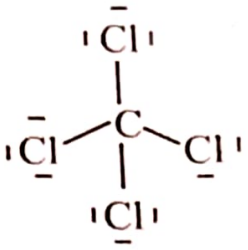
Solution

C : 1 × 4 = 4

Cl : 4 × 7 = 28

sum = 32

Pairs = 16

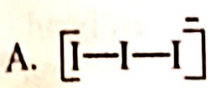
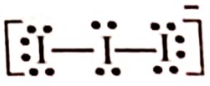
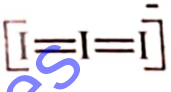


Lp = 12

bp = 4

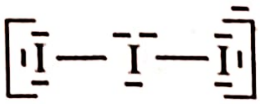
Answer :B

29. Which of the following Lewis structure is correct for tri iodide ion, I_3^-?

A.  B. C.  D. 

Solution

Total valence electrons = 3 × 7 + 1 = 22, pairs = 11



Answer: B

30. Draw the Lewis structure and determine the numbers of bonding and lone pairs.

a. NO^+

b. ClO_4^-

c. SCl_2

d. PH_4^+

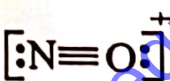
e. BrF_3

f. CO_3^{2-}

g. CH_3COOH

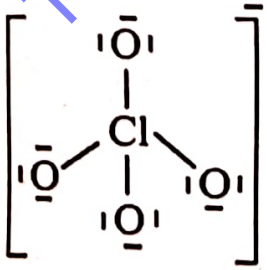
h. $COCl_2$

Solution

a. 

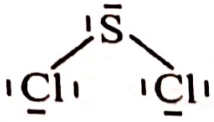
bp = 3

Lp = 2

b. 

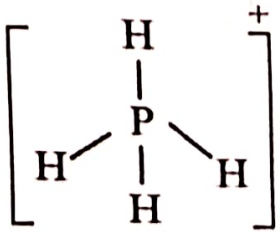
bp = 4

Lp = 12

c. 

bp = 2

Lp = 8

d. 

bp = 4

Lp = 0

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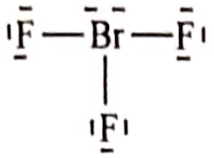
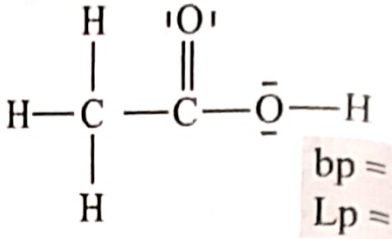
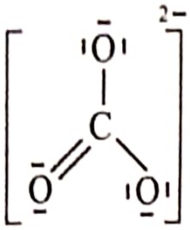
e. 

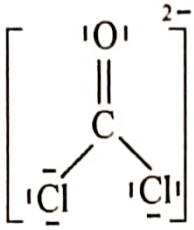
bp = 3
Lp = 11

g. 

bp = 8
Lp = 4

f. 

bp = 4
Lp = 8

h. 

bp = 4
Lp = 8

31. Arrange single,double and triple bonds in decreasing order of bond strength .

Solution

Triple bond > double bond > single bond

32. What is the attraction force in covalent bonding.

Solution

The attraction force in covalent bonding is the force between the shared electrons and the nuclei of the bonded atoms.

33. Determine the formal charges of each atom in nitrate ion (NO_3^-).

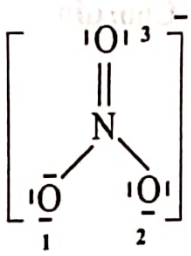
Solution

N : $1 \times 5 = 5$

O : $3 \times 6 = 18$

charge = $\frac{+1}{\text{sum} = 24}$

Pairs = 12



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$$N = 5 - \left(\text{O unshared element} + \frac{8}{2}\right)$$

$$= 5 - 4 = +1$$

$$O\text{-}1 = 6 - \left(6\,\text{unshared electrons} + \frac{2}{2}\right)$$

$$= 6 - (6 + 1) = -1$$

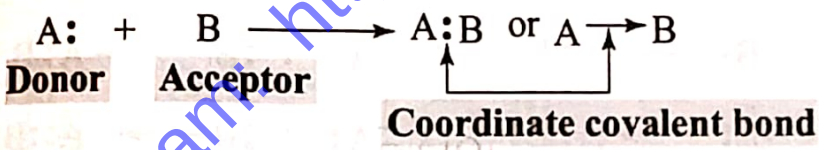
$$O - 2 = 6 - \left(6 + \frac{2}{2}\right) = -1$$

$$O - 3 = 6 - \left(4 + \frac{4}{2}\right) = 0$$

Sum of formula charges = +1 - 1 - 1+ 0 = -1 , thus the Lewis structure drawn is correct.

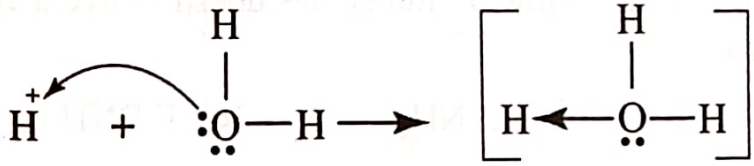
Coordinate Covalent Bond (Dative bond)

- ☑ It is also called **donor-acceptor bond**
- ☑ It is inter-atomic attraction resulting from the sharing of a lone pair of electrons from one atom with another atom.
- ☑ The shared electrons are obtained from one atoms and the other atom has vacant or unfulfilled orbital.



Examples

a) $H_3\overset{+}{O}$:



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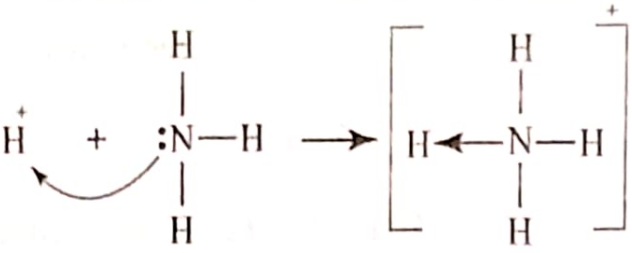
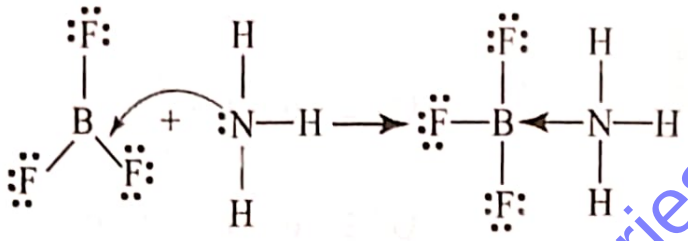
b) NH_4^+ :



c) F_3BNH_3 :



The arrow represent coordinate covalent bond

Illustrative Example

34. The type of bond formed when the shared electrons are contributed by one of the bond atom is

A. Covalent bond
C. Polar bond
B. Coordinate covalent bond
D. Ionic band

Answer: B

35. Which of the following shows the formation of coordinate covalent bond?

A. $A\cdot + B\cdot \longrightarrow A:B$
C. $A + :B \longrightarrow A:B$
B. $A: + \cdot\ddot{B}: \longrightarrow A:\ddot{B}:$
D. $A: + :B \longrightarrow A::B$

Answer: C

36. Which of the following molecules doesn't have a coordinate covalent bonds?

A. CO_2
B. NH_4
C. F_3BNH_3
D. H_3O^+

Answer: A

37. Identify the coordinate covalent bond in the following species by drawing Lewis structure.

a. HNO_3
b. CO
c. $POCl_3$

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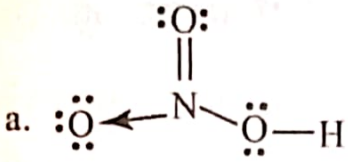
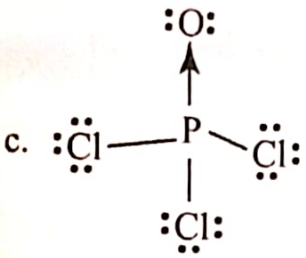
Solution

a. 

b.



The bold arrows represent the coordinate covalent bond

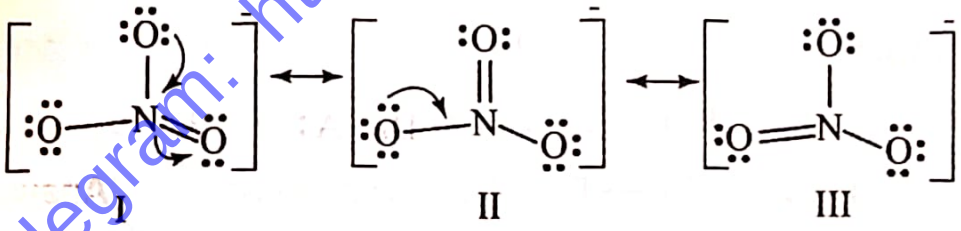
Resonance Structure

Resonance is a condition occurring when more than one valid Lewis structures can be written for a particular molecule due to the presence of multiples bonds.

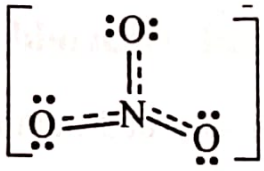
The various structures are called **resonance structures** and the actual species is called **resonance hybird.**

Example

Consider NO_3^-



I II III



IV

The structure I, II and III are resonance structure and IV is resonance hydrid.

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Exception to Octet Rule in Covalent Bonding

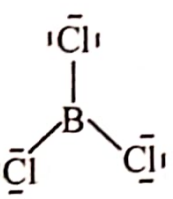
Covalent compounds that do not have eight shared electrons violate the octet rule.

1. Covalent Compounds Below Octet

These covalent compounds have less than eight electrons around the central atom.

Example

$BeCl_2$, BCl_3, $AlCl_3$ etc



: 4 electrons on Be

: 6 electrons on B

2. Covalent Compounds Above Octet

Such type covalent compounds have above eight electrons around the central atom.

Example

SF_4, SF_6 ,PCl_5 ,XeF_2, XeF_4, etc

XeF_2 : **10 electrons on Xe**

3. Covalent Compounds with odd Number of Electrons

All covalent compounds with odd number of electrons violate the octet rule.

Example

NO, NO_2 ,ClO_2, ClO_3 etc

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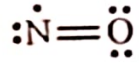
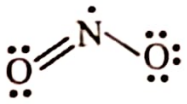
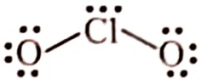
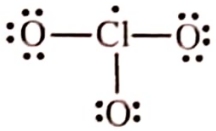
NO: 

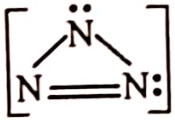
NO_2: 

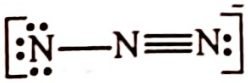
ClO_2: 

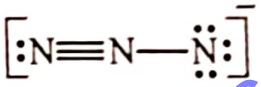
ClO_3: 

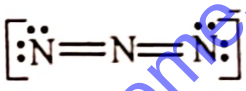
Illustrative Example

38. Which of the following is not a preferred resonating structure of azide, N_3^-?

A. 

B. 

C. 

D. 

Solution

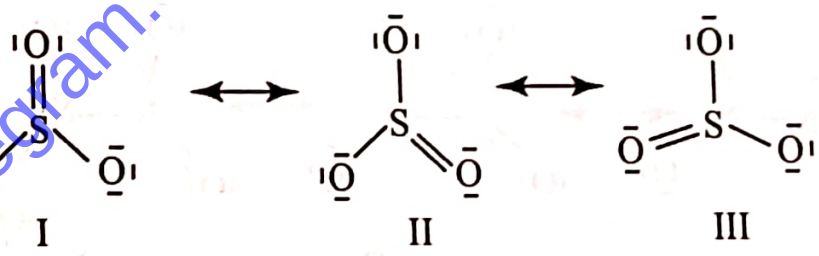
A cannot be resonance structure because sum of formal charges is not equal to -1 or total valence electrons is not even.

Answer: A

39. How many resonance structure does SO_3 have?

A. 1 B. 2 C. 3 D. 4

Solution



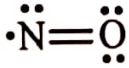
I II III

Answer: C

40. Which speices has Lewis structure that does not obey the octet rule?

A. NO B. CS_2 C. PF_3 D. HCN

Solution

NO:  does not obey the octet rule

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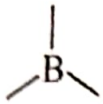
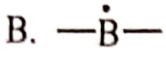
Answer :A

41. In which patter does the central atom B obey the octet rule?

A.  B.  C. D.

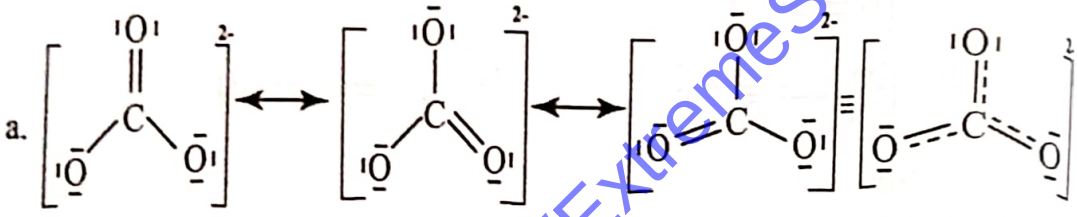
Answer :D

42. Draw the resonance structures of

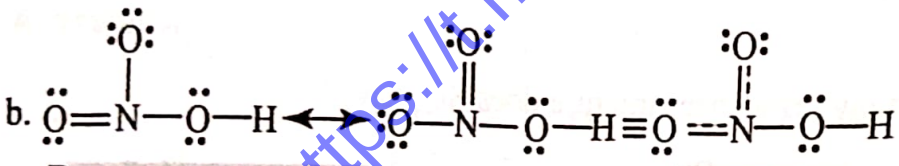
a. CO^{2-}_3 c. N_2O_4 e. NO_2^-

b. HNO_3 d. NO_2F f. C_6H_6

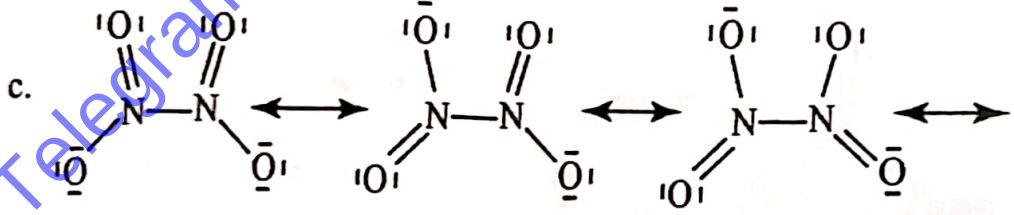
Solution

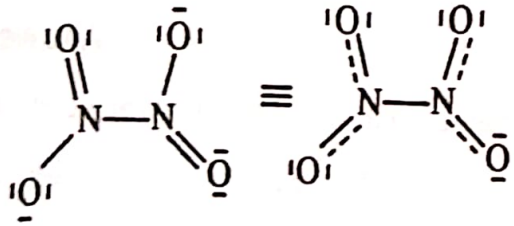
a. 

Resonance structures **Resonance hybrid**

b. 

Resonance structures **Resonance hybrid**

c. 



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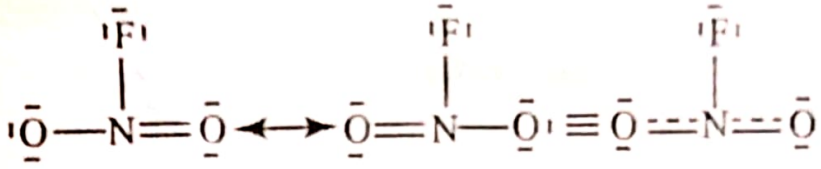
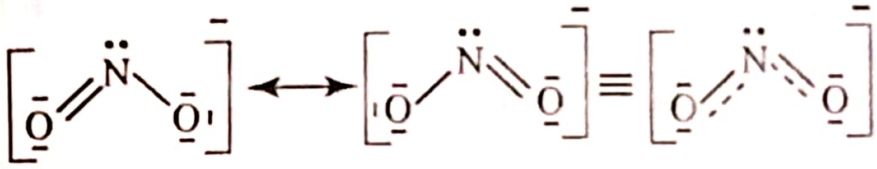
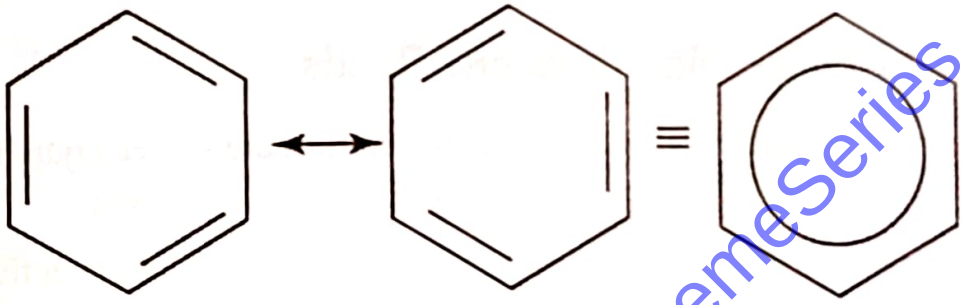
d. 

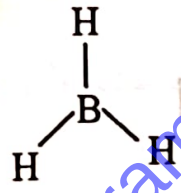
e. 

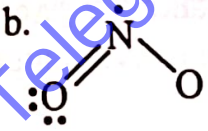
f. 

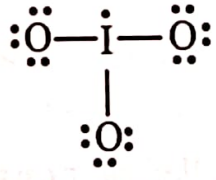
43. All of the following species violate the octet rule. Classify these species as below octet ,above octet or with odd number of electrons.

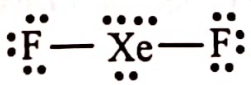
a. BH_3
b. NO_2
c. IO_3
d. XeF_2
e. ICl_4^-
f. AlH_3

Solution

a.  **below octet**

b.  **odd number of electrons**

c.  **odd number of electrons**

d.  **above octet**

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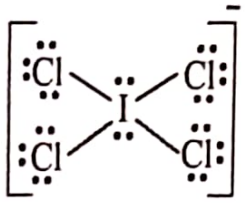
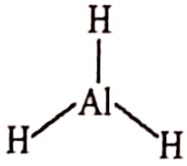
e.  **above octet**

f.  **below octet**

Polar and Non-polar Covalent Bonds

Non-polar covalent bond is a bond in which electrons are shared equally.

- ☑ It is a bond with a electro negativity difference less than **0.5**.
- ☑ It is a bond formed between elements that have similar electronegtivities.
- ☑ Molecules formed by non-polar covalent bond are called **non-polar molecules.**
- ☑ There are non-polar molecules with **polar covalent bonds** but with zero dipole moment.

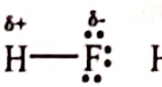
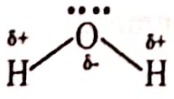
Example:

CO_2, CCl_4

Polar covalent bond : is bond formed when electrons are uneqally shared.

- ☑ It is a bond formed between elements that have different electronegativities.
- ☑ The more electronegtive elements carries a **partially negative charge(δ-)** and the less electron-negative elements carries a partially positive charge (δ^+)

Example

HF:  H_2O: 

- ☑ Molecules formed by polar covalent bonds are called **polar covalent molucules.**
- ☑ The measure of the polarity of a molecule is known as **dipolar moment(μ)**
- ☑ The greater the dipole moment,the more polar is the bond

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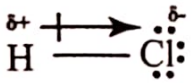
☑ The dipole moment or electronegative difference of a polar covalent compound is from 0.5 to 2.5.

☑ The dipole character of a molucule is offen reprsented by an arrow (⟶). This arrow points toward the negative charges center.

Example



☑ **Dipole moment** is defined as the product of the magnitude of separated charges and the distance between them.

$$\mu = q \cdot d$$

☑ The SI unit of μ is colomb meter(C.m) and the other unit is Debye(D).

$1D = 3.33 \times 10^{-30} C.m$

☑ As the electronegativits difference between the bonded atom increases,polarity and magitude of dipole -moment also increase.

Example

Polarity: HF > HCl >HBr > HI

Illustrative Examples

44. Which bond between the following groups of elements will be the most polar?

A. S—F B. S—Cl C. S—Br D. S—I

Solution

As the electro negativity difference increases,the polarity of the bond are increases.

Answer: A

45. Dichloroethlene,$C_2H_2Cl_2$, has the following Lewis structure. Which one has a dipole-moment of zero?

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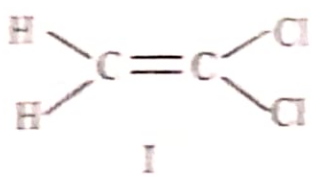
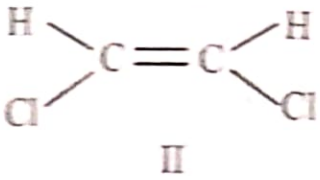
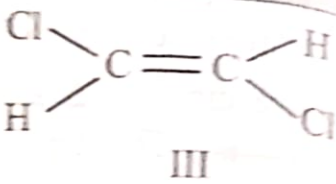


I



II



III

A. I B. II C. III D. II and III

Answer: C

46. Which of the following molecules is non-polar?

A. CS_2 B. SO_2 C. $CHCl_3$ D. SF_4

Solution

Draw the Lewis structure and predict the dipole moment using the vector method .If the resultant vector is zero,then μ will be zero and if it is different from zero,μ will be different from zero.

CS_2 $\ddot{S}=C=\ddot{S}$ $\vec{R} = 0 \Rightarrow \mu = 0$ thus

CS_2 is non - polar.

Answer: A

47. Which of the following compound has $\mu = 0$?

A. CCl_4 B. $CHCl_3$ C. HF D. NH_3

Answer: A

48. The molecule of which of the following substance has $\mu > 0$?

A. Water C. Carbon dioxide

B. Methane D. Nitrogen

Answer: A

49. Arrange the following molecules in decreasing order of polarity based on the given dipole moment?

Molecule	dipole moment(D)
I. HCl	1.07
II. H_2O	1.85
III. NF_3	0.23
IV. CO	0.12

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V. NH_3 1.47

Answer: II>V>I>III>IV

50. Classify the following molecules as polar or non-polar

a. $CHCl_3$ c. CH_3Cl e. PH_3 g. $TeCl_2$

b. CH_2Cl_2 d. CCl_4 f. BCl_3 h. PCl_5

Solution

a, c, e and g are polar covalent molecules.

b, d, f and h are non-polar covalent molecules.

51. Write three non-polar molecules with polar bonds.

Solution

CO_2 ,CCl_4 , BCl_3

52. Is there a polar covalent molecule with non-polar bond?

Solution

There is no polar molecule with non-polar bond

Properties of Covalent Compounds

1) They exist in three states of matter at room temperature
   - ➥ H_2 ,O_2 ,CH_4 ,CO_2, NH_3 → are gases
   - ➥ H_2O, CCl_4, H_6H_6 ,C_2H_5OH → are liquid
   - ➥ I_2 $C_{20}H_{42}$ $C_{22}H_{46}$ → are solids
2) They are volatile .
3) They have generally low boiling and melting points.
4) polar covalent compounds have higher melting and boling points than non polar covalent compounds of similar size.

Example:

$ICl > I_2$

5) Non polar compounds are non-electrolytes.
6) Some polar covalent compounds (acids) are electrolyte

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7) Polar conalent compounds are soluble in polar solvent but non- polar covalent compounds are solubel in non -polar solvents.

Illustrative Examples

53. Which of the following is a property of covalent compound? covalent compounds(ESSLCE...)

A. are generally insoluble in water.

B. are crystalline solid at room temperature

C. are non- volatile

D. have high melting and boiling points.

Solution

Most covalent compounds are non-polar. Thus ,they are generally insolubel in water.

Answer: A

54. Which of the following is a covalent moleucule?

A. NH_4^+ B. CO_3^{2-} C. PCl_5 D. NaCl

Solution

* NH_4^+ and CO^{2-}_3 are covalent molecular ion .

* PCl_5 is covalent molecule

* NaCl is ionic compound.

Answer: C

55. Which of the following has the lowest boiling point?

A. NH_3 B. H_2O C. HF D. HBr

Solution

The more polar the compound ,the higher is the boiling point.

Answer :D

56. Which one has higher boiling point? Explain

A. I_2 or I Cl B. H_2O or H_2S C. HF or HCl

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Solution

a. ICl because it is polar

b. H_2O because it is more polar than H_2S

c. HF because it is more polar than HCl.

Moleular Geometry/shape/

☑ The arragement of bonded atom in three dimentions is known as **molecular geometry.**

☑ **Electron pairs geometry or electron pair arrangment** is the arrangment of electron domains(bonding pairs and lone pairs) around the central atoms.

One set of electrons=one singel bond,one doubel bond,one triple bond,one lone pair or one electron

Types of Electron Pair Arrangments

There are five types of electron pair arrangment. These are;

1. Linear Electron Pair Arrangment

➥ Presence of two sets of electrons around the central atom.

Example

HCN ,CO_2 ,CS_2 ,$BeCl_2$, C_2H_2 etc

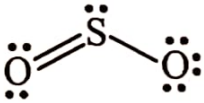
H—C≡N: There are two sets of electrons around carbon.

2. Trigonal Planar Electron Pair Arrngment

➥ Three sets of electrons around the central atoms.

Example

SO_3 , BCl_3 , SO_2 , O_3, C_2H_4 etc

SO_2: SO_2:  There are three sets of electrons around S.

3. Tetrahedral Electron Pair Arrangment

➥ Four sets of electrons around the central atoms.

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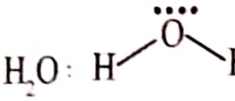
Example

H_2O, NH_3, CH_4, SiC_4, H_3O^+, SO_4^{2-} etc



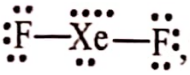
H_2O: There are four sets of electrons around O

4. Trigonal Bipyramid Eletron pair Arrangment

Presence of five sets of electrons around the central atom.

Example

PCl_5 ,ICl_3 ,XeF_2 ,SF_4, etc



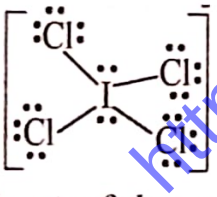
XeF_2: XeF_2 : There are five sets of electron around Xe.

5. Octahedral Electron Pair Arrangement

☑ Presence of sex sets of electrons around the central atom.

Example:

SF_6 ,BrF_5, ICl_4^- ,XeF_4 etc



ICl_4^-: **There are six sets of electrons around I**

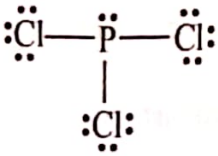
There are six sets of electrons around I

Illustrative Example

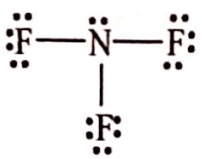
57. All of the following have tetrahedra electron pair arrangement. **EXCEPT**

A. PCl_3 B. NF_3 C. H_2O D. CO_2

Solution



PCl_3: **Tetrahedral**



NF_3: **Tetrahedral**

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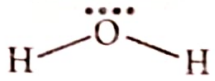
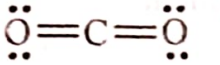
H_2O:  Tetrahedral

CO_2:  Linear

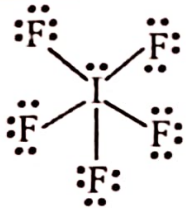
Answer :D

58. Determine the number of sets of electrons and identify the types of electron pair arrangements in the species below.

a. IF_5 c. SO_3^{2-} e. FbF_5

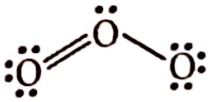
b. O_3 d. CS_2 f. $TiCl^{2-}_6$

Solution

a. 

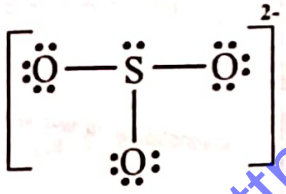
➥ Six sets electrons

➥ Octahedral electron pair arrangement

b. 

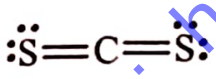
➥ Three sets of electron

➥ trigonal planar electron pair arrangement,

c. 

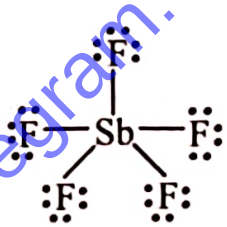
➥ Four sets of electrons

➥ Tetrahedral electron pair arrangement

d. 

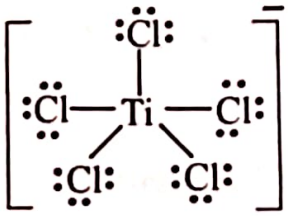
➥ Two sets electron

➥ Linear electron pair arrangement

e. 

➥ Five sets of electron

➥ Tirgonal bi pyramidal electron pair arrangement

f. 

➥ Six sets of electrons

➥ Octahedral electron pair arrangement.

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Molecular shape

- ☑ Molecular ion and molecules have three dimentional shapes.
- ☑ **Bond angle** is the angle between two atoms bounded to central atom in a molecule.
- ☑ Lewis can predict shape of molecules using Lewis structure and valence shell electron pair repulsion(VSEPR) model.
- ☑ The basis of the VSEPR model is that electron pair in the valence shell of an atoms repel one another.
- ☑ The repulsion force is in the order of:

Lp - Lp > Lp -bp > bp - bp

- ➥ Lp - Lp :lone pair - Lone pair repulsion.
- ➥ Lp - bp :lone pair - bonding pair repulsion.
- ➥ bp - bp :bonding pair - bonding pair repulsion.

Molecular shape is the same as electron pair arrangement if there is no lone pair on the central atom

1. Molecular Shape with two Sets of Electrons AX_2 Type

- ➥ Electron pair arrangement: linear
- ➥ Molecular shape : **linear**

- ➥ Bond angel:**180⁰**
- ➥ A : central atom, X :bonding set electron

Example

$BeCl_2$,CO_2 ,CS_2, HCN ,C_2H_2, etc

$BeCl_2$: :Cl—Be—Cl:

180°

2. Molecular Shape with Three Sets of Electrons

i. AX_3 type

- ➥ Electron pair arrangment: trigonal planar
- ➥ Molucular shape: **trional planar**

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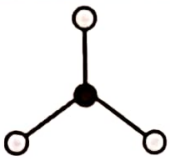
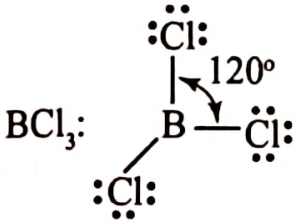


➥ Bond angel :120°

Example

BCl_3 ,$AlCl_3$, NO^-_3 ,CO^{2-}_3 .etc



ii. AX_2E type E: lone pair electrons

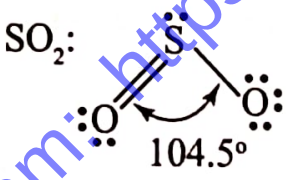
☑ Electron pair arrangement: trigonal planar

☑ Molecular shape: **bent ,angular or V - shape**

➥ Bond angel :**104.5⁰**

Example

SO_2 ,O_3 etc



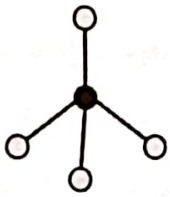
3. Molecular Shape with four Sets of Electrons

i. AX_4 type (4bp, 0LP)

☑ Electron pair arrangement :tetrahedral

☑ Molecular shape :**tetrahedral**

☑ Bond angel:**109.5⁰**



Example

CH_4 ,NH^+_4 ,$SiCl_4$, PH^+_4

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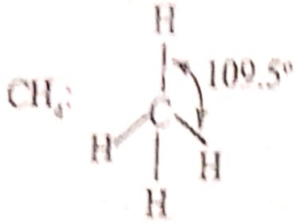


ii. AX_3E type(3bp,1Lp)

- ☑ Electron pair arrangement :tetrahedral
- ☑ Molecular shape :

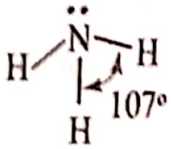
trigonal pyramid

- ☑ Bond angel: **107°**

Example

NH_3 ,PH_3 ,H_3O^+ ,PCl_3 NCl_3 ,etc

NH_3:



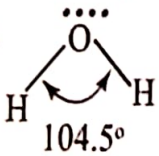
iii.AX_2E_2Type (2bp ,2Lp)

- ☑ Electron pair arrangement :tetrahedral
- ☑ Molecular shape: **bent, angular or V- shape**
- ☑ Bond angel: **104.5°**

Example

H_2O ,OF_2 ,NH_2^- etc

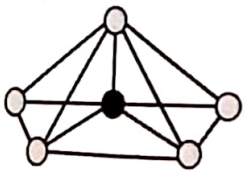
H_2O:



4. Molecular Shape with Five Sets of Electrons

i. AX_5 type(5Bp,0Lp)

- ➥ Electron pair arrangement: trigonal bi pyramid
- ➥ Molecular shape: **trigonal bipyramid**
- ➥ Bond angle : **90°,120°**



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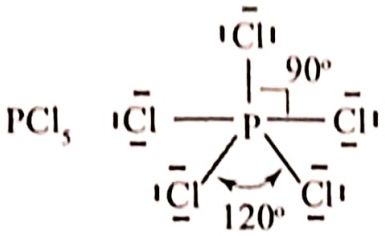
Example

PCl_5 ,SbF_5 $SiCl_5$ etc

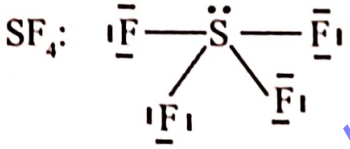


ii. AX_4E type(4bp,1Lp)

- Electron pair arrangement: trigonal bi pyramid
- Molecular shape : **see - saw**
- Bond angle: **90°**

Example

SF_4 ,ICl_4^+ , XeO_2F_2 etc

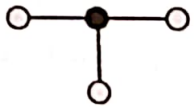


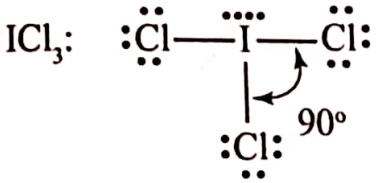
iii. AX_3E_2 type(3bp,2Lp)

- Electron pair arrangement: trigonal bi pyramid
- Molecular shape: **T - shape**
- Bond angle: **90°**

Example

ICl_3, BrF_3, ClF_3 etc.





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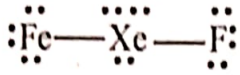
iv. AX_2E_3 type (2bp,3Lp)

- Electron pair arrangement = trigonal bi pyramid
- Molecular shape : **linear**
- Bond angle :**180°**

Example

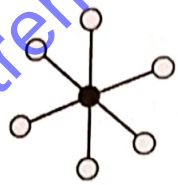
XeF_2, ICl_2, BrF_2 etc

XeF_2: 

5. Molecular Shape with six Sets of Electrons

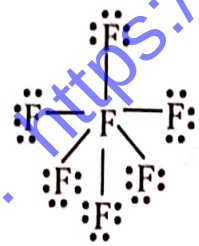
i. AX_6 type (6bp ,0LP)

- Electron pair arrangement = octahedra
- Molecular shape :**Octahedral**
- Bond angel :**90°, 180°**



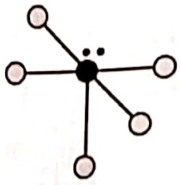
Example

SF_5, PCl^-_6, $SiCl^{2-}_6$ etc

SF_6: 

ii. AX_5E type(5bp ,1Lp)

- Electron pair arrangement: octahedral
- Molecular shape: **square pyramid**
- Bond angle: **90°**



Example

BrF_5 , ICl_5 , TeF^-_5 ,$SeCl^-_5$ etc

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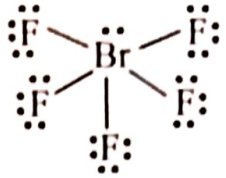
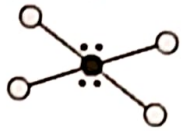
BrF_5:



iii. AX_4E_3 type(4bp, 2Lp)

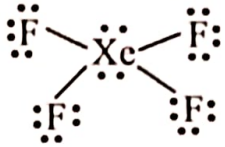
- Electron pair arrangement :octahedral
- Molecular shape : Square planar
- Bond angel :90°



Example

XeF_4 , ICl_4 , BrF_4^- etc

XeF_4:

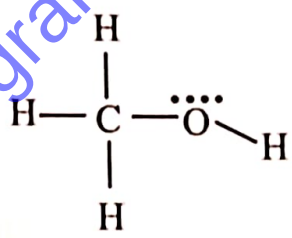


Molecular shape of Molecules with More than One Central Atoms

The molecular shape and electron pair arrangement of molecules with more than one central atoms can be predicted with respect to each central atom.

Example

CH_3OH



Molecular shape with respect to C : tetra hedral
O : V - shape

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Illustrative Example

59. The shape of H_3O^+ is

A. Tetrahedral
B. Pyramidal
C. trigonal planar
D. Angular

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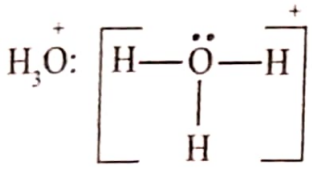
Solution



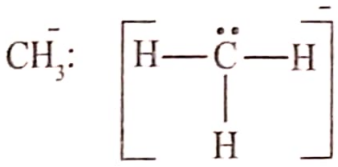
Molecular shape : trigonal pyramid

Answer: B

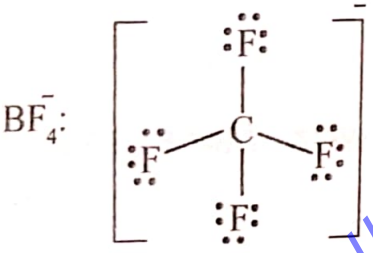
60. Which of the following species assumes planar structure ?

A. CH_3^- B. BF_4^- C. CH_3^+ D. CH_4

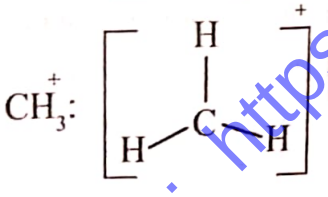
Solution



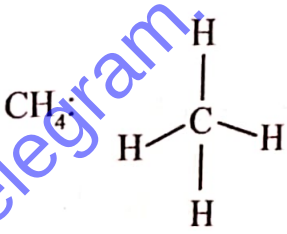
Trigonal pyramid



Tetrahedral



Trigonal planar



Tetrahedral

Answer: C

61. Which of the following does not have a tetrahedral atom?

A. SF_4 B. AH_4^- C. BF_4^- D. SiC_4

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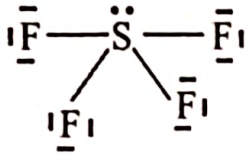
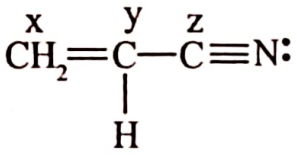
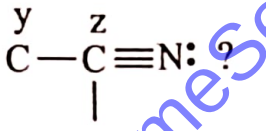
Solution

SF_4:  **See - saw shape**

Answer :C

62. A crylonitrile has the following Lewis structure with designation of X, Y and Z for each carbon atom.





What angel and geometry of

A. 109^0, tetrahedral
C. 180^0, linear
B. 120^0, trigonal pyramidal
D. 90^0 T - shaped

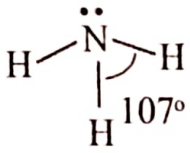
Answer :B

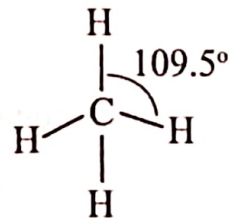
63. Which of the following species has the smallest H-X-H bond angel where X is central atoms?

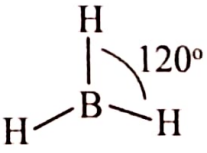
A. OH_2 B. NH_3 C. CH_4 D. BH_3

Solution

OH_2:

NH_3: 

CH_4: 

BH_3: 

Answer :A

64. Antimony (Sb) is a group VA elements. What will be the molecular geometry and number of lone pair electrons respectively that exist in

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the ion $SbCl_{+6}^{2-}$?

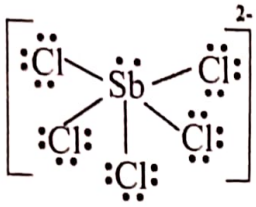
A. See - saw ,1

B. T - shape , 2

C. Linear , 3

D. Square pyramid, 1

Solution



Square pyramid, 1

Answer: D

65. The H - N - H bond angels in NH_2^-, NH_4^+ and NH_3 respectively are

A. 104.5°, 109.5° ,107°

B. 107°, 109.5° ,104.5°

C. 109.5°, 107° 104.5°

D. 109.5° 104.5° ,107°

Solution

NH_2^- =AX_3E_2 type .bond angel = 104.5°

NH_4^+ = AX_4 type , bond angel = 109.5°

NH_3 = AX_3E type bond angel =107°

Answer: A

66. Which of the following is not probable shape of a molecule?

A. $GeCl_4$, tetrahedra 1

B. HCN, bent

C. PCl_5, trigonal bi pyramid

D. BrF_3 T - shape

Solution

HCN : H - C ≡ N, linear shape:

Answer :B

67. Which reaction involves a change in electron pair geometry for the underlined atom ?

A. $\underline{B}F_3 + F^- \longrightarrow \underline{B}F_4^-$

B. $\underline{N}H_3 + H^+ \longrightarrow \underline{N}H_4^+$

C. $2\underline{S}O_2 + O_2 \longrightarrow 2\underline{S}O_3$

D. $H_2\underline{O} + H^+ \longrightarrow H_3\underline{O}^+$

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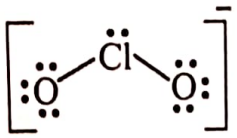
Answer: A

68. Use the VSEPR model to predict the electron pair arrangment and molecular shape of:

a. ClO^-_2 c. ICl^+_4 e. $SbCl_6$

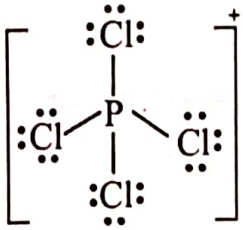
b. PCl^+_4 d. AsH_3 f. C_2H_2

Solution

a. 

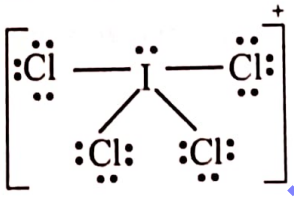
Shape: bent

electron pair arrangement :tetrahedral

b. 

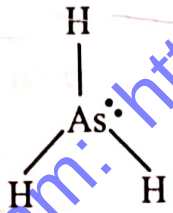
Shape : tetrahedral

electron pair arrangement :tetrahedral

c. 

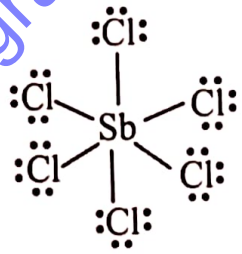
Shape: see - saw

electron pair arrangement: trigonal bi pyramid

d. 

Shape: trigonal pyramid

electron pair arrangement: tetrahedral

e. 

Shape: octahedral

electron pair arrangement: octahedral

f.

Shape: linear

electron arrangement: linear

69. Predict the shape of the following with respect to each central atom.

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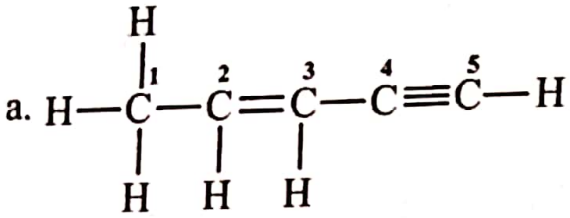
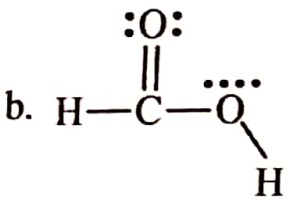
a. $CH_3-CH=CH-C\equiv CH$

b. HCOOH

Solution

a. 

- C - 1: tetrahedral
- C - 2 and C - 3: trigonal planar
- C - 4 and C - 5: linear

b. 

With respect to C: Trigonal planar

With respect to O :bent

70. What are the three aspects that minimize the repulsion between valence shell electron pair?

Solution

Lp - Lp , Lp - bp and bp - bp repulsions.

Intermolecular Forces in Covalent Compounds

Intermolecular forces are attractve forces between molecules.

There are three types of intermolecular forces in covalent compounds,

1. Dipole-dipole force
2. Hydrogen bond
3. London dispersion force

1. **Dipole forces are forces** exist between polar molecules ,that is between molecules that possess dipole moments.

- The higher the dipole moments,the stronger is the dipole-dipole force.

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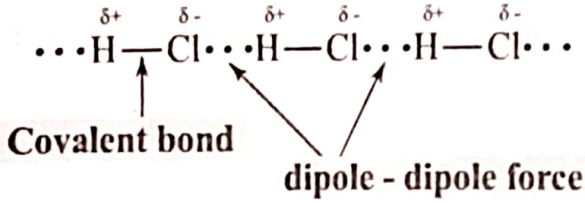
Example:



2. Hydrogen Bond

- It is particularly special type of dipole-dipole force.
- It is formed between molecules that have hydrogen-fluorine ,hydrogen-oxygen and hydrogen-nitrogen bonds

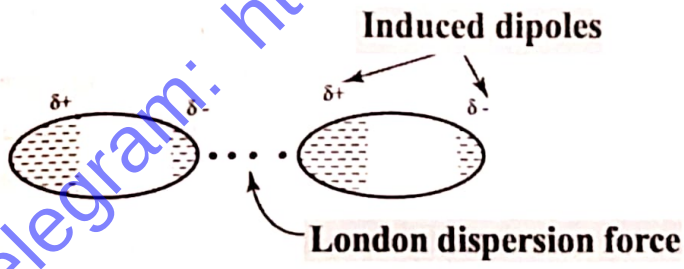
Example

HF , H_2O , NH_3 , CH_3OH , CH_3COOH , NH_2-CH_2-COOH

···H—F···H—F···H—F···

Hydrogen bond

3. **London dispersion forces** are forces that exist between non-polar molecules
   - London force is the weakest intermolecular force.
   - Non-polar molecules form induced dipoles. The attraction force between these induced-dipolse is **London force**



The strength of London force is mainly affected by size . As size of non-polar molecules increase, strength of London force also increases.

Example

$I_2 > Br_2 > Cl_2 > F_2$

$C_4H_{10} > C_3H_8 > C_2H_6 > CH_4$

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Illustrative Example

71. What are intermolecular force? They are force due to the attraction between .(ESSLCE)

A. Nuclei and electron pair

B. Cation and delocalized electrons

C. Molecules

D. Cations and anions

Solution

Intermolecular forces are forces between molecules.

Answer: C

72. How is hydrogen bonding formed? It is formed when a hydrogen atom is bonded to (ESSLCE)

A. highly electronegative elements,chlorine, oxygen and nitrogen

B. highly electronegative elements, fluorine oxygen and nitrogen

C. highly electronegative elements, fluorine and chlorine

D. highly electronegative elements , fluroine , oxygen and phosphorus

Solution

Hydrogen bonding is formed when hydrogen is bonded to the three most electronegative elements , fluorine, oxygen and nitrogen.

Answer :B

73. Which of the following explains,why at room temperature , I_2 is solid, Br_2 is liquid, and Cl_2 is gas

A. Ionic bonding

C. hydrogen bonding

B. Hybridization

D. London dispersion forc

Solution

As size increases ,strength of London dispersion force also increases. This has effect on the state of the non-polar molecules.

Answer: D

74. There is a strong covalent bond between the N-atoms in nitrogen gas

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N_2 .Why ,then does nitrogen have a such a low boiling point of -196?

A. The bond between the nitrogen-atoms is triple.

B. N is very electronegative,only next to F and O

C. The strong bond determines the boiling point of substance.

D. Boiling point is determined by intermolecular force ,which in this is weak as the molecule is non-polar.

Solution

N_2 is non-polar molecule tied togther with the weakest intermolecular force.

Answer :D

75. Of Br_2 , Ne, HCl and N_2 ,which likely to have the largest London dispersion force?

A. Br_2 B. Ne C. HCl D. N_2

Solution

Br_2 has the largest London dispersion forces due to its largest size.

Answer :A

76. Arrange the following in decreasing order of boiling ponit.

a. CF_4 ,CH_4,CCl_4, ICl_4

b. H_2O, HF, NH_3, H_2S

c. C_5H_{12} ,$C_{10}H_{22}$,C_8H_{18}

d. Br_2, Cl_2, I_2, F_2

Solution

a. $ICl_4 > CCl_4 > CF_4 > CH_4$

b. $HF > H_2O > NH_3 > H_2S$

c. $C_{10}H_{22} > C_8H_{18}$

d. $I_2 > Br_2 > Cl_2 > F_2$

77. N_2 and CO are isoelectronic. Which substance would you expect to have higher boiling point?

Solution

CO because it is polar but N_2 is non - polar.

78. Identify the types of intermolecular forces that are dominate in each substance below.

a. IF b. $CHCl_3$ c. CH_3NH_2

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d. HCOOH e. PCl_5 f. $AgNO_3$

Solution

a. dipole-dipole
b. dipole - diploe force
c. hydrogen bonding
d. hydrogen bonding
e. London force
f. Ion-dipole force because it is ionic compound

2.4 Metallic Bonding

The electrons in the outer energy level of the a metal move freely through the structure(they are delocalized forming a mobile "sea" of electrons)

The attraction force between free electrons of metals and their nuclei is called **metallic bond**

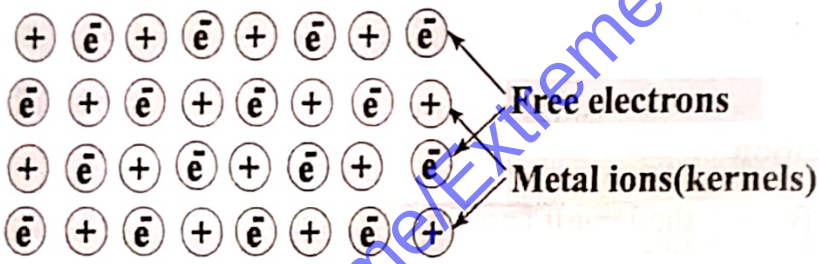

Fig 2.2 The Electron Sea Model

Factors Affecting Metallic Bond Strength

☑ There are two factors that affect the strength of metallic bond strength.

These are:

1. Size
2. Number of valence electrons

1. Size: As size increase,strength of metallic bond decreases because metal atoms are loosely packed

Example :

Li < Na < K < Rb :size

Li > Na > K > Rb :Metallic bond strength

2. Number of Valence Electrons

As the number of valence electrons increase,the strength of metallic bond also increase.

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Example

Na < Mg < Al→ metal bond strength

2.8.1 2,8,2 2,8,3→Number of valence electrons

Properties of Metals Related to Metallic Bonding

- ☑ They usually have high melting and boiling points due to the strong attraction between the positive metals ions and the mobile "sea"of electrons.
- ☑ They conduct electricity due to the mobile electrons within the metal structure.
- ☑ They are malleabel and ductile .If a force is applied to a metal ,rows of ions can slide over one another .They reposition themselves and the strong bonds reform.
- ☑ They have high densities because the atoms are very closely packed in a regular manner.

Illustrative Examples

79. In order to illustrative the structure and properties of a solid ,a student put hundred marble balls in to a box that represents metal cations and the empty space between and around the marble representing the electron sea. Which one of the following bonding is represented by this model?(ESSLCE)

A. Dative bonding
C. Ionic bonding
B. Metallic bonding
D. Covalent bonding

Solution

Metallic bond is attraction force between metal nuclei and delocalized electrons

Answer: B

80. Which of the following transmit electrcity in solid state ?

A. Water
C. Sodium chloride
B. Copper
D. Sulphur

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Solution

Metals transmit electrcity in solid state.

Answer :B

81. Which of the following has the hight melting point ?

A. Iodine B. Mercury C. Tungsten D. Bromine

Solution

Metal with the higest melting point is tungsten(3410^{0}C) and with lowest melting point is mercury(-39^{0}C)

Answer: C

82. Which of the following is not true of metallic bonding?

A. It gives rise to excellent electrical conductivity

B. Electrons are free to move throughout the structure

C. The strength of metallic bond increase down a group.

D. The strength of metallic bond affects the boiling point of metals

Solution

Down a group,size increase .As a result ,strength of metallic bond decreases.

Answer :C

83. Determine which one has higher melting point?

a. Li or Be b. Mg or Al c. K or Rb

Solution

As size increase ,metallic bond strength decrease

a. Li b. Mg c. Rb

84. Determine which one is more metallic element?

a. Li or Be b. Mg or Al c. K or Rb

Solution

As size increase , metallic character also increases

a. Li b. Mg c. Rb

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2.5 Chemical Bonding Theories in Covelent Bonding

There are two commonly used theories of chemical bonding based on the concept of quantum mechanical model.

1. Valence bond theory(VBT)
2. Molecular orbital theory(MOT)

1. Valence Bond Theory (VBT)

☑ According to VBT, atoms share electrons when a half-filled atomic orbital of one atom overlaps within an atomic orbital of another.

☑ If the half -filled orbitals overlap in head to head or end to end fashion, the type of covalent bond formed will be called **sigma bond(δ - bond)**

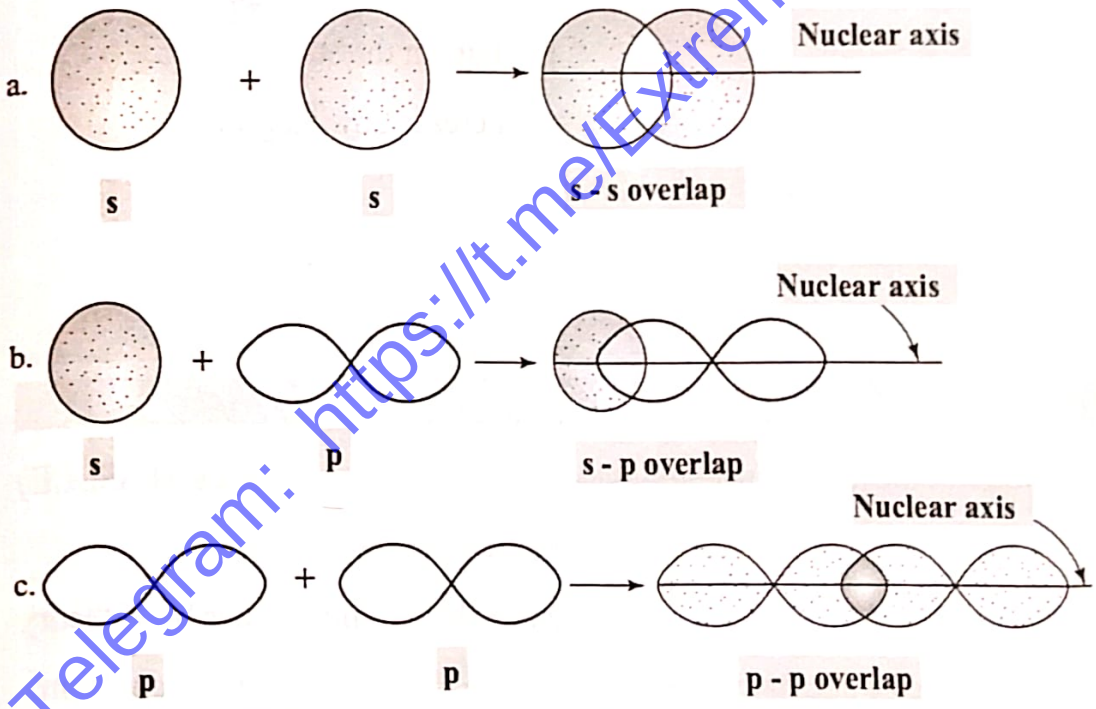

Fig 2.3 Sigma Bond Formation by s - s and p - p Half Filled Orbital Overlaps

☑ All s - orbital forms sigma bond only due to its spherical shape

☑ p - orbitals form sigma and pi bonds.

☑ In a sigma bond, electron density lies along the nuclear axis.

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☑ When half - filled orbitals overlap in a side to side or parallel way, the resulting bond is known as **pi bond (π bond)**

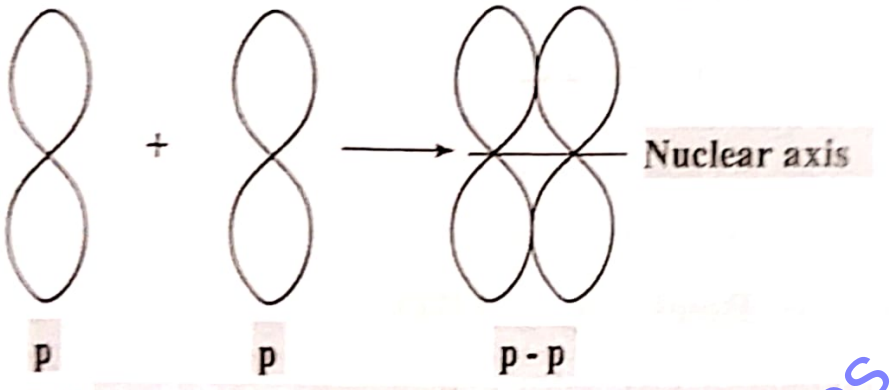

Fig 2.4 p – p side to side overlap for π – bond formation.

According to VBT, the higher the overlap, the stronger is the bond.

A δ - bond is a stronger than a π - bond

All single bonds are sigma bonds

A double bond =1δ + 1π

A triple bond =1δ + 2π

A single bond =1δ + 0π

Illustrative Examples

85. Which of the following belongs to chemical bonding theories? (ESSLCE)

A. Molecular orbital theory and kinetic molecular theory

B. Valence shell electron pair repulsion theory and valence bond theory

C. Kinetic molecular theory and valence shell electron pair repulsion theory.

D. Valence bond theory and molucular orbital theory.

Solution

The two bonding theories are valence bond theory and molecular orbital theory.

Answer :D

86. Which molecule listed below has two sigma (δ) and two pi bonds?

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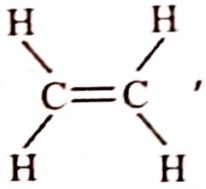
A. N_2 B. C_2H_4 C. N_2F_2 D. HCN

Solution

N_2: :N≡N: , $1\delta + 2\pi$, N_2F_2: :F—N=N—:F $3\delta + 1\pi$

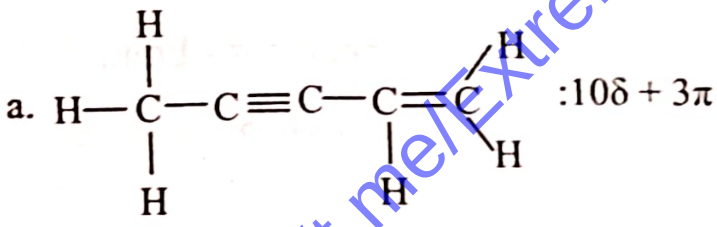
C_2H_4:  , $5\delta + 1\pi$, HCN: H—C≡N , $2\delta + 2\pi$

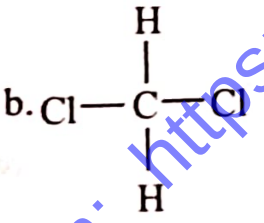
Answer: D

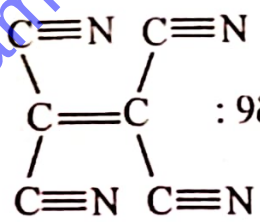
87. Determine the number of sigma and pi bonds in

a. $CH_3C{\equiv}C{-}CH{=}CH_2$ b. CH_2Cl_2 c. C_6N_4

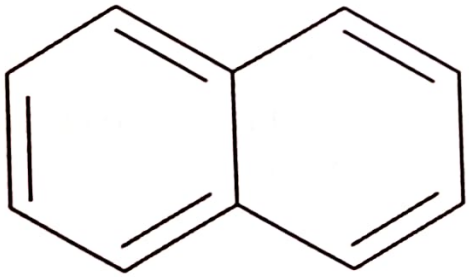
Solution

a.  :$10\delta + 3\pi$

b.  : $4\pi + 0\pi$

c.  : $9\delta + 9\pi$

88. Determine the number of sigma and pi bonds in the aromatic compound below



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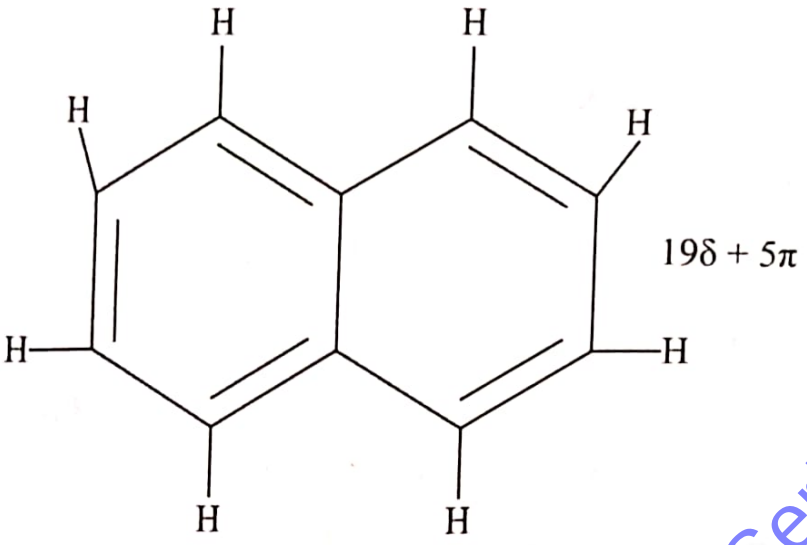
Solution



$19\delta + 5\pi$

Hybridization of Atomic Orbitals

☑ Although valence bond theory is useful and can explain more of our experimental observations than Lewis bond theory,it fails to explain the bonding in many of the molecules that are encounter.

☑ Based on VBT,an atom must have half-filled orbital to form a covalent bond .

☑ VBT failed to explain the bonding in $BeCl_2$, BCl_3 CH_4 and so on. The question can be answered by using the concept of hybridization.

☑ **Hybridization** is intermixing of two or more pure atomic orbitals of the same atom with almost the same energy to give same number of identical and degenrate new type of orbitals called **hybrid orbitals**

☑ The VBT was modified by **Linus Pauling** using the concept of hybirdization to explain the structure and bond angel of molecules.

☑ For atomic orbitals to undergo hybirdization:

1. The atomic orbitals of same atom with almost the same energy.
2. The fulfilled, half - filled or even empty orbitals can undergo hybridization.
   - ➥ The hybrid orbitals have the same energy and shape
   - ➥ The hybrid orbitals have one big lobe associated with a small lobe on the other side as shown below.

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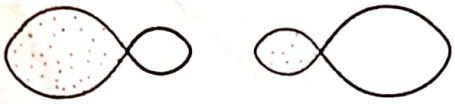


Fig 2.5 Shape of Hybrid Orbital

☑ The number of hybrid orbitals formed is equal to the number of pure atomic orbitals under going hybridization.

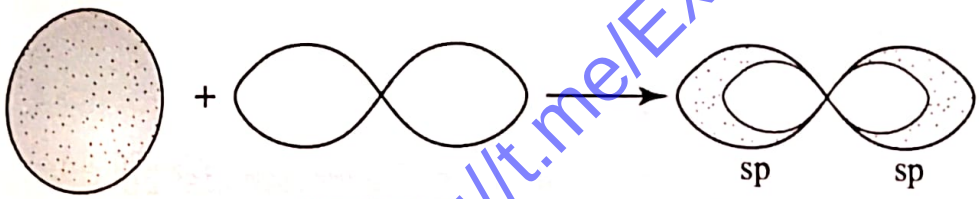
☑ The hybrid orbitals partcipate in the **sigma bond** formation with other atoms.

Hybdridization of s and p Orbitals

s - subshell has one orbital and p-sub shell has three orbitals.

a. sp - hybridization

- Intermixing of one 's'and one 'p'orbitals.
- It has 50% s and 50% p character.
- sp - hybird orbitals are arranged linearly by making 180^0 of angle

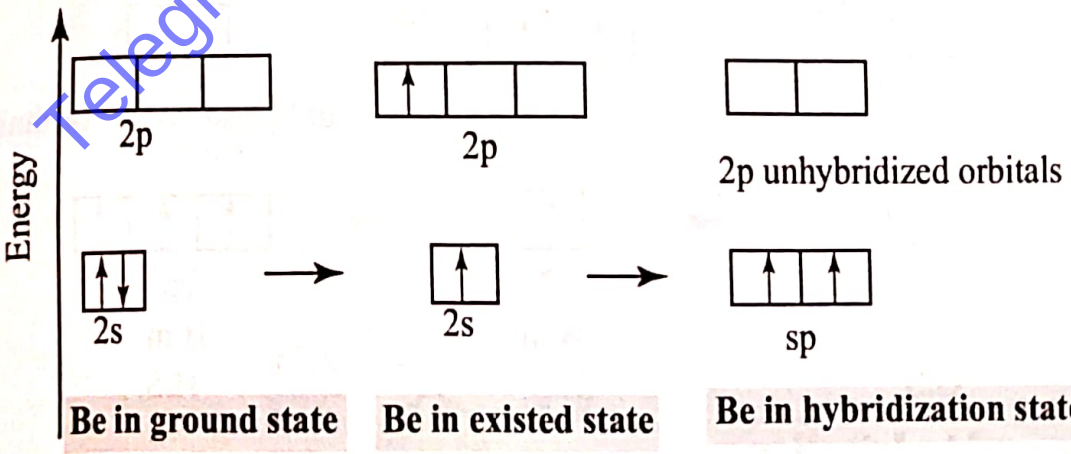


s - orbital **p - orbital** **Two sp hybrid orbitals**

Example

Berylium chloride ($BeCl_2$)

Be: $1s^2$ $2s^2$ $2p^0$,valence shell configuration:$2s^2$ $2p^0$



- Here ,berylium atom is in sp hybridization and forms a sigma bond

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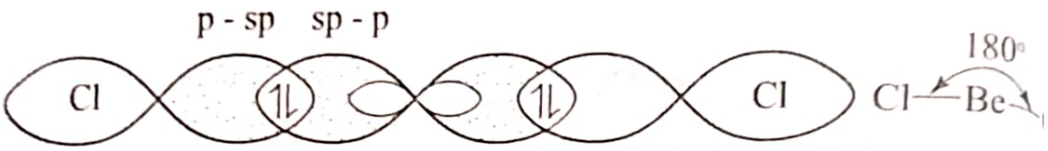
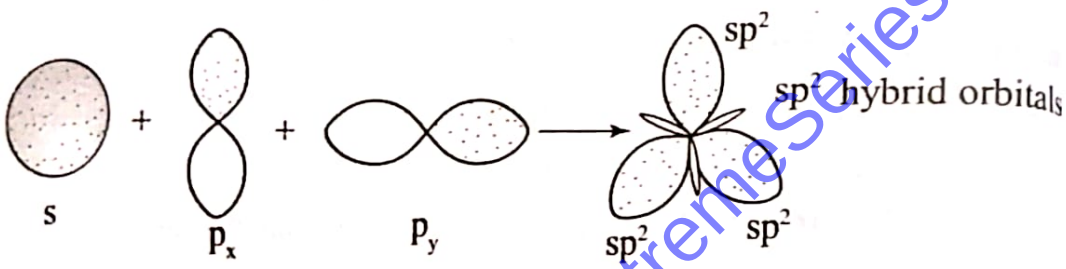
with the p - orbitas of the two chlorine atoms .



b. sp^2 - hybridization

- Intermixing of one 's' and two 'p' orbitals of equal energy.
- It contains 33% s and 67% p character
- The three sp^2 hybrid orbitals are oriented in trigonal planar symmet at angels of 120° to each other



Example

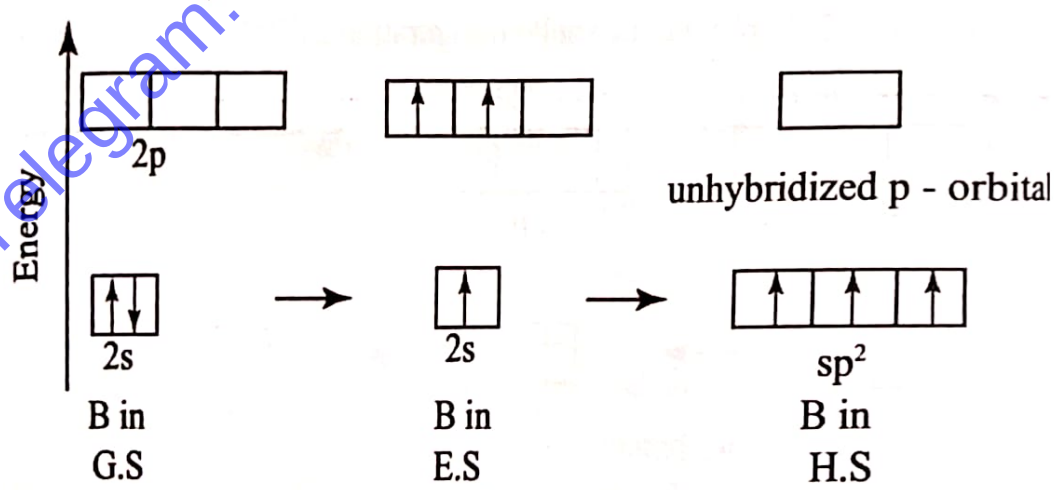
Boron hydride :BH_3

B : $1s^2\ 2p^2\ 2p^1$:valence shell configuration :$2s^2\ 2p^1$

G.S - ground state

E.S - excited state

H.S - Hydridization state



c. sp^3 hydrization

- one 's' and three 'p' orbitals of almost same energy intermix to form s

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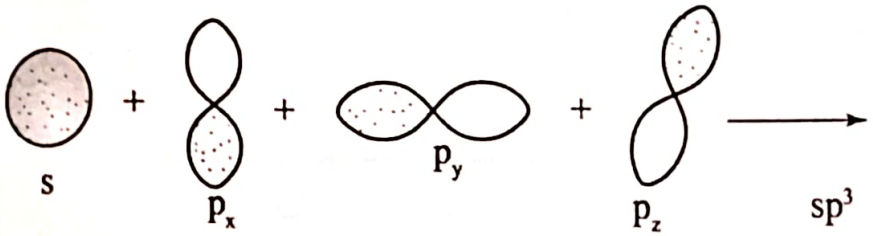
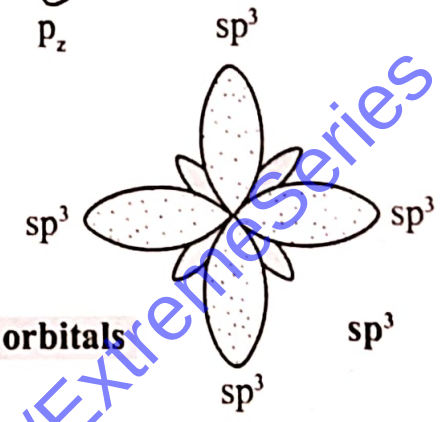
- Oriented in tetrahedral symmetry with 109.5^0
- 25%s and 75% p - character.

C :$1s^2\ 2s^2\ p^2$ valence shell configuration:$2s^2\ 2p^2$



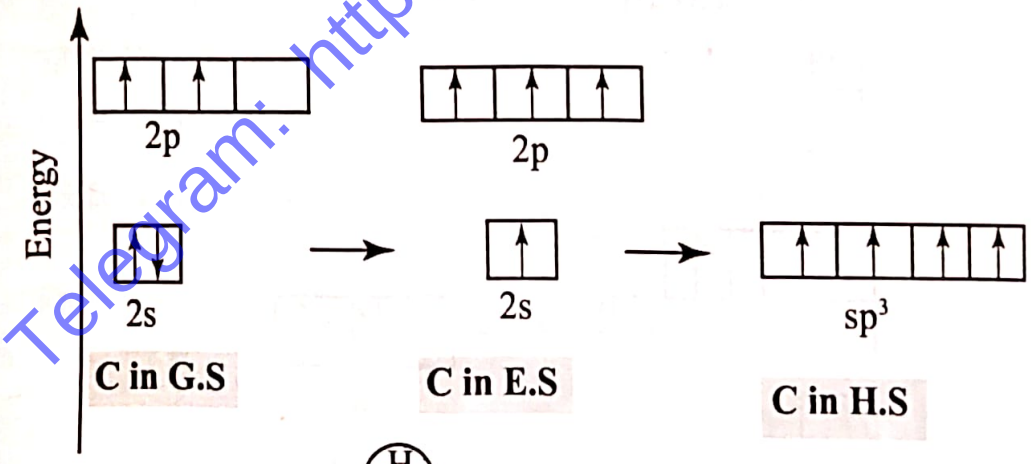


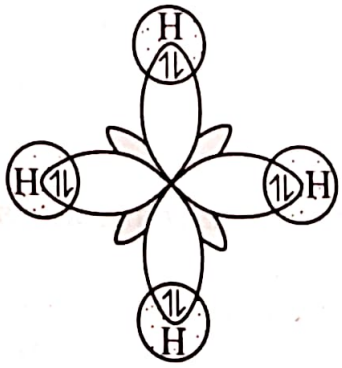
hybrid orbitals

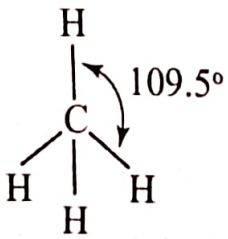
Example

Methane (CH_4)

C: $1s^2\ 2s^2\ 2p^2$: Valence shell configuration: $2s^2\ 2p^2$







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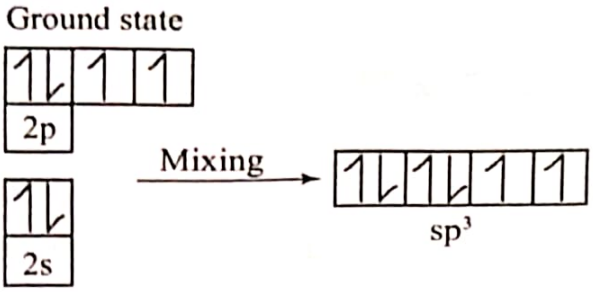
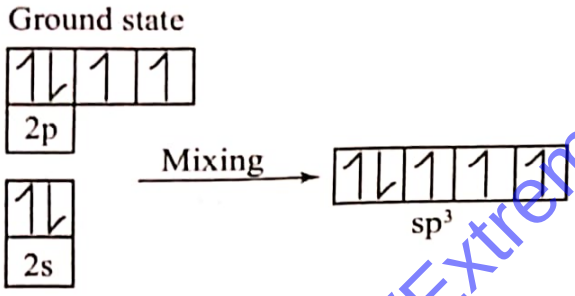
Illustrative Example

89. Which one of the following shows the hybridization Scheme of the central atom in H_2O?(ESSLCE)

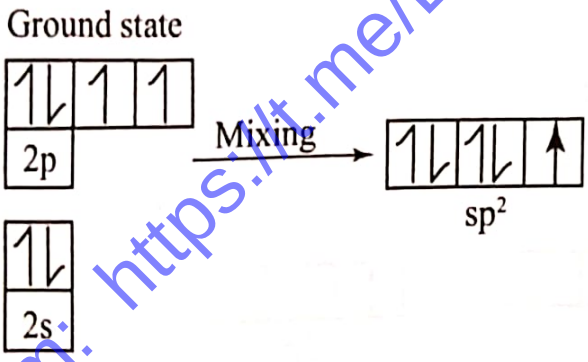
A.



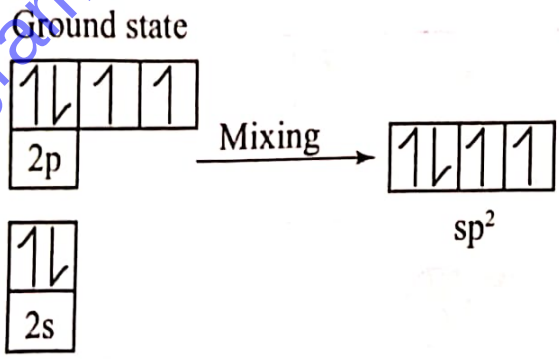
B.



C.



D.



Solution

H_2O : O : $1s^2\ 2s^2\ 2p^4$ valence shell configuration : $2s^2\ 2p^4$

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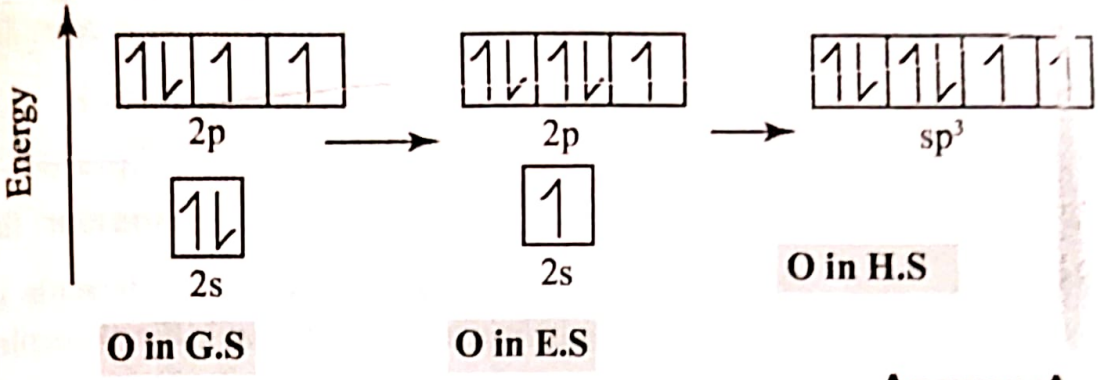


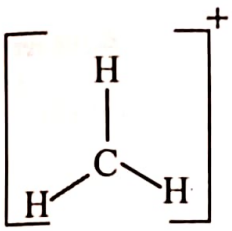
Answer :A

90. What is the hybrid state of carbon in methyl carbon cation(CH_3^+)?

A. sp^2 B. sp^2d C. sp^3 D. sp

Solution

Simply draw the Lewis structure and count the number of half-filled.



Three half-filled orbitals are required for C to overlap with the three H-atoms.

The hybrid state of C is sp^2

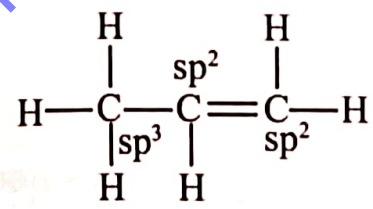
Answer: A

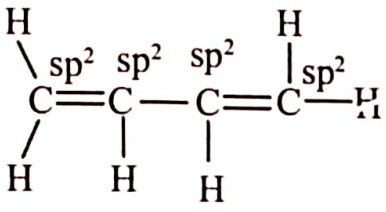
91. In which of the following molecules of carbon atom assumes sp and sp^2 hybrid states only?

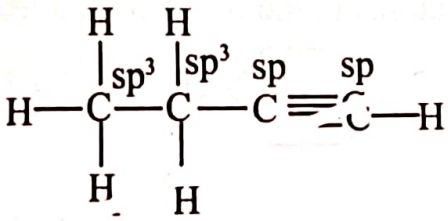
A. $CH_3—CH=CH_2$ C. $CH_2=CH—CH=CH_2$

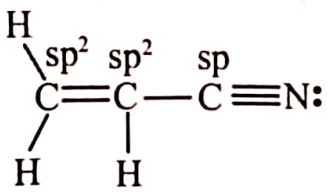
B. $CH_3—CH_2—C\equiv CH$ D. $CH_2=CH—CN$

Solution









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Answer: D

92. The bond angel around the atom which uses sp hybrid orbitals is

A. 120° B. 180° C. 109.5° D. 104.5°

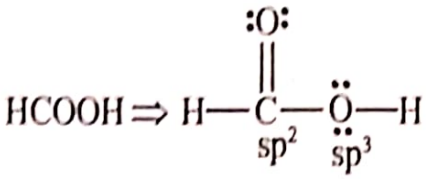
Answer: B

93. Formic acid ,which is released by ants,has a molecular formula of HCOOH. What are the possible hybridization that exist in the molecule?

A. sp^2 and sp^3 C. sp, sp^2 and sp^3

B. sp and sp^2 D. sp and sp^2

Solution



Answer :A

94. How many atomic orbitals are required for an sp^2 hybrdization ?

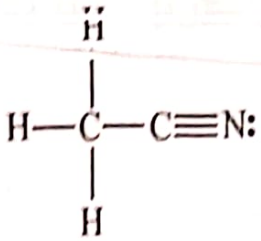
A. 2 B. 6 C. 4 D. 8

Solution

$sp^3 = s + p + p + p$ = 4 atomic orbitals are involved .

Answer :C

95. What is the hybridization of the carbon atom attached to nitrogen in acetonitrile shown below ?



A. sp

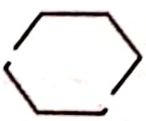
B. sp^2

C. sp^3

D. sp^4

Answer: A

96. Identify the hybridization of central atoms in the following molecules .

a. NH_3 b. BeH_2 c. 

d. $AlCl_3$

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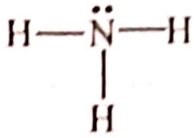
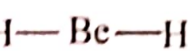
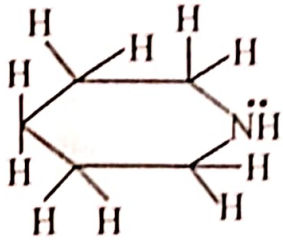
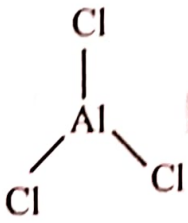
Solution

a. NH_3 :  N is sp^3 hybridized

b. BeH_2 :  Be is sp hybridized

c.  : Each carbon atom is sp^3 and N is sp^3 hybridized

d. $AlCl_3$:  Al is sp^2 hybridized

Hybridization of s,p and d Orbitals

This type of hybridization results by mixing s,p and d orbitals of similar energy.

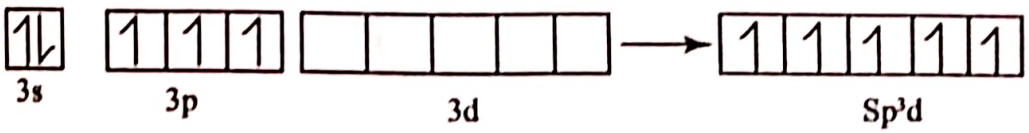
a. sp^3d hybridization

- Intermixing of one 's' ,three 'p' and one 'd' orbitals.
- It has 20% s, 60% p and 20%d character.
- Sp^3d hybrid orbitals are oriented in trigonal byipyramidal symmetry at an angel of 120% and 90%.

Example :

pCl^5

p : [Ne] $3s^2$ $3p^3$ $3d^0$



P in ground state

P in hybridization state

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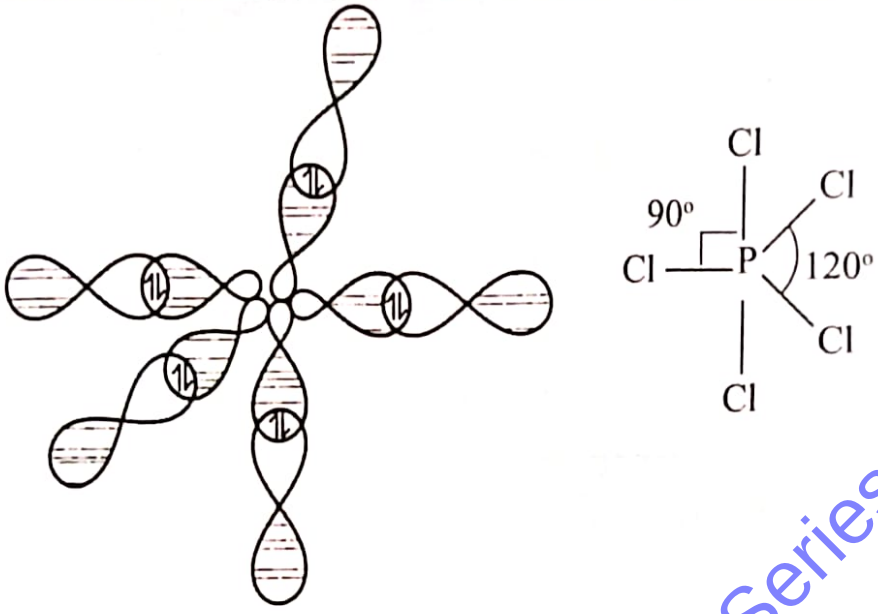


Fig 2.6 Trigonal Bipyramidal Structure of PCl_5

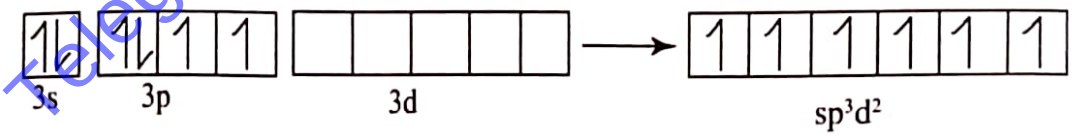
b. sp^3d^2 hybridization

- ☑ Intermixing of one 's' ,three 'p' and two 'd' orbitals of same energy by giving six identical and degenerate hybrid orbitals called sp^3d^2 hybridization.
- ☑ It contains 17%s, 50%p and 33%d character
- ☑ These six hybrid orbitlas are arranged in octahydral symmetry by making 90^0 angels to each other.

Example

Sulpher hexafluorid(SF_6)

S : [Ne] $3s^2\ 3p^4\ 3d^0$



S in ground state S in hybridization state

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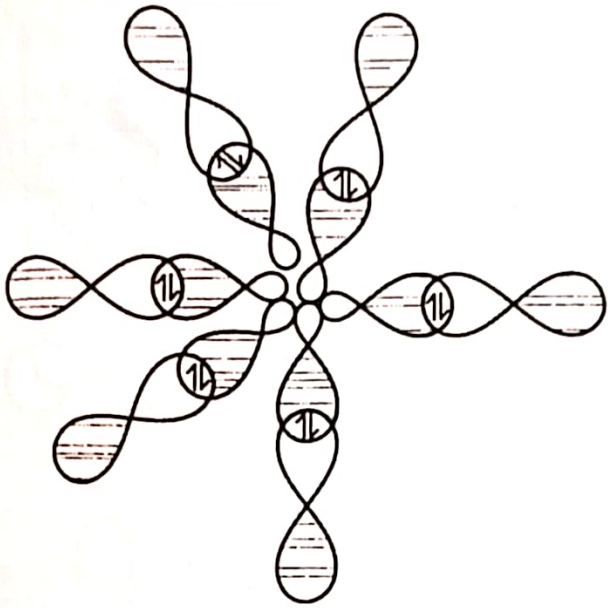


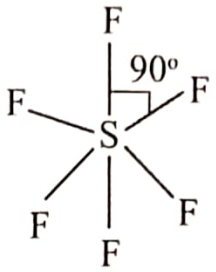


Fig 2.7 Octahedral structure of SF_6 with sp^3d^2 hybridization

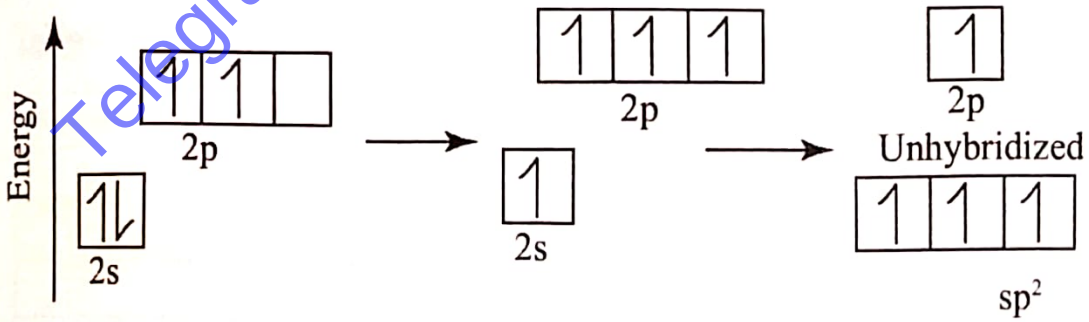
Hybridization and Multiple Bonds

a. Double Bond and Hybridization (sp^2)

☑ A double bond has one sigma and one pi bonds. For pi bond formation pure p - orbital is required. As a result, there will be unhybridized p - orbital.

Example

Ethene(C_2H_4) :C : $2s^2\ 2p^2$



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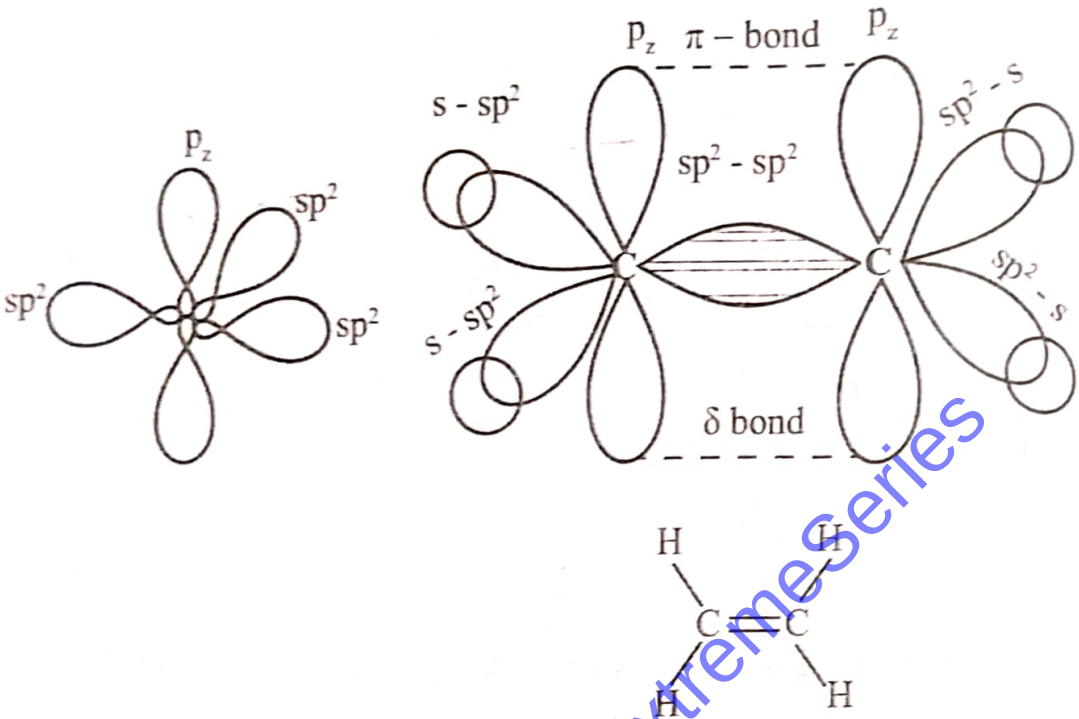

Fig 2.8 Trigonal Planar Structure of Ethene with sp^2 Hybridization

b. Triple Bond and Hybridization (sp)

A triple bond has one sigma and two Pi bonds. thus, two pure p - orbitals are required for this purpose.

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Example

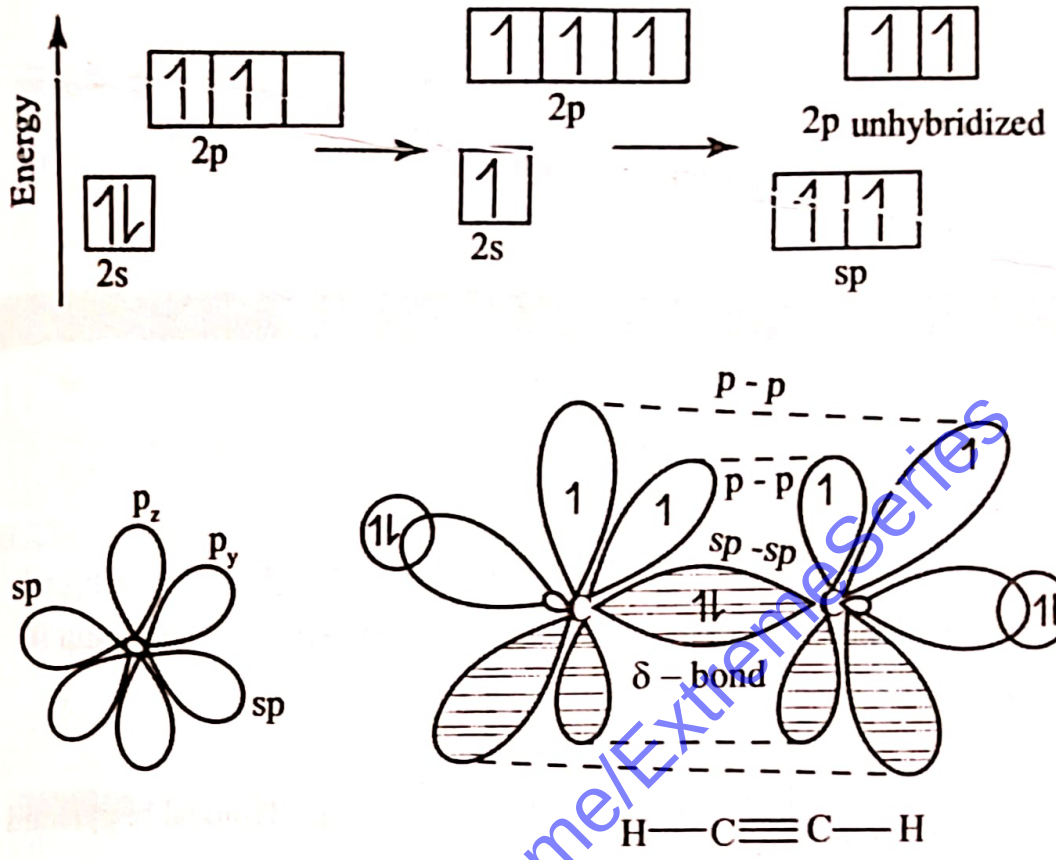

Fig 2.9 Linear Structure of Ethyne with sp Hybridization

Note: To know the hybridization of a central atom easily;

- ☑ Draw the Lewis structure and count the number of sets of electrons around the central atom.
- ☑ Number of set of electrons = Number of hybrid orbitals.

Table 2.2 Number of Sets of Electrons and Hybrid Orbitals on Central Atom

Number of Set of Electrons	Hybrid Orbitals
2	sp
3	sp^2
4	sp^3
5	sp^3d
6	$sp^3 d^2$

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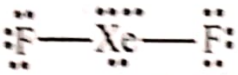
Example

Consider XeF_2

Lewis structure: 

Set of electrons around Xe = 5

Thus, hybridization of Xe is sp^3d

Illustrative Examples

97. In PCl_5 molecule

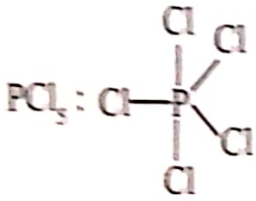
I. Hybridization state of P is sp^3d

II. Molecule is T - shaped

III. Bond angle is 180^0. The true statement /s/ is/ are

A. I only B. III only C. I and II D. II and III

Solution

PCl_5 : 

Hybridization sp^3d ; shape Trigonal bi pyramid

bond angle: 90^0, 120^0

Answer: A

98. The hybridization state of iodine in IF_5 is

A. sp^3d B. sp^3d^2 C. sp^3 D. sp^2

Solution

IF_5 :

Six sets of electrons around I. Thus the hybridization is sp^3d^2

Answer: B

99. Identify the hybridization of the central atom of

a. CO_2 c. SF_4 e. HCN

b. PF_6^- d. SO_3 f. ICl_3

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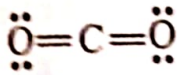
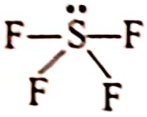
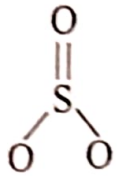
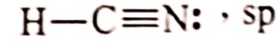
Solution

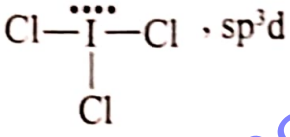
a.  · sp

b. · sp^3d^2

c.  · sp^3d

d.  · sp^2

e.  · sp

f.  · sp^3d

100. The following compounds have more than one central atoms. Identify the type of hybridization with respect to each central atom.

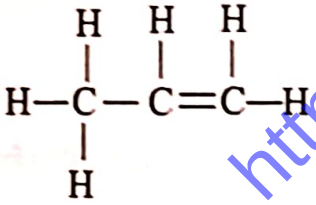
a. $CH_3—CH=CH_2$

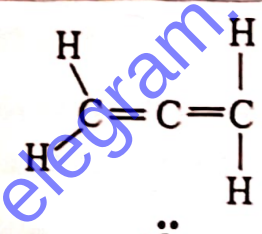
b. $H_2C=C=CH_2$

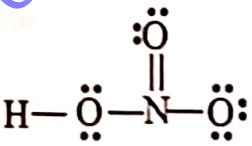
c. HNO_3

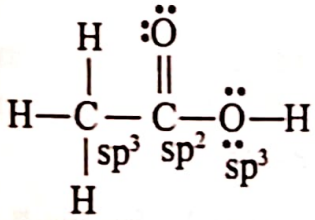
d. CH_3COOH

Solution

a.  **sp^3, sp^2, sp^2 from left to right**

b.  **sp^2, sp, sp^2 from left to right**

c.  O: sp^3, N: sp^2

d. 

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101. Identify the type of orbital overlap for the formation of the compound below.

a. BH_3 c. C_2H_2 e. PCl_5

b. $BeCl_2$ d. SF_6 f. CCl_4

Solution

a. sp^2 - s d. sp^3d^2 - p

b. sp - p e. sp^3d - p

c. sp - sp, sp - s and p - p f. sp^3 - p

102. What is the relationship between s - character and bond length?

Solution

As s - character increases, bond length decreases.

2. Molecular Orbital Theory (MOT)

☑ Based on MOT, the atomic orbitals involved in bonding actual combine to form **bonding and anti bonding molecular orbitals.**

☑ **Bonding molecular orbital (BMO)** is formed by addition of atomi orbitals.

☑ It is represented by δ and π.

☑ **Anti bonding molecular orbital (ABMO)** is formed by subtraction atomic orbitals.

☑ It is represented by δ^* and π^*

☑ BMO_s have lower energy than the original atomic orbitals.

☑ $ABMO_s$ have higher energy than the original atomic orbitals.

Principles of MOT

1. Orbitals are conserved. The total number of molecular orbitals equa the total number of atomic orbitals that contributed.

2. Conservation of energy is obeyed. Bonding molecular orbitals ha lower energy than the average of the contributing atomic orbital. A bonding molecular orbitals have higher energy.

3. When assigning electrons to molecular orbitals the **Pauli exclusi** principle and **Hund's rule** are still applied.

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Rule 1: For species that have 14 and below 14 electrons: the order of filling is

$$(\delta_{1S})\ (\delta_{1S}^*)\ (\delta_{2S})\ (\delta_{2S}^*)\ (\pi 2py = \pi 2pz)\ (\delta_2 px)\ \text{------}$$

Example C_2 : electrons $= 2 \times 6 = 12$

$$(\delta 1s)^2\ (\delta 1s^*)^2\ (\delta 2s)^2\ (\delta 2s^*)^2\ (\pi 2py)^2 = (\pi 2pz)^2$$

Rule 2: For species that have above 14 electrons

$$(\delta_{1S})\ (\delta_{1S}^*)\ (\delta_{2S})\ (\delta_{2S}^*)\ (\delta 2p\times)\ (\pi 2py = \pi 2pz)\ (\pi 2py^* = \pi 2pz^*)\ (\delta 2px^*)$$

Example O_2 : electrons $= 2 \times 8 = 16$

$$(\delta 1s)^2(\delta 1s^*)^2(\delta 2s)^2(\delta 2s^*)^2(\delta 2p^*)^2(\pi 2py^*)^2=(\pi 2pz^*)^2(\pi 2pz^*)^1=(\pi 2pz^*)^1$$

π_{2Py} and π_{2Pz}
$\pi_{2Py}{}^*$ and $\pi_{2Pz}{}^*$ } are degenerate orbitals

4. The maximum number of electrons in a molecular orbital is two.
5. In ground state configurations, electrons enter molecular orbitals of same energies single before they pair up.
6. When two atomic orbitals overlap in end to end way they form sigma molecular orbital (δ)
7. When two atomic orbitals overlap in side to side way they form Pi molecular orbital

A molecular orbital diagram shows the relative energy and number of electrons in each MO and the atomic orbitals from which they form.

Example

B_2: electrons $= 2 \times 5 = 10$

Electron configuration: $(\delta 1s)^2\ (\delta 1s^*)^2\ (\delta 2s)^2\ (\delta 2s^*)^2\ (\pi 2py)^1 = (\pi 2pz)^1$

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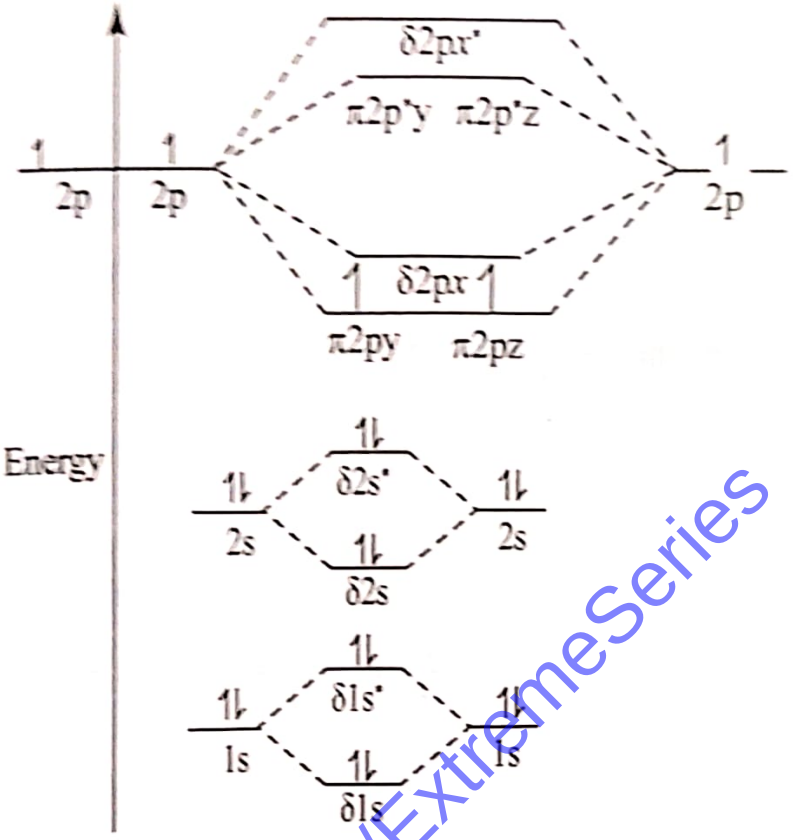

Fig 2.10 Molecular Orbital Diagram and Bonding the Molecule of B_2

Bond order (BO) is one half the differenced between the number of bonding electrons (n_b) and anti bonding electrons (n_a).

$$BO = \frac{1}{2}(n_b - n_a)$$

- ☑ BO = 0, implies the species does not exist.
- ☑ BO > 0, implies the species does exist.
- ☑ As BO increases, stability and bond strength increase but bond length decreases.
- ☑ **MOT** is better than the previous theories because it describes some important properities such as **paramagnetic and diamagnetic properties.**
- ☑ A species with unpaired electrons exhibits **paramagnetic property** (attracted by an external magnetic field)

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Example:

All species with odd number of electrons and some species with even number of electrons (B_2, O_2) are paramagnetic secies.

- A species in which all the electrons are paired exhibits **diamagnetic species** (not attracted by magnetic field)

Example

- Most species with even number of electrons are diamagnetic species.
- For example, consider N_2
  a. Write electron configuration based on MOT.
  b. Determine its bond order
  c. Is it paramagnetic or diamagnetic molecule?

Solution

a. Electrons $= 2 \times 7 = 14$

$(\delta 1s)^2 (\delta 1s^*)^2 (\delta 2s)^2 (\delta 2s^*)^2 (\pi 2py)^2 = (\pi 2pz)^2 (\delta 2px)^2$

b. $n_b = 2 + 2 + 2 + 2 = 10$, $n_a = 2 + 2 = 4$

$$BO = \frac{1}{2}(n_b - n_a) = \frac{1}{2}(10 - 4) = 3$$

c. It is diamagnetic because it has all paired electrons.

Illustrative Examples

103. Which among the following species is most stable?

A. He_2 B. H_2^+ C. He^+_2 D. H_2

Solution

$$He_2 : (\delta 1s)^2 \ (\delta 1s^*)^2, \ BO = \frac{1}{2}(2 - 2) = 0$$

$$H_2^+ : (\delta 1s)^1, \ BO = \frac{1}{2}(1 - 0) = 0.5$$

$$He_2^+ : (\delta 1s)^2 \ (\delta 1s^*)^1, \ BO = \frac{1}{2}(2 - 1) = 0.5$$

$$H_2 : (\delta 1s)^2, \ BO = \frac{1}{2}(2 - 0) = 1$$

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H_2 is the most stable because it has the highest bond order.

Answer: D

104. Which of the following molecule is paramagnetic?

A. F_2 B. O_2 C. N_2 D. H_2

Solution

$F_2(e = 18)$: $(\delta 1s)^2 (\delta 1s^*)^2 (\delta 2s)^2 (\delta 2s^*)^2 (\pi 2px)^2 (\pi 2py)^2 = (\pi 2pz)^2 (\pi 2pz^*)^2 = (\pi 2py^*)^2$

$O_2(e = 16)$: $(\delta 1s)^2 (\delta 1s^*)^2 (\delta 2s)^2 (\delta 2s^*)^2 (\pi 2px)^2 (\pi 2py)^2 = (\pi 2pz)^2 (\pi 2pz^*)^1 = (\pi 2py^*)^1$

$N_2(e = 14)$: $(\delta 1s)^2 (\delta 1s^*)^2 (\delta 2s)^2 (\delta 2s^*)^2 (\pi 2py)^2 = (\pi 2pz)^2 (\delta 2px)^2$

$H_2 (e = 2) : (\delta 1s)^2$

Answer: B

105. The species having highest bond order is

A. O_2 B. O^-_2 C. O^+_2 D. O^{2-}_2

Solution

O_2: $BO = \frac{1}{2}(10 - 6) = 2$ O_2^+: $BO = \frac{1}{2}(10 - 5) = 2.5$

O_2^-: $BO = \frac{1}{2}(10 - 7) = 1.5$ O_2^{2-}: $BO = \frac{1}{2}(10 - 8) = 1$

Answer: C

106. What would happen to the O_2 molecule upon ionization to O^+_2?

A. Bond length and energy will increase

B. Bond length and energy will decrease

C. Bond length increases and bond energy decreases

D. Bond length decreases and bond energy increases

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Solution

$$O_2 \longrightarrow O_2^+$$

BO = 2 BO = 2.5

$$\Rightarrow BO\ \alpha\ \frac{\text{Bond energy}}{\text{bond length}}$$

Answer: D

107. Which of the following is the least stable?

A. CN B. CN^- C. CN^+ D. C^+_2

Answer: D

108. Draw molecular orbital diagram for

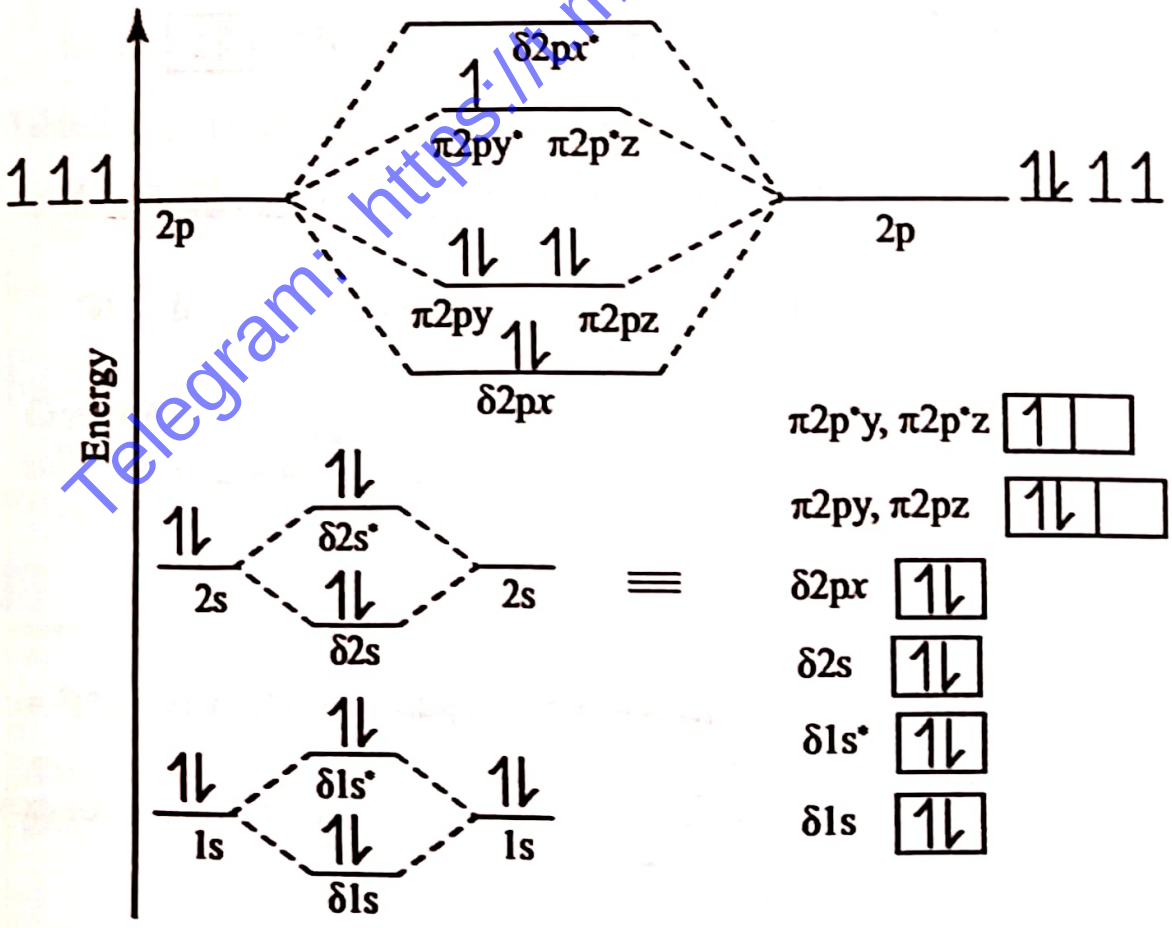
a. NO b. C_2

Solution

a. First write electron configuration based on MOT

NO: electrons = 7 + 8 = 15

$(\delta 1s)^2 (\delta 1s^*)^2 (\delta 2s)^2 (\delta 2s^*)^2 (\delta 2px)^2 (\pi 2py)^2 = (\pi 2pz)^2 (\pi 2py^*)^1$



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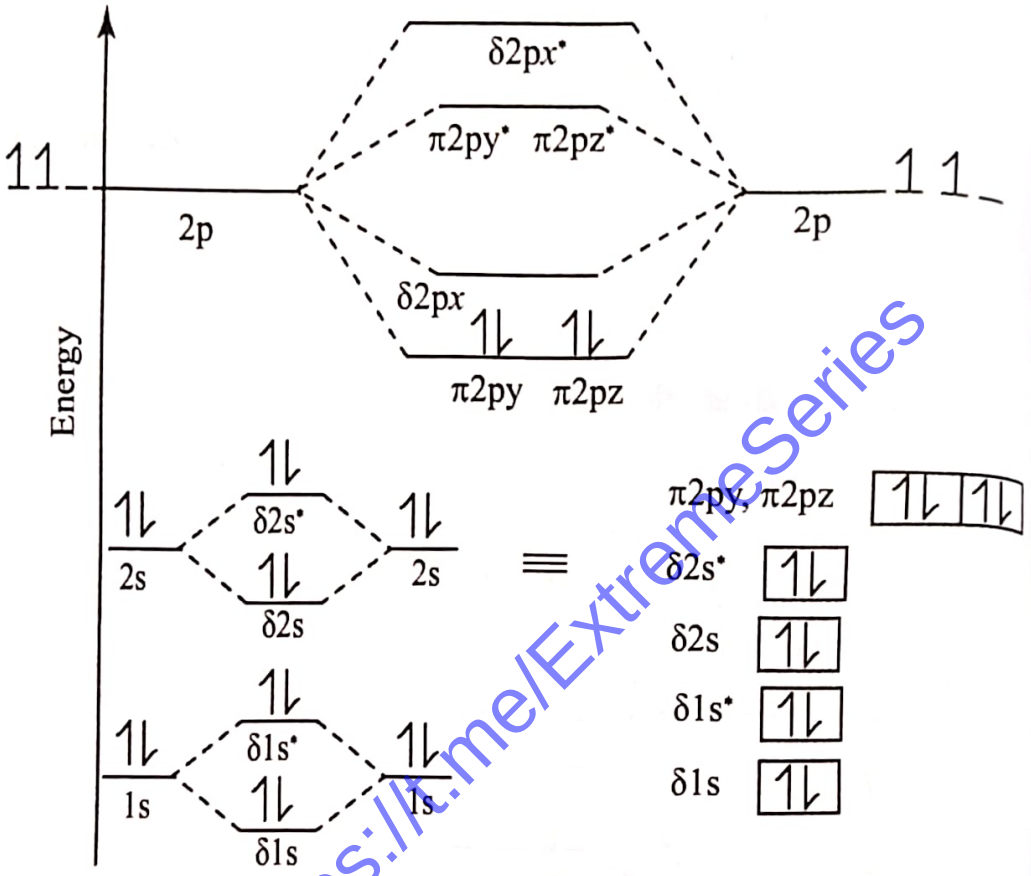
b. C_2: (electrons = 12)

$(\delta 1s)^2\ (\delta 1s^*)^2\ (\delta 2s)^2\ (\delta 2s^*)^2\ (\pi 2py)^2 = (\pi 2pz)^2$



109. Identify those which exist or do not exist by determining their bond order.

a. Li_2 b. Ne_2 c. NO^- d. CN^+

Solution

a. Li_2 : e = 6 $(\delta 1s)^2\ (\delta 1s^*)^2\ (\delta 2s^*)^2$, BO = $\frac{1}{2}(4-2) = 1$

$\therefore Li_2$ is stable

b. Ne_2 (e = 20)

$(\delta 1s)^2\ (\delta 1s^*)^2\ (\delta 2s)^2\ (\delta 2s^*)^2 (\delta 2px)^2\ (\pi 2py)^2 = (\pi 2pz)^2 (\pi 2py^*)^2 =$

$(\pi 2pz^*)^2\ (\delta 2p^*)^2$, BO = $\frac{1}{2}(10-10) = 0$, Thus, Ne_2 does not exist

c. NO^- : e = 7 + 8 1 = 16

$(\delta 1s)^2\ (\delta 1s^*)^2\ (\delta 2s)^2\ (\delta 2s^*)^2 (\delta 2px)^2\ (\pi 2pz)^2 = (\pi 2py)^2 (\pi 2py^*)^2 =$

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$(\pi 2pz^*)^1\ (\pi 2py^*)^1$: BO $= \frac{1}{2}(10 - 6) = 2$, Thus, NO^- is stable

d. CN^+ : e = 6 + 7 - 1 = 12

$(\delta 1s)^2\ (\delta 1s^*)^2\ (\delta 2s)^2\ (\delta 2s^*)^2\ (\pi 2px)^2 = (\pi 2pz)^2$

BO $= \frac{1}{2}(8 - 4) = 2$, Thus, CN^+ is stable

110. Write the central ideas of molecular orbital theory.

Solution

- Atomic orbitals of all atoms mathematically combine to form molecular orbitals
- Electrons then populate the molecular orbitals and are delocalized across entire molecule.

2.6 Types of Solids

Depending on the arrangement of atoms, there are two types of solids. **Theses are**

1. Amorphous
2. Crystalline solid

Table 2. 3 Difference Between Amorphous and Crystalline Solids

Type of solid	Properties	Example
Crystalline solids	➥ Consist of individual crystal each with regular shape. ➥ Cleave to give flat, well defined faces ➥ Melt at different temperature	➥ Metals ➥ Ionic compounds ➥ Molecular crystals ➥ Covalent crystal
Amorphous solids	➥ Have no well defined shape ➥ Break to give curved or irregular faces ➥ Soft melt over a range of temperature	➥ Paraffin ➥ Asphalt ➥ Rubber ➥ Glass

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☑ A crystal is a piece of solid with regular shape.

☑ The basic repeating structure unit of crystalline solid is called a **unit cell**

☑ There are four type of crystalline solids. These are

1. Ionic crystal
2. Covalent crystal
3. Metallic crystal
4. Molecular crystal

Table 2.4 The Four Types of Crystals and their Properties

Type of crystal	Nature of bonding force	Properties	Example
Ionic crystals	➥ Coulombic and electrostatic forces of attraction (ionic bond)	➥ Brittle, high meting point and conduct electricity in molten or aqueous state	NaF, NaCl, AgCl, LiF CsCl
Metallic crystals	➥ Positive ions in a sea of electrons held by electrical attraction (Metallic bond)	➥ High melting and boiling points ➥ High thermal and electrical conductivity	➥ Na, Cu Au
Covalent crystals	➥ Sharing of electrons (covalent bond)	➥ Hard, non - conductor, high meting point	➥ diamond, SiC, ZnS AgI, SiO_2
Molecular crystals	➥ Dipole - dipole and van der waals	➥ Low melting point, volatile and soft	➥ Solid CO_2

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Illustrative Examples

111. Which of the following is covalent network crystal?

A. NaCl B. Cu C. SiC D. $C_{12}H_{22}O_{11}$

Answer: C

112. Solid A is brittle and conducts electricity in molten state A is:

A. Ionic crystal
B. Metallic crystal
C. Molecular crystal
D. Covalent crystal

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Answer: A

113. All of the following are crystalline solids, **EXCEPT**

A. Potassium chloride
C. Rubber
B. Dry ice
D. Iron

Solution

Rubber is an amorphous solid

Answer: C

114. Which of the following is true about crystalline solids?

A. They melt over a range of temperature
B. They have no well defined shape
C. They break and give regular faces
D. They soften when heated

Answer: C

115. Identify the attractic forces in the four types of crystals

Solution

- Ionic crystal: Ionic bond
- Covalent network crystal: covalent bond
- Metallic crystal: metallic bond
- Molecular crystal: dipole - dipole force, hydrogen bond and London dispersion force.

Solve Problems

116. What type of electrons participate in bond formation?

Solution: Valence electrons

117. Name one ionic compound that contains only

a. Metal and non - metal
b. Non - metals
c. a poly atomic cation and a poly atomic anion

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Solution

a. KBr - Potassium bromide

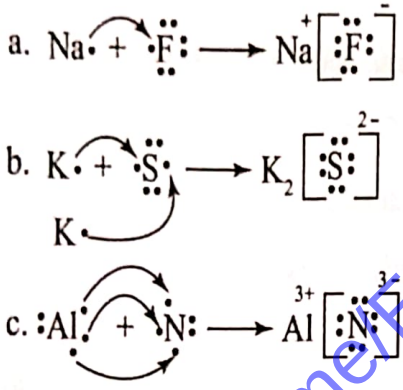
b. NH_4Cl - Ammonium chloride

c. NH_4NO_3 - Ammonium nitrate

118. Use Lewis dot symbols to show the transfer of electrons between

a. Na and F b. K and S c. Al and N

Solution



119. Write the formula of

a. Calcium nitride

b. Aluminum sulphate

c. Iron (III) carbonate

d. Ammonium phosphate

Solution

a. Ca_3N_2 b. $Al_2(SO_4)_3$ c. $Fe_2(CO_3)_3$ d. $(NH_4)_3PO_4$

120. Give the chemical name for

a. Cr_2O_3 b. Li_3P c. $HgCl_2$ d. $PbSO_4$

Solution

a. Chromium (III) oxide

b. Lithium phosphide

c. Mercury (II) chloride

d. Lead (II) sulphate

121. For each of the following pairs of ionic compounds, specify which compound has higher lattice energy.

a. KCl of CaO

b. LiBr or NaBr

c. Mg_3N_2 or Na_3N

d. AlF_3 or MgF_2

Solution

a. CaO b. LiBr c. Mg_3N_2 d. AlF_3

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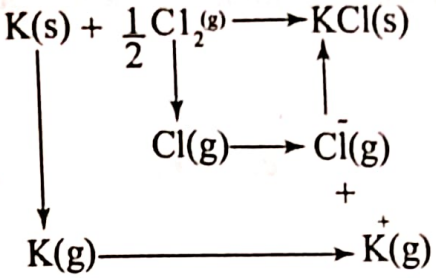
122. What is the type of interaction force in ionic compounds responsible for their solubility?

Solution

Ion - dipole force

123. Draw the Born - Haber cycle for the formation of potassium chloride (KCl).

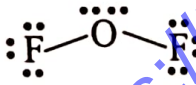
Solution

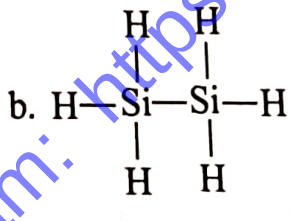


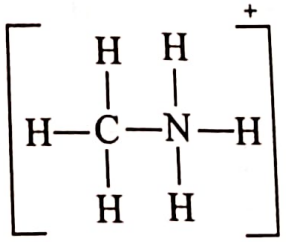
124. Draw the Lewis structures for the following molecules or molecular ions and determine the number of lone pairs on the central atom.

a. OF_2 b. Si_2H_6 c. $CH_3NH^+_3$

Solution

a.  2 Lp

b.  0 Lp

c.  0 Lp

125. Draw the Lewis structure for:

a. HCOOH b. H_2O_2 c. ClO^-_3 d. H_2SO_4

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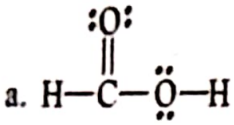
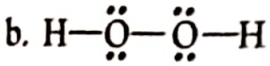
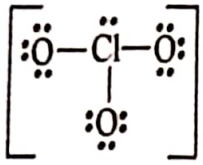
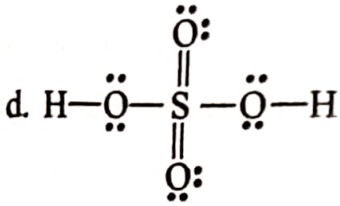
Solution

a. 

b. 

c. 

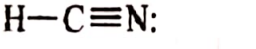
d. 

126. Draw the Lewis structure and determine the formal charge of each atom.

Solution

a. HCN b. NO^+ c. NO_2^- d. ClF_2^+

Solution

a. HCN : 

Formal charge for: H: $(1 - \left(0 + \frac{1}{2} \times 2\right) = 0$

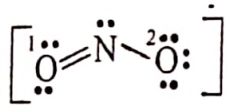
C: $(4 - \left(0 + \frac{1}{2} \times 8\right) = 0$

N: $(5 - \left(2 + \frac{1}{2} \times 6\right) = 0$

b. $\overset{+}{NO}$:

Formal chang for: N: $(5 - \left(2 + \frac{1}{2} \times 6\right)) = 0$

O: $(6 - \left(2 + \frac{1}{2} \times 6\right)) = +1$

c. 

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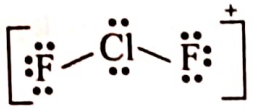
Formal charge for: O - 1: $(6 - \left(4 + \frac{1}{2} \times 4\right)) = 0$

$$N : (5 - \left(2 + \frac{1}{2} \times 6\right)) = 0$$

$$O - 2: (6 - \left(6 + \frac{1}{2} \times 2\right) = -1$$

d. ClF_2^+ : 

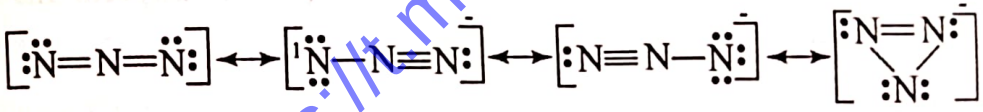
Formal charge: $F = (7 - \left(6 + \frac{1}{2} \times 2\right) = O$

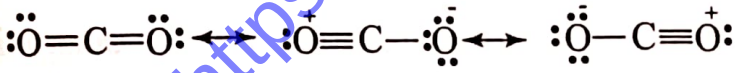
$$Cl = \left(7 - \left(4 + \frac{1}{2} \times 4\right) = +1\right.$$

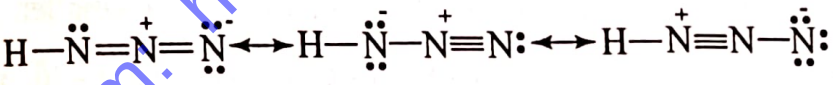
127. Draw all possible structure of:

a. N_3^- b. CO_2 c. HN_3

Solution

a. 

b. 

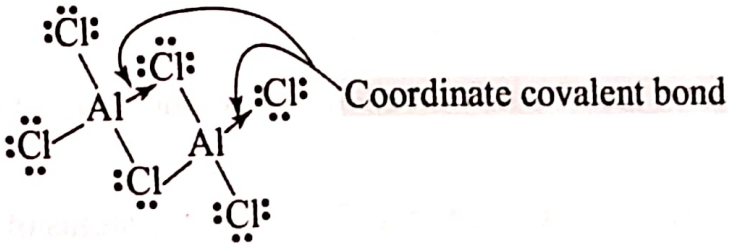
c. 

128. In the gas phase, aluminum chloride exists as a dimmer (a unit of two) with the formula Al_2Cl_6.

a. Draw the Lawis structure

b. Indicate the coordinate covalent bond in the molecule.

Solution

a) 

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b) The two arrows represent the coordinate covalent bonds.

129. What will happen to the energy of the system when a chemical bond is formed?

Solution

Energy of the system decreases.

130. PH_3 and H_2S are isoelectronic and their dipole moments are 0.56D and 0.95D respectively. Which substance has a higher boiling point?

Solution

H_2S because it has higher dipole moment.

131. Sulphur forms both SF_2 and SF_4 molecules but oxygen forms only OF_2. Explain.

Solution

Sulphur expands its octet because it is found in period 3 and has vacant 3d orbitals but not oxygen.

132. The bond angle of H_2O is 104.5^0 and H_2S is 92^0. Explain this difference.

Solution

The larger size of S - atom as compared to O - atom minimizes electron repulsions and allow the bond angle to be smaller.

133. The bond energy for H - A, H - B, H - C and H - D are 436, 565, 297 and 427 respectively in kJ/mol. Arrange these molecules in increasing order of boiling point.

Solution

As bond energy increases, bond strength also increases. As a result, boiling also increases.

H - C < H - D < H - A < H - B

134. Four atoms are arbitarily labelled A, B, C and D. Their electronegative are as follows:

A = 3.8, B = 3.3, C = 2.8 and D = 1.2. If the atoms of these elements form the molecules AB, AC, AD and BD, how would you arrange these

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molecules in increasing order of polarity?

Solution

As the electro negativity gap between the bonded atoms increase, polarity also increases.

AB = 3.8 - 3.3 = 0.5 BD = 3.3 - 1.2 = 1.1

AC = 3.8 - 2.8 = 1.0 ∴ AB < AC < BD < AD

AD = 3.8 - 1.2 = 2.6

135. Predict the molecular shape of the following species.

a. $AlCl_3$ d. AlH_4^- g. IF_5

b. $ZnCl_2$ e. IF_2^- h. IF_4^-

c. $TeCl_4$ f. IF_3 i. $SeCl_6$

Solution

a. Trigonal planar d. Tetrahedral g. Square pyramidal

b. Angular e. Linear h. Square planar

c. See - saw f. T - shape i. Octaheral

136. Arrange the following bonds in decreasing order of polarity

a. P—H , H—O , H—Cl b. O—F , H—F , Cl—F

Solution

a. H—O > H—Cl > H—P b. H—F > Cl—F > O—F

137. List properties of water that results from hydrogen bonding.

Solution

- High boiling point
- High heat of vaporization
- High specific heat, high ionic conductance of hydronium and hydroxide ions.

138. What type of atom or molecule has induced dipoles?

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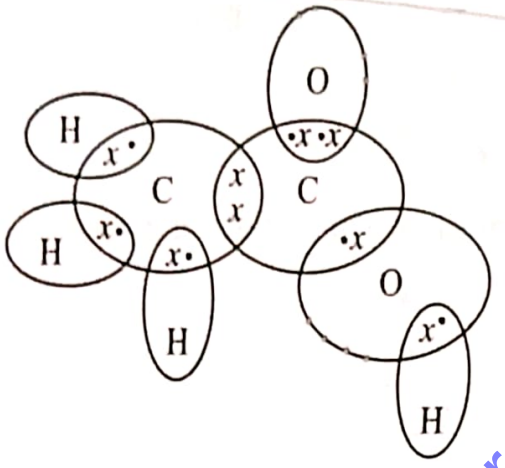
Solution

An atom or a molecule that is temporary polarized.

139. The diagram shows the arrangment of the outer electrons only in a molecule of ethanoic acid.



a. Name the different element found in this compound

b. What is the total number of atoms present in this compound?

c. Between which two atoms is there a double covalent bond?

d. How many single covalent bonds each carbon have?

Solution

a. Carbon, hydrogen and oxygen

b. Eight (4H + 2O + 2C)

c. Between carbon and oxygen that cantain two pairs of shared electrons.

d. From left to right, the first four and the second carbon two.

140. What are the preconditions for the formation of covalent bond according to valence bond theory?

Solution

- When a half - filled orbital of one atom overlaps with a half - filled orbital of another atom.
- The total number of electrons in both orbitals is no more than two.

141. What is hybridization?

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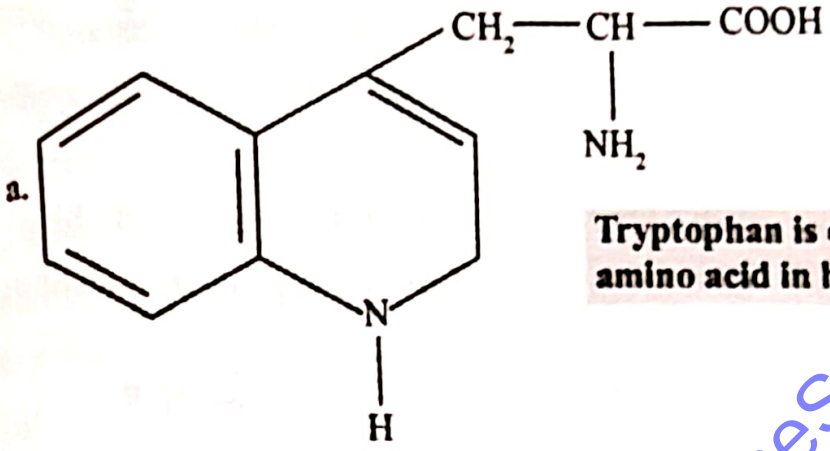
Solution

It is an imaginary mixing process in which the orbitals of an atom rearrange themselves to form new atomic orbitals called hybrid orbitals.

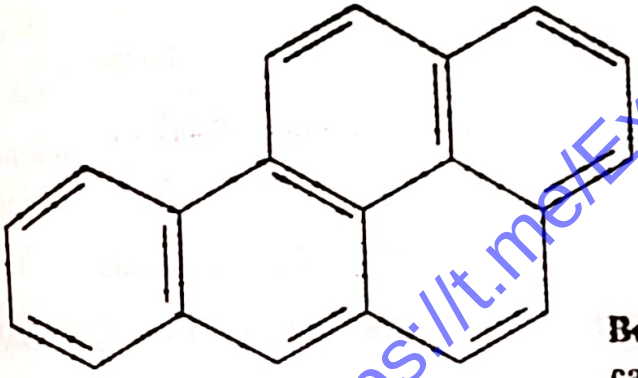
142. Determine the number of sigma and pi bonds in the compounds below

a.



Tryptophan is one of the 20 amino acid in human body

b.



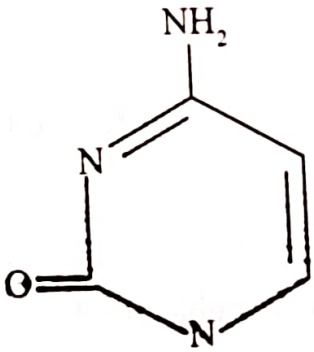
Benzo (α) pyrene - is a potent carcinogent found in coal and cigarette smoke

Solution

a. 31δ and 5δ bonds

b. 35δ and 10π bonds

143. What is the hybridization of C and N in this molecule?



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Solution

☑ All the carbon atoms are sp^2 hybridized

☑ The nitrogen with double bond is sp^2 hybridized. The other two N atoms are sp^3 hybridized.

144. How many orbitals do the following hybrid orbitals contain?

a. sp^2 b. sp^3 c. sp^3d d. sp_3d^2

Solution

a. Three b. Four c. Five d. Six

145. How many p - atomic orbitals are required to generate a set of the following hybrid orbitals?

a. sp b. sp^2 c. sp^3d

Solution

a. One b. Two c. Three

146. Compare the concepts of hybrid and molecular orbitals.

Solution

Hybrid Orbitals	Molecular Orbitals
* All have same energy	* BMO has lower energy ABMO
* One central atom	* More than one central atom
* Same number	* Same number

147. Arrange the following species in order of increasing stability Li_2, Li_2^+, Li_2^-

Solution

$$Li^+_2 = Li^-_2 < Li_2$$

148. Determine the bond order C—N , C=N and C≡N bonds.

Solution

C—N: BO = 1 because it is single bond

C=N: BO = 2 because it is double bond

C≡N: BO = 3 since it is triple bond

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149. The order of bond length of O - O bond in O_2, H_2O_2 and O_3 is ____

Solution

$H_2O_2 > O_3 > O_2$

Supplementary Exercise

1. Multiple Choice Items

1. The nature of bonding in CH_4 and NaH respectively

   A. Ionic, covalent
   C. Covalent, ionic
   B. Ionic, metallic
   D. Metallic, ionic

2. The number of electrons being shared in oxygen molecule is

   A. 4 B. 2 C. 6 D. 8

3. Which of the following compound is most ionic?

   A. NaCl B. Na F C. KCl D. KF

4. A molecule AX_4 has a square planar shape. The number of non - bonding pair on the central atom is

   A. 1 B. 0 C. 2 D. 3

5. Which one is not present in CS_2 molecule?

   A. a π bond
   C. a dative bond
   B. a δ bond
   D. Lone pairs

6. The shape of the molecular structure which will have the bond angle of 120^0 around the central atom is:

   A. Triangular planar
   C. Bent shape
   B. Square planar
   D. Tetrahedral shape

7. All of the following are paramagnetic. EXCEPT

   A. O_2 B. C_2 C. NO D. CN^+

8. Which of the following have identical bond order?

   I. CN^- II. CN^+ III. O^-_2 IV. NO^+

   A. I and IV
   C. III and IV
   B. II and III
   D. II and IV

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9. What is the correct sequence of stability?

A. $O_2^+ > O_2^- > O_2$ C. $O_2 > O_2^- > O_2^+$

B. $O_2^+ > O_2 > O_2^-$ D. $O_2^- > O_2^+ > O_2$

10. The hydrogen bond is strongest in

A. O—H···S C. S—H···O

B. F—H···F D. F—H···O

11. How many resonance structure can be drawn for the nitrite ion (NO_2^-)

A. 1 B. 2 C. 3 D. 4

12. In which of the following species does the central atom not obey the octet rule?

A. ClO_2^- B. CO_2 C. HCN D. ICl_4^-

13. The change in hybridization (if any) of the Al atom in the following reaction is

$$Al Cl_3 + Cl^- \longrightarrow Al Cl_4^-$$

A. sp to sp^2 C. sp to sp^3

B. sp^2 to sp^3 D. sp^2 to sp^2

14. The structure of ClO_3F is:

A. Tetrahedral C. Square - Planar

B. Trigonal - planar D. Trigonal bipyramid

15. The bond formed when ammonia reacts with aluminum chloride is:

A. Ionic bond C. Metallic bond

B. Covalent D. Dative bond

16. Only one has linear geometry, which one?

A. OF_2 B. COS C. H_2O D. CH_4

17. Which one of the following represent the correct electron - pair geometry and molecular geometry of Cl_2O?

A. Linear, linear C. Trigonal planar, T - shape

B. Tetrhedral, bent D. Square planar, octahedral

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18. Which of the following has the shortest bond length?

A. O_2 B. N_2H_2 C. N_2 D. F_2

19. A neutral molecule having the general formula AB_3 has two unshared pair of electrons on A. What is the hybridization of A?

A. sp B. sp^2 C. sp^3 D. sp^3d

20. Given CF_4, CS_2, PH_3, PCl_5 and SF_6. Which of the following does not describe any of the molecules above?

A. Linear C. Square planar

B. Octahedral D. Tetrahedral

21. Which of the following molecules has the greater number of lone pair of electrons?

A. CCl_4 B. NCl_3 C. BCl_3 D. $BeCl_2$

22. Which of the following contains both ionic and dative bond?

A. HCl B. NH_4Cl C. NaCl D. CCl_4

23. Identify the compound formed by an overlap of sp^3 and s orbitals?

A. BeH_2 B. BCl_3 C. CH_4 D. PCl_3

24. Which one of the following statements is correct?

A. Valence bond model explain molecular shape from the interaction of atomic orbitals.

B. In Lewis model, electrons repel one another and a pair of bonding electrons lies far apart as possible.

C. Lewis model predicts the shapes of molecules based on electron - pair repulsions around a central atom.

D. Valence bond model simply shows the linkage between atoms and the presence of lone pairs.

25. Which statement describes the formation of a covalent bond?

A. Shared between non - metallic atoms

B. Transferred from non - metallic atoms to metallic atoms.

C. Shared between metallic atoms.

D. Transferred from metallic atoms to non - metallic atoms.

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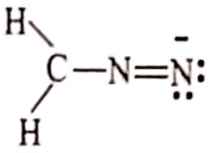
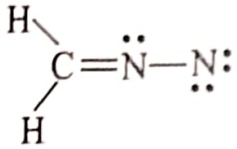
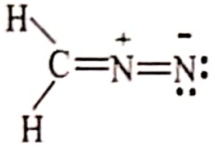
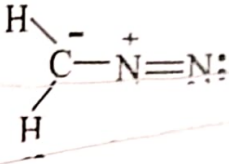
26. Which of the following is a correct Lewis structure for diazomethane (CH_2N_2)?

A. 

C. 

B. 

D. 

27. Unequal distribution of electrons between the bonded atoms results

A. Ionic bond
C. Non - polar covalent bond
B. Polar covalent bond
D. Dative bond

28. If an electron is lost by one atom and completely captured by another. What is this type of bond called?

A. Ionic bond
C. Metallic bond
B. Covalent bond
D. Hydrogen bond

29. The weakest bond among the following is:

A. Polar bond
C. Metallic bond
B. Covalent bond
D. Hydrogen bond

30. Which of the following has least melting point?

A. $CaCl_2$ B. CaF_2 C. $CaBr_2$ D. CaI_2

II. Short answer questions

31. Which one in each pair is expected to exhibit hydrogen bonding?

a. $CH_3\ CH_2OH$ or $CH_3\ CH_3$
c. $(CH_3)_2\ NH$ or $(CH_3)_3\ N$
b. $CH_3\ SH$ or $CH_3\ NH_2$
d. $CH_3\ COOH$ or $HCOOCH_3$

32. Which element in the periodic table form ionic bond?

33. Write the Lewis structure for

a. Barium sulphide
b. Aluminum fluoride

34. Which compound has stronger ionic bonding?

a. NaF or MgF_2
b. MgO or Al_2O_3

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35. Which compound has stronger attraction between ions?

a. LiF or NaF b. CaF_2 or KF

36. Write two cations that obey the pseudonoble gas configuration.

37. Determine the total number of valence electrons for

a. COS b. NO^-_3 c. H_3O^+

38. Explain the difference between intramolecular forces and intermolecular forces.

39. List the three types of attractive forces that exist between natural molecules.

40. To obey the octet rule atoms _____ , _____ and _____

41. Which one is more stable?

a. NO^+ or NO

b. CO or CO^+

c. CN or CN^-

d. N_2 or N^-_2

42. Write the types of orbitals that form the following molecular shape

a. Octahedral

b. linear

c. Trigonal bi pyramid

d. Trigonal planar

e. Tetrhedral

43. Show the direction of the bond moment or dipole moment for the bonds below.

a. H - F b. N - H c. N - O d. C - Cl

44. Classify the following as conduction conductor or non - conductor

a. NaCl (1)

b. Al(s)

c. NaCl(s)

d. Al(1)

e. NaCl (aq)

f. $C_{12}H_{22}O_{11}$(aq)

45. Which intermolecular forces is dominant in the following substances.

a. IF_3 b. Ar c. $NH_2 - CH_2 - COOH$

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Answer to Supplementary Exercise on Unit 2

Part I

1. C	5. C	9. B	13.B	17.B	21.A
2. A	6. A	10.B	14.A	18.C	22.B
3. D	7. B	11.B	15.D	19.D	23.C
4. C	8. A	12.D	16.B	20.C	24.A

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25. A 26. B 27. B 28. A 29. D 30. D

Part II

31. a. CH_3CH_2OH b. CH_3NH_2 c. $(CH_3)_2NH$ d. CH_3COOH

32. Elements on the left (metals) and the right (non- metals) sides of the periodic table.

33. a. Ba^{2+} $[:\ddot{S}:]^{2-}$ b. Al^{3+} $[:\ddot{F}:]^{-}_{3}$

34. a. MgF_2 c. Al_2O_3

35. b. LiF b. CaF_2

36. Ag^+, Zn^{2+}

37. a. 16 b. 24 c. 8

38.

Intra molecular forces (Chemical bonds)	**Inter molecular forces**
* Forces in the molecule	* Forces between molecules
* Stronger	* Weaker
* Related to chemical properties	* Related to physical properties

39. Dipole - dipole force, hydrogen bond and London force.

40. Lose electrons, gain electrons and transfer electrons

41. a. NO^+ b. CO c. CN^- d. N_2

42. a. sp^3d^2 b. sp c. sp^3d d. sp^2 e. sp^3

43. a. $H \longrightarrow F$ b. $N \longleftarrow H$ c. $N \longrightarrow O$ d. $C \longrightarrow Cl$

44. a. conductor b. conductor c. non - conductor
d. conductor e. conductor f. non - conductor

45.
a. Dipole - dipole force
b. London dispersion force
c. Hydrogen bonding

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UNIT 3

Physical States of Matter

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3.1 Introduction

- ☑ In all the discussions of this unit any material object whether simple or complex will be referred as **matter.**
- ☑ **Matter** is any thing that has mass and occupies space.
- ☑ Matter exists in three main states - gas, liquid and solid.
- ☑ The main difference in the three states lies in the arrangement of atoms, molecules or ions constituting the matter.

1. Solids

- ☑ They have definite shape and volume.
- ☑ Compactness of the molecules is responsible for the definite shape and volume of solids.
- ☑ The stronger cohesive forces in a solid are responsible for their higher density, hardness, strength and resist the effect of heating with the exception of ice.
- ☑ They occur at low temperature and high pressure

Example

Salt, sugar, metals, stone etc

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2. Liquids

- ☑ They have no definite shape but have definite volume.
- ☑ They occur at intermediate temperature and pressure
- ☑ They are intermediates between solids and gases.
- ☑ They flow readily (are fluids)

Example

Water, blood, milk, oil etc.

3. Gases

- ☑ They have no definite shape and volume.
- ☑ The molecules in gases are not orderly arranged.
- ☑ The molecules in a gas are far apart and are moving at a very high s in all directions.
- ☑ Gases fill completely any container in which they are placed.
- ☑ They are fluids.
- ☑ They occur at high temperature and low pressure.

Example

➥ Oxygen, Nitrogen, carbondioxde etc.

Note: Gases and liquids are **fluids**. because they flow readily.

➥ Solids and liquids are **condensed state** since they have hig density than gases.

4. Plasma

- ☑ It is often called "the fourth state of matter"
- ☑ It occurs at very high temperature.
- ☑ Heating a gas will form a **plasma** - a soup of positively charged parti
- ☑ It is ionized gas.
- ☑ No material can occur in the plasma state.

Note: The four states of matter are inter convertible eithe by heating or by cooling.

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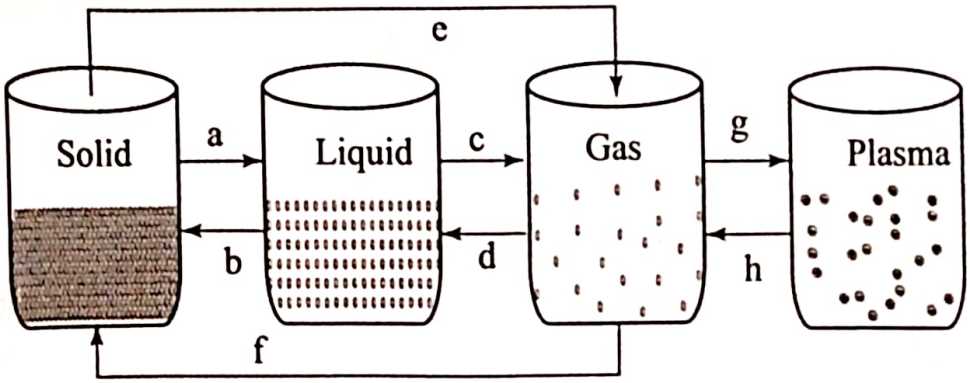

Fig 3.1 Particle Arrangrnment and Inter Conversion Between Solid, liquid, Gas and Plasma States.

a. Melting or fusion
c. Evaporation or vaporization
e. Sublimattion
g. Ionization
} occur by heating

b. Freezing or solidification
d. Condensation
f. Deposition or resublimation
h. Deionization
} occur by cooling

Illustrative Examples

1. Which of the following exists at very high temperature?

A. Solid B. Liquid C. Gas D. Plasma

Answer: D

2. Which two states of matter are fluids?

A. Solid and liquid
B. Solid and gas
C. Liquid and gas
D. Gas and plasma

Solution

Liquid and gas are fluids because they flow readily.

Answer: C

3. Which of the following is not true about gases?

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A. Particles of gases are highly discorded

B. Gases diffuse spontaneously into each other

C. Gases are lighter than solids and liquids

D. Gases have lower energy than solids.

Answer: D

4. Arrange solid, liquid and gaseous states in increasing order of:
   a. Compressibility
   b. Strength of forces that hold particles together
   c. Density
   d. Kinetic energy (motion of particles)

Solution

a. Solid < liquid < gas
b. Gas < liquid < solid
c. Gas < liquid < soil
d. Solid < liquid < Gas

5. _____ is evaporation with out melting.

Answer: Sublimation

6. Why there is no material in the plasma state?

Solution

This is due to the extreme temperature.

7. Water exists in three states of matter. List them

Solution

Ice - solid, water and water vapour.

3.2 The kinetic Theory of Matter

☑ **The kinetic theory** helps to explain the way in which matter behaves.

☑ The evidence is consistent with the idea that all matter is made up of tiny **particles.**

☑ This theory explains the physical properties of matter in terms of the movement of its constituent particles.

☑ **The main points of the theory are:**

1. All matter is made up of tiny, moving particles, invisible to the naked

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eye, different substances have different types of particles (atoms, molecules or ions)

2. The particles move all the time. The higher the temperature, the faster they move on average.
3. Heavier particles move more slowly than lighter ones at a given temperature.
4. The difference between solid, liquid and gas is the energy they contain and motion of the particles.
5. The temperature of a substance is the measure of the average kinetic energy of the particles.
6. The attractive forces between the particles are **called intermolecular forces.** The closer the particles the stronger is the attractive forces.

The Kinetic Molecular Theory of Gases

1. Gas molecules are very apart and have high freedom of motion.
2. Collisions between molecules of gases is perfectly elastic.
3. The attractive force between gas molecules is almost negligible.
4. The volume of the gas is negligible compared to the total volume of the gas.
5. The average kinetic energy of the gas is directly proportional to the absolute temperature of gas. As temperature increases, average kinetic energy of particles also increases.

$$KE = \frac{1}{2} mV^2$$

KE - Kinetic energy

m - mass of particle of gas

V - Velocity of particle

Illustrative Examples

8. **Which of the following assumptions of the kinetic molecular theory of gases explains the high compressibility of gases? (ESSLCE)**

   A. Particles are always in random motion.

   B. Different particles have different speeds.

   C. The actual volume of gas molecules is negligible.

   D. There is no force of attraction between gas molecules.

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Solution

There is a lot of space between gas particles. The total volume of the molecules is extremely very small compared to the volume of the gas.

Answer: C

9. Which of the following statements is True about gases? (ESSLCE)

A. The volume of the gas is equal to the volume of its container

B. When a gas sample is heated at constant pressure, its volume increases.

C. The volume of a gas is kept constant as pressure increases.

D. The particles of gases are closely packed together.

Answer: B

10. According to kinetic theory of matter

A. All matter is made of small particles that has space between them.

B. Particles in a liquid have less energy than in a solid.

C. Particles in gases are held together by strong attractive forces.

D. Particles in solid move freely than in a liquid.

Solution

Energy of particles: gas > liquid > solid

Freedom of motion: gas > liquid > solid

Compressibility: gas > liquid > solid

Attractive force: Solid > liquid > gas

Answer: A

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3.3 The Gas Laws

Gas laws relate volume, pressure, temperature and number of mole.

I. Volume (V)

- It is the space occupied by an object.
- The SI unit of volume is cubic meter (m^3)

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- Other units of volume are liter (L), cubic decimeter (dm^3), milliter (mL), cubic centimeter (cm^3)

Relationship between units volume

$1m^3 = 10^3L = 10^3dm^3 = 10^6mL = 10^6\ cm^3$

$1L = 1dm^3 = 10^3mL = 10^3cm^3$

$1mL = 1cm^3 = 10^{-3}\ L = 10^{-3}\ dm^3 = 10^{-6}\ m^3$

Molar volume: is the volume of **1 mole** of any gas at standard temperature (0^0 C or 273K) and pressure (1atm). **V = 22.4L**

Example

Convert $0.065m^3$ into L, dm^3, mL and cm^3

Solution

$$0.065m^3 = 0.065\cancel{m^3} \times 10^3 \frac{L}{m^3} = 65L = 65dm^3$$

$$0.065m^3 = 0.065 \times 10^6 cm^3 = 6.5 \times 10^4\ cm^3 = 6.5 \times 10^4\ ml$$

2. Pressure (P)

- It is force divided by area

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} \qquad \boxed{P = \frac{F}{A}}$$

- The SI unit of pressure is **Pascal (Pa)**

$\boxed{1Pa = 1N/m^2}$

- Other units of pressure are atmosphere (atm), torr, millimeter of mercury (mmHg), centimeter of mercury (cmHg).
- Relationship between units of pressure

1 atm = 76 cmHg = 760 mmHg = 760 torr = 101.3 kpa

1 cm Hg = 10 mm Hg = 10 torr

1 mm Hg = 1 torr

1kpa = 1000 Pa

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Example

Convert 1520mmHg into atm

Solution

1 atm = 760 mmHg

? = 1520 mm Hg

$$\Rightarrow 1520\ \text{mmHg} = \frac{1\text{atm} \times 1520\text{mmHg}}{760\text{mmHg}} = 2\text{atm}$$

3. Temperature (T)

- ☑ **Temperature** is the measure of the average kinetic energy of particle
- ☑ It is the measure of hotness and coldness of an object.
- ☑ Its SI unit is **Kelvin.**
- ☑ The lowest attainable temperature is **- 273. 15°C or 0 K.** It is called **absolute Zero.**
- ☑ The Kelvin and degree celsius are related by the formula:

$$K = {}^{0}C + 273.15$$
$${}^{0}C = K - 273.15$$

For calculation we may omit 0.15.

Example

Convert human body temperature (37°C) in to Kelvin

Solution

$K = {}^{0}C + 273 = 37 + 273 = 310\ K$

Relationship Between Degree Celcius and Degree Fahrenheit (°F)

The freezing and boiling points of water are 0°C (32°F) and 100°C (212°F respectively. Subtract the smallest value from both sides and by rati method find the relationship.

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$$\Rightarrow \frac{^{0}C - 0}{^{0}F - 32} = \frac{100 - 0}{212 - 32}$$

$$\frac{^{0}C}{^{0}F - 32} = \frac{100}{180} = \frac{5}{9}$$

$$\Rightarrow \boxed{\begin{array}{l} ^{0}C = \frac{5}{9}(^{0}F - 32) \\ ^{0}F = \left(\frac{9}{5} \times {}^{0}C\right) + 32 \end{array}}$$

Example

Convert 50°C in to °F

Solution

$$^{0}F = \frac{9}{5} \times {}^{0}C + 32 = \frac{9}{5} \times 50 + 32 = 122^{0}F$$

4. Mole (n)

1 mole = 6.02×10^{23} particles (atoms, molecules, ions, formula units)

1 mole = 1 gram - atom = atomic mass = 6.02×10^{23} atoms

1 mole = 1 gram - molecule = molecular mass = 6.02×10^{23} molecules

1 mole = 1 gram - formula = Formula mass = 6.02×10^{23} formula units

Example

1 mole Na = 1 gram - atom Na = 23g Na = 6.02×10^{23} atoms of Na

1 mole H_2O = 1gram - molecule H_2O = 18g H_2O

= 6.02×10^{23} molecules of H_2O

1 mole NaCl = 1 gram - formula NaCl = 58. 5g Na Cl

= 6.02×10^{23} formula units of Na Cl

$$\boxed{\begin{array}{l} \text{Number of mole} = \frac{\text{given mass}}{\text{Molar mass}} \\ n = \frac{m}{M} \end{array}}$$

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Example

Calculate the number of mole of 360g of glucose ($C_6H_{12}O_6$). (Atomic mass: C = 12, O = 16, H = 1)

Solution

m = 360g, M = 6 × 12 + 12 × 1 + 6 × 16 = 180g/ mol

$$n = \frac{m}{M} = \frac{360g}{180g/mol} = 2mol$$

Illustrative Examples

11. All of the following quantities represent the same amount, **EXCEPT**.

A. A mole
B. A gram - molecule
C. a formula unit
D. a gram - atom

Solution

A mole = a gram - atom = a gram molecule

= a gram formula = 6.02×10^{23}

$$\text{A fromul unit} = \frac{1}{6.02 \times 10^{23}}$$

Answer: C

12. -40^0 F _____ K

A. 200 B. 300 C. 273 D. 233

Solution

➥ $-40^0F = -40^0C$

➥ $K = {}^0C + 273 = -40 + 273 = 233$ K

Answer: D

13. Which one is not equal to 0.1d m^3

A. 0.1L B. 100m1 C. 0.1m^3 D. 100cm^3

Solution

$$0.1dm^3 = 0.1L = 0.1dm^3 \times 1000 \frac{m_l}{dm^3} = 100m1 = 100cm^3$$

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$$0.1dm^3 = 0.1dm^3 \times 0.001 \frac{m^3}{dm^3} = 0.0001m^3$$

Answer: C

14. The lowest attainable temperature is

A. -273.15^0C B. 273.15^0C C. ^0C D. -100^0C

Answer: A

15. Convert the following into atmosphere.

a. 100cmHg b. 38mmHg c. 1000 kpa

Solution

a. latm = 76 cmHg

? = 100 cm Hg

$$\Rightarrow 100cm\,Hg = \frac{1atm \times 100cm\,Hg}{76mmHg} = 1.32\ atm$$

b. latm = 760 mmHg

? = 380mm Hg

$$\Rightarrow 380mm\,Hg = \frac{1380mm\,Hg \times 1atm}{760mmHg} = 0.5\ atm$$

c. latm = 101.3 kpa

? = 1000kpa

$$1000\ kpa = \frac{latm \times 1000\ kpa}{101.3kpa} = 9.87\ atm$$

16. Convert 25^0C in to

a. K b. ^0F

Solution

a. $K = {}^0C + 273 = 25 + 273 = 298K$

b. $^0F = \frac{9}{5} \times 25 + 32 = 77\ ^0F$

17. Convert the following in to Kelvin.

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a. -173^0C b. 212^0F

Solution

a. $K = {}^0C + 273 = -173 + 273 = 100K$

b. $212^0F = 100^0C$, $K = 100 + 273 = 373K$

18. Calculate the mass of 1.5mol of H_3PO_4

Solution

$$M = 98\ g/mol, m = n \times M = 1.5mol \times 98\ \frac{g}{mol} = 147g$$

19. Determine the number of molecules of 24g of CH_4

Solution

$$M = 16g/mol,\ n = \frac{m}{M} = \frac{24g}{16g/mol} = 1.5mol$$

$$\text{Number of molecules} = n \times 6.02 \times 10^{23}\ \frac{\text{molecules}}{mol}$$

$$= 1.5\ mol \times 6.02 \times 1023\ \frac{\text{molecules}}{mol}$$

20. 10g of a compound has 0.25 mol. Find its molar mass.

Solution

$$M = \frac{m}{n} = \frac{10g}{0.25mol} = 40g/mol$$

Gas Laws

. Boyle's law : The volume - pressure Relationship

oyle's law states that " the volume of a fixed amount of gas maintained t constant temperature is inversely proportional to the gas pressure"

athematically: $V\ \alpha\ \frac{1}{P}$ (at constant temperature)

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PV = constant (K)

$K_1 = P_1 V_1$, $K_2 = P_2 V_2$ but K is constant

$\Rightarrow K_1 = K_2 \Rightarrow \boxed{P_1 V_1 = P_2 V_2}$

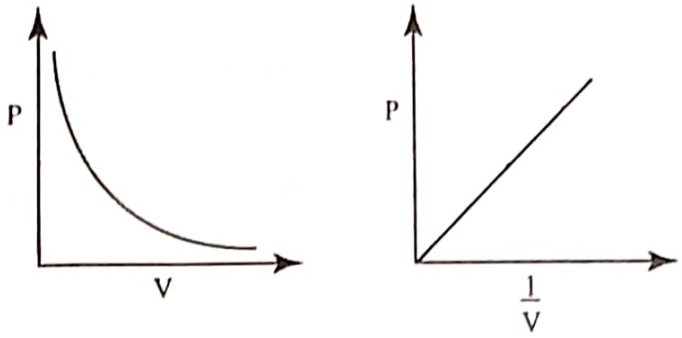

Fig 3.2 Graphs showing Variation of the Volume of a Gas Sample with the Pressure Exerted at Constant Temperature.

Example

In the decomposition of hydrogen peroxide, if the volume of oxygen gas collected is $100cm^3$ at 1×10^5 pa. What would be the volume of the gas at a pressure of 2×10^5 pa constant temperature.

Solution

$V_1 = 100cm^3$, $P_1 = 1 \times 10^5$ pa

$V_2 = ?$ $P_2 = 2 \times 10^5$ pa

Based on Boyle's law, if the pressure doubles the volume halves and vice - versa.

$$\Rightarrow V_2 = \frac{1}{2} \times V_1 = \frac{1}{2} \times 100cm^3 = 50cm^3 \text{ or}$$

By using the formula

$$P_1 V_1 = P_2 V_2 \Rightarrow V_2 = \frac{P_1 V_1}{P_2} = \frac{1 \times 10^5 \text{ pa} \times 100cm^3}{2 \times 10^5 \text{ pa}} = 50cm^3$$

Illustrative Examples

21. Which of the following statements correctly describes Boyle' law?(ESSLCE)

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A. The volume of a gas at a given temperature is proportional to the square root of the applied pressure

B. The volume of gas at a given temperature is inversely proportional to the square root of the applied pressure.

C. The volume of a gas at a given temperature is proportional to the applied pressure.

D. The volume of gas at a given temperature is inversely proportional to the applied pressure.

Solution

Boyle's Law: $V \alpha \frac{1}{P}$ (at constant T)

Answer: D

22. A sample of gas has a volume of 183 cm^3 and a pressure of 740 mmHg. If the volume is changed to 116cm^3, assuming the temperature is held constant, what will be the new pressure?

A. 856 mm Hg C. 28.7 mm Hg

B. 469 mm Hg D. 1167mm Hg

Solution

$P_1 = 740$ mmHg, $V_1 = 183$ cm^3, $P_2 = ?$, $V_2 = 116 cm^3$

$$P_2 = \frac{P_1 V_1}{V_2} = \frac{740 mmHg \times 183 \cancel{cm^3}}{116 \cancel{cm^3}} = 1167 \text{ mmHg}$$

Answer: D

23. By what factor must the pressure be increased to compress 8L of nitrogen to 2.5L?

Solution

$$P_2 = \frac{V_1 P_1}{V_2} = \frac{8 \cancel{L} \times P_1}{2.5 \cancel{L}} = \frac{16}{5}$$

Thus, the pressure increases by a factor of $\frac{16}{5}$

24. A gas in a balloon exerts a pressure of 1 atm. If the temperature is kept constant, at what pressure will the volume of the gas double?

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Answer : $P_2 = 0.5atm$

25. A gas occupies a volume of 5L at 300 torr. The pressure increases to 400torr at constant temperature, what is the new volume?

Solution

$$V_2 = \frac{P_1 V_1}{P_2} = \frac{300 \text{ torr} \times 5L}{400 \text{torr}} = 3.75L$$

2. Charles' Law: The Volume - temperature Relationship

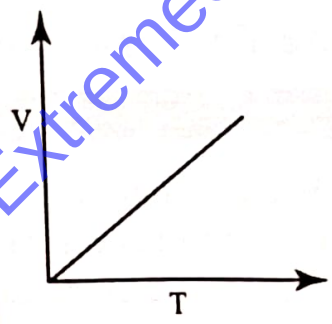
Charles' law states that the volume of a fixed mass of gas is directly proportional to its absolute temperature, if the pressure is kept constant.

Mathematically: $V \alpha T$ (at constant pressure)

$$\frac{V}{T} = K \text{ (constant)}$$

$$K_1 = \frac{V_1}{T_1}, \; K_2 = \frac{V_2}{T_2}, \text{ but K is constant}$$

$$\Rightarrow K_1 = K_2 \Rightarrow \boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2} \text{ or } V_1 T_2 = V_2 T_1}$$



Example

A gas syringe contains $50cm^3$ of oxygen gas at 20^0C. If the temperature was increased to 45^0C, What would be the volume occupied by this gas, assuming constant pressure throughout?

Solution

$V_1 = 50cm^3, T_1 = {}^0C + 273 = 20 + 273 = 293K$

$V_2 = ?, T_2 = {}^0C + 273 = 45 + 273 = 318\,K$

$$V_2 = \frac{T_2 V_1}{T_1} = \frac{318K \times 50cm^3}{293K} = 54.3cm^3$$

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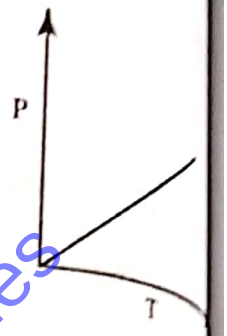
3. Gay Lussac's law: The Temperature - pressure Relationship

Gay Lussac's Law states that "the pressure of a fixed amount of a gas is directly proportion to Kelvin temperature at constant volume."

Mathematically: $P \alpha T$ (at constant Volume)

$$\frac{P}{T} = K \text{ (constant)}$$



$K_1 = \frac{P_1}{T_1}$, $K_2 = \frac{P_2}{T_2}$ but $K_1 = K_2$ since it is constant. Thus, $\boxed{\frac{P_1}{T_1} = \frac{P_2}{T_2} \text{ or } P_1 T_2 = P_2 T_1}$

Note: Gay Lussac's law is Simply charles' law

Illustrative Examples

26. Which of following gas laws states the volume of a fixed amount of a gas maintained at constant pressure is directly proportional to the absolute temperature of the gas?

A. Combined gas law
B. Avogadro's law
C. Boyle's law
D. Charles' law

Solution

$V \alpha T$ (at constant pressure): Charles' law

Answer: D

27. If you heat a 5L balloon from a temperature of 25°c to 50°c, its new volume will be:

A. 4.61L B. 2.5L C. 10L D. 5.42L

Solution

$V_1 = 5L$, $T_1 = 25 + 273 = 298$ K, $T_2 = 50 + 273 = 323K$

$V_2 = ?$, $V_2 = \frac{T_2 V_1}{T_1} = \frac{323K \times 5L}{298K} = 5.42L$

Answer: D

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28. A sample of gas is sealed in a 150ml flask at 127°C and 380mmHg. What will be its pressure if the gas is cooled to 27°C at constant volume?

Solution

$P_1 = 380mmHg, T_1 = 127^0C = 127 + 273 = 400K$

$P_2 = ?, T_2\ 27^0C = 27 + 273 = 300K$

$$P_2 = \frac{P_1 T_2}{T_1} = \frac{380\,mmHg \times 300\cancel{K}}{400\cancel{K}} = 285mmHg$$

29. At constant pressure, by what factor of its volume will a quantity of a gas change if the temperature changes from - 73°C to 127°C?

Solution

$T_1 = -73^0C = 200K, T_2 = 127^0C = 400K$

$$V_2 = \frac{T_2 V_1}{T_1} = \frac{400K \times V_1}{200K} = 2V_1,$$ thus the volume doubles

30. A given mass of oxygen occupies $20dm^3$ at 47°C. Determine its final temperature in °C when the gas occupies $8dm^3$ of volume at constant pressure.

Solution

$V_1 = 20dm^3, T_1 = 47^0C = 47 + 273 = 320K$

$V_2 = 8dm^3, T_2 = ?$

$$T_2 = \frac{V_2 T_1}{T_2} = \frac{8dm^3 \times 320K}{20dm^3} = 128K$$

$T_2 = K - 273 = 128 - 273 = 145^0C$

31. At what temperature will the volume of the gas be

a. Doubled? b. Tripled

c. Halved if the initial temperature is 327°C?

Solution

T_1 [illegible]27°C = 600K

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a) $V_2 = 2V_1$, $T_2 = \frac{V_2\ T_1}{V_1} = \frac{2V_1 \times 600K}{V_1} = 1200K$

b) $V_2 = 3V_1$, $T_2 = \frac{V_2\ T_1}{V_1} = \frac{3V_1 \times 600K}{V_1} = 1800K$

c) $V_2 = \frac{1}{2}V_1$, $T_2 = \frac{V_2\ T_1}{V_1} = \frac{\frac{1}{2}V_1 \times 600K}{V_1} = 300K$

32. What would be the pressure of the gas, originally at 100kpa i temperature is lowered from 30°C to 20°C at constant volume?

Answer: P_2 = 96 . 7k

33. The pressure of a confined gas is doubled. Calculate the temperat if the original temperature was 0°C. What is the change in temperature?

Solution

$P_2 = 2P_1$, $T_1 = 0^0C = 237K$

➥ $T_2 = 2T_1 = 2 \times 273K = 546K$

$\Delta T = T_2 - T_1 = 546 - 273 = 273K$ or

$\Delta T = T_2 - T_1 = 273 - 0 = 273^0C$

The value of ΔT is same in Kelvin and degree celcius scales.

4. The Combined Gas Law: Temperature, Volume and Pressure Relationship

➥ It is the combination of Boyle's law and Charles' law

$V \alpha \frac{1}{P}$ (Boyle's law)

$V \alpha T$ (Charles' law)

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$V \alpha \frac{T}{P}$ (Combined gas law)

$\frac{PV}{T} = K$ (Conctant)

$K_1 = \frac{P_1 V_1}{T_1}$, $K_2 = \frac{P_2 V_2}{T_2}$, but $K_2 = K_1$

$\Rightarrow \boxed{\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}}$ or $\boxed{T_2 P_1 V_1 = P_2 V_2 T_1}$

Illustrative Examples

34. If the conditions on $166cm^3$ of gas at 93000kpa and 15^0C are increased by 7kpa and 5^0C, what will be the new volume of the gas in liters?

A. 0.16gL B. 160L C. 206L D. 20.6L

Solution

$V_1 = 166cm^3 = 0.166L$, $V_2 = ?$

$T_1 = 15^0C = 288K$, $T_2 = 5^0C + 15^0C = 293K$

$P_1 = 93000kpa$, $P_2 = 7 + 93000 = 93007$ kpa

$$V_2 = \frac{T_2 P_1 V_1}{P_2 T_1} = \frac{293K \times 93000 \text{ kpa} \times 0.166L}{93007 \text{ kpa} \times 288K} = 0.169L$$

Answer: A

35. A sample of gas kept in a 2.0L container at 50^0C exerts a pressure of 700torr. If the volume is halved and temperature is tripled, what will be the final pressure of the gas?

A. 2100torr B. 4200torr C. 1600torr D. 2500torr

Solution

$V_1 = 2L$, $V_2 = \frac{1}{2} \times V_1 = \frac{1}{2} \times 2L = 1L$

$T_1 = 50^0c = 323K$, $T_2 = 3T_1 = 969K$

$$P_2 = \frac{P_1 V_1 T_2}{V_2 T_1} = \frac{2L \times 700 \text{ torr} \times 969K}{1L \times 323K} = 4200torr$$

Answer: B

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36. A gas collected at 227°C has a volume of 20dm³ and a pres 800torr. What would the temperature of the gas be if the vol changed to 500mℓ at standerd pressure?

Solution

$V_1 = 20dm^3$, $V_2 = 500m\ell = 0.5dm^3$

$P_1 = 800torr$, $P_2 = 1atm = 760$ torr (standard condition)

$$T_2 = \frac{P_2 V_2 T_1}{P_1 V_1} = \frac{0.5dm^3 \times 760\,torr \times 500K}{800torr \times 20dm^3} = 11.88K$$

37. A certain gas sample of gas occupies 44.3L at 298K and 21 Calculate its volume at STP.

Solution

$V_1 = 44.8L$, $T_1 = 298\ K$, $P_1 = 210$ kpa

$V_2 = ?$, at STP, $T_2 = 237K$, $P_2 = 1$ atm $= 101.3$kpa

$$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1} = \frac{44.8L \times 210kpa \times 273K}{101.3kpa \times 298K} = 85.1L$$

38. The volume of a gas at 17°C and 1500 torr is 500ml. Calculate its volume at STP.

Solve by your own! $V_2 = 929$ ml

5. Avogadro's law: The volume - Number of Mole Relationship

Avogadro's law states that "At the same temperature and pressure e volume of different gases contain the same number of molecules" reverse of this statement is also true.

Mathematically: $V \alpha n$ (at constant temperature and pressure)

$$\frac{V}{n} = K \text{ (Constant)}$$

$$\frac{V_1}{n_1} = K_1, \frac{V_2}{n_2} = K_2 \text{ but K is constant}$$

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$$K_1 = K_2 \Rightarrow \boxed{\frac{V_1}{n_1} = \frac{V_2}{n_2} \text{ or } V_1 n_2 = V_2 n_1}$$

Example

A 5L container held 2moles of a gas. What volume will be needed to hold 4moles of the gas at the same temperature and pressure?

Solution

$V_1 = 5L, n_1 = 2mol, n_2 = 4mol, V_2 = ?$

➥ If n doubles, V also doubles

➥ $V_2 = 2V_1 = 2 \times 5L = 10L$

or

$$V_2 = \frac{n_2 \times V_2}{n_1} = \frac{4mol \times 5L}{2mol} = 10L$$

6. The Ideal Gas Equation

☑ This equation is derived from **Boyle's law , Charles'** law and **Avogadro's law.**

☑ Boyle's law: $V \alpha \frac{1}{P}$ (at constant T and n)

☑ Charles' law: $V \alpha T$ (at constant P and n)

☑ Avogadro's law: $V \alpha n$ (at constant T and P)

☑ Ideal gas law: $V \alpha \frac{nT}{P}$

$\boxed{PV = nRT}$ ------ Ideal gas equation

R - universal gas constant

At standard temperature and pressure (STP)

1mole of ideal gas at STP occupies 22.4L

$T = 0^0C = 273K$, $P = 1atm$

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$$R = \frac{PV}{nT} = \frac{1atm \times 22.4L}{1mol \times 273K} = 0.082 \frac{atm.L}{mol.K}$$

$PV = nRT$, $n = \frac{m}{M}$, m = given mass

M = molar mass

$\Rightarrow PV = \frac{mRT}{M}$,

$P = \frac{mRT}{VM}$, $p = \frac{m}{V}$, p = density

$$\boxed{P = \frac{pRT}{M}}$$

☑ **An ideal gas** is a hypothetical gas whose pressure volume - temperatu behavior can be completely accounted for ideal gas equation.

☑ The molecules of ideal gas do not attract or repel one another and the volume is negligible compared to the volume of the container.

☑ The ideal gas equation can be applied to determine density or mola mass of gaseous substance.

Illustrative Examples

39. " Equal volume of all gases contain the same number of molecules of gases" This law is

A. Charles' law
C. Ideal gas law
B. Avogadro's law
D. Combined gas law

Answer : B

40. Which one of the following mathematical expressions is derived from the combination of Boyles' law, Charles' law and Avogadro's law? (ESSLCE)

A. $V \propto \frac{PT}{nT}$ B. $V \propto \frac{nP}{T}$ C. $V \propto \frac{T}{nP}$ D. $V \propto \frac{nT}{P}$

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Solution

$V \alpha \frac{1}{P}$ (Boyle's law)

$V \alpha T$ (Charles' law)

$V \alpha n$ (Avogadro's law)

$V \alpha \frac{nT}{T}$ (Ideal gas law)

Answer: D

41. What is the molar volume of a gas? (ESSLCE)

A. The volume of a gas at STP (T = 0^0C and P = 1kpa) which is equal to 22.4L/mol

B. The volume of a gas at STP (T = 25^0C and P = 1Kpa), which is equal to 22.4L/mol.

C. The volume of a gas at STP (T = 0^0C and P = 1atm), whic is equal to 22.4L /mol

D. The volume of a gas at STP (T = 25^0C and P = latm), which is equal to 22.4L/ mol.

Solution

Molar volume is the volume of one mole of any gas STP (T = 0^0C or 273K and P = latm), which is equal to 22.4L.

Answer: C

42. Which of the following statements is true about an ideal gas? It is a hypothetical gas that obeys the gas law.

A. Closely at high temperature and low pressure

B. Closely at high temperature and high pressure

C. At low temperature and low pressure

D. At low temperature and high pressure

Answer: A

43. What is the volume of 2.8 grams of carbon monoxide at 27^0C and 0.82 atm?

A. 3L B. 30L C. 0.27L D. 27L

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Solution

$R = 0.082 \frac{atm.L}{mol.K}$, M (CO) = 28g/mol, m= 2.8g

$T = 27^0C = 300K$, P = 0.82 atm

$PV = \frac{mRT}{M}$

$$\Rightarrow V = \frac{mRT}{MP} = \frac{2.8g \times 0.082 \frac{atm.L}{mol.K} \times 300L}{28g/mol \times 0.82\ atm} = 3L$$

Answer: A

44. Equal volume of different gases under the same conditions of temperature and pressure have equal _____

A. Number of atoms
C. Number of molecules
B. Density
D. Molar mass

Answer: C

45. A 60L balloon is required to b filled with helium gas at 37^0C and a pressure of 8.2 atm. What mass of helium is required?

Solution

V = 60L, $T = 37^0C = 310K$, P = 8.2 atm, M = 4 g/mol

$PV = \frac{mRT}{M} \Rightarrow PVM = mRT$

$$\Rightarrow m = \frac{PVM}{RT} = \frac{8.2atm \times 60L \times 4g/mol}{0.082 \frac{atm.L}{mol.K} \times 310K} = 80g$$

46. Unknown gas has a density of 1.17 g/L at STP. Determine the molar mass of the unknown gas.

Solution

At STP, $T = 0^0C = 273K$, P = latm

$PV = \frac{pRT}{M} \Rightarrow$

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$$\Rightarrow M = \frac{pRT}{P} = \frac{1.17\,g/L \times 0.082\,\frac{atm.L}{mol.K} \times 273\,K}{1\,atm}$$

$$M = 26g/mol$$

47. Calculate the volume of 8g of CH_4 at 227°C and under a pressure of 760 torr.

Solution

$$V = \frac{mRT}{MP} = \frac{8g \times 0.082\,\frac{atm.L}{mol.K} \times 500\,K}{16g/mol \times 1\,atm} = 20.5L$$

48. Compute the approximate density of methane, CH_4 at 20°C and 6atm.

Solution

$$PM = \rho RT$$

$$\Rightarrow \rho = \frac{PM}{RT} = \frac{6atm \times 16g/mol}{0.082atm \times 293K} = 4g/L$$

49. An electric vacuum tube was sealed off during manufacture at pressure of 2×10^{-3} torr at 25°C. Its volume is 596ml, calculate the number of gas molecules remaining in the tube.

Solution

$$n = \frac{PV}{RT} = \frac{\frac{0.002}{760}\,atm \times 0.596L}{0.082\,\frac{atm.L}{mol.K} \times 298K} = 6.42 \times 10^{-8}\,mol$$

$$\text{Number of molecules} = n \times 6.02 \times 10^{23}\,\frac{\text{molecules}}{mol}$$

$$= 6.42 \times 10^{-8}\,mol \times 6.02 \times 10^{23}\,\frac{\text{molecules}}{mol}$$

$$= 3.86 \times 10^{16}\,\text{molecules}$$

50. What is the molecular mass of a gas whose density is 3.5 g/L at 27°C and a pressure of 2atm?

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Solve by your own! M = 44 g/mol

7. Graham's Law of Diffusion: Rate of Diffusion Molar Mass or Density Relationship.

- ☑ It is also called **Graham's law of effusion.**
- ☑ **Diffusion** is the random motion of gases in moving from one another.
- ☑ It is the tendency of a gas to mix spontaneously with and speed throughout another gas.
- ☑ **Effusion** is the term used to describe the passage of a gas through a tiny orifice in to an evacuated chamber.

Graham's law of effusion /diffusion/ relates the rate at which a gas diffuses (or effuses) to the type of molecule in the gas. It can be expressed as follows:

The rate of effusion of a gas is inversely proportional to the square root of its molar mass or density.

Mathematically.

$$\text{Rate of gas} \propto \sqrt{\frac{1}{\text{molar mass of gas}}}$$

or

$$\text{Rate of gas} \propto \sqrt{\frac{1}{\text{Density of gas}}}$$

For two gases A and B

$$\frac{\text{Rate A}}{\text{Rate B}} = \sqrt{\frac{M_B}{M_A}} = \sqrt{\frac{\rho_B}{\rho_A}}$$

- ➥ M_A - molar mass of gas A. ρ_A - density of A
- ➥ M_B - molar mass of gas B . ρ_B - density of gas B

The smaller the molar mass or density the higher is the rate of diffusion or effusion.

Example

$$\text{Rate } H_2 > \text{Rate } N_2 > \text{Rate of } O_2$$

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Illustrative Examples

51. Graham's law refers to

A. Boiling points of gases
B. Gaseous diffusion
C. Gas compression problems
D. Volume change of gases

Answer: B

52. Which one has the highest molecular speed?

A. O_2 B. N_2 C. CO_2 D. CH_4

Solution

The smaller the molar mass, the higher is the molecular speed of the gas.

Answer: D

53. If an unknown gas diffuses 1.73 times faster than krypton, what is the molar mass of the unknown gas? (MKr = 83.80 g/mol)

A. 28.0 g /mol
B. 63. 0 g /mol
C. 110g /mol
D. 251 g/mol

Solution

Let the unknown gas be X.

Rate X = 1.73 × rate of K_r

$$\frac{\text{Rate X}}{\text{Rate } K_r} = \sqrt{\frac{MKr}{Mx}}$$

or

$$\frac{1.73 \text{ rate } K_r}{\text{rate } K_r} = \sqrt{\frac{83.80}{Mx}}$$

$$(1.73)^2 = \left(\sqrt{\frac{83.80}{Mx}}\right)^2 \Rightarrow Mx = 28.0\,g/mol$$

Answer: A

54. Two gases A and B were applied to the two ends of a 40cm long - glass tube. The two gases met at a distance of 30cm the side where B is applied, which statement is true about those gases?

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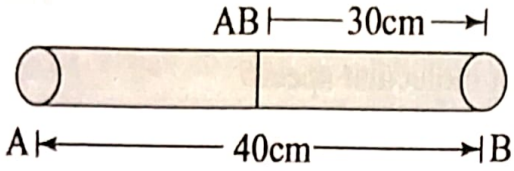
A. The molar mass of A is bigger than the molar mass of B

B. The rate of diffusion of A is bigger than the rate of diffusion

C. The density A is smaller than the density of B.

D. The rate of B is smaller than the rate of diffusion of A

Solution



➥ The rate of A is lower than rate of B

➥ Density and molar mass of A is higher than that of B.

Answer

55. Gas A diffuses twice as fast as gas B. Gas B has a molecular weigh 60. What is the molar mass of gas A?

Solution

Rate A = 2 × rate B

$$\frac{\text{Rate A}}{\text{Rate B}} = \sqrt{\frac{MB}{MA}} \Rightarrow \frac{2\ \text{Rate } \cancel{B}}{\text{Rate } \cancel{B}} = \sqrt{\frac{60}{MA}}$$

$$\Rightarrow (2)^2 = \left(\sqrt{\frac{60}{MA}}\right)^2, \quad MA = 15\ \text{g/mol}$$

56. Helium and argon are monoatomic and their masses are 4 and respectively. What is the rate of diffusion of helium?

Solution

$$\frac{\text{Rate He}}{\text{Rate Ar}} = \sqrt{\frac{MAr}{MHe}} = \sqrt{\frac{40}{4}} = \sqrt{10} = 3.16$$

$\Rightarrow$ Rate He = 3.16 × rate Ar

57. Arrange the following gases in increasing order of rate diffusion.
SO_3, CO_2, N_2, CH_4, NH_3,

Answer: $SO_3 < CO_2 < N_2 < NH_3 < CH_4$

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3.4 The Liquid State

- A Liquid can be described as a form of matter that has a definite volume and takes the shape of its container.
- Its molecules are more ordered than that of gases.

Phase Changes in Liquids

- Liquids undergo two types of changes. These are, evaporation and condensation.
- **Evaporation /Vaporization/** is the process of changing liquid into gas.
- It is an endothermic process.
- **Heat of vaporization (ΔH_{vap})** is the amount of energy necessary to change liquid to gas at constant temperature and pressure.
- **Condensation** is the process of converting gas into liquid.
- It is the reverse of evaporation and is exothermic process.
- **Heat of condensation (ΔH_{cond})** is the amount of heat released when a gas is converted in to liquid.
- ΔHvap and ΔHcond have same magnitude but opposite sign.

$$\text{Liquid} \underset{\text{Condensation}}{\overset{\text{Evaporation}}{\rightleftharpoons}} \text{Gas}, \quad \boxed{\Delta H_{vap} = -\Delta H_{cond}}$$

The liquid and gaseous state are in dynamic equilibrium if the rate of evaporation is equal to the rate of condensation.

- **Rate of evaporation** is the speed at which liquids vaporize.
- It is affected by temperature, surface area, intermolecular forces and types of liquids.

1. **Temperature**: As temperature increases, the rate of evaporation also increases.
2. **Surface area:** As surface area increases, rate of evaporation also increases.
3. **Intermolecular forces:** As strength of intermolecular forces increases, rate of evaporation decreases.
4. **Types of liquids:** Liquids are classified into two based on their tendency to evaporate.

i. **Volatile liquids** - are liquids that have high tendency to evaporate.

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- They have weak intermolecular forces between their particles.
- They have high vapour pressure.

Example

- Benzene, ethanol, diethyl ether etc.

ii. **Non - volatile liquids** are liquids that have low tendency to evapora at room temperature .

- They have relatively strong intermolecular forces between their particles.
- They have low vapour pressure.

Example

Sulphuric acid, mercury, molten salts etc.

☑ **Vapour pressure** is the pressure exerted by vapour

☑ **Boiling** occurs when a liquid is heated in an open container and start to vaporize from every parts of its body. It is the conversion of liqui within as well as the surface of liquid. However, during evaporation only molecules at the surface escapes into the vapour state.

☑ **Boiling point** is the temperature at which the vapour pressure of the liquid is equal to the external atmospheric pressure.

☑ **Normal boiling point** is the temperature at which the external pressure equal to one atmosphere .

☑ The boiling point of water decreases as altitude increases.

Illustrative Example

58. Which of the following statements explains why water vapour condenses as temperature decreases?

A. Vapor molecules are pulled together by the force of gravity.

B. Attractive forces between vapor molecules overcome molecular motion.

C. Collisions between vapor molecules become elastic

D. Vapor molecules chemically react when they collide

Solution

As vapor condenses, temperature decreases and attraction forces

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between particles overcome kinetic energy.

Answer: B

59. A substance has a freezing point of -38°c and a boiling point of 356°C. At what temperature would this substance be in its liquid state?

A. 80°C B. -50°C C. 375°C D. -100°C

Solution

Liquid state occurs between freezing and boiling points.

Answer: A

60. Which sample of carbon dioxide has a definite volume but no definite shape?

A. $CO_2(g)$ B. $CO_2(s)$ C. $CO_2(1)$ D. None

Answer: C

61. Which of the following factors does not have an effect on the rate of evaporation of liquid ?

A. Surface area C. Temperature

B. Volume of liquid D. Intermolecular force

Answer: B

62. All of the following are volatile liquids. **EXCEPT**

A. CH_3OH B. C_2H_5OH C. C_6H_6 D. H_2O

Answer: D

63. What happens at the boiling point of a liquid?

Solution

The vapour pressure over the liquid equals the pressure surrounding the liquid.

64. The boiling points of three liquid A,B and C are 50°C, 78°C and 100°C respectively. Identify the liquid with the highest vapour pressure.

Solution

Liquid A because it has the lowest boiling point.

65. Why polar liquids have lower vapour pressure than non - polar liquids?

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Solution

The strength of intermolecular force in polar liquids is higher than in non - polar liquids.

3.5 The Solid State

- The solid state exists at low temperature and high pressure
- The particles in solids are more ordered than that of liquid.

Phase Changes in Solid

- Solids undergo four phase changes. These are melting, freezing, sublimation and deposition.
- **Melting or fusion** is the process of converting solid into liquid.
- It is an endothermic process.
- **Freezing or solidification or crystallization** is the conversion of liquid to solid.
- It is an exothermic process.

$$\text{Solid} \underset{\text{Freezing}}{\overset{\text{Melting}}{\rightleftharpoons}} \text{Liquid}$$

- **Heat of fusion** (ΔH_{fus}) is the amount of heat required to convert solid into liquid.
- **Heat of solidification or crystallization** (ΔH_{cry}) is the amount of energy released when liquid converted in to solid.
- ΔH_{fus} and ΔH_{cry} have same magnitude and oppositive sign

$$\boxed{\Delta H_{fus} = -\Delta H_{cry}}$$

- **Sublimation** is the direct change of solid to gas.
- It is evaporation without melting and is an endothermic process.
- **Deposition** is the direct change of gas to solid.
- It is exothermic process.
- **Heat of sublimation** (ΔH_{sub}) is the amount of heat required to convert solid to liquid.

$$\boxed{\Delta H_{sub} = \Delta H_{fus} + \Delta H_{vap} = -\Delta H_{dep}}$$

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Phase and Energy Changes in Solids

☑ **Phase** is a homogeneous part of a system in contact with other parts of the system but separated from them by a well - defined boundary.

☑ **Phase changes** are transformations from one phase to another, occur when energy (usually in the form of heat) is added or removed.

☑ Phase changes are physical changes.

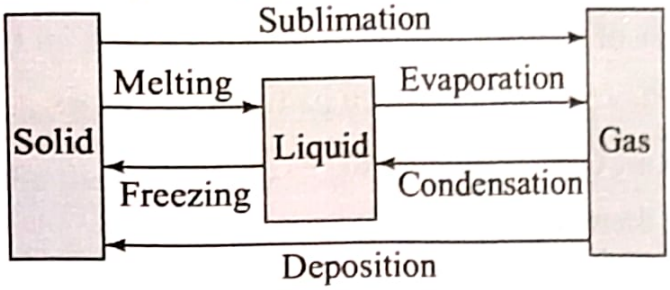

Fig 3.3 The Various Phase Changes that a Substance Undergo.

☑ The changes that take place in physical state of a substance as th temperature is varied can be represented by a plot of temperature versu time curve called **heating curve**.

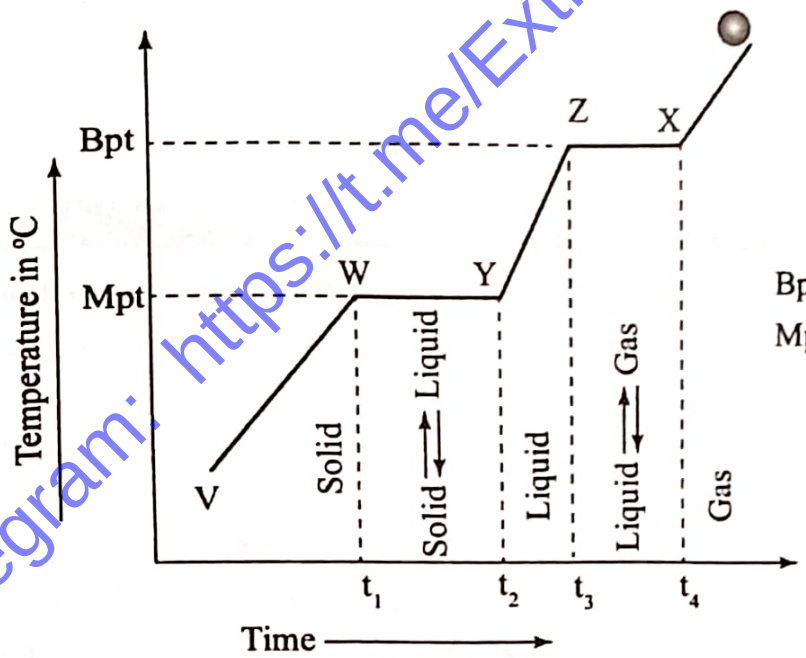

Fig 3.4 Heating Curve

☑ Region V to W : temperature of solid rises

☑ Region W to Y : temperature remains constants (melting point)

☑ Region Y to Z: temperature of liquid rises.

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☑ Region Z to X : temperature remains constant (boiling point)

☑ Above region X : temperature of gases rises.

☑ **For conversion of solid ⟶ Liquid ⟶ Gas**

- ➥ Temperature increases.
- ➥ Order of particles decreases
- ➥ Kinetic energy of particles increases
- ➥ Strength of attraction force between particles decreases.

☑ **For conversion of: Gas ⟶ Liquid ⟶ solid**

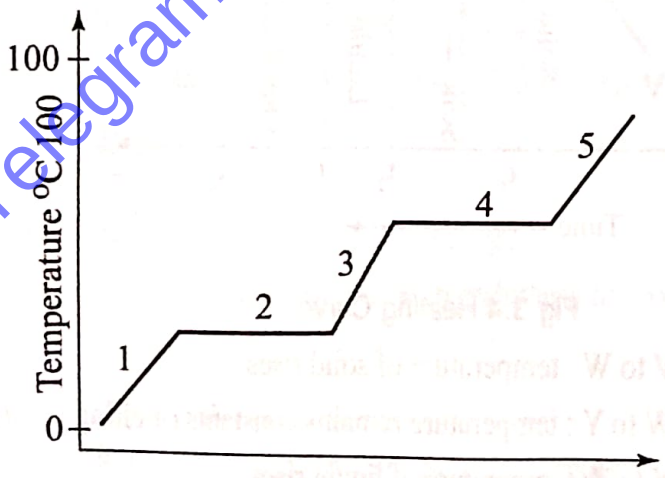
- ➥ Temperature decreases.
- ➥ Order of particles increases.
- ➥ Kinetic energy decreases
- ➥ Strength of intermolecular force of attraction increases.

Note: The following exist at the same temperature

- ➥ Melting point and freezing point
- ➥ Boiling point and condensation point
- ➥ Sublimation point and deposition point

Illustrative Examples

66. The heating curve shown below shows a substance in the solid, liquid and gas phase. Which segment represents only the liquid phase? (ESSLCE)



A. 3 B. 5 C. 1 D. 2

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Solution

1: Solid, 2: Solid ⇌ liquid, 3: liquid, 4: liquid ⇌ gas, 5: gas

Answer: A

67. The amount of heat released when one mole of liquid is converted to a solid at its freezing point is:

A. Molar heat of vaporization
B. Molar heat of sublimation
C. Molar heat of solidification
D. Molar heat of fusion

Answer: C

68. The direct change of solid to gas is

A. Sublimation
B. Deposition
C. Fusion
D. Freezing

Solution

Sublimation: solid ⟶ gas.

Answer: A

69. Which of the following is wrong?

A. $\Delta H_{fus} = -\Delta H_{cry}$
B. $\Delta H_{vap} = -\Delta H_{cond}$
C. $\Delta H_{sub} = \Delta H_{fus} + \Delta H_{vap}$
D. $\Delta H_{dep} = \Delta H_{sub}$

Answer: D

70. Write three solids that vaporize without melting

Solution

Iodine, solid carbon dioxide, Ammonium chloride, Naphthalene.

71. Classify the following changes of state as endothermic or exothermic process.

a. Sublimation
b. Deposition
c. Fusion
d. Freezing

Solution

- a and c are endothermic processes
- b and d are exothermic processes.

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Solved problems

72. What are the basic difference between three states of matter?

Solution

The energy content they have and the motion of the particles.

73. When 200L of gas at constant pressure is heated from 0^0C to 100^0C. the volume must be multiplied by

Solution

$V_1 = 200L, T_1 = 0^0C = 273K, T_2 = 100^0C = 373K, V_2 = ?V_1$

$V_2 = \frac{T_2 V_1}{T_1} = \frac{373}{273} V_1$, thus the volume is multiplied by $\frac{373}{273}$

74. A sample of nitrogen gas kept in a container of volume 2.3L and a temperature of 32^0C exerts a pressure of 4.7 atm. Calculate the number of moles of gas present.

Solution

$V = 2.32 \quad T = 32^0c = 305K, P = 4.7atm$

$$n = \frac{PV}{RT} = \frac{4.7atm \times 2.3L}{0.082 \frac{atm.L}{mol.K} \times 305K} = 0.43mol$$

75. A 2L Dumas bulb contains n moles of N_2 at 0.5atm pressure at T.K on addition of 0.01 mol of O_2, it is necessary to maintain the same pressure calculate n and T.

Solution

$PV = nRT, n = \frac{PV}{RT}$, after the addition

of O_2, $(n + 0.01) = \frac{PV}{RT} = \frac{0.5 \times 2}{0.082 \times 28}$

Solving for n, $n = 0.0332mol$

$$T = \frac{PV}{nR} = \frac{0.5 \times 2}{0.0332 \times 0.082} = 367K$$

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76. One mole of an ideal gas occupies 12L at 25°C. What is the pressure of the gas.

Solution

$$P = \frac{nRT}{nR} = \frac{1mol \times 0.082 \frac{atm.L}{mol.K} \times 298K}{12\ L} = 2.04atm$$

77. Explain why a helium weather balloon expands as it rises in the air Assume that the temperature remains constant.

Solution

A helium weather balloon expands as it rises in the air because of decrease in pressure.

78. What volume would 5.6 moles of sulfur hexafluoride (SF_6) gas occupy if the temperature and pressure of the gas are 128°C and 9.4 atm?

Answer: V = 19.6L

79. Calculate the density of air at 27°C and 745 torr if its molar mass is 30 g /mol.

Solution

$$T = 27°C = 300K,\ P = 745\ torr = 0.98atm$$

$$\rho = \frac{PM}{RT} = \frac{0.98 \times 30}{0.082 \times 300} = 1.2\ g/L$$

80. A sample of gas occupies 500ml at 40°c and a pressure of 250mmHg. What is the volume if the temperature is raised to 60°c and pressure is raised to 300mmHg.

Solution

$$V_2 = \frac{P_1\ V_1\ T_2}{T_1\ P_2} = \frac{200mmHg \times 500ml \times 333K}{313K \times 300mmHg} = 443.3ml$$

81. Calculate the number of molecules of hydrogen gas in a 1.0L sample of gas at STP

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Solution

$$n = \frac{PV}{RT} = \frac{1L \times 1atm}{0.082 \frac{atm.L}{mol.K} \times 273.K} = 0.045mol$$

Number of molecules = $n \times 6.02 \times 10^{23} = 0.045 \times 6.02 \times 10^{23}$
$= 2.71 \times 10^{22}$ molecules.

82. Determine the mass of oxygen that occupies a volume of 2.8L at STP.

Solution

$$m = \frac{MPV}{RT} = \frac{32 \frac{g}{mol} \times 2.8L \times 1atm}{0.082 \frac{atm.L}{mol.K} \times 273K} = 4g$$

83. What volume of NH_3 will be produced when 180ml of H_2 reacts with 60ml of N_2 according to the following equation:
$3H_2(g) + N_2(g) \longrightarrow 2NH_3(g)$?

Assume constant T and P for reactants and products.

Solution

$$3H_2(g) + N_2(g) \longrightarrow 2NH_3(g)$$

$$VT(H_2) = \frac{180ml}{3} = 60ml$$

$$VT(N_2) = \frac{60mol}{1} = 60ml$$

Both H_2 and N_2 are limiting reactants. Solve the volume of NH_3 by volume relationship.

$$3H_2(g) + \overset{60mol}{N_2}(g) \longrightarrow 2\overset{x}{N}H_3(g)$$
$$1 \qquad\qquad 2$$

$$60ml = \frac{x}{2} \Rightarrow x = V(NH_3) = 120ml$$

84. Arrange the following gases in decreasing order of rate of diffusion.

$NO_2, N_2O, NO, N_2O_3, N_2O_5$

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Solution

$$NO > N_2O > NO_2 > N_2O_3 > N_2O_5$$

85. If $140cm^3$ of methane diffuses in to air in 72 seconds. How long will it take $210cm^3$ of sulfur dioxide (SO_2) to diffuse under the same condition?

Solution

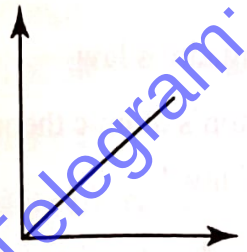
$$\frac{\frac{V_{CH_4}}{t_{CH_4}}}{\frac{V_{SO_2}}{t}} = \sqrt{\frac{M_{CH_4}}{M_{SO_2}}}$$

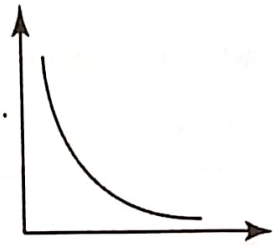
$$\frac{\frac{140}{72}}{\frac{210}{t}} = \sqrt{\frac{16}{64}} \Rightarrow t = 216 \text{seconds}$$

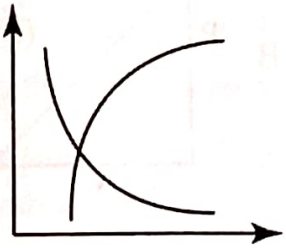
Supplementary Exercise

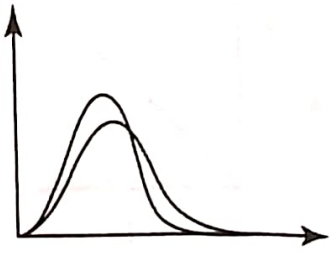
Part I. Multiple choice items

1. When 200ml of a gas at constant pressure is heated, its volume
   A. Increases C. Remains unchanged
   B. Decrease D. None
2. Which is a graphic depiction of Boyle's law?

A. 

B. 

C. 

D. 

3. Which one has no definite shape and volume?

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A. H_2O B. O_2 C. NaCl D. Fe

4. Which two states of matter are condensed states?

A. Solid and liquid C. Liquid and gas

B. Solid and gas D. Plasma and gas

5. If the temperature in Kelvin of a confined gas sample is double the volume is held constant, what will happen to the pressure?

A. It will decrease by half C. It will increase by four

B. It will be doubled D. It will remain same

6. A bottle of cold beverage taken only of a refrigerator often covered drops of water. The process that leads to the formation of those of water is:

A. Boiling C. Freezing

B. Condensation D. Sublimation

7. The average kinetic energy of gas molecules is directly proportion

A. The number of moles of the gas

B. The volume occupied by the gas

C. The absolute temperature of the gas

D. The total pressure of the gas

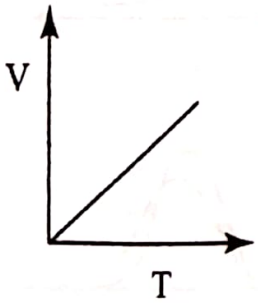
8. Which of the following gives a compressive explanation about phy states of substances?

A. Kinetic theory of matter C. Avagadro's law

B. Le-chatler's principle D. Dalton's atomic theory

9. Which of following graph shows the Charles' law?

A.



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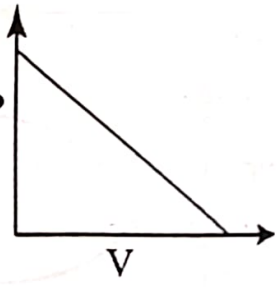

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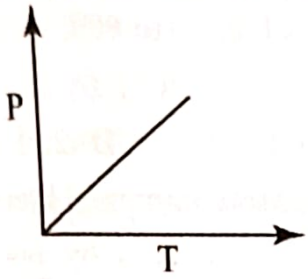
C.



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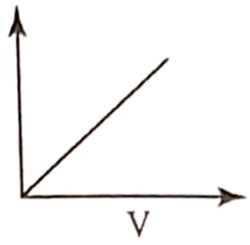

10. At what temperature in degree Celsius, does 3L of gaseous oxygen weigh 12g if the pressure is 20 torr?

A. -127 B. -203 C. -146 D. -270

11. Which of the following vaporize without melting?

A. Molten NaCl C. Ammonium chloride

B. Sugar D. Condensation

12. The direct change of gas to solid is:

A. Sublimation C. Evaporation

B. Deposition D. Condensation

13. If 3moles of oxygen gas was placed inside a 500ml container, how many moles of oxygen can be placed in 125ml vessel assuming pressure and temperature remain same?

A. 0.75mol B. 0.25mol C. 0.5mol D. 1.2mol

14. Which of the following is commonly known as dry ice?

A. Solid water C. Solid oxygen

B. Solid carbon dioxide D. Solid nitrogen

15. Which one has the least rate of diffusion?

A. H_2 (g) B. CH_4(g) C. N_2(g) D. SO_2(g)

16. Identify the correct statement about gases

A. Gas particles are highly disordered when compared with that of liquids and solids.

B. Energy is absorbed when a gas changes to liquid

C. The closer the particles in gases are the smaller is the force of attraction between them.

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D. Particles in gases vibrate about a fixed position

17. If you heat a 5L balloon from a temperature of 25°C to 50°C , its volume will be

A. 10L B. 5.42L C. 4.61L D. 2.5L

18. By what factor must the pressure be increased to compress 8 liters nitrogen to 4 liters?

A. 2 B. 4 C. 16 D. 8

19. During melting of a solid, all are true. **EXCEPT**

A. Intermolecular force of attraction remain same

B. Two phases are in equilibrium

C. Temperature remains same in the process

D. Energy change is involved in the process

20. What is the volume at STP of a gas whose volume is $200cm^3$ at 27 and 700mmHg pressure?

A. $203cm^3$ B. $167.6cm^3$ C. $230cm^3$ D. 181.3

II. Workout Questions

21. Express the following in atmosphere

a. 300cmHg b. 480 torr c. 1 bar

22. Calculate the volume which 4L of gas at0°c will occupy at 100°C a the same pressure

23. Sample of gas at 1.20 atm and 27°C is heated at constant pressur 57°C. Its final volume is 4.75L. What was its original volume?

24. When 2L of a gas at 760 torr and 27°C is expanded to 3L at 77°C. W the final pressure be?

25. A mass of gas at 80°C and 785 torr occupies 350ml. What volume the gas occupy at STP?

26. Calculate the mass of 6L of ammonia gas at STP.

27. Determine the volume occupied by 4g of oxygen at STP

28. Determine the approximate molecular weight of a gas if 560ml wei

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1.80g at STP.

29. Compare the rate of diffusion of CH_4 and SO_3

30. At 120°C, the pressure of a sample of nitrogen is 1.07atm. What the pressure be at 205°C, assuming constant volume?

Answer to supplementary Exercise.

Part I

1. A	5. B	9. A	13. A	17. B
2. B	6. B	10. D	14. B	18. A
3. B	7. C	11. C	15. D	19. A
4. A	8. A	12. B	16. A	20. B

Part II

21. a. 0.395 atm - 3.95 atm b. 0.632 atm

22. 5.5L

23. 4.32L

24. 557.3torr

25. 279ml

26. 4.569

27. 2.8L

28. $72 \frac{g}{mol}$

29. rate $CH_4 = \sqrt{5} \times$ rates SO_3

30. 1.3atm

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UNIT 4

Chemical Kinetics

4.1 Introduction

Chemical reaction is a process in which one or more substances are changed into one or more new substances. Some chemical reactions are fast and others are slow.

Example of Fast Reactions

- Reactions of alkali metals (Na, K) with water.
- Striking a match
- Precipitation reaction
- Burning of petrol in the car engine
- Ionic reactions
- Burning of magnesium ribbon
- Explosion of gun powder
- Neutralization reaction of strong acid with strong base. etc

Example of Slow Reactions

- Fermentation
- Rusting of iron
- Ripening fruit

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- Formation of coal
- Combination of hydrogen and oxygen to give water
- Weathering of rocks
- Cheese making e t c

The rates at which chemical reactions occur are just as important to you as they to the industrialist and the chemical engineers. At home you might be interested in the rate at which you can bake bread or cake or boil an egg. Therefore, studying chemical kinetics is important and interesting

Chemical kinetics deals with the study of the rate or speed of chemical reactions and with the elucidation of their mechanism.

Illustrative Examples

1. All of the following are slow reactions. **EXCEPT**

   A. Combination of hydrogen and oxygen to give water in the absence of catalyst

   B. Weathering of rocks to yield soils

   C. Rusting of iron

   D. Reaction between potassium and water

   Answer : D

2. Chemical kinetics is mainly concerned on

   A. Initial and final states of chemical reaction

   B. The energy changes that takes place in reactions

   C. The rate and mechanism of reactions.

   D. None of the above.

   Solution

   Chemical kinetics is manly concerned on the rate and mechanism of reactions.

   Answer : C

3. What is the meaning of the word **kinetics** ?

   Solution

   Kinetic comes from the Greek word kinetikos, meaning of motion

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which in turn traces to the verb kinein , meaning to move.

4. From a fundamental point of view,there are two important aspects to any process of change. What are they ?

Solution

- The extent to which a change had taken place .
- The time taken to accomplish the change.

4.2 Rate of Reaction

The rate / speed / of a chemical reaction is defined in exactly the same way that we define the speed of a car or of any other moving object.

$$\text{Speed} = \frac{\text{distance travelled}}{\text{time required for the travel}} = \frac{\text{change in position}}{\text{time required for the change}}$$

$$\text{Rate of reaction} = -\frac{\text{change in concentration of reactant}}{\text{a particular time interval}} = \frac{-\Delta C}{\Delta t}$$

$$\text{or Rate of reaction} = \frac{\text{change in concentration of product}}{\text{a particular time interval}} = \frac{\Delta C}{\Delta t}$$

For the general reaction : $A \rightarrow B$

$$\text{Rate} = \frac{-\Delta[A]}{\Delta t} = \frac{-([A]_f - [A]_i)}{t_f - t_i}$$

$$\text{Rate} = \frac{\Delta[B]}{\Delta t} = \frac{[B]_f - [B]_i}{t_f - t_i}$$

- The subscript f and i are final and initial states.
- [ ] is concentration usually expressed in mole per liter (mol /L).
- The SI unit of rate of reaction is **mole per liter second (M/s) or mol . L^{-1}. S^{-1}**
- Rate is scalar quantity, that is why there is a **negative sign** infront of the change in concentration of reactants
- As time goes, concentration of reactants **decreases** and concentration of products **increases.**

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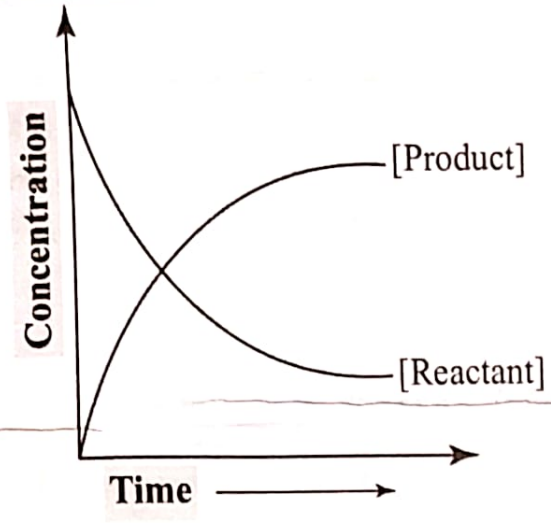

Fig 4.1 Plot Showing the Concentration of Reactant and Product as a Function of Time.

Rate and Stoichiometry

Stoichiometry deals with quantitative relationship between reactants, products or reactants and products.

For a hypothetical reaction : $aA + bB \rightarrow cC + dD$, rate is expressed interms of reactants and products as follows :

$$\text{Rate} = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = \frac{1}{c}\frac{\Delta[C]}{\Delta t} = \frac{1}{d}\frac{\Delta[D]}{\Delta t}$$

To write the rate of one in terms of another use the following relation ships

$$\text{Rate A} = \frac{a}{b}\frac{\Delta[B]}{\Delta t} = \frac{-a}{c}\frac{\Delta[C]}{\Delta t} = \frac{-a}{d}\frac{\Delta[D]}{\Delta t}$$

$$\text{Rate B} = \frac{b}{a}\frac{\Delta[A]}{\Delta t} = \frac{-b}{c}\frac{\Delta[C]}{\Delta t} = \frac{-b}{d}\frac{\Delta[D]}{\Delta t}$$

$$\text{Rate C} = \frac{-c}{a}\frac{\Delta[A]}{\Delta t} = \frac{-c}{b}\frac{\Delta[B]}{\Delta t} = \frac{c}{d}\frac{\Delta[D]}{\Delta t}$$

$$\text{Rate D} = \frac{-d}{a}\frac{\Delta[A]}{\Delta t} = \frac{-d}{b}\frac{\Delta[B]}{\Delta t} = \frac{d}{c}\frac{\Delta[C]}{\Delta t}$$

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Example

Give the reaction : $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

a. Write the rate of the reaction in terms of reactants and products.

b. If the rate of formation O_2 is 2×10^{-3} M/s, then find :

i. Rate of decomposition of N_2O_5

ii. Rate of formation of NO_2.

Solution

a. Rate $= -\frac{1}{2}\frac{\Delta[N_2O_5]}{\Delta t} = \frac{1}{4}\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$

b. i. Rate $N_2O_5 = -\frac{\text{coefficient } N_2O_5}{\text{cofficient } O_2} \times \text{rate } O_2$

$= \frac{2}{1} \times 2 \times 10^{-3} = 4 \times 10^{-3}$ M/s

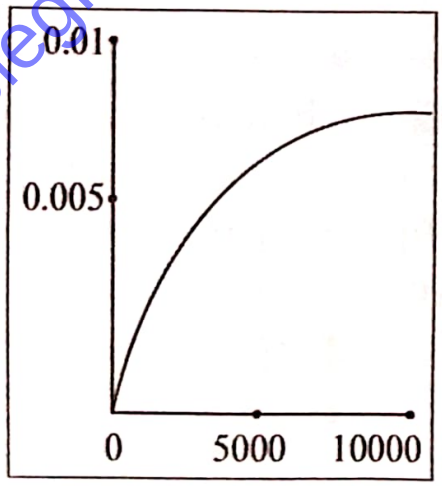
ii. Rate $NO_2 = \frac{\text{coefficient } NO_2}{\text{coefficient } O_2} \times \text{rate } O_2$

$= \frac{4}{1} \times 2 \times 10^{-3} = 8 \times 10^{-3}$ M/s

Illustrative Example

5. Which one of the following graphs illustrates the rate change in terms of change in reactant concentration with time ? (ESSLCE)

A.



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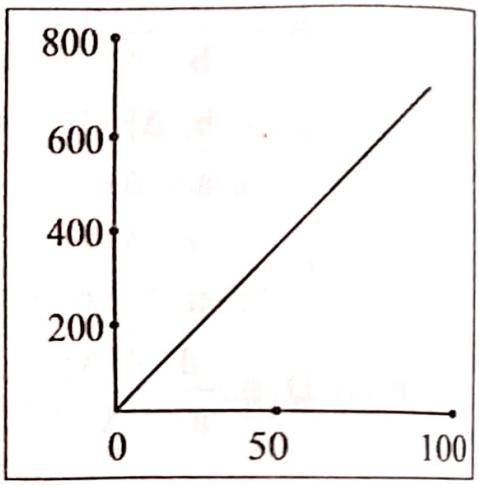

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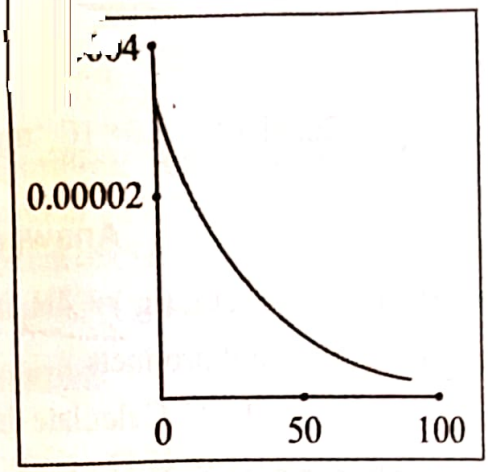
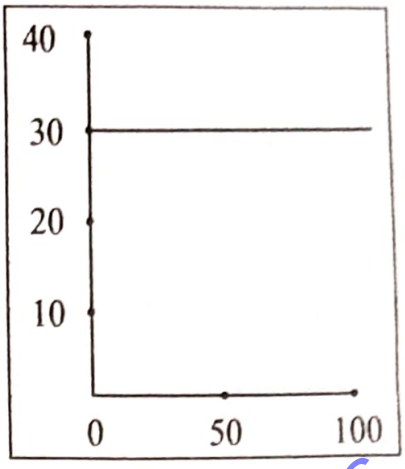
C.



D.



Solution

Concentration of reactants decreases with time.

Answer : C

6. Which of the following expressions is correct for the rate of a reaction below ?

$$6H^+ + BrO_3^- + 5Br^- \longrightarrow 3Br_2 + 3H_2O$$

A. $\frac{\Delta[Br^-]}{\Delta t} = 5\frac{\Delta[H^+]}{\Delta t}$

B. $\frac{\Delta[Br^-]}{\Delta t} = \frac{5}{6}\frac{\Delta[H^+]}{\Delta t}$

C. $\frac{\Delta[Br^-]}{\Delta t} = \frac{6}{5}\frac{\Delta[H^+]}{\Delta t}$

D. $\frac{\Delta[Br^-]}{\Delta t} = 6\frac{\Delta[H^+]}{\Delta t}$

Solution

$$\frac{\Delta[Br^-]}{\Delta t} = \frac{5}{6}\frac{\Delta[H^+]}{\Delta t} = \frac{-5}{3}\frac{\Delta[Br_2]}{\Delta t} = \frac{-5}{3}\frac{\Delta[H_2O]}{\Delta t}$$

Answer : B

7. Given the reaction : $2KClO_3 \rightarrow KCl + 3O_2$ .If the rate of decomposition of $KClO_3$ at a certain time is determined to be 2.2×10^{-2} mol / L. s . What is the rate of formation of O_2 ?

A. 1.1×10^{-2} B. 2.2×10^{-2} C. 3.3×10^{-2} D. 4.4×10^{-2}

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Solution

$$\text{Rate } O_2 = \frac{\text{coeff.}O_2}{\text{coeff.}KClO_3} = \frac{3}{2} \times 2.2\times10^{-2} = 3.3\times10^{-2} \text{mol/L}$$

Answer : c

8. Given the reaction : $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$

   a. Write rate expression in terms of reactants and products.

   b. If the rate of formation of CO_2 is 0.30. mol / L . s . Calculate the rate of disappearance of C_3H_8, O_2 and appearance of H_2O.

Solution

a. $$\text{Rate} = \frac{-\Delta[C_3H_8]}{\Delta t} = -\frac{1}{5}\frac{\Delta[O_2]}{\Delta t} = \frac{1}{3}\frac{\Delta[CO_2]}{\Delta t} = \frac{1}{4}\frac{\Delta[H_2O]}{\Delta t}$$

b. $$\text{Rate } C_3H_8 = \frac{1}{3} \times \text{rate } CO_2 = \frac{1}{3} \times 0.30 = 0.1 \text{ mol/L.s}$$

$$\text{Rate } O_2 = \frac{5}{3} \times \text{rate } CO_2 = \frac{5}{3} \times 0.30 = 0.5 \text{ mol.L}^{-1} .s^{-1}$$

$$\text{Rate } H_2O = \frac{4}{3} \times \text{rate } CO_2 = \frac{4}{3} \times 0.30 = 0.4 \text{ mol.L}^{-1}.s^{-1}$$

9. What is the unit of rate of a reaction if the unit of time is minute ?

Solution

$$M / min = mol / L . min = mol. L^{-1} . min^{-1}$$

10. At some point in the reaction : $4A + B \rightarrow 2C$, $[A]_o = 0.5M$, at a time 10min later $[A] = 0.4$ M. What is the rate of the reaction during this time interval expressed in mol. $L^{-1} . s^{-1}$?

Solution

$$[A]_i = 0.5M,\ [A]_f = 0.4M,\ t_i = 0,\ t_f = 10min = 600s$$

$$\text{Rate} = -\frac{1}{4}\frac{\Delta[A]}{\Delta t} = \frac{1}{4}\frac{([A]_f - [A]_i}{t_f - t_i}$$

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$$= -\frac{1}{4}\left(\frac{0.4 - 0.5}{600 - 0}\right)\frac{M}{s} = 4.17 \times 10^{-5}\ \text{mol. L}^{-1}\ .\ \text{s}^{-1}$$

11. Thiosulfate ion is oxidizing by iodine according to the following reaction : $2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$ If in a certain experiment, 0.006 mol of $S_2O_3^{2-}$ is consumed in 1.0 L of each second, at what rates are $S_4O_6^{2-}$ and I^- produced in this solution ?

Solution

$$[S_2O_3^{2-}] = \frac{n}{v} = \frac{0.006\,\text{mol}}{1\text{L}} = 0.006\ \text{mol/L}$$

$$\text{Rate } S_2O_3^{2-} = -\frac{\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{1}{2} \times \left(\frac{0.006\,\text{mol/L}}{15\text{se}}\right) = 3\times10^{-3}\text{M/s}$$

$$\text{Rate } I^- = \frac{2}{2} \times \text{rate } S_2O_3^{2-} = 3 \times 10^{-3}\ \text{M/s}$$

$$\text{Rate } S_4O_6^{2-} = \frac{1}{2} \times \text{rate } S_2O_3^{2-} = 1.5 \times 10^{-3}\ \text{M/s}$$

12. Define rate of reaction

Solution

Rate of reaction is expressed in terms of decrease of one of concentration of reactant or increase concentration of product.

Determination of Rate of Reaction

- ☑ To determine a rate of reaction, we need to measure changes in concentration over time.
- ☑ A change in time can be measured with a **stop watch** or other limiting device.
- ☑ An accurate, quantitative estimate of rate of reaction at any given time can be obtained from the **slope of tangent line** from the concentration time graph at that particular time.

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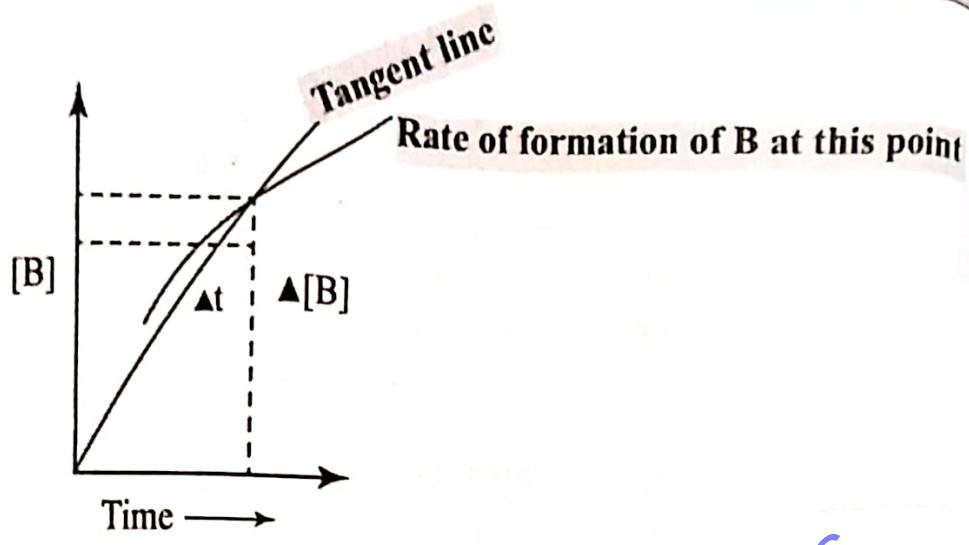

Fig 4.2 Estimation of the Rate of a Reaction Based on Δ [B] with Time

☑ The rate of the reaction is greatest at the beginning and slows down with time.

Some of the methods used to determine the concentration of reactants or products are :

- Amount of precipitate formed
- Color change
- Gain or loss of weight
- Increase or decrease in volume or pressure of gases

Instantaneous rate is the rate of reaction determined from the slope of tangent line to a concentration time curve.

Illustrative Example

13. Given the concentration of reactant versus time graph. At which point the rate of a reaction is maximum.

A. I
B. II
C. III
D. IV

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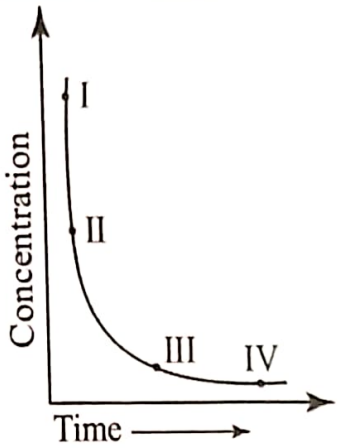

Solution

The rate of the reaction is maximum at the beigining of the reaction

Answer : A

14. How can the rate of a reaction be measured ?

Solution

By following the loss of reactant or formation of product.

4.3 Factors That Affect the Rate of Reaction

The main factors are :

- Nature of reactants
- Surface area of the reactants
- Temperature at which the reaction is carried out
- Consentration of reactants and gas pressure
- Use of a catalyst

Collision Theory

For a reaction to occur :

i. **Reactant** particles need to collide with one another
   - Collisions that result the desired products are known as **succesful** or **effective collisions.**

ii. **Reactants** should have proper orientation
   - This leads to the breaking of existing bonds and forming of the new bonds

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iii. The energy of reactants should be greater than or equal to activation energy (E_a)

➥ **Activation energy (Ea)** is the energy which reactants over come their particles collide, to react successfully and form products.

According to Collision Theory :

$$\text{Rate } \alpha \frac{\text{Number of collisions}}{\text{Second}}$$

1. Nature of Reactant

➥ More reactive reactant - faster rate

➥ Less reactive reactant - slower rate

Example

$2Na(s) + 2H_2O(1) \rightarrow 2NaOH(aq) + H_2(g)$ ----- fast

$Ca(s) + 2H_2O(1) \rightarrow Ca(OH)_2(aq) + H_2(g)$ ------slow

Reason : Na is more reactive than Ca

2. Surface Area

➥ The surface area of a solid reactant can be increased by powdering or grinding.

➥ The higher the surface area, the faster is the rate of the reaction because the number of collisions between the reacting species rises.

Example

$2HCl(aq) + CaCO_3$ (Powdered) $\rightarrow CaCl_2(aq) + CO_2(g) + H_2O(1)$ ----fast

$2HCl(aq) + CaCO_3$ (Lump) $\rightarrow CaCl_2$) aq) $+ CO_2(g) + H_2O(1)$ --- slow

Reason : powdered $CaCO_3$ has higher surface area than lump.

➥ The rate at which the two reactions occur can be found by :

➥ The volume of the CO_2 gas which is produced, or

➥ The loss in mass of the reaction mixture with time

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Surface area affects the rate of heterogeneous reactions (reactions involving solid reactant).

Example

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

3. Temperature at which Reaction is Carried out

- Why do you think food is stored in a refrigerator ?

The reason is that the rate of decay is **slower at lower temperatures.**

- Generally, as temperature increases, the rate of the reaction also increase because average kinetic energy of particles increases that lead to rise the number of collisions.

Number of collision α Temperature

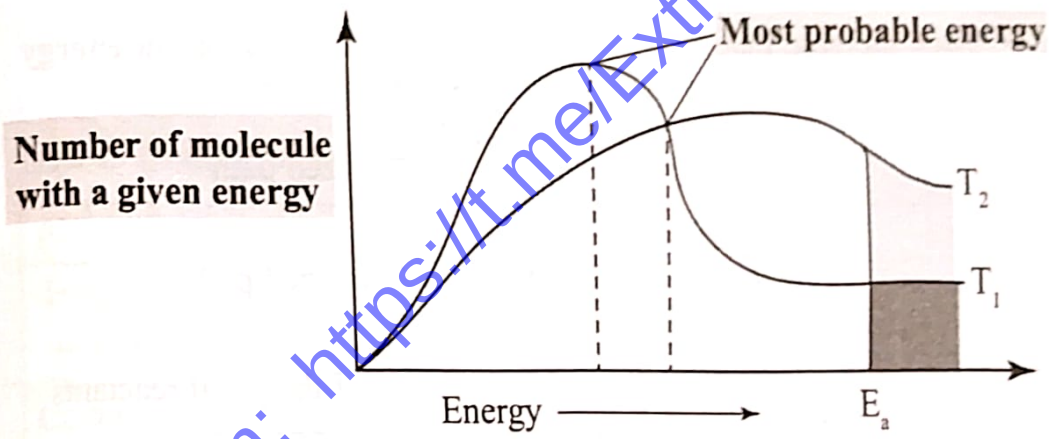

Fig 4.3 Boltzmann Distribution at Temperatures, T_1 and T_2

This graph indicates that at higher temperature T_2 :

- There are fewer particles with lower energy
- The most probable energy is higher
- More particles have energy greater than or equal to activation energy (E_a)

At higher temperature a bigger proportion of particles have sufficient energy and hence the rate of the reaction increases.

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4. Concentration of Reactants and Gas Pressure

- This factor has effect on the rate of reactions involving reactants in solution form.

Example

$$Zn(s) + H_2SO_4(1M) \rightarrow ZnSO_4(aq) + H_2(g) \text{ --- slow}$$

$$Zn(s)\, H_2SO_4(2M) \rightarrow ZnSO_4(aq) + H_2(g) \text{ --- fast}$$

- For a reaction in gas phase, increasing **pressure** increases the rate of a reaction.

Number of collision α concentration of reactants

5. Effects of Catalysts

- Catalysts speed up chemical reactions without being permanentry changed themselves.
- They increase the rate of reaction by **lowering activation energy** (E_a).

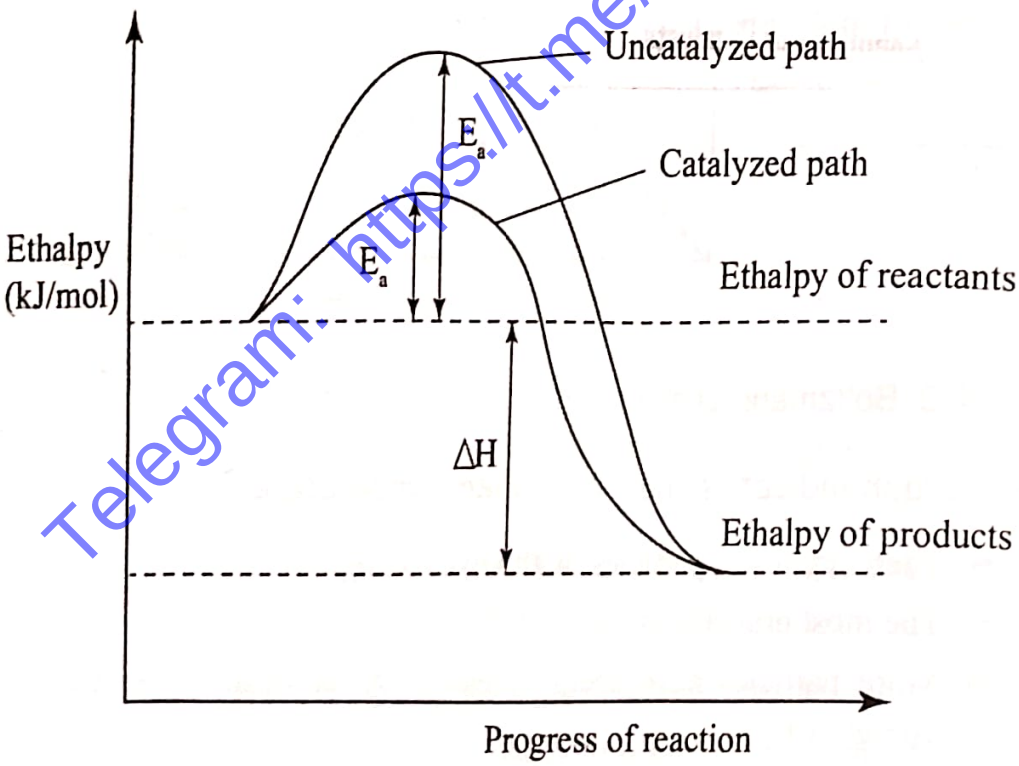

Fig 4.4 Activation Energy in the Presence and Absence of a Catalyst.

- In the presence of catalyst, the position of E_a moves to the left,

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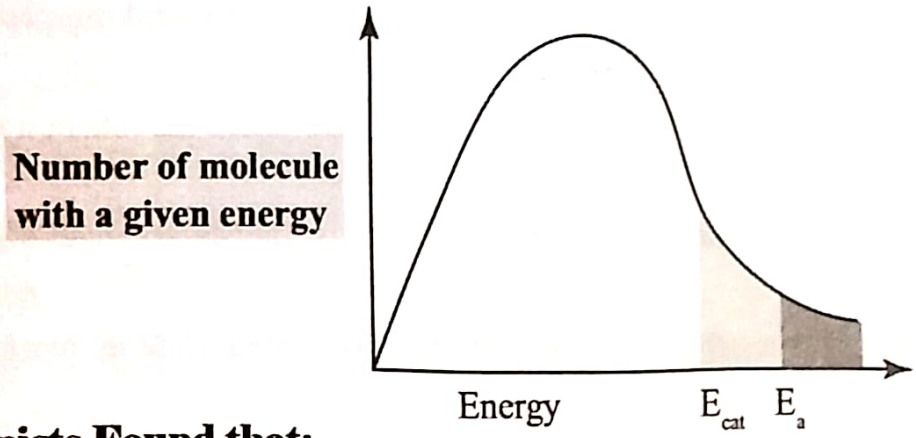
asing the proportion equal to activation energy.



Chemists Found that:

- A small amount of catalyst will produce a large amount of chemical change.
- Catalysts remain unchanged chemically after a reaction has taken place, but they can change physically. For example, a finer manganese (IV) oxide powder is left behind after the decomposition of hydrogen peroxide.
- Catalysts are very specific to a particular chemical reaction.

Table 4.1 Examples of Catalysts

Process	Catalyst
Haber process ➥ For the manufacture of ammonia	Iron (Fe)
Contact process ➥ For the manufacture of sulphuric acid	Vanadium (V) oxide (V_2O_5)
Ostwald process ➥ Oxiation of ammonia to give nitric acid	Platinum (Pt)
Hydrogenation of oil ➥ To make margarines	Nickel (Ni)

- **Enzymes** are protein molecules produced in living cells.
- They are used by living organisms as catalysts to speed up many reactions going on inside them.

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- They are biological catalysts
- They are specific in that each chemical reaction taking place has a different enzyme catalyst.
- **Inhibitors** are negative catalysts that slow down the rate of a reaction by increasing activation energy.
- Based on the phase of the reactions, catalysts are classified into two :

i. **Homogeneous catalyst** is catalyst that has similar state as reactant and products.

Example

$$HCOOH\ (aq) + CH_3OH\ (aq) \xrightarrow{H^+} HCOOH_3\ (aq) + H_2O\ (l)$$

ii. **Heterogeneous catalyst** is catalyst that has different state.

Example

$$N_2(g) + 3H_2\ (g) \xrightarrow{Fe\ (s)} 2NH_3\ (g)$$

Illustrative Example

15. In a kinetic experiment a student placed crystals of iodine in a closed reaction vessel, introduced a given quantity of hydrogen gas, and collected data to calculate what would be the effect obtained on the rate of the reaction, if the student takes more amount of iodine from that used in takes more amount of iodine from that used in the first experiment ? (ESSLCE)

A. The rate of the reaction will be constant

B. The formation of hydrogen iodide will be decreased.

C. It lowers the rate of the reaction

D. It speeds up the rate of the reaction

Solution

Increasing amount of solid has no effect on concentration because solids have constant concentration. As a result, the rate remains same

Answer : A

16. Which one of the following could not be consider as a factor influencing

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the rate of chemical reaction ? (ESSLCE)

A. Temperature of reactants

B. The rate constant of the reaction

C. Surface area of reactants

D. Presence of catalyst.

Solution

Rate constant does not affect the rate the reaction because it is a constant.

Answer : B

17. How does a catalyst increase the rate of a reaction? (ESSLCE)

A. By lowering activation energy

B. By changing the equilibrium constant

C. By forming an intermediate complex

D. By increasing the activation energy

Solution

Catalysts increasing the rate of the reaction by lowering activation energy and changing reaction mechanism.

Answer : A

18. If we increase the concentration of a reactant, what happens to the collisions between particles ?

A. There are more collisions

B. There are fewer collisions

C. There are the same number of collisions, but they have less energy

D. There are the same number of collisions, but they have more energy.

Solution

Concentration α Number of collisions.

Answer : A

19. A homogeneous liquid reaction mixture is often heated to increase

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the rate of reaction. This is best explained by the fast that raising th temperature ?

A. Increase the heat of reaction

B. Decrease the energy of activation

C. Increase the vapor pressure of the liquid

D. Increase the average kinetic energy of the reactants.

Answer : D

20. The minimum amount of energy that the reacting molecules must possess at the time of collisions in order to produce affective collisions is

A. Activation energy

B. Potential energy

C. Free energy

D. Internal energy

Answer : A

21. The value of activation energy is primary determined by _____.

Solution

Chemical nature of reactants and products.

22. The catalyst remains unchanged at the end of the reaction in terms of ____ and _____

Solution

Chemical composition and quantity or amount.

23. What is the difference between positive and negative catalysts ?

Solution

Positive catalysts

➥ Speed up the rate of a reaction

Negative catalysts

➥ Slow down the rate of a reaction

24. Mention some effects of temperature that can be experienced in your daily life.

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Solution

- Food is cooked by raising temperature.
- Food is preserved by lowering temperature.
- During surgery, the body of the patient should be at low temperature to slow down the rates of biochemical reactions

25. Give five ways used to increase reaction rates.

Solution

- Increase concentration of reactants.
- Raising temperature of reactants.
- Increasing the pressure of gaseous reactants.
- Increasing surface area of solid reactants
- Using positive catalysts.

Solved problems

26. What is the difference between rate and rate constant ?

Solution

- Rate is a variable proportional to the concentration of reactants but rate constant is a number which depends on the temperature only but not on concentration.

27. Write some fast and slow reactions.

Solution

Fast Reactions	Slow Reactions
Combustion of fuel	Souring of milk
Firing of gun	Fermentation
Precipitation reactions	Oxidation of copper
Neutralization reactions	Geographical reactions

28. In the reaction $2X \longrightarrow Y$, concentration of X was found to be 0.54M at t = 50 s and 0.42 M at t = 62s , what is the average rate of the reaction during this time interval ?

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Solution

$[X]_i = 0.54\,M,\ [X]_f = 0.42\,M \quad t_i = 50\,s\ ,\ t_f = 62\,s$

$$\text{Rate} = \frac{-1}{2}\frac{\Delta[X]}{\Delta t} = -\frac{1}{2}\left(\frac{[X]_f - [X]i}{t_f - t_i}\right)$$

$$= -\frac{1}{2}\left(\frac{0.42-0.54}{62-50}\right) = 0.005\,M\,s^{-1}$$

29. What happens to the rate of a reaction for gaseous reactants if you increase the pressure ?

Solution The rate of the reaction also increase.

30. Write three heterogeneous catalyst used in industrial production of some substances.

Solution

Process	Catalyst
➥ Haber process (Production of ammonia)	Iron
➥ Contact process (Production of H_2SO_4)	vanadium (V) oxide
➥ Hydrogenation of oil (Production of fat)	Nickel

31. What are the three pre - conditions for a reaction to occur ?

Solution

- ➥ Reactants must collide.
- ➥ Reactants should have proper orientation
- ➥ Reactants energy must be greater than or equal to activation energy.

32. Successful processes using enzymes need to ensure that __,____ and _

Solution

- ➥ The enzyme is able to function for long periods of time by optimising the environment.

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- The enzyme is not lost by trapping it on the surface of an inert soli
- Continuous processes occur rather than batch processes.

33. $Br_2(aq) + HCOOH(aq) \rightarrow 2Br^-(aq) + 2H^+(aq) + CO_2(g)$

Time (s)	$[Br_2]$, M
0.00	0.0120
50.0	0.0101
100.0	0.00846
150.0	0.00710

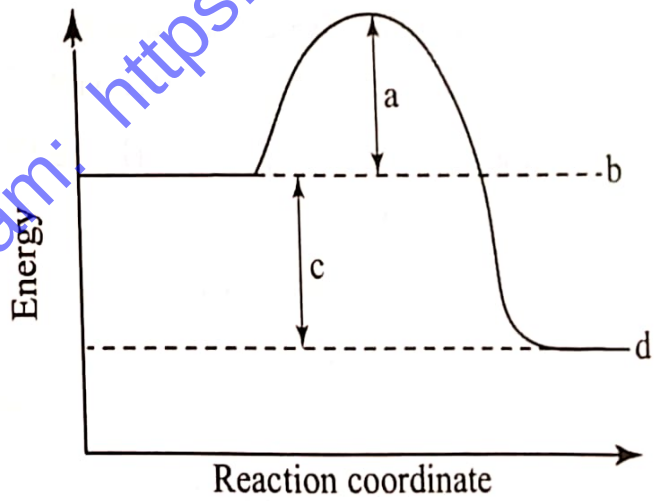
Calculate the average rate over the first 50 seconds time interval.

Solution

$$\text{Rate} = \frac{-\Delta[Br_2]}{\Delta t} = -\left(\frac{0.0101 - 0.0120}{50 - 0}\right)$$

$$= 3.80 \times 10^{-5}\ M/s$$

34. Given the energy diagram below. Describe the terms represented by the letters, a, b, c and d.



Solution

a. Activation energy (Ea)
b. Energy of reactant (H_r)
c. Energy change of reaction (ΔH)
d. Energy of product (H_p)

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$\Delta H = H_p - H_r$

Supplementary Exercise

I. Multiple choice items

1. One of the steps in the manufacture of nitric acid is the oxidation of ammonia shown in this equation

   $4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$

   If gaseous water appears at a rate of 0.25 mol. L^{-1} min^{-1}, at what rate does ammonia disappear ?

   A. 0.0040 mol. L^{-1} . min^{-1}
   C. 0.038 mol. L^{-1} . min^{-1}
   B. 0.017 mol. L^{-1} min^{-1}
   D. 0.150 mol. L^{-1} . min^{-1}

2. Which of the following is true about the factors affecting rate of reaction ?

   A. As surface area of solid reactants increases, the rate of the reaction also increases.

   B. As concentration of reactants increases, the rate of any order of reaction increase.

   C. As temperature increases, the rate of gaseous reaction also increase.

   D. A catalyst always speed up the rate of the reaction.

3. Which of the following is not heterogeneous reaction ?

   A. $3Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2(g)$

   B. $C(s) + O_2(g) \longrightarrow CO_2(g)$

   C. $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$

   D. $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

4. An increase in concentration

   A. is related to the number of collisions directly

   B. is related to the number of collisions inversely

   C. has no effect on the number of collisions

   D. none of the above.

5. At the beginning of a reaction, the reaction rate for the reactants is :

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A. Largest, then decreasing

B. Largest and remains constant

C. Smallest then increasing

D. Smallest and remains constant

6. Which of the following is true for the reaction

$$N_2 + 3H_2 \rightarrow 2NH_3$$

A. $\frac{\Delta[N_2]}{\Delta t} = 3\frac{\Delta[H_2]}{\Delta t}$

B. $-\frac{2}{3}\frac{\Delta[H_2]}{\Delta t} = \frac{\Delta[NH_3]}{\Delta t}$

C. $\frac{-\Delta[N_2]}{\Delta t} = +\frac{1}{3}\frac{\Delta[H_2]}{\Delta t}$

D. $-3\frac{\Delta[H_2]}{\Delta t} = 2\frac{\Delta[NH_3]}{\Delta t}$

7. Which of the following reaction is the fastest ?

A. Rusting of iron

B. Burning of sulphur

C. Transition of monoclinic sulphur to rhombic sulfur

D. Precipitation of barium sulphate by mixing solution of sodium sulphate and barium chloride

8. The rate of reaction increases with increase in temperature. This is mainly because of :

A. Increase in collision frequency

B. Decrease in activation energy

C. Change in mechanism of the reaction

D. Increase in fraction of molecules having energy equal to or greater than there should energy.

9. Which of the following changes in the presence of the catalyst ?

A. Energy of reactants

B. Energy products

C. Energy of reaction

D. Activation of energy

10. Which of the following does not affect the rate of a chemical reaction ?

A. Enthalpy of reaction

B. Surface area

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C. Concentration of reactants D. Temperature

11. The addition of inhibitors to a reaction

A. Increase the rate of the reaction

B. Changes the entropy of the reaction

C. Decrease the rate of the reaction

D. The rate of the reaction is unaffected

12. Which of the following is true about the reaction below ?

$2A(g) + B(g) \rightarrow 3C(g)$

A. Rate of dis appearance of C is three times rate of disappearance of A

B. Rate of dis appearance of B is twice that of rate of disappearance of A

C. Rate of dis appearance of A is twice rate of appearance of C

D. Rate of dis appearance of B is half of that of disappearance of A

13. The negative slope of the tangent to the reactant concentration versus time is

A. The instantaneous rate of consumption of reactant

B. The instantaneous rate of formation of product

C. The average rate of consumption reactant

D. The average rate of formation of product

14. For the reaction : $2SO_2 + O_2 \rightarrow 2SO_3$, the rate of formation of SO_3 is 100g/min. The rate of disappearance of O_2 is

A. $100g.min^{-1}$ C. $200g \cdot min^{-1}$

B. $50\ g.\ min^{-1}$ D. $75\ g.\ min^{-1}$

15. _____ is defined as the ratio of change in concentration of reactants or products to the change in time

A. Average rate C. In the grated rate

B. Instantaneous rate D. Rate law.

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II. Short answer question

16. Given the reaction : $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$ The initial concentration of $N_2O_5(g)$ is 2M and it is 1.5M after 20 minutes. The rate of formation of NO_2 is ___.

17. The change in molar concentration of either reactants or products at an instant of time or in infinitesimally small interval of time is ___

18. Give the reaction : $2H_2O_2 \rightarrow 2H_2O + O_2$, the rate of formation of O_2 is 50g per minute. Determine the rate of decomposition of H_2O_2.

19. List the five factors that affect the rate of a reaction.

20. Write rate expression for the reaction below in terms of reactants and products.

$$2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(g)$$

Answer to Supplementary Exercise

Part I

1. B	5. A	9. D	13. A
2. A	6. B	10. A	14. B
3. D	7. D	11. C	15. B
4. A	8. A	12. D	

Part II

16. 0.05 mol . L^{-1} min^{-1}

17. Instantaneous speed

18. 100g . min^{-1}

19. Naturare of reactants

- ➥ Surface area of solid reactant
- ➥ Temperature of reactants
- ➥ Concentration / pressure of reactants.
- ➥ Catalysts.

20. Rate $= -\frac{1}{2}\frac{\Delta[C_2H_6]}{\Delta t} = \frac{-1}{7}\frac{\Delta[O_2]}{\Delta t} = \frac{1}{4}\frac{\Delta[CO_2]}{\Delta t} = \frac{1}{6}\frac{\Delta[H_2O]}{\Delta t}$

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UNIT 5

Chemical Equilibrium

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5.1 Introduction

Some reaction proceed to completion but other reactions do not proceed to completion.

- Reactions that go to completion are called **irreversible reactions.**
- Chemical reactions that do not go to completion are known as **reversible reactions.**

Table 5.1 The Difference Between Irreversible and Reversible Reaction

Irreversible reaction	Reversible reaction
➥ One way reaction (→) ➥ Goes to completion ➥ Reactants form products ➥ The amount of limiting reactant that left is neglible **Example** $2Na + 2H_2O \rightarrow 2NaOH + H_2$ $NaOH + HCl \rightarrow NaCl + H_2O$	➥ Two ways reaction ($\rightleftharpoons$) ➥ does not go to completion ➥ Reactants form products and products give back the original reactants. **Example** $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

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In reversible reactions $aA + bB \rightleftharpoons cC + dD$

- The reaction that proceeds from left to right direction is known as **forward reaction**:- $aA + bB \rightarrow cC + dD$
- The reaction that goes from right to left is called backward or reverse **reaction:-** $cC + dD \rightarrow aA + bB$

5.2 Reversible Reactions and Equilibrium

Equilibrium is a condition in which the forward and backward rates in a physical or a chemical system are equal.

It is attained or reached when the rate of forward reaction is equal to the rate of backward

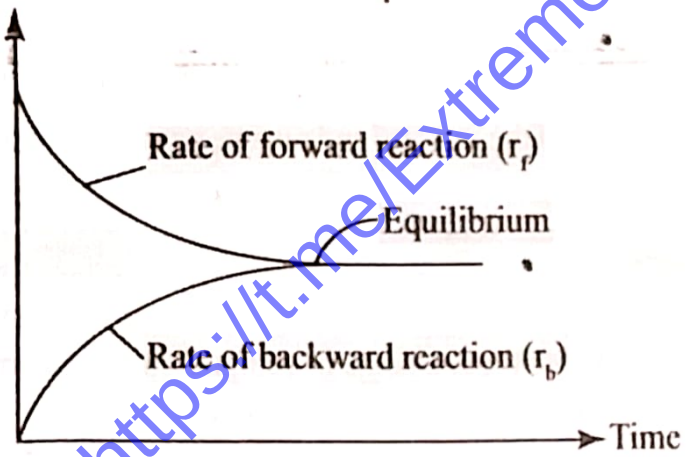

Fig 5 . 1. Rate versus Time graph for Reversible Reaction

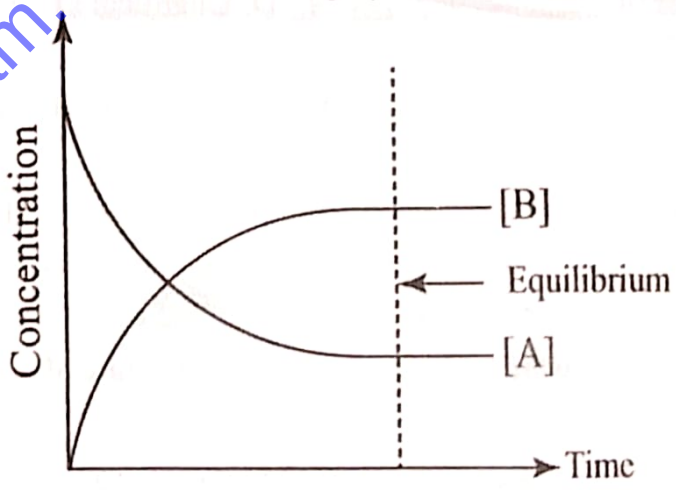

Fig 5 . 2 Concentration Versus Time Graph for the Hypothetical Reaction $A \rightleftharpoons B$

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Characterstics of Chemical Equilibrium

It is reached when forward rate is equal to backward rate.

It is dynamic meaning the reaction does not stop at equilibrium.

It is applicable in reversible reaction

The concentration of reactants is not necessarily equal to the concentration of products.

5. Entropy of the universe is zero at equilibrium, ($\Delta S_{univ} = 0$)

Gibb's free energy change is zero, ($\Delta G = 0$)

The value of cell potential is zero ($E_{cell} = 0$)

The value of equilibrium constant is one ($K_{eq} = 1$)

Macro - properties such as temperature, pressure, volume and concentration should be constant.

The change in concentration of reactants and products is constant.

The rate constant of the forward and reverse reactions are not necessarily equal.

Illustrative Examples

A forward and reverse reaction continues to take place at equal rate in microscopic level without change in composition at macroscopic level. This statement is the definition of (ESSLCE)

A. The law of mass action
B. Degree of freedom
C. Dynamic equilibrium
D. Chemical kinetics

Answer: C

Which of the following correctly describes a chemical reaction at equilibrium?

A. The concentration of reactants and products are equal
B. The change in concentration of reacts and products is constant.
C. The rate of forward reaction is less than the rate of reverse reaction.
D. The rate of forward reaction is greater than the rate of reverse reaction.

Solution

- At equilibrium, rates of forward and reverse reactions are equal.
- Concentration of products is not necessary equal concentration of reactant.

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Answer: B

3. What is the difference between reversible and irreversible reactions? (ESSLCE)

A. In reversible reactions, the reaction goes to completion.

B. In irreversible reactions, there is a mixture of products and reactants.

C. In reversible reactions, there is equilibrium between reactants and products.

D. In irreversible reactions, there is equilibrium between reactants and products.

Solution

Reversible Reaction	Irreversible Reaction
➥ Does not go to completion	➥ Goes to completion
➥ There is a mixture of reactants and products	➥ No mixture of reactants and products
➥ There is equilibrium between reactants and products	➥ No equilibrium between reactants and products

Answer: C

4. Which of the following is wrong about chemical equilibrium?

A. ΔS univ = 0

B. $\Delta C = 0$

C. $K_{eq} = 1$

D. $E_{cell} = 1$

Solution

At equilibrium, $\Delta G = \Delta S univ = E_{cell} = 0$, $K_{eq} = 1$

Answer: D

5. For the reversible reaction; $N_2 + 3H_2 \rightleftharpoons 2NH_3$. Write the forward and backward reactions

Solution

Forward reaction: $N_2 + 3H_2 \rightarrow 2NH$

Backward reaction: $2NH_3 \rightarrow N_2 + 3H_2$

6. List the macroscopic properties that should be constant at equilibrium

Solution

Temperature, Concentration, Pressure and Volume

7. Given the rate versus time graph below. Write the terms represented with numbers 1 to 3.

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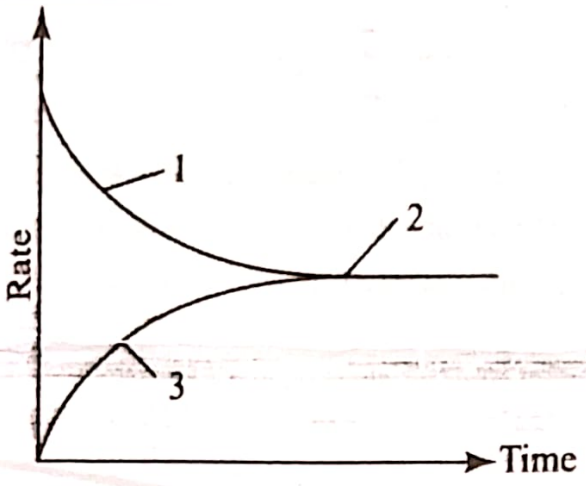


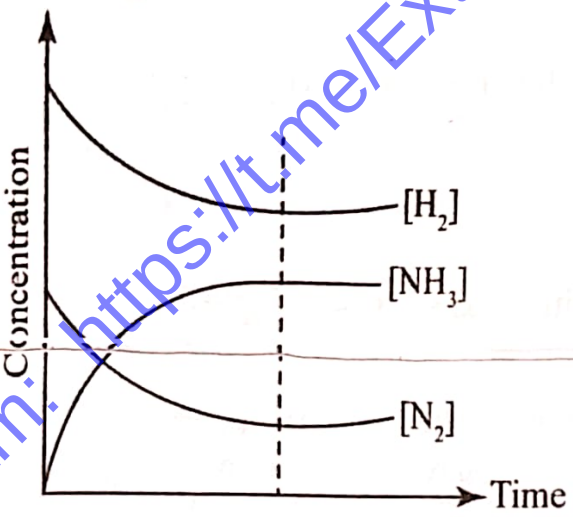
Solution

1. Forward rate 2. Equilibrium 3. Backward rate

8. Draw the concentration time graph for the reaction: $N_2 + 3H_2 \rightarrow 2NH_3$

Solution

The concentration of H_2 should be above the concentration of N_2 because of its higher coefficient



9. Write some physical process that are associated with equilibrium condition

Solution

- Saturate solution: rate of dissolution = rate of crystallization
- The vapour pressure of liquid in a closed vessel: rate of evaporation is equal to rate of condensation.

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5.3 Equilibrium Expression and Equilibrium Constant

Law of mass action or law of concentration action is a law that characterized a dynamic equilibrium

It states that for a reversible reaction at equilibrium at a given condition, equilibrium constant is equal to the product of the amount of substance formed divided by the amount of reactants each raised to the power of the corresponding coefficients.

For a hypothetical reaction: $aA + bB \rightleftharpoons cC + dD$

$r_f \acute{\alpha} [A]^a [B]^b \Rightarrow r_f = K_f [A]^a [B]^b$

$r_b \acute{\alpha} [C]^c [D]^d \Rightarrow r_b = K_b [C]^c [D]^d$

At equilibrium: $r_f = r_b$

$\Rightarrow K_f [A]^a [B]^b = K_b [C]^c [D]^d$

$\Rightarrow$ Rearranging the equation results

$$\frac{K_f}{K_b} = \frac{[C]^c [D]^d}{[A]^A [B]^b}$$

The ratio of K_f to K_b is known as **equilibrium constant (K_{eq})**

$$K_{eq} = \frac{K_f}{K_b}$$

:- r_f - rate of forward reaction

r_b - rate of backward reaction

K_f - rate constant for the forward reaction

K_b - rate constant for the backward reaction

If the amount of reactants and products are expressed in concentration, K_{eq} will be represented by K_c

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

If the amount of reactants and products is expressed interms of partial pressure (for gaseous substance), K_{eq} will be expressed with K_p.

$$K_P = \frac{P_C^{\,c} \cdot P_D^{\,d}}{P_A^{\,a} \cdot P_B^{\,b}}$$

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Example

Write K_c and K_p expression for the reaction:

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Solution

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$K_P = \frac{(PNH_3)^2}{(PN_2)(PH_2)^3}$$

To Write K_c or K_p Follow the Rules Below

- The equilibrium products concentrations appear in numerator.
- The equilibrium reactant (s) concentrations appear in denominator
- The equilibrium concentration of each species is raised to its coefficient.
- Pure solids and liquids or solvents in dilute solution do not appear in the equilibrium constant expression because their concentration is constant.
- K_p expression is written for gaseous substances.

Note:

i. A chemical equilibrium that has species in the same state or phase is called homogeneous equilibrium

Example

$H_2(g) + F_2(g) \rightleftharpoons 2HF(g)$

$CH_3COOH(eq) + H_2O(1) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$

ii. A chemical equilibrium that contains species in different states or phases is known as **heterogeneous equilibrium.**

Example

$Fe_2O_3(s) + 3CO(g) \rightleftharpoons 2Fe(s) + 3CO_2(g)$

Illustrative Examples

10. The conventional equilibrium constant expression (K_c) for the reaction: $2ICl(s) \rightleftharpoons I_2(s) + Cl_2(g)$ is

A. $\frac{[I_2][ICl]}{[ICl]^2}$ B. $\frac{[I_2][Cl_2]}{2[ICl]}$ C. $[Cl_2]$ D. $\frac{[I_2]+[Cl_2]}{2[ICl]}$

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Solution

Solids are not included in K_c expression

$\Rightarrow K_c = [Cl_2]$ because ICl and I_2 are solids

Answer: C

11. What is the equilibrium constant expression for the following reaction?

$2Hg(g) + O_2(g) \rightleftharpoons 2HgO(s)$

A. $K_c = \frac{1}{[Hg]^2[O_2]}$ B. $K_c = [Hg]^2[O_2]$

C. $K_c = \frac{[HgO]^2}{[Hg]^2}$ D. $K_c = \frac{[2HgO]}{[2Hg][O_2]}$

Answer: A

12. For the reaction A → B and B → A, the rate constants are $8 \times 10^2\ s^{-1}$ and $5.2 \times 10^4\ s^{-1}$ respectively. What is the value of the equilibrium constant for the equilibrium $A \rightleftharpoons B$?

A. 1.54×10^{-3} B. 1.54×10^{-2} C. 6.5×10^{-3} D. 6.5×10^{-2}

Solution

$$K_c = \frac{K_f}{K_b} = \frac{[B]}{[A]} = \frac{8 \times 10^2 s^{-1}}{5.2 \times 10^4\ s^{-1}} = 1.54 \times 10^{-2}$$

Answer: C

13. If the equilibrium concentrations found in the reaction:

$2A(g) + B(g) \rightleftharpoons C(g)$ are: $[A] = 2.4 \times 10^{-2}M$, $[B] = 4.6 \times 10^{-3}M$ and $[C] = 6.2 \times 10^{-3}M$, then what is the value of K_c?

A. 1.3×10^{-4} B. 2.3×10^3 C. 56 D. 5.8×10^2

Solution

$$K_c = \frac{[C]}{[A]^2[B]} = \frac{6.2 \times 10^{-3}}{(2.4 \times 10^{-2})(4.6 \times 10^{-3})} = 2.4 \times 10^3$$

Answer: B

14. At a certain temperature for the reaction;

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$, the equilibrium concentrations are, $[NO] = 0.52M$, $[O_2] = 0.24M$ and $[NO_2] = 0.18M$.

What is the value of K_c at this temperature?

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Solution

$$K_c = \frac{[NO_2]^2}{[NO]^2[O_2]} = \frac{(0.18)^2}{(0.52)^2(0.24)} = 0.54$$

15. At 295K, a 3L flask contains 7mol HI, 4mol H_2 and 0.6mol I_2 in equilibrium. What is the value of K_c at this temperature?

$H_2(g) + I_2 \rightleftharpoons 2HI(g)$

Solution

First determiner the molarity of each species.

$$[HI] = \frac{n_{HI}}{V} = \frac{7mol}{3L} = 2.33M$$

$$[H_2] = \frac{nH_2}{V} = \frac{4mol}{3L} = 1.3M$$

$$[I_2] = \frac{nI_2}{V} = \frac{0.6mol}{3L} = 0.2M$$

$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2.33)^2}{(1.33)(0.2)} = 20.47$$

16. The value K_c for the reaction; $COCl_2(g) \rightleftharpoons CO(g) +$ is 2.5×10^{-10} at 300K. Starting with pure $COCl_2(g)$ in a closed vessel, if the concentration of CO(g) is 1×10^{-5}M at equilibrium, what would be the concentration of $COCl_2(g)$?

Solution

Since the coefficient ration is 1:1, at equilibrium $[CO] = [Cl_2] = 1 \times 10^{-5}$M

$$K_c = \frac{[CO][Cl_2]}{[COCl_2]}$$

$$[COCl_2] = \frac{[CO][Cl_2]}{K_c}$$

$$[COCl_2] = \frac{(1 \times 10^{-5})\ (1 \times 10^{-5})}{(2.5 \times 10^{-10})} = 0.40M$$

17. Write K_c expression for

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a. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

b. $NH_3(g) + H_2S(aq) \rightleftharpoons NH^+_4(aq) + HS^-(aq)$

c. $HF(g) + H_2O(1) \rightleftharpoons H_3O^+(aq) + F^-(aq)$

d. $Mg(s) + 2HCl(aq) \rightleftharpoons MgCl_2(aq) + H_2(g)$

e. $Ag^+(aq) + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq)$

f. $PCl_5(s) \rightleftharpoons PCl_3(1) + Cl_2(g)$

Solution

a. $K_c = [Ag^+][Cl^-]$

b. $K_c = \frac{[NH_4^+][HS^-]}{[NH_3][H_2S]}$

c. $K_c = \frac{[H_3O^+][F^-]}{[HF]}$

d. $K_c = \frac{[MgCl_2][H_2]}{[HCl]^2}$

e. $K_c = \frac{[[Ag(NH_3)_2]^+]}{[Ag+][NH_3]^2}$

f. $K_c = [Cl_2]$

18. Write K_p expression for

a. $H_2(g) + F_2(g) \rightleftharpoons 2HF(g)$

b. $CO_2(g) + CaO(s) \rightleftharpoons CaCO_3(s)$

c. $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$

Solution

a) $K_p = \frac{(PHF)^2}{(PH_2)(PF_2)}$

b) $K_p = \frac{1}{PCO_2}$

c) $K_p = \frac{(PNO)^4\ (PH_2O)^6}{(PNH_3)^4\ (PO_2)^5}$

19. Classify the following as homogeneous or heterogeneous equilibrium.

a. $H_2CO_3(aq) + H_2O(1) \rightleftharpoons HCO^-_3(aq) + H_3O^+(aq)$

b. $NH_3(g) + H_2O(1) \rightleftharpoons NH^+_4(aq) + OH^-(aq)$

c. $Zn(s) + 2HCl(aq) \rightleftharpoons ZnCl_2(aq) + H_2(g)$

d. $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$

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Solution

- a and d are homogeneous equilibrium.
- b and c are heterogeneous equilibrium.

20. Given the reaction at equilibrium A (s) + 2B (g) $\rightleftharpoons$ AB_2 (g). If the partial pressure of B and AB_2 are 4atm and 3atm respectively at equilibrium, then find K_p.

Solution

$$K_p = \frac{PAB_2}{(PB)^2} = \frac{3}{(4)^2} = 0.188$$

Units of K_c and K_p

A. Units of K_c

Consider the general reaction $aA + bB \rightleftharpoons cC + dD$

$$K_P = \frac{[C]^c[D]^d}{[A]^a[B]^d}$$

$$\Rightarrow \text{Unit of } K_C = \frac{M^c\ M^d}{M^a\ M^d} = \frac{M^{c+d}}{M^{a+d}} = M^{(c+d)-(a+d)}$$

$$\boxed{\text{Unit of } K_C = (\text{mol/L})^{\Delta n}}, \Delta n = (c+d) - (a+b)$$

Example

Determiner the unit of Kc for the reaction:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

Solution

$\Delta n = 2 - (1 + 2) = -1$

Unit of $K_C = (\text{mol/L})\ \Delta n = (\text{mol/L})^{-1} = \text{L. mol}^{-1}$

B. Units of K_p

Assume the unit of partial pressure is atmosphere (atm).

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$$aA + bB \rightleftharpoons cC + dD$$

$$K_p = \frac{(Pc)^c\,(PD)^d}{(PA)^a\,(PB)^b}$$

$$\text{Unit of } K_p = \frac{(atm)^c\,(atm)^d}{(atm)^a\,(atm)^d} = (atm)^{(c+d)-(a+b)}$$

$$\boxed{\text{unit of } K_p = (atm)^{\Delta n}}, \Delta n = (c+d) - (a+b)$$

Example

Find the unit of K_p for the reaction

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$$

Solution

$\Delta n = (1 + 3) - 2 = 2$

$\therefore$ Unit of $K_p = (atm)^{\Delta n} = (atm)^2$

Relationship Between K_p and K_c

Consider a general gaseous reaction:

$$aA + bB \rightleftharpoons cC + dD$$

$$K_c = \frac{[C]^c\,[D]^b}{[A]^a\,[B]^b} \quad \text{----(1)}$$

From ideal gas law, we know that

$$PV = nRT$$

$$P = \frac{n}{V}RT$$

$P = MRT$, because $M = \frac{n}{V}$

$$\left.\begin{aligned} PC &= [C]RT \\ PD &= [D]\,RT \\ PA &= [A]\,RT \\ PB &= [B]RT \end{aligned}\right\} \text{-------(2)}$$

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$$K_p = \frac{(Pc)^c\,(PD)^d}{(PA)^a\,(PB)^d} \quad \text{——3}$$

Ralete (2) and (3)

$$K_P = \frac{([C]RT)^c\,([D]RT)^d}{([A]RT)^a\,([B]RT)^b}$$

$$K_P = \frac{[C]^c\,[D]^d}{[A]^a\,[Bd]} = \frac{(RT)^{c+d}}{(RT)^{a+b}} = \frac{[C]^c\,[D]^d}{[A]^a\,[B]^d}(RT)^{\Delta n} \quad \text{——(4)}$$

Relating (1) and (4)

$$\boxed{K_P = K_c\,(RT)^{\Delta n}},\ \Delta n = (c + d) - (a + b)$$

$$R = 0.082\ \frac{atm.L}{mol.K}$$

T = temperature in Kelvin

i. $\Delta n = 0 \Rightarrow K_p = K_c$

ii. $\Delta n > 0 \Rightarrow K_p > K_c$

iii. $\Delta n < 0 \Rightarrow K_p < K_c$

Illustrative Examples

21. For the reaction: $CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$, the unit of K_c is

A. mol/L B. $(mol/L)^{1/2}$ C. $(mol/L)^{-1/2}$ D.1

Solution

$$\Delta n = 1 - \left(1 + \frac{1}{2}\right) = \frac{-1}{2}$$

unit of $K_c = (mol/L)^{-1/2} = L^{1/2}.mol^{\frac{-1}{2}}$

Answer: C

22. Consider the reactions all given bellow are in gaseous state. For which of the following reactions is K_p less than K_c? (ESSLCE)

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A. $N_2O_4 \rightleftharpoons 2NO_2$ C. $2HI \rightleftharpoons H_2 + I_2$

B. $2SO_2 + O_2 \rightleftharpoons 2SO_3$ D. $N + O_2 \rightleftharpoons 2NO$

Solution

A. $\Delta n = 2 - 1 = 1 \Rightarrow K_p > K_c$ because $\Delta n > 0$

B. $\Delta n = 2 - (2 + 1) = -1 \Rightarrow K_p < K_c$ because $\Delta n < 0$

C. and D: $\Delta n = 2 - 2 = 0 \Rightarrow K_p = K_c$ because $\Delta n = 0$

Answer: B

23. For the reaction: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ K_p is equal to

A. K_c B. $K_c RT$ C. $K_c (RT)^2$ D. $K_c (RT)^{-1}$

Solution

$\Delta n = (1 + 0) - (0) = 1$

$K_p = K_c (RT)^{\Delta n} = K_c(RT)^1 = K_c RT$ **Answer: B**

24. Which of the following reversible reactions has the same K_p and K_c values?

A. $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

B. $2CO_2(g) + H_2O(g) \rightleftharpoons 2O_2(g) + CH_2CO(g)$

C. $MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g)$

D. $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

Solution

$K_p = K_c$ if $\Delta n = 0$

Answer: B

25. Determiner the unit K_c for

a. $2HF(g) \rightleftharpoons H_2(g) + F_2(g)$

b. $CaCO_3(s) \rightleftharpoons Ca^{2+}(aq) + CO^{2-}_3(aq)$

Solution

a. $\Delta n = (1 + 1) - 2 = O$

Unit of $K_c = (mol/L)^{\Delta n} = (mol/L)^0 = 1$

b. $\Delta n = (1 + 1) - 0 = 2$

Unit of $K_c = (mol/L)^{\Delta n} = (mol/L)^2 = mol^2 . L^{-2}$

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26. Find the unit of K_p for the following reactions. Assume the unit of pressure is torr.

a. $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

b. $H_2O(g) + C(s) \rightleftharpoons CO(g) + H_2(g)$

c. $2NO(g) + 5H_2(g) \rightleftharpoons 2NH_2(g) + 2H_2O(1)$

Solution

a. $\Delta n = 2 - (2 + 1) = -1 \therefore$ unit of $K_p = torr^{-1}$

b. $\Delta n = (1 + 1) - (1 + 0) = 1 \Rightarrow$ unit of $K_p = torr$

c. $\Delta n = (2 + 0) - (5 + 2) = -5 \Rightarrow$ unit of $K_p = torr^{-5}$

27. The equilibrium constant K_c, for the reaction $I2(g) \rightleftharpoons 2I(g)$ is 4×10^{-5} mol. L^{-1} at 727^0C. Calculate K_p at the given temperature

Solution

$T = 727^0c = 727 + 273 = 1000K$

$\Delta n = 2 - 1 = 1$

$$K_p = K_c (RT)^{\Delta n} = K_c\, RT = 4 \times 10^{-5}\ mol.L^{-1} \times 0.082 \frac{atm.L \times 1000K}{mol.K}$$

$K_p = 3.28 \times 10^{-3}$ atm

28. K_p for the reaction below at 573K is 11.5 atm. What is the value of K_c at this temperature. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Solution

$\Delta n = (1 + 1) - 1 = 1$

$K_p = K_c (RT)^{\Delta n} = K_c RT$

$$\Rightarrow K_c = \frac{K_P}{RT} = \frac{11.5atm}{0.082 \frac{atm.L}{mol.K} \times 573K} = 0.245\, mol.L^{-1}$$

29. For the reaction: $A(g) \rightleftharpoons 2B(g)$ at equilibrium. 0.4mol of A was introduced in 2L in vessel and allowed to decompoae until it reached equilibrium with B. The concentration of A at equilibrium was found to be 0.02M at 127^0C. Calculate K_p for this reaction at the same temperature.

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Solution

$$[A] = \frac{0.4\,mol}{2\,L} = 0.2\,M$$

$$[A]converted = [A]_0 - [A] = 0.2 - 0.02 = 0.18\,M$$

since the coefficient ration of A to B is 1:2

$$[B]eq = 2 \times 0.18\,M = 0.36\,M$$

$$K_c = \frac{[B]^2}{[A]} = \frac{(0.36\,M)^2}{(0.02\,M)} = 6.44\,mol.L^{-1}$$

$$K_p = K_c\,RT = 6.48\frac{mol}{L} \times 0.082\,\frac{atm.L}{mol.K} \times 400K$$

$$K_p = 212.5\,atm$$

30. Determiner K_c from the given K_p value

$Sb_2S_3(s) + 3H_2(g) \rightleftharpoons 2Sb(s) + 3H_2S(g)$, $K_p = 0.43$ at 713K

Solution

$$\Delta n = 3 - 3 = 0 \Rightarrow K_p = K_c = 0.43$$

Solubility Product (K_{sp}) for Salts

Generally, for sparingly soluble salt M_aX_b salt that dissociates at equilibrium

$$M_aX_b(s) \rightleftharpoons a\,M^{b+} + bX^{a-}$$

$$K_{sp} = [M^{b+}]^a\,[X^{a-}]^b$$

Example

The solubility product for

a. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

$K_{sp} = [Ag^+]\,[Cl^-]$

b. $Ca_3(PO_4)_2(s) \rightleftharpoons 3\,Ca^{2+}(aq) + 2PO^{3-}_4(aq)$

$K_{sp} = [Ca^{2+}]\,[PO_4^{3-}]^2$

Solubility Product for Weak Base

$$Fe(OH)_3(s) \rightleftharpoons Fe^{3+}(aq) + 3OH^-(aq)$$

$$K_p = [Fe^{3+}]\,[OH^-]^3$$

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Factors Affecting Equilibrium Constant

1. Temperature

As temperature increases the value of equilibrium constant also increases and vice - versal

2. The form of balanced Chemical Equations which Describe the Equilibrium

a. When we reverse an equation, the new equilibrium constant is the reciprocal of the original

$$A \rightleftharpoons B,\ K_1$$

$$B \rightleftharpoons A,\ K_2 = \frac{1}{K_1}$$

Example

$$A \rightleftharpoons B,\ K_1 = 10$$

$$B \rightleftharpoons A,\ K_2 = \frac{1}{K_1} = \frac{1}{10} = 0.1$$

b. When the whole reaction is multiplied by a constant,n , the new equilibrium constant is the original the power of n.

$$A \rightleftharpoons B,\ K_1$$

$$nA \rightleftharpoons nB,\ K_2 = (K_1)^n$$

c. When we reverse and multiply the equation by a constant n, the new equilibrium constant is, the reciprocal of the original the power of n.

$$A \rightleftharpoons B,\ K_1$$

$$nB \rightleftharpoons nA,\ K_2 = \left(\frac{1}{K_1}\right)^n$$

Example

$$2A \rightleftharpoons B,\ K_1 = 4$$

$$\frac{1}{2}B \rightleftharpoons A,\ K_2 = \left(\frac{1}{K_1}\right)^{1/2} = \left(\frac{1}{4}\right)^{1/2} = 0.5$$

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d. When two or more equations are added to produce a net equation, the equilibrium constant for the net equation is the product of the equilibrium constants for the added equations.

$$A + C \rightleftharpoons E + F, K_1$$

$$E + F \rightleftharpoons C + D, K_2$$

$$A \rightleftharpoons D, K = K_1 \times K_2$$

Example

Given the reactions at equilibrium

$$2A \rightleftharpoons B, K_1 = 3$$

$$C \rightleftharpoons 2B, K_2 = 4$$

Find equilibrium constant for the reaction $C \rightleftharpoons 4A$.

Solution

You may not get the answer directly by multiplication. You may reverse or multiply by a constant to find the required equilibrium constant.

Multiply by 2 and reverse the first and the second remains same

$$2B \rightleftharpoons 4A, K_1 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

$$C \rightleftharpoons 2B, K_2 = 4$$

$$C \rightleftharpoons 4A, K = K_1^1 \times K_2 = 4 \times \frac{1}{9} = \frac{4}{9}$$

Illustrative Examples

31. The solubility product K_{sp} for the sparingly soluble salt Ag_2CrO_4 is 4×10^{-12}. What is the solubility of the salt?

A. 2×10^{-6}M B. 1×10^{-12}M C. 1×10^{-4}M D. 2×10^{-3}M

Solution

$$Ag_2CrO_4\,(s) \rightleftharpoons 2Ag^+\,(aq) + CrO^{2-}_4(aq)$$

$$K_{sp} = [Ag^+]^2\left[CrO_4^{2-}\right]$$

since the coefficient ratio of Ag^+ to CrO^{2-}_4 is 2 : 1, thus

$[Ag^+] = 2x$ and $[CrO^{2-}_4] = x$

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$\Rightarrow 4 \times 10^{-12} = (2x)^2 (x) = 4x^3$

Solving for x, $x = 1 \times 10^{-4}$ mol/L

Therefore, the solubility of Ag_2CO_4 is 1×10^{-4} M to give $[Ag^+]$ of 2×10^{-4}M and $\left[CrO_4^{2-}\right]$ concentration of 1×10^{-4}M.

Answer: C

32. Which of the following is the most soluble salt?

A. $BaSO_4$, $K_{sp} = 1.1 \times 10^{-10}$
B. $CaCO_3$, $K_{sp} = 8.7 \times 10^{-9}$
C. BaF_2, $K_{sp} = 1.7 \times 10^{-7}$
D. CdS, $K_{sp} = 8 \times 10^{-28}$

Solution

As the value of K_{sp} increases, solubility of salt also increases.

Answer: C

33. $SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$, $K_{eq} = 0.25$ Find the value of K_{eq} at the same temperature for the reaction:

$2SO_2(g) + 2NO_2(g) \rightleftharpoons 2SO_2(g) + 2NO(g)$

A. 0.25 B. 0.062 C. 0.50 D. 16

Solution

The reaction is multiplied by 2.

$\Rightarrow K_{eq} = (0.25)^2 = 0.0625$

Answer: B

34. The following equilibrium constants were determined at 300°c.

$2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g)$, $K_1 = 4 \times 10^{18}$

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, $K_2 = 4 \times 10^{-31}$. What will be the equilibrium constant at 300°C for the gaseous reaction of

$2N_2O(g) \rightleftharpoons 4NO(g)$?

A. 3.2×10^{-12} B. 2.4×10^{-13} C. 6.4×10^{-43} D. 1.6×10^{-61}

Solution

Multiply the second by 2

$2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g)$, $K_1 = 4 \times 10^{18}$

$2N_2(g) + 2O_2(g) \rightleftharpoons 4NO(g)$, $K_2^1 = (K_2)_2 = (4 \times 10^{-31})^2 = 1.6 \times 10^{-61}$

$2N_2O(g) + O_2(g) \rightleftharpoons 2NO(g)$, $K = K_1 \times K_2^1 = 6.4 \times 10^{-43}$

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Answer: C

35. Which of the following mathematical relationship between K, K_1 and K_2 is correct?

$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g), K$

$Fe(s) + CO_2(g) \rightleftharpoons FeO(s) + CO(g), K_1$

$Fe(s) + H_2O(g) \rightleftharpoons FeO(s) + H_2(g), K_2$

A. $K = K_1 + K_2$ C. $K = K_1/K_2$

B. $K = K_1 \times K_2$ D. $K = K_2/K1$

Solution

The first reaction is obtained by reversing the third.

$Fe(s) + CO2(g) \rightleftharpoons FeO(s) + CO(g), K_1$

$FeO(s) + H_2(g) \rightleftharpoons Fe(s), + H_2O(g), 1/K_2$

$H_2(g) + CO_2(g) \rightleftharpoons H_2O(g), + CO(g), K = K_1/K_2$

Answer: C

36. Given the reactions at equilibrium

$SO_2(g) + \frac{1}{2} CO_2(g) \rightleftharpoons SO_3(g), K_1$

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g), K_2$

37. Which of the following equilibrium expression is correct?

A. $K_1 = K_2$ B. $K_2 = K_1$ C. $K_2 = \frac{1}{K_1^2}$ D. $K_2 = \frac{1}{K_1}$

Solution

The first reaction is reversed and multiply by 2. Thus,

$$K_2 = \frac{1}{K_1^2}$$

Answer: C

38. Given the equilibrium constant values for the following reactions

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i. $N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons N_2O(g),\ K_1 = 3.4 \times 10^{-18}$

ii. $N_2O_4(g) \rightleftharpoons 2NO_2(g),\ K_2 = 4.6 \times 10^{-3}$

iii. $\frac{1}{2}N_2(g) + O_2(g) \rightleftharpoons NO_2(g), K_3 = 4.1 \times 0^{-9}$

Determine K_c for the reaction $2N_2O(g) + 3O_2(g) \rightleftharpoons 2N_2O_4(g)$

Solution

- Reverse (i) and multiply by 2
- Reverse (ii) and multiply by2
- Multiply (iii) by 4.

$$2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g), K_1^1 = \left(\frac{1}{K_1}\right)2 = 8.65 \times 10^{34}$$

$$4NO_2(g) \rightleftharpoons 2N_2O_4(g),\ K_2^1 = \left(\frac{1}{K_2}\right)^2 = 4.73 \times 10^{4}$$

$$2N_2(g) + 4O_2(g) \rightleftharpoons 4NO_2(g), K_3^1 = (K_3)^4 = 2.83 \times 10^{-34}$$

$$2N_2O(g) + 3O_2(g) \rightleftharpoons 2N_2O_4(g),\ K = K_1^1 \times K_2^1 \times K_3^1$$

$$K = 8.65 \times 10^{34} \times 4.73 \times 10^{4} \times 2.83 \times 10^{-34} = 1.16 \times 10^{6}$$

39. At a fixed, elevated temperature, the following equilibrium constants are determined

$CoO(s) + H_2(g) \rightleftharpoons Co(s) + H_2O(g),\ K_1 = 67$

$CoO(s) + CO(g) \rightleftharpoons Co(s), + CO_2(g),\ K_2 = 490$

What is K at the same temperature for

$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$

Solution

Reverse the second.

$$\Rightarrow K = \frac{K_1}{K_2} = \frac{67}{490} = 0.14$$

40. What is the main factor that affects the value of equilibrium constant?

Answer: Temperature

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41. ...e K_p for the reaction:

$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) + \frac{1}{2}Br_2(g) \rightleftharpoons NOBr(g)$ from the following informations.

i. $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$, $K_p = 2.4 \times 10^{30}$

ii. $NO(g) + \frac{1}{2}Br_2(g) \rightleftharpoons NOBr(g)$, $K_p = 0.28$

Solve by your own: $K_p = 1.81 \times 10^{-16}$

42. Given the reaction: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, $K_c = 64$ find K_c for:

a. $2HI(g) \rightleftharpoons H_2(g) + (g)$

b. $\frac{1}{2}H_2(g) + \frac{1}{2}I_2(g) \rightleftharpoons HI(g)$

c. $HI(g) \rightleftharpoons \frac{1}{2}H_2(g) + \frac{1}{2}I_2(g)$

Solution

a. $\frac{1}{64}$ b. 8 c. $\frac{1}{8}$

43. What is the molar solubility of $Fe(OH)_3$ in a solution that is buffer at $[OH^-] = 3.16 \times 10^{-11}$M at 25^0c? ($K_{sp}(Fe(OH)_3) = 4 \times 10^{-38}$)

Solution

$Fe(OH)_3(s) \rightleftharpoons Fe^{3+}(aq) + 3OH^-(aq)$

$K_{sp} = [Fe^{3+}][OH^-]^3$

$$[Fe^{3+}] = \frac{K_{sp}}{[OH^-]^3} = \frac{4 \times 10^{-38}}{(3.16 \times 10^{-11})^3} = 1.26 \times 10^{-6}M$$

Thus, the molar solubility of $Fe(OH)_3$ is 1.26×10^{-6}M

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Application of Equilibrium Constant

There are two applications of equilibrium constant

1. Position of Equilibrium

$K_{eq} > 1$: formation of products is favoured (the equilibrium shifts to the right).

$K_{eq} < 1$: formation of reactants is favoured (between 0 and 1) (the equilibrium shifts to the left).

$K_{eq} = 1$: formation of products and reactants is favored equally (the reaction is at equilibrium).

Example

- $CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$, K = 14

  Formation of $COCl_2$ is favoured. because K > 1
- $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$, $K = 1.6 \times 10^{-21}$

  Formation of CO and Cl_2 is favoured because K < 1
- $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, K_p = 1atm 1200K
- Formation of reactant and products is favoured equally, since Kp is 1

2. Extent of Reaction

A very large numerical value of K_{eq} signifies that the forward reaction goes to completion or nearly so.

Example

$2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(1)$, $K_p = 1.4 \times 10^{93}$ at 293K

A very small numerical value of K_{eq} shows that the extent of reaction is very small, that is forward reaction does not occur to any significant extent.

Example

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = 1.9 \times 10^{-23}$ at 298k

Reaction Quotient (Q)

It is the ratio of the product concentration to the concentration of reactants, each raised to its stoichiometric coefficients in the chemical equations.

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$aA + bB \rightleftharpoons cC + dD$

$$Qc = \frac{[C]^c\,[D]^d}{[A]^a\,[B]^b},\ Qp = \frac{(PC)^c\,(PD)^d}{(PA)^a\,.\,(PB)^b}$$

Application of Reaction Quotient

Q > K, the equilibrum shifts to the left

Q < K, the equilibrium shifts to the right

Q = K, the reaction is at equilibrium

Illustrative Examples

44. Which of the following reactions has the greatest extent of reaction?

A. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g),\ K_c = 3.6 \times 10^8$

B. $N_2O_4(g) \rightleftharpoons 2NO_2(g),\ K_c = 5 \times 10^{-3}$

C. $CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g),\ K_c = 13.8$

D. $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g) + Cl_2(g),\ K_c = 4.7\ 10^{-4}$

Solution

The higher the value of K_c, the greater is the extent of reaction.

Answer: A

45. For which of the following reactions, formation of reactants is highly favoured?

A. $2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g),\ K_c = 4.45 \times 10^{-24}$

B. $2NOBr(g) \rightleftharpoons 2NO(g) + Br_2(g),\ K_c = 1.69 \times 10^{-2}$

C. $N_2O_4(g) \rightleftharpoons 2NO_2(g),\ K_c = 2.1 \times 10^{-4}$

D. $CO(g) + Cl_2(g) \rightleftharpoons CoCl_2(g),\ K_c = 14$

Solution

Formation of reactions is favoured highly for smaller value of K_{eq}

Answer: A

46. At 445°c, K_c for the following reactions is 0.020.

$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$. A mixture of H_2, I_2 and HI in a vessel at 445°c has the following concentrations: $[HI] = 2M$, $[I_2] = 0.50M$ and $[I_2] = 0.10M$. Which of the following statements concerning the reaction quotient, Q_c is true for the above

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system?

A. Q_c is less than K_c more Hi will be produced

B. Q_c is greater than K_c; more HI will be produced

C. Q_c is less than K_c; more H_2 and I_2 Will be produced

D. Q_c is grater than K_c ; more H_2 and I_2 will be produced

Solution

$$Q_c = \frac{[H_2][I_2]}{[HI]^2} = \frac{(0.50)(0.10)}{(2)^2} = 0.0125$$

$\Rightarrow Q_c < K_c$, formation of H_2 and I_2 is favored

Answer: C

47. The equilibrium constant is 0.09 at 295K for the reaction, $H_2O(g) + Cl_2O(g) \rightleftharpoons 2HOCl(g)$. For which of the following sets of conditions is the system at equilibrium?

A. A 1L flask contains 1mol of HOCl, 0.10mol of Cl_2O and 0.10mol of H_2O

B. A 2L flask contains 0.084 mol of HOCl, 0.080mol of Cl_2O and 0.98 mol of H_2O.

C. A 3L flask contains 0.25mol of HOCl, 0.0010mol of Cl_2O and 0.56mol of H_2O

D. A4L flask contains 0.25mol of HOCl, 0.001mol of Cl_2O and 0.56 mol of H_2O

Solution

The system is at equilibrium if $Q_c = K_c$

A. $$[HOCl] = \frac{nHOCl}{V} = \frac{1mol}{1L} = 1M$$

$$[Cl_2O] = \frac{nCl_2O}{V} = \frac{0.10mol}{1L} = 0.10M$$

$$Q_c = \frac{[HOCl]^2}{[Cl_2O][H_2O]} = \frac{(1)^2}{(0.2)(0.1)} = 100$$

$\therefore Q_c > K_c$, the reaction shifts to the right.

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B. $[HOCl] = \frac{nHOCl}{V} = \frac{0.084}{2L} = 0.042M$

$[Cl_2O] = \frac{nCl_2O}{V} = \frac{0.08mol}{2L} = 0.04M$

$[H_2O] = \frac{nH_2O}{V} = \frac{0.98mol}{2L} = 0.45M$

$Q_c = \frac{[HOCl]^2}{[Cl_2O][H_2O]} = \frac{(0.042)^2}{(0.04)(0.45)} = 0.09$

∴ The reaaction is at equilibrium because $Q_c = K_c$

C. $[HOCl] = \frac{0.25mol}{3L} = 0.083M$

$[Cl_2O] = \frac{0.0010mol}{3L} = 0.0003M$

$[H_2O] = \frac{0.56mol}{3L} = 0.187M$

$Q_c = \frac{[HOCl]^2}{[H_2O][Cl_2O]} = \frac{(0.083)^2}{(0.0003)(0.187)} = 122.8$

$Q_c > K_c$, the reaction shifts to the left.

D. $[HOCl] = \frac{0.25mol}{4L} = 0.0625M$

$[Cl_2O] = \frac{0.001mol}{4L} = 0.00025M$

$[H_2O] = \frac{0.56mol}{4L} = 0.14M$

$Q_c = \frac{[HOCl]^2}{[Cl_2O][H_2O]} = \frac{(0.0625)^2}{(0.00025)(0.14)} = 111.6$

$Q_c > K_c$, the reaction shifts to the left.

Answer: B

48. At 373K, $K_p = 5 \times 10^{-5}$ atm for the reaction $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$. For equal $P(CO) = P(COCl_2) = P(Cl_2) = 1$atm placed in a flask. Which of the following is true at equilibrium?

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A. The total pressure of the system must remains at 3atm.

B. P(CO) and $P(Cl_2)$ must decrease from their initial values.

C. P(CO) and $P(Cl_2)$ must increase from their initial values.

D. $P(COCl_2)$ remains constant.

Solution

$$Q_p = \frac{P_{CO} \times P_{Cl2}}{PCOCl_2} = \frac{1atm \times 1atm}{1atm} = 1atm$$

$Q_c > K_c$, the equilibrium shifts to the left As a result, the P(CO) and $P(Cl_2)$ must decrease.

Answer: B

49. For the reaction: $PCl_5(g) \rightleftharpoons PCl_3(q) + Cl_2(g)$, Kc = 0.0454 at 261°C. If a vessel is filled with these gases such that the initial concentrations are; $[PCl_3]$ = 0.50M, $[Cl_2]$ = 0.20M and $[PCl_5]$ = 4.50M. In which direction will a net change occur?

Solution

$$Q_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{(0.50)(0.20)}{(4.50)} = 0.022$$

$Q_c < K_c$, the net change occurs from left to right

50. The equilibrium constant for the reaction: $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is 4×10^{-4} at 2000K. In the presence of a catalyst, equilibrium is attained ten times faster. What is the value of equilibrium constant, in the presence of catalyst at 2000K?

Solution

$K = 4 \times 10^{-4}$ because catalysts do not affect the value equilibrium constant.

51. Arrange the following in decreasing order of extent of reaction.

I. $A + B \rightleftharpoons 2C$, $K_c = 1 \times 10^{-5}$

II. $X \rightleftharpoons Y + Z$, $K_c = 2 \times 10^{3}$

III. $E + F \rightleftharpoons 3D$, $K_c = 5 \times 10^{-4}$

Solution

II > III > I

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5.4 Calculating the Equilibrium Concentrations

This topic is very important to calculate the pH of weak acid and weak bases.

Follow the following steps to determine the equilibrium concentrations of reactants and products.

1. Determine the direction of net change by comparing Q and K_{eq} is necessary.
2. State the initial concentrations and let x be change in concentration or partial pressure.
3. Express equilibrium concentration or partial pressure in terms of x.
4. Write equilibrium constant expression.
5. Substitute equilibrium concentration or partial pressure in to K_c or K_p expression and solve for x.
6. Check the equilibrium concentrations or partial pressure by solving for Kc and K_p.

Example

Initially the concentration of PCl_5 is 0.02M. Find the equilibrium concentration of all species.

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, $K_c = 1 \times 10^{-2}$

Solution

	$PCl_5(g)$	$\rightleftharpoons$	$PCl_3(g)$	+	$Cl_2(g)$
Initial conc, M	0.02		0		0
Change in conc. M	$-x$		$+x$		$+x$
Equilibrium conc. M	$0.02-x$		x		x

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

$$1 \times 10^{-2} = \frac{x \cdot x}{0.02 - x} \Rightarrow x^2 + 10^{-2}x - 2 \times 10^{-4} = 0$$

Solve for x by using quasratic formula

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$$x = \frac{-b \pm \sqrt{b2 - 4ac}}{2a}, \; a = 1, \; b = 10^{-2}, \; c = -2 \times 10^{-4}$$

$$x = \frac{-10^{-2} \pm \sqrt{(10^{-2})2 - (4 \times 1 \times (-2 \times 10^{-4})}}{2 \times 1}$$

$x = 1 \times 10^{-2}$ or $x = -2 \times 10^{-2}$ but cobcentration cannot be negative.

Thus, the valu of x is 1×10^{-2}.

At equilibrium.

$[PCl_5] = 2 \times 10^{-2} - x = 2 \times 10^{-2} - 1 \times 10^{-2} = 1 \times 10^{-2}M = 0.01M$

$[PCl_3] = [Cl_2] = 1 \times 10^{-2}M = 0.01M$

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} \Rightarrow 1 \times 10^{-2} = \frac{(1 \times 10^{-2})(1 \times 10^{-2})}{(1 \times 10^{-2})} = 1 \times 10^{-2}$$

Illustrative Examples

52. One mole of SO_3 was placed in a 2L vessel at a certain temperature. The following equilibrium was established in the vessel

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$$

At equilibrium, the vessel was found to contain 0.5mol of SO_3. The value of K would be

A. 0.25 B. 0.5 C. 0.125 D. 1.0

Solution

	$2SO_3(g)$	$\rightleftharpoons$	$2SO_2(g)$	+	$O_2(g)$
Initial conc M	0.02		0		0
Change in conc. M	-2x		+2x		+ x
Equilibrium conc. M	1-2x		2x		x

$\Rightarrow 1 - 2x = 0.5 \Rightarrow x = 0.25$

$\therefore nSO_3 = nSO_2 = 0.5mol, nO_2 = 0.25mol$

$$[SO_3] = [SO_2] = \frac{0.5mol}{2L} = 0.25M$$

$$[O_2] = \frac{0.25mol}{2L} = 0.125M$$

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$$K_c = \frac{[SO_2]^2[O_2]}{[SO_3]^2} = 0.125$$

Answer: C

53. When 1mole of N_2 and 1mol of H_2 is enclosed in 5L vessel and the reaction is allowed to attain equilibrium, it is found that at equilibrium there is x mole of N_2. The number of mole of NH_3 formed would be;

A. 2x B. 2(1 - x) C. 2(1 + x) D. 1 - x

Solution

	$N_2(g)$	+	$3H_2(g)$	$\rightleftharpoons$	$2NH_2$
Initial	1		1		
Change in	-x		-3x		+ 2x
Equilibrium	1- x		1- 3x		2x

At equilibrium, $nN_2 = x \Rightarrow nN_2 = 1 - x$

$\Rightarrow nNH_3 = 2x = 2(1 - x)$

Answer: B

54. At a certain temperature $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, only 50% HI is dissociated at equilibrium. The equilibrium constant is

A. 0.25 B. 3.0 C. 1.0 D. 0.5

Solution

Using the coefficient of the species in the reaction:

$[HI] = 50\% \times 2 = 1M$

$[H_2] = [I_2] = 50\% \times 1 = 0.5M$

$$K_c = \frac{[H_2][I_2]}{[HI]^2} = \frac{0.5 \times 0.05}{(1)^2} = 0.25$$

Answer: A

55. When 0.5mol of N_2O_4 is placed in a 4L reaction vessel and heated to 400K, 80% of the N_2O_4 decomposes to NO_2 gas as follows: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, What will be the value of K_p, in units of pressure, at 400K for this reaction?

A. 2.62 B. 13.12 C. 52.48 D. 16.20

Solution

80% N_2O_4 Converted means 80% of 5mol of N_2O_4 is changed to NO_2.

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$$\Rightarrow \frac{80}{100} \times 0.5\text{mol} = 0.4\text{mol}$$

$$[N_2O_4]0 = \frac{n}{V} = \frac{0.5\text{mol}}{4L} = 0.125M$$

$$[N_2O_4] \text{ left} = \frac{0.4\text{mol}}{4L} = 0.1M$$

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

Initial conc.	0.125	0
Change in conc.	-x	+2x
Equilibrium conc.	0.125-x	2x

$[N_2O_4]eq = 0.125\text{-}x = 0.125 - 0.1 = 0.025M$

$[NO_2] = 2x = 2 \times 0.1 = 0.2M$

$$K_c = \frac{[NO_2]^2}{[N_2O_4]} = \frac{(0.2)^2}{(0.025)} = 1.6\frac{\text{mol}}{L}$$

$$K_p = K_c(RT)^{\Delta n} = 1.6\frac{\text{mol}}{L} \times 0.082\frac{\text{atm.L}}{\text{mol.k}} \times 400K$$

$K_P = 52.48\text{atm}$

Answer: C

56. In the reaction: $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $K_{eq} = 100$.
What will be the concentration of O_2, if the concentration of SO_2 is the same as the concentration of SO_3?

Solution

$$K_{eq} = \frac{[SO_2]^2}{[SO_2]^2[O_2]}$$

$$K_{eq} = \frac{1}{[O_2]} \text{ because } [SO_2] = [SO_3]$$

$$[O_2] = \frac{1}{K_{eq}} = \frac{1}{100} = 0.01M$$

57. The equilibrium constant for the following reaction:
$3H_2(g) + CO(g) \rightleftharpoons CH_4(g) + H_2O(g)$ is 3.39 at 1200K.
A system at equilibrium has [CO] = 0.1522M, $[H_2]$ = 0.1566m and

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$[H_2O] = 0.0478M$. What is the $[CH_4]$?

Solution

	$C_2H_6(g)$	$\rightleftharpoons$	$H_2(g)$	+	$Cl_2(g)$
Initial conc	0.05		0		0
Change in conc	$-x$		$+x$		$+x$
Equilibrium conc	$0.05 - x$		x		x

$$K_c = \frac{x \cdot x}{0.05 - x} \Rightarrow \frac{x^2}{0.05 - x} = 1.02 \times 10^{-19}$$

For very small value of K_c, the reaction will not proceed far to the right. So the value of x is too small and you can make an approximation, $0.05 - x \approx 0.05$

$$1.02 \times 10^{-19} = \frac{x^2}{0.05} \Rightarrow x = \sqrt{0.05 \times 1.02 \times 10^{-19}} = 7.1 \times 10^{-11} M$$

➥ At equilibrium,

$[H_2] = [C_2H_4] = 7.1 \times 10^{-11}M$ $[C_2H_6] = 0.05 - 7.1 \times 10^{-11} = 0.05M$

58. Given the reaction: $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$, $K_p = 0.108$ atm^2 at 25^0C. What is the total gas pressure at equilibrium?

Solution

At equilibrium, $P(NH_3) = P(H_2S)$, because the coefficient ratio is one to one.

$$K_p = P(NH_3) \times P(H_2S)$$

$$P(NH_3) = P(H_2S) = \sqrt{KP} = \sqrt{0.108} = 0.33atm$$

$$P_{Total} = P(NH_3) + P(H_2S) = 0.33 + 0.33 = 0.66 \text{ atm}$$

59. Starting with 1atm of $SO_3(g)$, determine the partial pressure of each gas at equilibrium and the total pressure for the reaction:

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g), K_p = 3.2 \times 10^{-5}$$

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Solution

	$2SO_2H_6(g)$	$\rightleftharpoons$	$2SO_2(g)$	+	$O_2(g)$
Initial partial pressure	1		0		0
Change in pressure	$-2x$		$+2x$		$+x$
Equilibrium pressure	$1-2x$		$2x$		x

$$K_p = \frac{(PSO_2)^2 (PO_2)}{(PSO_3)^2}$$

$$3 \times 10^{-5} = \frac{(2x)^2 (x)}{(1-2x)^2}$$

Since K_p is small value use approxiation method

$1 - 2x \approx 1$

$$3 \times 10^{-5} = \frac{4x^3}{1} \Rightarrow x = 0.02$$

At equilibrium: $P(SO_2) = 2x = 2 \times 0.02 = 0.04$ atm

$PO_2 = 1 - 0.02 = 0.98$ atm

$P_{Total} = P(SO_3) + P(SO_2) + P(O_2) = 0.98 + 0.02 + 0.04 = 1.04$atm

60. For the reaction: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = 1.15$atm at 1073K. If 40g of $CaCO_3$ was put into a 10L container and heated to 1073K, what percent of $CaCO_3$ remains undecomposed.

Solution

$$K_p = P(CO_2) = 1.16\text{atm}$$

$$n(CO_2) = \frac{PV}{RT} = \frac{1.16\text{atm} \times 10\text{L}}{0.082 \frac{\text{atm.L}}{\text{mol.K}} \times 1073\text{K}} = 0.40\text{mol}$$

$$n(CaCO_3)(\text{initial}) = \frac{m}{M} = \frac{40\text{g}}{100\text{g/mol}} = 0.40\text{mol}$$

$$\%\ CaCO_3 \text{ converted} = \frac{nco_2}{ncaco_3} \times 100 = \frac{0.132}{0.40} \times 100 = 33\%$$

Thus, 100 - 33 = 67% remains undecomposed

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5.5 Factors Affecting the Position of Chemical Equilibrium

Le chatelier's principle can be used to explain the results of applying any change of condition (stress) on a system in equilibrium.

It states that if a stress is placed upon a system in equilibrium, the equilibrium is displaced in the direction that counteracts the effect of the stress.

1. Effect of Changing the Concentrations

- Increasing concentration of reactants or decreasing concentration of products shifts the equilibrium to the **right** (formation of products is favored)
- Decreasing concentration of reactants or increasing concentration of products shifts the equilibrium to the left (formation of reactants is favored)

Example

- $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
- Increasing $[SO_2]$ and $[O_2]$ or decreasing $[SO_3]$ shifts the equilibrium to the right
- Removing SO_2 and O_2 or increasing $[SO_3]$ shifts the equilibrium to the left.

Adding solid substance does not disturb the position of equilibrium because the concentration of solid is constant. However, removing solids shifts the equilibrium to the direction in which they are removed.

Example

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

- Addition of $CaCO_3(s)$ = No effect
- Removing of $CaO(s)$ = shift to the right

2. Effect of Temperature on Equilibrium

If the temperature of a system in equilibrium is raised, the equilibrium is shifted in the direction that **absorbs heat (endothermic side)**

Lowering the temperature of the system in equilibrium shifts the equilibrium to the **exothermic side**

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If the forward reaction is endothermic, the reverse becomes exothermic and vice - versal.

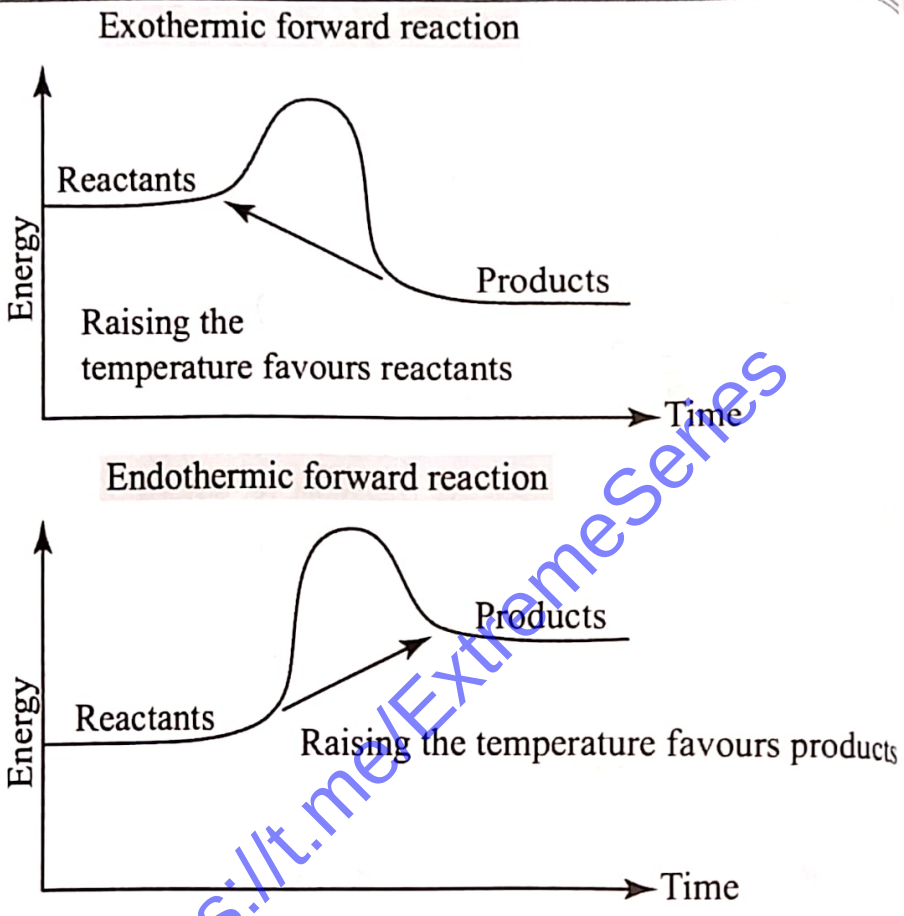

Fig 5 . 3 The Effect of Increasing Temperature on the Proportions of Reactants and products in the Equilibrium Mixture for Exothermic and Endothermic Reactions

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Table 5.2 The Effect of Temperature Changes on Chemical Equilibria

Nature of forward reaction (sign of ΔH)	Change in temperature	Shift in the positions of equilibrium	Effect on value of K_c
Endothermic ($\Delta H > 0$)	Increase	To the right	K_c increases
Endothermic ($\Delta H > 0$)	Decrease	To the left	K_c decreases
Exothermic ($\Delta H < 0$)	Increase	To the left	K_c decreases

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Exothermic ($\Delta H < 0$)	Decrease	To the right	K_c increases

Note that the shift in equilibrium as a result of temperature is actually a change in the value of equilibrium constant.

Example

- $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + heat$
- Forward reaction is exothermic
- Backward reaction is endothermic
- Increasing temperature shifts to the left.
- Lowering temperature shifts to the right.

3. Effect of Pressure and Volume on Equilibrium

Pressure and volume affect the position of equilibrium for

i. a reaction involving gases.

ii. a reaction in which the number of moles of reactants is not equal number of moles of products.

Increasing pressure or decreasing volume shifts the equilibrium to the **smaller number of moles.**

Lowering pressure or increasing volume shifts the equilibrium to the **larger number of moles.**

Reason: $V \alpha \frac{1}{P}$ (Boyle's law)

$V \alpha n$ (Avoabadro's law)

Example

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_2(g)$

$n_{reactants} = 2 + 1 = 3mol$ $n_{product} = 2mol$

Increasing pressure or decreasing volume shifts the equilibrium to the right.

Lowering pressure or increasing volume shifts the equilibrium to the left.

4. Addition of Inert Gas

It shifts the equilibrium to the **larger number of moles** of gaseous species.

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If an inert gas is added at constant volume there will be no change on the position of equilibrium

Note

- Adding catalyst to a reaction mixture speeds up both the forward and reverse reaction.
- Equilibrium is a chieved more rapidly but the equilibrium is not disturbed by the catalyst.

Illustrative Examples

61. Consider this equilibrium system

$CO(g) + Fe_3O_4(s) \rightleftharpoons CO_2(g) + 3FeO(s)$

Which of the following disturbances shift the equilibrium position to the right? (ESSLCE)

A. Adding $Fe_3O_4(s)$
B. Removing FeO (s)
C. Removing CO(g)
D. Removing $CO_2(g)$

Solution

The equilibrium shifts to the right by:

- adding CO (g)
- by removing $CO_2(g)$ or FeO (s)

Answer: B

62. In which of the following system will the position of equilibrium shift to the left up on an increase in pressure, but to the right up on an increase in pressure, but to the right up on an increase in temperature?

A. $CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$, $\Delta H < 0$
B. $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$, $\Delta H < 0$
C. $C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$, $\Delta H > 0$
D. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H < 0$

Solution

Increase in pressure to the smaller number of molecules and increase in temperature to the endothermic side.

Answer: C

63. In what manner will increase of pressure affects the following equilibrium?

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$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$

A. Shifts in the forward direction
B. Shifts in the reverse direction
C. Increase in the yield of hydrogen
D. No effect

Solution

Increasing pressure shift the equilibrium to the smaller number of moles.

Answer: B

64. For the reaction: $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$, at a given the temperature the equilibrium amount of $CO_2(g)$ can be increased by:

A. Adding inert gas
B. Adding a suitable catalyst
C. Decreasing the volume of container
D. Increasing the amount of CO (g)

Answer: D

65. The hydrogen used in Haber process is made by the following reactions. $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$, $\Delta H = +206kJ$, which of the following sets of conditions will favor the formation of H_2?

A. Low pressure and high temperature
B. Low pressure and low temperature
C. High pressure and low temperature
D. High pressure and high temperature

Answer: A

66. Which change will increase the amount of $SO_3(g)$ at equilibrium?

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H = -197$ KJ/mul

I. Increasing temperature
II. Decreasing volume
III. Adding a catalyst

A. I only B. II only C. I and II D. II and III

Answer: B

67. Given the reaction at equilibrium

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$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O(1)$, $\Delta H > O$.

Explain the effect of the following on the position of equilibrium.

a. Adding $NaHCO_3(s)$
b. Removing $Na_2CO_3(s)$
c. Increasing temperature and volume
d. Lowering temperature and pressure
e. Increasing concentration of $CO_2(g)$
f. Adding inert gas
g. Adding catalyst.

Solution

a. No effect because it is solid reactant
b. Shifts the equilibrium to the right
c. Both shifts to the right
d. Unpredictedtable because they shift in opposite direction
e. Shifts to the left
f. Shifts to the right
g. Has no effect

68. Consider the reaction: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ in a closed container at equilibrium at a constant temperature. What will be effect of adding more PCl_5 on the equilibrium concentration of $Cl_2(g)$?

Answer: Increases

Optimum Conditions

- They are conditions that yield maximum amount of products.
- They are conditions that shift the equilibrium to the right.
- They favour formation of products.

i. Concentration

- Increasing concentration of reactants or removing products.

ii. Temperature

- For endothermic forward reaction, **increasing temperature**
- For exothermic forward reaction, **lowering temperature**

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iii. Pressure

For $\Delta n > 0$, lowering pressure

For $\Delta n < 0$, raising pressure

iv. Addition of Inert Gas for $\Delta n > 0$

5. Application of Chemical Equilibrium

There are many applications of chemical equilibrium two of them are given below.

I. Production of Ammonia (NH_3)

In the Haber process, nitrogen and hydrogen in the correct proportions (1:3) are pressurized to approximately 200 atmospheres and passed over a catalyst of freshly produced, finely divided iron at a temperature of between 350°C and 500°C

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g),\ \Delta H = -92kJ/mol$$

Formation of NH_3 is favored at low temperature and high pressure.

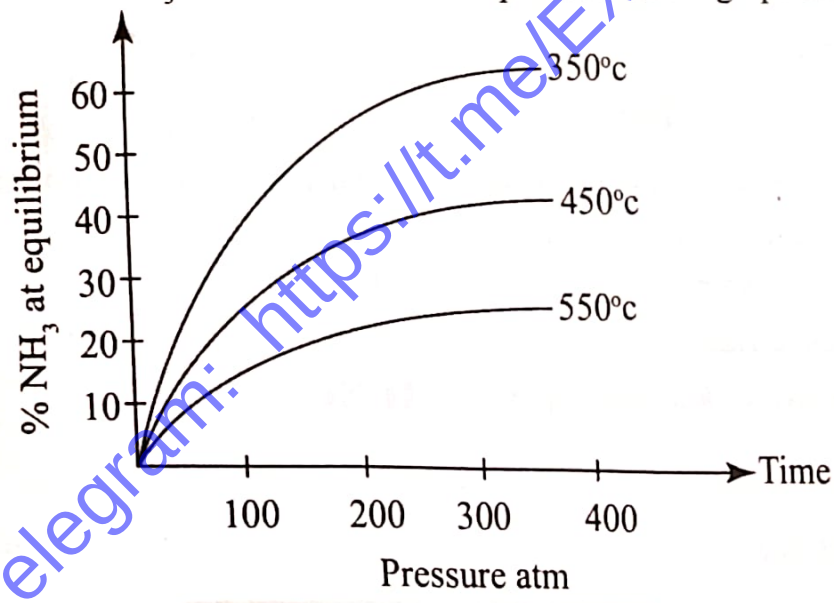

Fig 5 . 4 Yields from the Haber Precess

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A Flow Chart for Haber Process

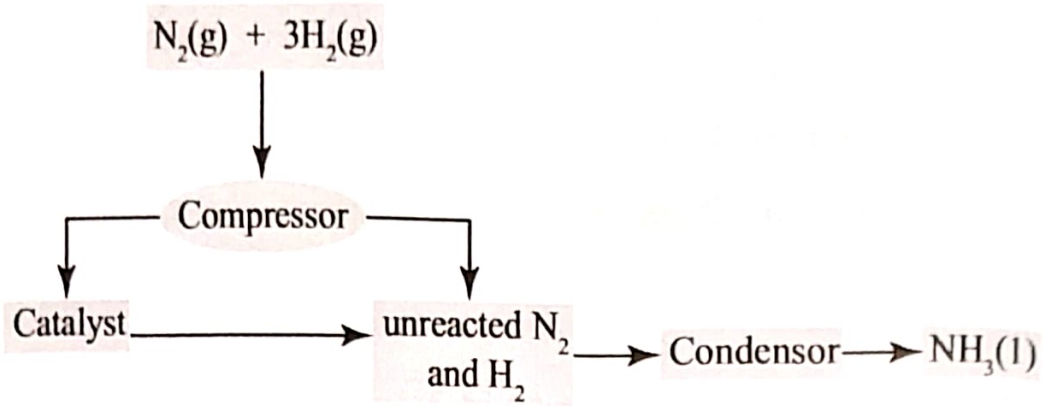

Fig 5 . 5 Flow chart for production of Ammonia by Haber process

Physical Properties of Ammonia

- It is alkaline colorless gas.
- It has Pungent smell.
- It is lighter than air and extremely soluble in water.
- Liquefies on cooling under pressure
- It causes tears in eyes.

Uses of Ammonia (NH_3)

- It is used to make nitrogeneous fertilizers such as urea, ammonium nitrate and ammonium sulphate
- It is used to make nitric acid by Ostwald process
- It is used as refrigerant.

2. Production of Sulphuric Acid (H_2SO_4)

Sulphuric acid is industrially manufactured by

Contact Process

There are four steps in the contact process

1. Burning of sulfur

$$S + O_2 \longrightarrow SO_2$$

2. Burning of sulfurdioxide

$$2SO_2(g) + O_2(g) \rightleftharpoons^{V_2O_5(g)} 2SO_3(g) + 196kJ/mol$$

Formation of $SO_3(g)$ is favored by:

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- Lowering temperature
- Increasing pressure
- Removing SO_3
- Increasing concentration of SO_2 and O_2

3. Reaction of sulphutrioxide with sulphuric acid

$$SO_3(g) + H_2SO_4(1) \rightarrow H_2S_2O_7(1)$$

oleum, Fumining sulphutic acid

4. Dissolving oleum in water

$$H_2S_2O_7(1) + H_2O(1) \rightarrow 2H_2SO_4(1)$$

- Sulphuric acid is also called **oil of vitriol.**

Combustion Chamber , Buring of sulfur

↓

Combustion Chamber , formation of SO_3

↓

Combustion tower , SO_3 is absorbed in $H_2S_2O_7$

↓

Hydration of oleum , Production of H_2SO_4

Fig 5 . 6 Flow Chart for Contact Process

The sulphur used in the contact process is industrially manufactured by **Frasch process** from the underground deposit.

Uses of Sulphuric Acid

It is used to make drugs, dyes, paints, sulphates, detergents, nitric acid hydrochloric acid.

It is used in storage batteries as electrolytes

- It is used to clean metal surfaces before electroplating
- It is dehydration and oxidizing agents.
- It is used to make fertilizers.

Illustrative Examples

69. All of the following are optimum conditions for the reaction,

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$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$ + heat. **EXCEPT**

A. Increasing pressure
B. Lowering pressure
C. Addition of inert gas
D. Removing $CO_2(g)$

Answer: C

70. If the following reaction is at equilibrium all of the following favor formation of NH_3. **EXCEPT**

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + 92kJ/mol$

A. Increasing pressure
B. Decreasing temperature
C. Adding N_2 and H_2
D. Increasing volume

Answer: D

71. The catalyst that provides surface area for SO_2 and O_2 to collide in the contact process is:

A. MnO_2 B. Fe C. V_2O_5 D. Ni

Solution

V_2O_5 provides surface area for SO_2 and O_2 to make contact. That is why it is called contact process.

Answer: C

72. Ammonia is commercially manufacture by

A. Contact process
B. Haber precess
C. Oswald process
D. Frasch process

Solution

Contact process : Sulphuric acid

Haber process: Ammonia

Ostwald process: Nitric acid

Frasch process: Sulphur

Answer: B

73. The correct order of steps in contact process are

I. $2SO_2 + O_2 \rightarrow 2SO_3$
II. $S + O_2 \rightarrow SO_2$
III. $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$
IV. $SO_3 + H_2SO_2 \rightarrow H_2S_2O_7$

A. I, II, III, IV
B. II, I, IV, III
C. III, I, II, IV
D. IV, III, I, II

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Answer: B

74. Why is SO_3 not directly combined with water to make sulphuric acid?

Solution

Because the reaction cannot be controlled easily and creates a fog of sulphuric acid.

75. Write the main uses of:

a. Ammonia (NH_3) b. Sulphuric (H_2SO_4)

Solution

a. Ammonia is mainly used to make fertilizers and nitric acid.

b. Sulphuric acid is the most important chemical in industry as raw material, catalyst, oxidizing and dehydrating agents.

76. Which step in the contact process is directly related with the application of chemical equilibrium?

Solution

Step 2: Combustion of sulphur dioxide

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) + \text{Heat.}$$

Solved problems

77. K_p/K_c for the reaction: $CO_2(g) \rightleftharpoons CO(g) + \frac{1}{2}O_2(g)$ is _____

Solution

$$\Delta n = \left(1 + \frac{1}{2}\right) - 1 = \frac{1}{2}$$

$$K_P = K_c\ (RT)^{\Delta n}$$

$$\Rightarrow \frac{K_p}{K_c} = (RT)^{1/2} = \sqrt{RT}$$

78. If for a reaction $A(g) + 2B(g) \rightleftharpoons 3C(g) + D(g)$ is 0.05 atm at 1000K. Its K_c value is _____.

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Solution

$$\Delta n = (3+2) - (2+1) = 1$$

$$K_p = K_c\ (RT)^{\Delta n}$$

$$K_c = \frac{K_p}{(RT)^{\Delta n}} = \frac{K_p}{RT} = \frac{0.05 atm}{0.082 \frac{atm.L}{mol.K} \times 1000K} = 6.1 \times 10^{-4} \frac{mol}{L}$$

79. The initial pressure of $COCl_2$ is 100 torr. The total pressure of the system becomes 150 torr. When the equilibrium $COCl_2(g) \rightleftharpoons CO(g) + Cl(g)$ is attained at constant temperature. What is the value of K_p of the reaction?

Solution

$$COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$$

Initial P (torr):	100	0	0
Change P(torr):	- x	+x	+x
Equilibrium P(torr)	100 - x	x	x

$$P_{Total} = P_{COCl_2} + P_{CO} + P_{Cl_2}$$

$$150 = 100 - x + x + x$$

$$x = 150 - 100 = 50$$

$$\therefore P_{COCl_2} + P_{CO} + P_{Cl_2} = 50 torr$$

$$K_p = \frac{PCO.\ PCl_2}{PCOCl_2} = \frac{50 \times 50}{50} = 50 torr$$

80. A quantity of AB_2 was heated in a 10L vessel 500K, $2AB_2(g) \rightleftharpoons A(g) + 2B(g)$. At equilibrium the vessel contains, 0.10molAB_2, 0.2 mol A and 0.03 mol B Calculate K_c and K_p

Solution

$$[AB_2] = \frac{n}{V} = \frac{0.10mol}{10L} = 0.01M$$

$$[A] = \frac{n}{V} = \frac{0.2mol}{10L} = 0.02M$$

$$[B] = \frac{n}{V} = \frac{0.3mol}{10L} = 0.03M$$

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$$= \frac{[A][B]^2}{[AB_2]} = \frac{(0.02)(0.03)^2}{0.01} = 1.8 \times 10^{-3} \text{ mol/L}$$

$$K_c[RT]^1 = 1.8 \times 10^{-3} \frac{\text{mol}}{\text{L}} \times 0.08 \frac{\text{atm.L}}{\text{mol.K}} \times 500\text{K}$$

0.0738atm

81. ...ntration of CrO_4^{2-} ions in a saturated solution of silver chromate is 2×10^{-4}M. What is the solubility product of the salt.

Solution

$$Ag_2CrO_4(s) \rightleftharpoons 2Ag^+(aq) + CrO_4^{2-}(aq)$$

$$[Ag^+] = 2 \times [CrO^{2-}_4] = 2 \times 2 \times 10^{-4}\text{M} = 4 \times 10^{-4}$$

$$K_{sp} = [Ag^+]\,[CrO^{2-}_4] = (4 \times 10^{-4})^2 \times 2 \times 10^{-4} = 3.2 \times 10^{-11}$$

82. Initially 1molA, 1.5mol B and 0.4mol of C are placed in a 1.0L reaction vessel and allowed to come to equilibrium. If the final concentration of C is 0.80M, then find the value of equilibrium constant for the reaction $A + B \rightleftharpoons 2C$.

Solution

$$[A]_o = \frac{1\text{mol}}{1\text{L}} = 1\text{M}$$

$$[B]_o = \frac{1.5\text{mol}}{1\text{L}} = 1.5\text{M}$$

$$[C]_o = \frac{0.4\text{mol}}{1\text{L}} = 0.4\text{M}$$

	A +	B ⇌	$2C$
Initial:	1	1.5	0.4
Change	-x	-x	+2x
Equilibrium:	1-x	1.5-x	2x + 0.4

$\Rightarrow 2x + 0.4 = 0.80\text{M}, \quad x = 0.2$

At equilibriium ; $[A] = 1 - x = 1 - 0.2 = 0.8\text{M}$

$[B] = 1.5 - x = 1.5 - 0.2 = 1.3\text{M}, [C] = 0.8\text{M}$

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$$K_c = \frac{[C]^2}{[A][B]} = \frac{(0.8)^2}{(1.3)(0.8)} = 0.615$$

83. Given the equilibrium the equilibrium constant values if the following reactions 25°C.

i. $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$, $K_c = 1.1 \times 10^{81}$

ii. $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$, $K_c = 4.4 \times 10^{32}$

Find the value of K_p at 25°C for the reaction:

$2H_2O(g) + 2Cl_2(g) \rightleftharpoons 4HCl(g) + O_2(g)$

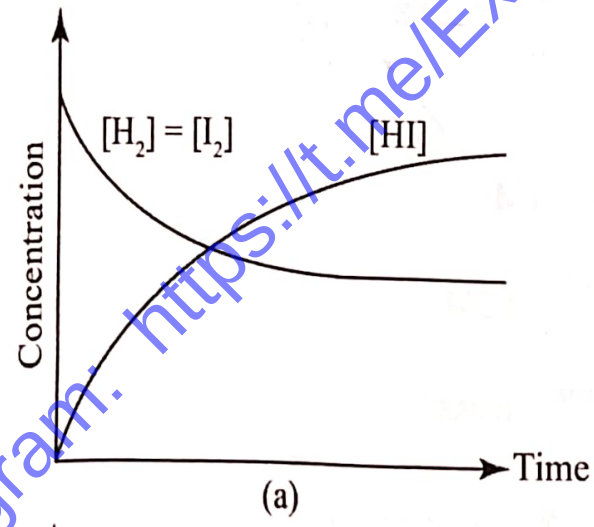
Answer: $K_p = 4.3 \times 10^{-15}$ atm

84. Draw the concentration - time graph for the reactions below at equilibrium.

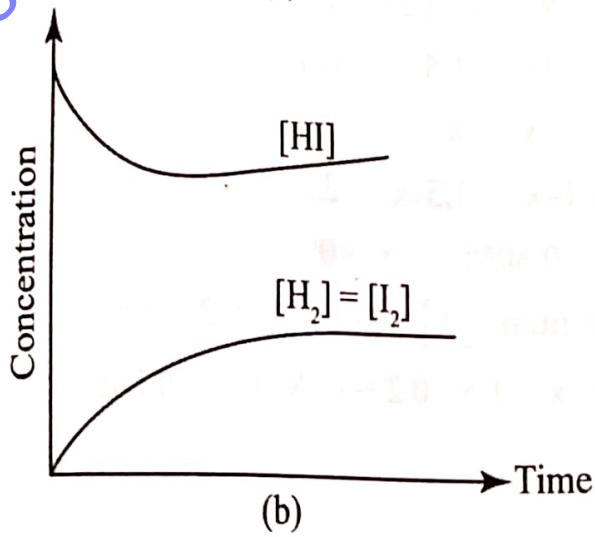
i. $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

ii. $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

Solution



(a)



(b)

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85. Write the basic features of dynamic equilibria

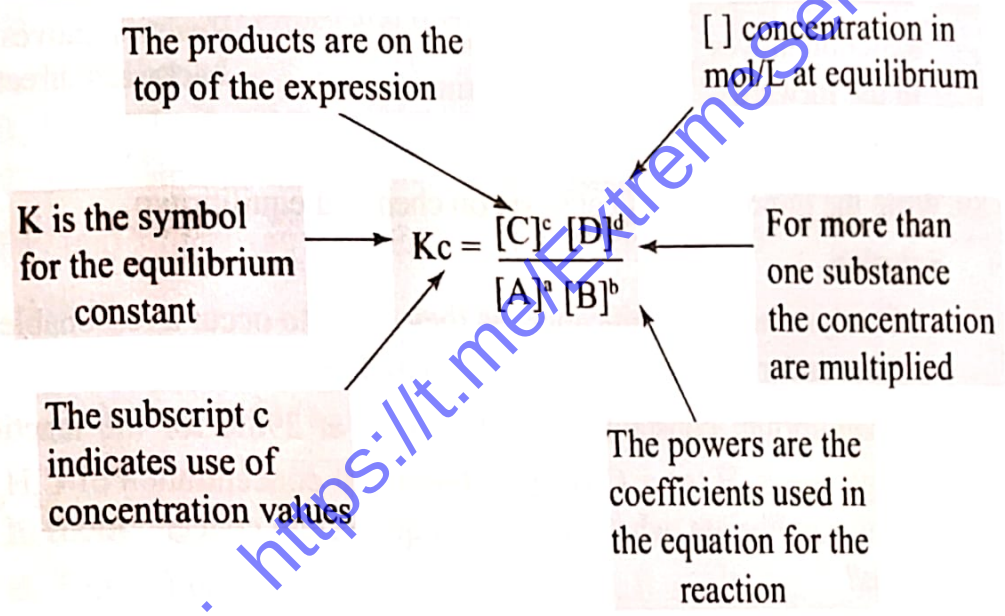
Solution

- A dynamic equilibria can only be established in a closed system.
- The amounts of reactants and products remain constant.
- The equilibrium is dynamic, not static
- Forward rate equals backward rate.
- The equilibrium position can be achieved from either directions.

86. Describe equilibrium constant expression for the general reaction below.

$$aA + bB \rightleftharpoons cC + dD$$

Solution



87. Write K_c expression for

a. $2CrO_4^{2-}(aq) + H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + OH^-(eq)$

b. $Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$

c. $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

Solution

a. $$K_c = \frac{[Cr_2O_7^{2-}][OH^-]}{[CrO_4^{2-}]^2[H^+]}$$

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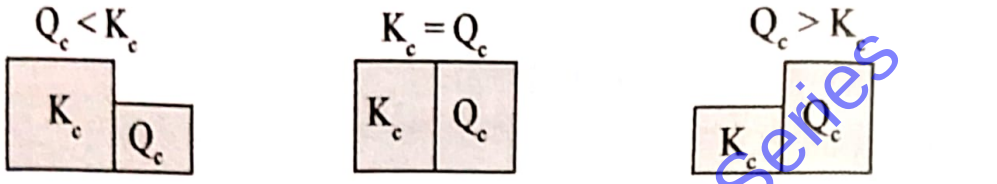
b. $K_c = \frac{[[Fe(SCN)]^{2+}]}{[Fe^{3+}][SCN^-]}$

c. $K_c = \frac{[NO_2]^2}{[NO]^2[O_2]}$

88. Explain the application of reaction quotient using diagrams.

Solution



Reaction moves in the forward direction

Reaction is at equilibrium

Reaction moves in backward direction

89. Write the importance of catalysts on chemical equilibrium

Solution

Catalysts are of significance, as they allow to occur a reasonable rates under milder, and more economic, conditions.

90. The equilibrium constant, K_c is 1×10^{-19} at 298K for the reaction: $C_2H_6(g) \rightleftharpoons H_2(g) + C_2H_2(g)$ If the initial concentration of C_2H_6 is 0.1M at equilibrium, what will be the equilibrium concentrations of all species?

Solve by approximation method.

$[C_2H_4] = [H_2] = 1.0 \times 10^{-10}M$

$[C_2H_6] = 0.1M$

91. Equilibrium constant expressions do not contain concentration terms for solids or liquids of a single component. Explain why?

Solution

The concentration of the single component with a pure solid or liquid phase cannot change and the activities of pure solids and liquids are set equal to one.

92. Arrange the following reaction in order of their increasing tendency to proceed toward completion

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a. $[illegible] + 3O_2(g) \rightleftharpoons 2N_2(g) + 6H_2O(g)$, $K_c = 1 \times 10^{28}$

b. $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, $K_c = 5 \times 10^{-31}$

c. $2HF(g) \rightleftharpoons H_2(g) + F_2(g)$, $K_c = 1 \times 10^{-13}$

d. $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$, $K_c = 4.7 \times 10^{-4}$

Answer: b < c < d < a

Supplementary Exercise

I. Multiple Choice items

1. Which statements are correct for a reaction at equilibrium?

   I. The forward and reverse reactions both continue

   II. The concentrations of reactants and products are equal

   III. The rates of forward and reverse reactions are equal

   A. I and II only
   C. II and III only
   B. I and III only
   D. I,II and III

2. Which changes to the equilibrium will make the solution go yellow?

   $Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$ $\Delta H > 0$

   Yellow red

   I. Lowering temperature

   II. Adding $FeCl_3(aq)$

   III. Adding $[Fe(SCN)]^{2+}(aq)$

   A. I and II only
   C. II and III only
   B. I and III only
   D. I, II and III

3. Which statement describes and explains the conditions that favour the formation of hydrogen iodide?

   $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, $K_c = 60$ at 350^0C, $K_c = 47$ at 445^0c

   A. Increasing pressure
   C. Removing H_2 and I_2
   B. Lowering temperature
   D. Increasing temperature

4. Which of the following is a correct statement for the effect of a catalyst on reversible chemical reaction?

   A. It allows the chemical reaction to reach equilibrium more quickly
   B. It only increases the rate of the forward reaction
   C. It increases the activation energy of the reaction
   D. It increases the amount of product.

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5. For the gaseous reaction, the equilibrium constant expression is:

$$K_c = \frac{[NO]^4[H_2O]^6}{[NH_3]^4[O_2]^5}$$

Which equation corresponds to this equilibrium expression?

A. $4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_2O$

B. $2NH_3 + \frac{5}{2}O_2 \rightleftharpoons 2NO + 3H_2O$

C. $2NO + 3H_2O \rightleftharpoons 2NH_3 + \frac{5}{2}O_2$

D. $4NO + 6H_2O \rightleftharpoons 2NH_3 + 5O_2$

6. The equilibrium constant expression, K_p for the reaction:
$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$ is

A. $\frac{PCH_3OH}{P(CO)(PH_2^2)}$ B. $PCH_3OH.PCO.PH_2^2$

C. $\frac{PCO.PH_2^2}{PCH_3OH}$ D. $\frac{2PH_2.PCO}{PC_3OH}$

7. Given the reaction at equilibrium:

$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g), \Delta H < 0$

Which of the following sets of conditions would give the best equilibrium yield of ethanol?

A. 100°C, 3atm C. 50°C, 4atm

B. 75°C, 2atm D. 25°C, 10atm

8. The larger the value an equilibrium constant. K_c

A. The larger is the rate of the reaction

B. The larger is the concentration of products at equilibrium

C. The more the exothermic reaction

D. The slower the rate of the reaction

9. The equilibrium constant for the reaction

$2HI(g) \rightleftharpoons H_2(g) + I_2(g), K_c = 0.04$

What would be the value of the equilibrium constant for the reaction

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below at the same temperature?

$$\frac{1}{2}H_2(g) + \frac{1}{2}I_2(g) \rightleftharpoons HI(g)$$

A. 0.02 B. 0.04 C. 5 D. 4

10. Which statement about chemical equilibria implies they are dynamic?
   A. The position of equilibrium constantly changes
   B. The rates of forward and backward reactions change
   C. The reactants and products concentrations continue to change
   D. The reactants and products continue to react.

11. What is the effect of an increase of temperature in the yield and the equilibrium constant for the following reaction respectively?

$2H_2(g) + CO(g) \rightleftharpoons CH_3OH(1), \Delta H = -128kJ$

   A. Increases, increases C. Decreases, decrease
   B. Increase decreases D. Decreases, increases

12. 0.50 mol of $I_2(g)$ and 0.50 mol of $Br_2(g)$ are pleased in a closed flask. The following equilibrium is established. $I_2(g) + Br_2(g) \rightleftharpoons 2IBr(g)$ The equilibrium mixture contains 0.80 mol of IBr (g). What is the value of K_c?

   A. 0.64 B. 6.4 C. 64 D. 2.6

13. For a reaction at equilibrium

   A. $Q = K$ B. $Q > K$ C. $Q < K$ D. $Q = 1$

14. For the reaction, $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$ $K_c = 1.00$ at 1100K. Which of the following must always be true concerning this reaction at 1100K?
   A. $2[CO] = [H_2O] = [CO_2] = [H_2]$
   B. $[CO] \times [H_2O] = [CO_2][H_2]$
   C. $[CO] = [H_2O]$ and $[CO_2] = [H_2]$
   D. $[CO] \times [H_2O] = [CO_2] = [H_2] = 1.00$

15. In which of the following the yield of the product in the reaction, $V(g) + \text{heat} \rightleftharpoons A(g) + B(g)$, would be higher?
   A. Low pressure, high temperature
   B. Low pressure, low temperature
   C. High pressure, low temperature

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D. High pressure, high temperature

16. For which of the following reactions, $K_C > K_P$?

A. $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$

B. $H_2(g) + CO_2(g) \rightleftharpoons CO(g) + H_2O(g)$

C. $Ni(s) + 4CO(g) \rightleftharpoons Ni(CO)_4(g)$

D. $MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g)$

17. For the reaction, $H^+(aq) + OH^-(aq) \rightleftharpoons H_2O(1)$, $K_f = 1.4 \times 10^{11}$ and $K_b = 2.5 \times 10^{-5}$ at 250^0c, the which one of then following is the value of K_c for this process?

A. 5.6×10^{15} B. 1.8×10^{-16} C. 1.4×10^{11} D. 3.5×10^{6}

18. The unit of K_c for the reaction $H_3A(aq) \rightleftharpoons 3H^+(aq) + A^{3-}$ is

A. mol. L^{-1} B. $mol^2. L^{-2}$ C. $mol^{-1}. L$ D. $mol^{-2}. L^2$

19. Which one of the following does not affect the position of equilibrium?

A. Temperature C. Pressure

B. Concentration D. Catalyst

20. For the equilibrium: $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$, $K_p = 16.75$atm at 500^0c . What is the partial pressure of CO_2 that is in equilibrium with CO whose partial pressure is 0.50atm

A. 0.0299 atm C. 0.25 atm

B. 0.0149 atm D. 0.05 atm

21. 4.5mol each of H_2 and I_2 are heated in a 10L sealed vessel. At equilibrium 3mol of HI were found. What is the value of equilibrium constant?

A. 1.0 B. 0.1 C. 10 D. 0.01

22. Consider the following equilibrium

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$, What happens if the volume is increased by adding nitrogen gas?

A. K_c increases

B. The equilibrium shifts to the right

C. The equilibrium shifts to the left

D. The position of the equilibrium is not affected

23. Consider equilibrium: $H_2(g) + CO_2(g) \rightleftharpoons H_2O(g) + CO(g)$ for which ΔH for the reaction = + 41 kJ. How is the amount of H_2 affected in the system when it is subjected to external stress?

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A. Addition of CO_2 increases the amount of H_2

B. Increase in temperature decreases the amount of H_2

C. Addition of a catalyst increases the amount of H_2

D. Decrease in volume, increases the amount of H_2

24. Consider the reaction, $PCl_5(s) \rightleftharpoons PCl_3(1) + Cl_2(g)$ in a closed container at equilibrium at a constant temperature. What will be the effect of adding more $PCl_5(s)$ on the equilibrium concentration of $Cl_2(g)$?

A. Decreases
C. Increases
B. No change
D. Can not be predicted

25. For which of the following cases, the reaction tends to go to completion?

A. $K = 1$ B. $K > 1$ C. $K < 1$ D. $K >> 1$

II. Short Answer Item

26. Write the equilibrium expression for each of the following reactions

a. $2H_2S(g) + 3O_2(g) \rightleftharpoons 2H_2O(g) + 2SO_2(g)$

b. $2N_2O_5(g) \rightleftharpoons 4NO_2(g) + O_2(g)$

c. $2N_2O(g) + 3O_2(g) \rightleftharpoons 4NO_2(g)$

27. Iodine monochloride reacts with chlorine in a reversible reaction to form iodine trichloride.

$ICl(1) + Cl_2(g) \rightleftharpoons ICl_3(g)$

Brown yellow

The equilibrium mixture contains both the brown liquid and the yellow solid.

a. What would you see if more chlorine was added to the mixture?

b. What would you see if the chlorine was allowed to leave the equilibrium mixture?

c. What would be the effect, if any, of increasing the pressure?

d. When the mixture is heated the yellow solid disappears. What can you deduce about the forward reaction?

28. The conversion of oxygen to ozone has very small equilibrium constant.

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$\frac{3}{2} O_2(g) \rightleftharpoons O_3(g), K_c = 2.5 \times 10^{-29}$ Caculate K_c for :

a. $3O_2(g) \rightleftharpoons 2O_3(g)$ b. $2O_3(g) \rightleftharpoons 3O_2(g)$

29. Lead sulphide can be converted to metallic lead in the following series of reactions

$2PbS(s) + 3O_2(g) \rightleftharpoons 2PbO(s) + 2SO_2(g), K_1$

$PbO(s) + C(s) \rightleftharpoons Pb(1) + CO(g), K_2$

$PbO(s) + CO(g) \rightleftharpoons Pb(1) + CO_2(g), K_3$

If the over all reaction is

$2PbS(s) + 3O_2(g) + C(s) \rightleftharpoons 2Pb(1) + CO_2(g) + 2SO_2(g)$

What is the equilibrium constant for the over all reaction interms of K_1, K_2 and K_3

30. 1mol of Br_2 is placed in 1L vessel and heated to 175°C, a temperature at which the halogen molecules dissociate to Br atoms. $Br_2(g) \rightleftharpoons 2Br(g)$, if $Br_2(g)$ is 1% dissociated at this temperature, calculate K_c.

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Answer to Supplementary Exercise on Units

Part I

1. B	6. A	11.C	16.C	21.A
2. B	7. D	12.C	17.A	22.B
3. B	8. B	13.A	18.B	23.B
4. A	9. C	14.B	19.D	24.B
5. A	10.D	15.A	20.B	25.D

Part II

26. a) $K_c = \frac{[H_2O]^2[SO]^2}{[H_2S]^2[O_2]^3}$, $K_p = \frac{(PH_2O)^2(PSO_2)^2}{(PH_2S)^2(PO_2)^3}$

b) $K_c = \frac{[NO_2]^4[O_2]}{[N_2O_5]^2}$, $K_p = \frac{(PNO_2)^4(PO_2)}{(PN_2O_5)^2}$

C) $K_c = \frac{[NO_2]^4}{[N_2O]^2[O_2]^3}$, $K_p = \frac{(PNO_2)^4}{(PN_2O)^2(PO_2)^3}$

27. a. Yellow
b. Brown
c. No effect
d. It is exothermic reaction

28. a. 6.25×10^{-58}
b. 1.54×10^{57}

29. $K_1 \times K_2 \times K_3$

30. 4.04×10^{-4}

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UNIT 6

Some Important Oxygen - Containing Organic Compounds

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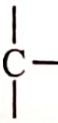
6.1 Introduction

- Organic compounds obtained by replacing hydrogen with oxygen are known as **oxygen - containing organic compounds.**
- Alcohols, ether, aldehydes, ketones, carboxylic acids, esters and fats and oils are some of the oxygen - containing organic compounds.

Table 6.1 Oxygen - containing Organic Compounds, their General Formula, Functional Groups and Examples

Organic Compound	General Formula	Functional Group	Example
Alcohol	R—OH or ROH		$CH_3—CH_2—OH$

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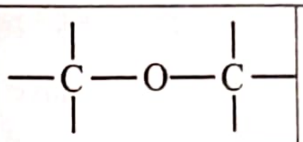
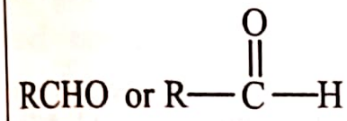
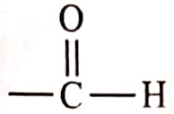
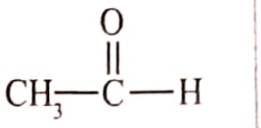
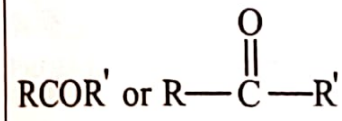
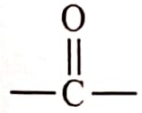
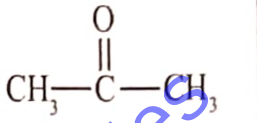
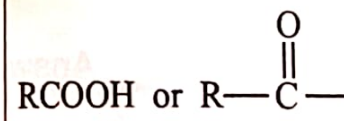
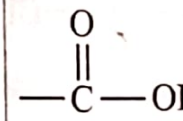
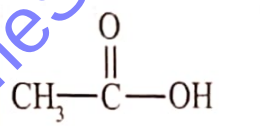
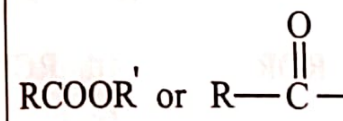
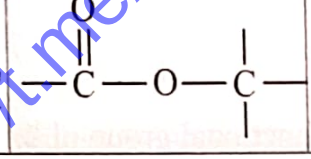
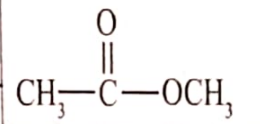
Ether	R—O—R' or ROR'		$CH_3—O—CH_3$
Aldehyde	RCHO or 		
Ketone	RCOR' or 		
Carboxylic acid	RCOOH or 		
Ester	RCOOR' or 		

Note:

With the same carbon number

i. Alcohol and ether
ii. Aldehyde ketone
iii. Carboxylic acid and ester

} are isomers

Illustrative Examples

1. Which of the following is not derivative hydrocarbon?

A. $NH_2—CH_2—CH_3$
B. $CH_3—CH_3$
C. $CH_3—CH_2OH$
D. $CH_3—COOH$

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Solution

Derivative hydrocarbons have other atoms out of hydrogen and carbon.

Answer: B

2. What is the functional group in CH_3COCH_3?

A. Aldehyde B. Alcohol C. Ketone D. Ether

Solution

To identify the functional group easily, draw the detail structural formula.

$$CH_3COCH_3 \Rightarrow CH_3-\overset{\overset{O}{\|}}{C}-CH_3$$

Answer: C

3. Write the general formula of

a. Alcohol b. Ketone c. Ether d. Aldehyde

Solution

a. ROH b. RCOR' c. ROR' d. RCHO

4. Identify the functional group of the following.

a. $CH_3CH_2COO\,CH_2CH_3$
b. $HO\,CH_2\,CH_2\,CH_2\,CH_2\,CH_3$
c. $CH_3\,CH_2\,CH_2\,CH_2\,COOH$
d. $CH_3\,CH_2\,COCH_2\,CH_3$
e. $CH_3\,CH_2OCH_2\,CH_2CH_3$
f. $CH_3\,CH_2\,CH_2\,CH_2\,CHO$

Solution

a. Ester c. Carboxylic acid e. Ether
b. Alcohol d. Ketone f. Aldehyde

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6.2 Alcohols and Ethers

I. Alcohols

- Alcohols are oxygen containing derivative hydrocarbons.
- They are common known as **Alkanols** since the last letter of alkanes "- e" replaced with '- ol'

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➥ They are represented by the general formula R—OH where **R** is alkyl group.

Classification of Alcohols

1. Based on the Number of Hydroxyl Groups (- OH) , Alcohols can be Classified as follows

i. **Monohydric alcohols** - contain one hydroxyl group.

➥ Their general formula is **ROH**

Example

CH_3CH_2OH, $CH_3—\underset{\displaystyle OH}{\underset{|}{CH}}—CH_2—CH_3$ etc

ii. **Dihydric alcohols** - have two hydroxyl groups

➥ Their general formula is $R—\underset{\displaystyle OH}{\underset{|}{CH}}—CH_2—OH$

Example

$\underset{\displaystyle OH}{\underset{|}{CH_2}}—\underset{\displaystyle OH}{\underset{|}{CH_2}}$, $CH_3—\underset{\displaystyle OH}{\underset{|}{CH}}—CH_2—OH$ etc

iii. **Trihydric alcohols** are alcohols containing three hydroxyl groups.

➥ Their general formula is $R—\underset{\displaystyle OH}{\underset{|}{CH}}—\underset{\displaystyle OH}{\underset{|}{CH}}—OH$

Example

$\underset{\displaystyle OH}{\underset{|}{CH_2}}—\underset{\displaystyle OH}{\underset{|}{CH}}—\underset{\displaystyle OH}{\underset{|}{CH_2}}$

iv. **Polyhydric alcohols** are alcohols containing three or more hydroxyl groups.

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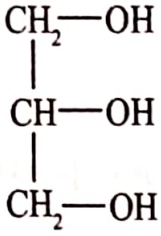
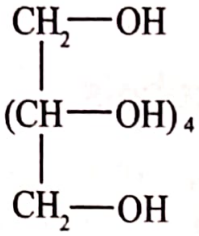
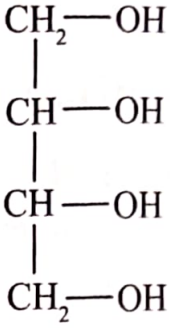
Example



Glycerol



Aditol

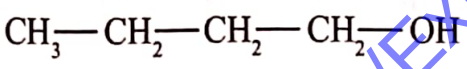


1, 2, 3, 4 - butantetralol

2. Based on Carbon - carbon Band, Alcohols can be Classified in to Two:

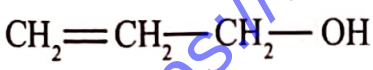
i. **Saturated alcohols** - are alcohols with all carbon - carbon single bonds.

Example



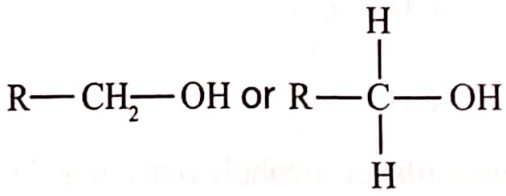
ii. **Unsaturated alcohols** - have carbon - carbon double or triple bonds.

Example



Classification of Monohydric Alcohols

1. **Primary alcohol** is an alcohol in which the - OH bearing carbon is bonded to one other carbon atom.

General Formula: 

Example

2. **Secondary alcohol** the - OH bearing carbon is bonded to two other carbon atoms

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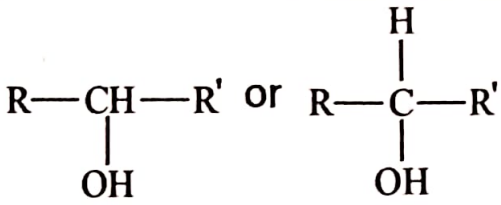
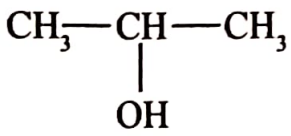
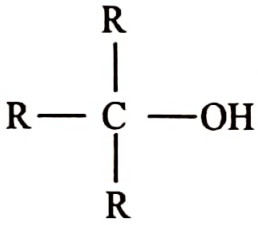
General Formula:  or

Example

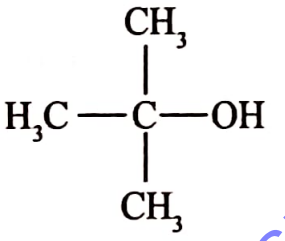


3. **Tertairy alcohol** is an alcohol in which the -OH containing carbon is bonded to other three carbon atoms

General Formula:

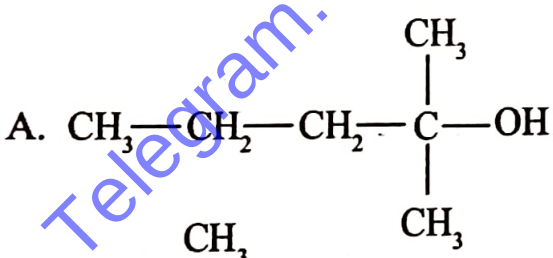


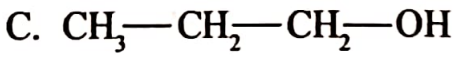
Example

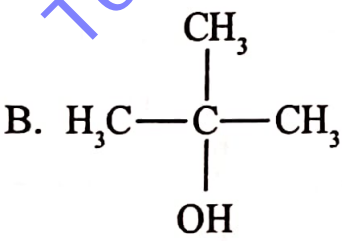


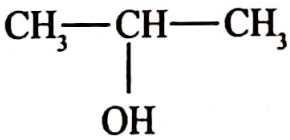
Illustrative Examples

4. Which of the following is secondary alcohol?

A. 

C. 

B. 

D. 

Solution

In secondary alcohol, the OH is attached to secondary carbon.

Answer: D

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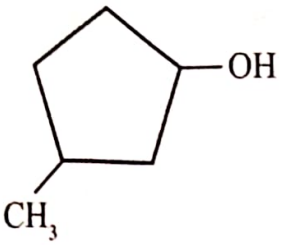
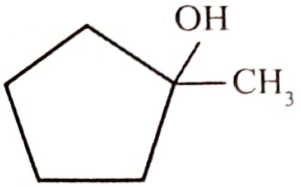
5. Which of the following is tertairy alcohol?

A. 

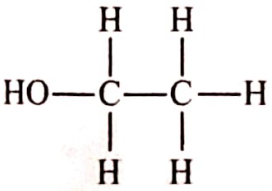
C. 

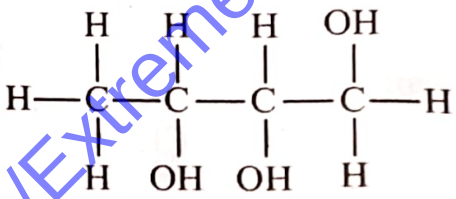
B. CH_3OH

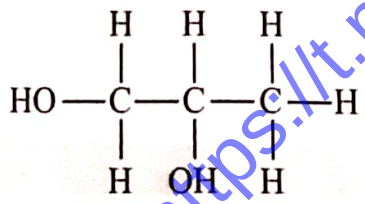
D. $CH_3CH_2CH_2OH$

Answer: C

6. Classify the following alcohols as monohydric, dihydric or trihydric alcohol.

a. 

b. 

c. 

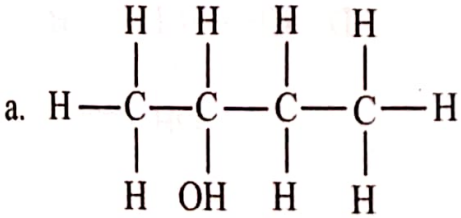
Solution

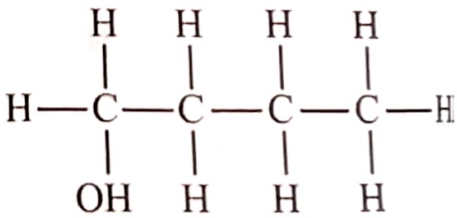
a. Monohydric alcohol

b. Trihydric alcohol

c. Dihydric alcohol

7. Classify the following as primary, secondary or tertiary alcohol.

a. 

b. 

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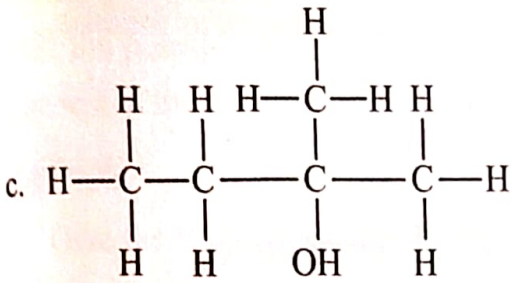
c. 

Solution

a. Secondary alcohol

b. Primary alcohol

c. Tertiary alcohol

8. Write two examples of unsaturated alcohols

Solution

a. $CH_3—CH=CH_2—CH_2—OH$

b. $HC\equiv C—CH_2—CH_2—\underset{\underset{OH}{|}}{CH}—CH_3$

Nomclature of Alcohols

A. Common Name

➥ The common name of simple monohydric alcohol is the name of alkyl group followed by the word alcohol.

Example

CH_3OH - Methyl alcohol

$CH_3\ CH_2\ OH$ - Ethyl alcohol

$CH_3\ CH_2CH_2OH$ - n - propyl alcohol

$CH_3—\underset{\underset{OH}{|}}{CH}—CH_3$ - Iso - propyl alcohol

$CH_2=CHOH$ - Vinyl alcohol

$CH_2=CH—CH_2—OH$ - Allyl alcohol

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* The common name of dihydric alcohol can be named as followings

Base name + -ylene + glycol

Example

$HO—CH_2—CH_2—OH$: Ethylene glycol

$CH_3—\underset{\underset{OH}{|}}{CH}—\underset{\underset{OH}{|}}{CH_2}$: Propylene glycol

The common trihydric alcohol with three carbon - atoms has a common name of **glycerol**.

$\underset{\underset{OH}{|}}{CH_2}—\underset{\underset{OH}{|}}{CH}—\underset{\underset{OH}{|}}{CH_2}$: Glycerol

B. IUPAC Name of Alcohols

➥ The IUPAC name of an alcohol is formed by dropping the '- e' in the name of hydrocarbon and then ending with ' - ol'

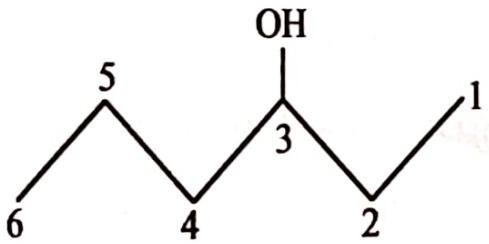
Example

CH_4 - Methane · CH_3OH - Methanol

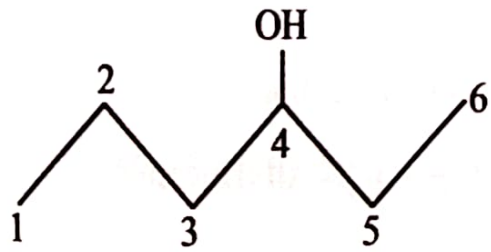
$CH_3—CH_3$ - Ethane · CH_3CH_2OH - Ethanol

➥ The longest chain is numbered to give the carbon bearing the - OH group the lowest number.

Example



Correct numbering



Wrong numbering

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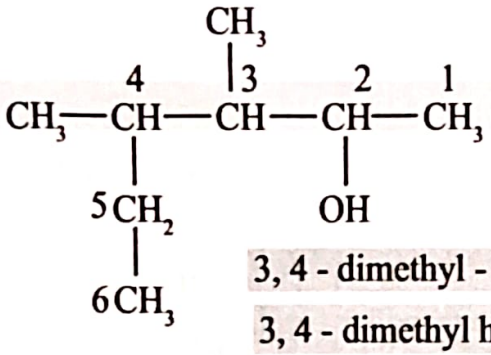
- State the position of - OH group.
- The rest of the rules are similar with hydrocarbons.

Example

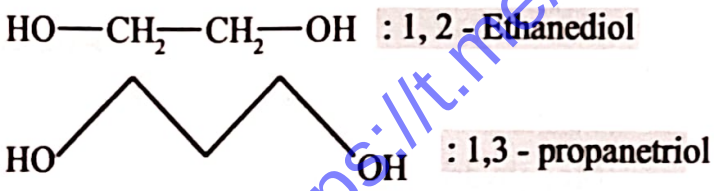
Give the IUPAC name of the alcohol with the formula



3, 4 - dimethyl - 2 - hexanol or
3, 4 - dimethyl hexan - 2 - ol

- For dihydric alcohols, the position of - OH groups, the name of alkane followed by - diol'.

Example

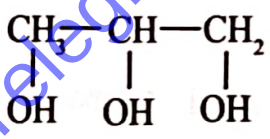


: 1, 2 - Ethanediol
: 1,3 - propanetriol

- For trihydric alcohols,

Position of - OH groups + name of alkane + - diol.

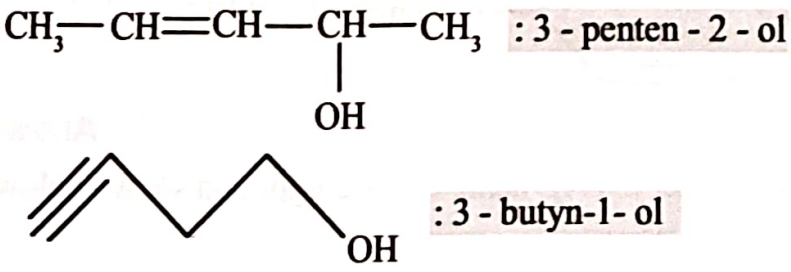
Example



: 1, 2, 3 - propanetriol

- For unsaturated alcohols, state the position of double or triple bond.

Example



: 3 - penten - 2 - ol
: 3 - butyn-1- ol

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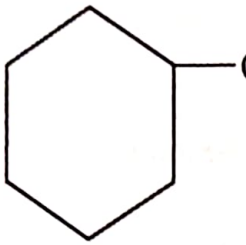
➥ For cyclic alcohol



Cyclo hexanol

Illustrative Examples

9. Which of the following represent the general formula of monohydric alcohols? (ESSCLE)

A. $C_nH_{n-2}(OH)_2$ C. $C_nH_{2n+1}OH$

B. $C_nH_{2n-2}OH$ D. $C_nH_{2n}OH$

Solution

In monohydric alcohols, one - OH group replaces one hydrogen of alkanes

Answer: C

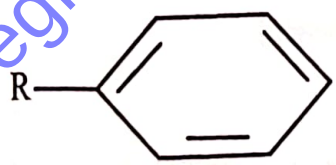
C_nH_{2n+2} = $C_nH_{2n+1}OH$

Alkane Monohydric alcohol

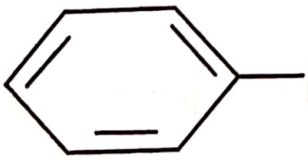
10. Which functional group containing compound is named using the suffix "- 0ℓ" ? (ESSLCE)

A. Aryl B. Carboxyl C. Phenyl D. Hydroxyl

Solution



Aryl R—COOH - Carboxyl



Phenyl R—OH - Hydroxyl

Answer: D

11. What is the correct IUPAC name of the compound shown below?

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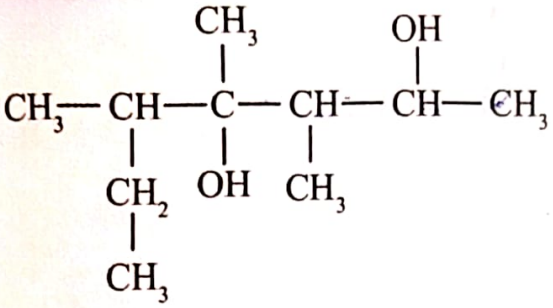


A. 3,4,5 - trimethyl heptan - 2,4 - diol

B. 3,4,5 - trimethyl heptane - 4, 6 - diol

C. 5 - ethyl - 3, 4 - dimethyl hexane - 2, 4 - diol

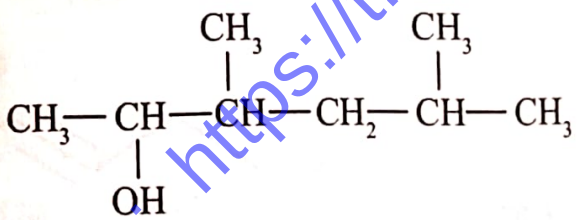
D. 2 - ethyl - 3, 4 - dimethyl hexane - 2, 4 - diol

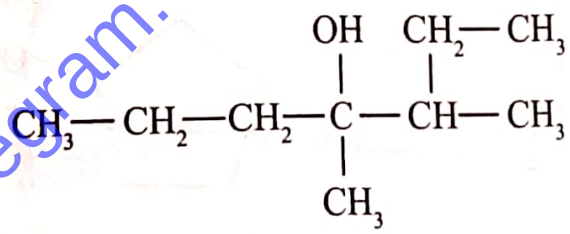
Solution

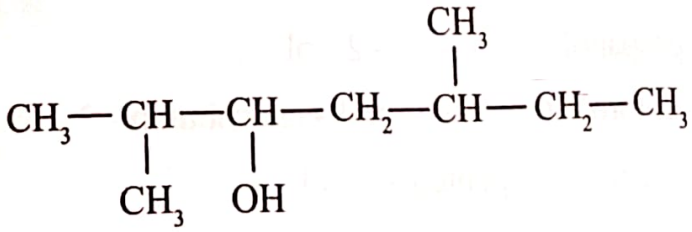
Start counting from the right which is nearer to the hydroxyl group.
3, 4, 5 - trimethyl heptane - 2, 4 - diol.

Answer: A

12. Which of the following is the correct structure for the hydrocarbon having the IUPAC name: 3, 5 - dimethyl - 3 - heptanol?

A. 

B. 

C. 

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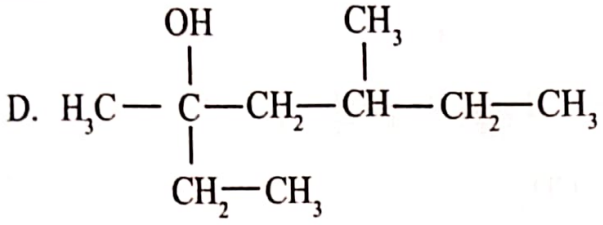
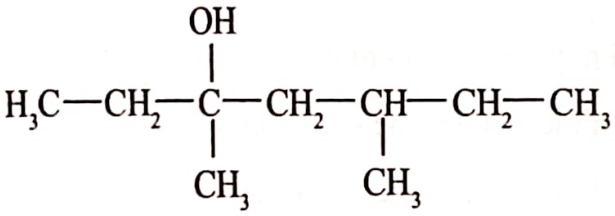
D. 

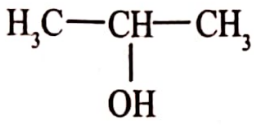
Solution

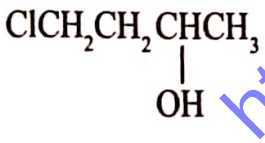
Start by drawing skeletal structure

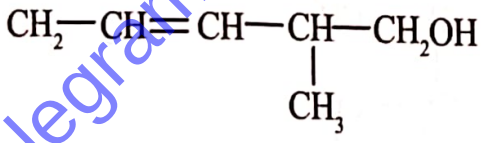


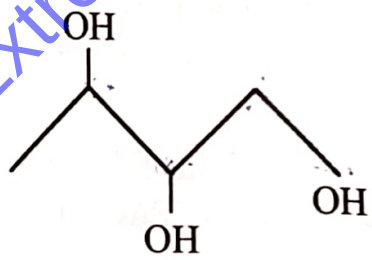
Answer: D

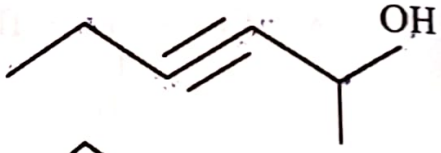
13. Write the IUPAC name of the following alcohols

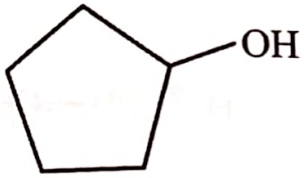
a. 

b. 

c. 

d. 

e. 

f. 

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Solution

a. 2 - propanol or propan - 2 - ol

b. 4 - chloro - 2 - butanol or 4 - chorobutan - 2 - ol

c. 2 - methyl - 3 - penten - 1 - ol

d. 1,2,3 - butanetriol

e. 3 - hexyn - 2 - ol

f. Cyclopentanol

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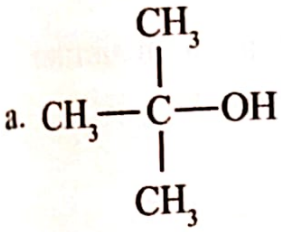
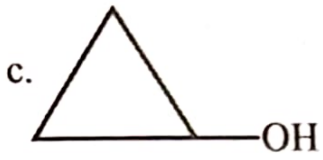
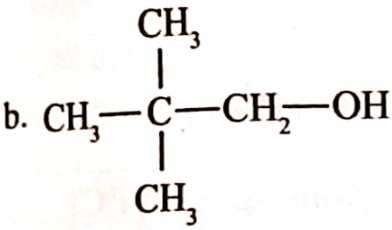
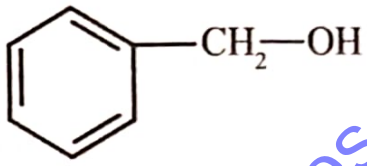
14. Write the common name for

a. 

c. 

b. 

d. 

Solution

a. t - butyl alcohol

b. Neopentyl alcohol

c. Cyclopropyl alcohol

d. Benzyl alcohol

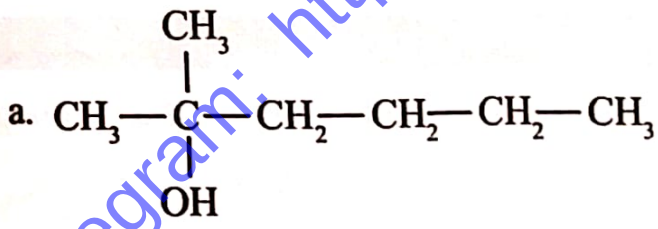
15. Draw the structure of

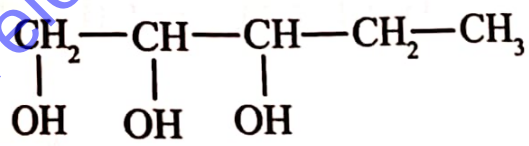
a. 2 - Methyl - 2 - hexanol

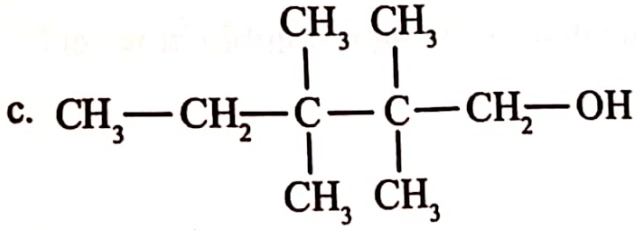
b. 1,2,3 - pentanetriol

c. 2, 2 , 3, 3 - teramethyl - 1 - Pentanol

Solution

a. 

b. 

c. 

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Physical Properties of Alcohols

- Alcohol have higher boiling point than hydrocarbons with similar mass due to the hydrogen bond in alcohols

Example

Boiling point of $CH_3CH_2OH > CH_3CH_3$

- As carbon increases, boiling point also increases.

Table 6.2 Some Members of Alcohols and their Boiling Points

Alcohol	Formula	Boiling point (^0C)
Methanol	CH_3OH	64
Ethanol	CH_3CH_2OH	78
1 - Propanol	$CH_3CH_2CH_2OH$	97
1 - Butanol	$CH_3CH_2CH_2CH_2OH$	117

- Boiling point of polyhydric alcohol > monohydric alcohol
- Boiling point of 1^0 alcohol > 2^0 alcohol > 3^0 alcohol of similar carbon number because boiling point decreases with increase in branches.
- Lower member alcohols (methanol and ethanol) are completely miscible with water. However, solubility in water dereases with carbon number.

Illustrative Examples

16. Which of the following organic compounds has the highest boiling point?

A. $CH_3CH_2CH_2OH$

B. $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

C. $CH_3-CH_2-CH_3$

D. $CH_2=CH-CH_3$

Answer: A

17. Which of the following alcohols is the most soluble in water?

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A. $CH_3—CH_2—CH_2—OH$

B. $CH_3—\underset{\displaystyle OH}{\underset{|}{CH}}—CH_3$

C. $CH_2(OH)—CH(OH)—CH_2(OH)$

D. $CH_3—CH(OH)—CH_2(OH)$

Solution

The more the hydroxyl groups in a compound the higher is the solubility in water.

Answer: C

18. Arrange the following alcohol in decreasing order of boiling point.
    a. n - Butyl alcohol
    b. Iso butyl alcohol
    c. t - butyl alcohol

Solution

Draw the structures and decide your answer

a. $CH_3—CH_2—CH_2—OH$

b. $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2OH$

c. $CH_3—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}—OH$

c < b < a as branching increases, boiling point decreases

19. Arrange the following alcohols in decreasing order of solubility in benzene.

I. 1 - pentanol II. 1 - octanol III. 1 - butanol

Solution

The solubility of alcohols in non - polar solvents such as benzene increases with carbon number.

II > I > III

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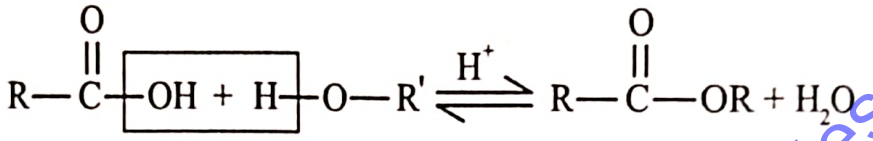
Chemical Properties of Alcohols

➥ Alcohols undergo two type of reactions

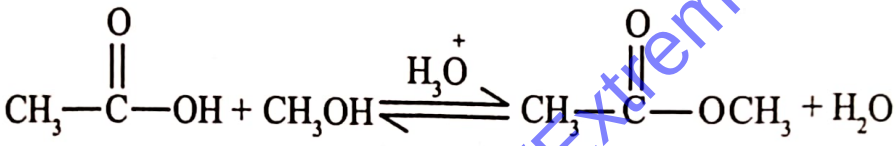
I. Reactions of Alcohols Involving Oxygen - hydrogen (O - H) Bond Cleavage

i. **Esterification** is the reaction between alcohol and carboxylic acid to yield ester and water.



Carboxylic acid alcohol Ester

Example

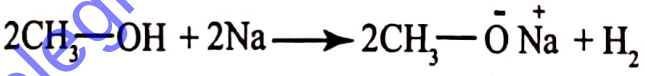


Acetic acid methanol methyl acetate

ii. Reaction of Alcohols with Active Metals

Alcohol + active metal ⟶ metal alkoxide + Hydrogen

Example



iii. Oxidation of Alcohols

a. Oxidation of Primary Alcohols Yields Aldehyde, and then Carboxylic

[O] = oxidizing agent ($KMnO_4$ or $K_2Cr_2O_7$ or CrO_3 or H_2SO_4)

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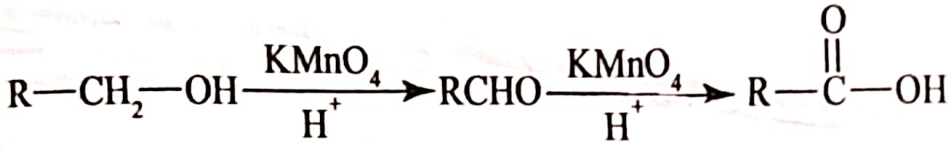
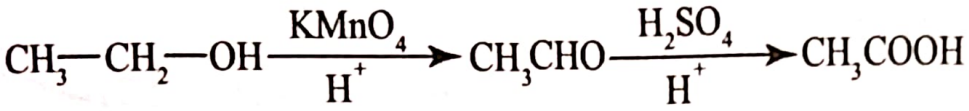
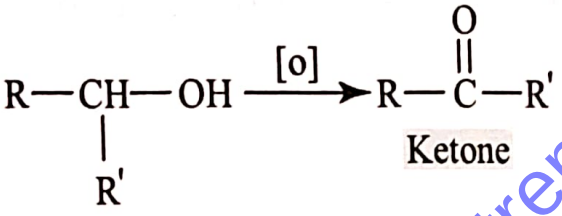




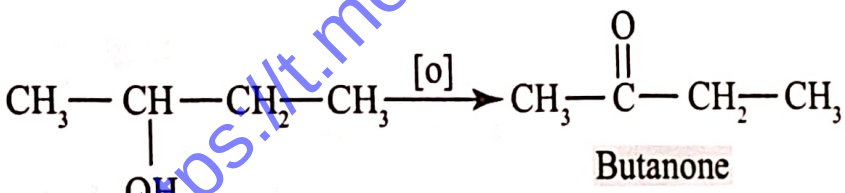
Ethanol Ethanal Ethanoic acid

b. Oxidation of Secondary Alcohols Gives Ketone



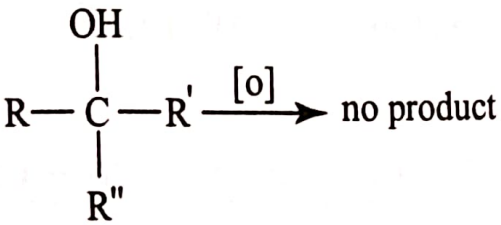
2° alcohol Ketone

Example



2 - Butanol Butanone

c. Oxidation of 3° Alcohols Yields no Product Under Normal Conditions.



2 - Butanol

Note: Tertairy alcohols undergo oxidation under drastic conditions and yields a mixture of carboxylic acids.

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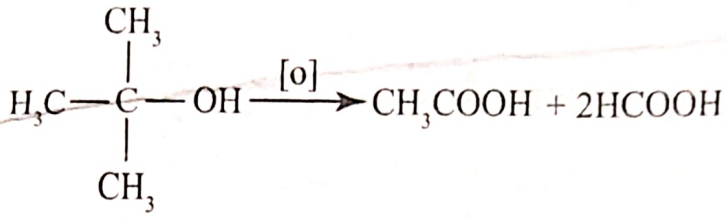
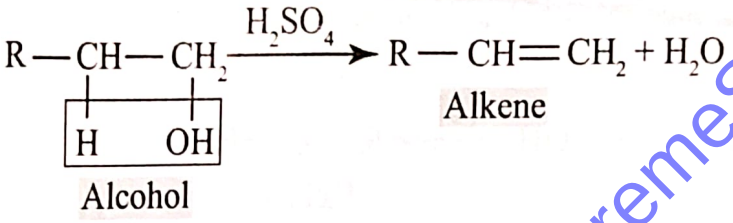
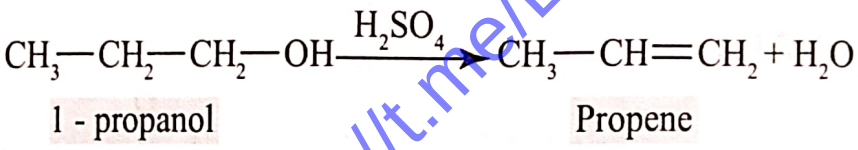


2. Reactions of Alcohols Involving Carbon - Oxygen (C - O) Bond Cleavage

i. **Dehydration of alcohols** - removal of water molecules using sulfuric acid.



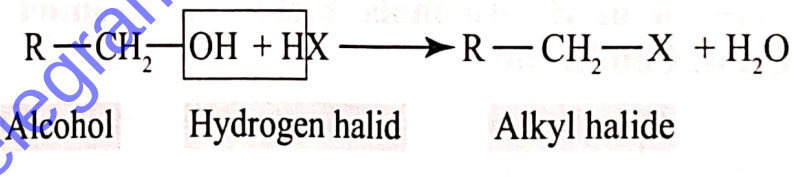
Example



Note: Ease of dehydration:

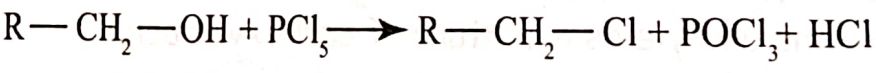
3^0 alcohol > 2^0 alcohol > 1^0 alcohol

ii. **Conversion of alcohols to alkyl halide**



Example

iii. **Reaction of alcohols with phoshorus chloride**



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Example

$$CH_3—CH_2—OH + PCl_5 \longrightarrow CH_3—CH_2—Cl + POCl_3 + HCl$$

Illustrative Examples

20. All of the following reactions involve O - H bond cleavage. EXCEPT

A. $CH_2\ COOH + CH_3\ OH \longrightarrow$

B. $CH_3\ CH_2\ OH + Na \longrightarrow$

C. $CH_3\ CH_2\ OH \xrightarrow{H_2SO_4}$

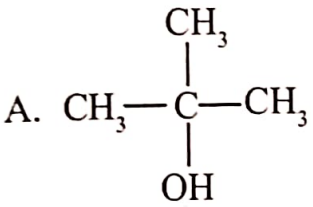
D. $CH_3\ CH_2\ OH \xrightarrow{[O]}$

Solution

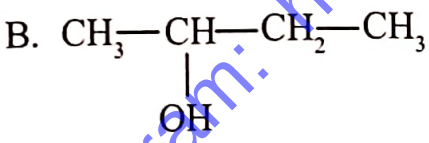
Dehydration of alcohols is reaction of alcohols that involves O—C bond cleavage.

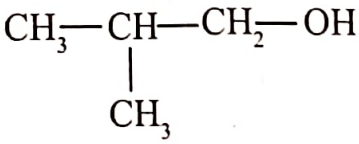
Answer: C

21. Which of the following alcohols is the easiest to be dehydrated?

A. 

C. $CH_3CH_2CH_2CH_2OH$

B. 

D. 

Solution

Tertairy alcohol is easily dehydrated.

Answer: A

22. Which of the following alcohols yields ketone up on oxidation?

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A. $CH_3—CH_2—OH$

C. $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2—OH$

B. $CH_3—\underset{\displaystyle OH}{\underset{|}{CH}}—CH_3$

D. $CH_3—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}—OH$

Solution

Secondary alcohols result ketones up on oxidation

Answer: B

23. Complete and balance the following

a. $CH_3—\overset{\displaystyle O}{\overset{\|}{C}}—OH + CH_3—CH_2—OH \xrightarrow{H^+}$

b. $CH_3—CH_2—CH_2—OH + Na \longrightarrow$

c. $CH_3—CH_2—CH_2—OH \xrightarrow{[o]}$

d. $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—OH \xrightarrow{H_2SO_4}$

e. $CH_3OH + HCl \longrightarrow$

f. $CH_3CH_2CH_2OH + PCl_5 \longrightarrow$

Solution

a. $CH_3—\overset{\displaystyle O}{\overset{\|}{C}}—OH + CH_3—CH_2—OH \xrightarrow{H^+} CH_3—\overset{\displaystyle O}{\overset{\|}{C}}—OCH_2—CH_3 + H_2O$

b. $2CH_3—CH_2—CH_2—OH + 2Na \longrightarrow 2CH_3CH_2CH_2\bar{O}\,\overset{+}{Na} + H_2$

c. $CH_3CH_2CH_2OH \xrightarrow{KMnO_4/H^+} CH_3CH_2CHO \xrightarrow{KMnO_4/H^+}$

$CH_3—CH_2—\overset{\displaystyle O}{\overset{\|}{C}}—OH$

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d. $CH_3-\underset{\underset{CH_3}{|}}{CH}-OH \xrightarrow{H_2SO_4} CH_2=CH-CH_3 + H_2O$

e. $CH_3OH + HCl \longrightarrow CH_3Cl + H_2O$

f. $CH_3CH_2CH_2OH + PCl_5 \longrightarrow CH_3CH_2CH_2Cl + POCl_3 + HCl$

Preparation of Alcohols

There are different methods used to prepare alcohols.

1. Acid - catalyzed Hydration of Alkenes.

An alcohol is formed when water adds to the double bond in an alkene

$$R-CH=CH_2 + H_2O \xrightarrow{H^+} R-\underset{\underset{OH}{|}}{CH}-CH_3$$

Alkene — Alcohol

Example

$$CH_3CH=CH_2 + H_2O \xrightarrow[H_2SO_4]{} CH_3\underset{\underset{OH}{|}}{CH}CH_3$$

Propene — 2 - Propanol

➥ The alcohol that results is from Markovnikov addition of water to the double bond.

2. Hydrolysis of Ester

Hydrolysis is the split of water molecule into H^+ and OH^-

➥ An ester undergoes hydrolysis reaction to yield alcohol and organic salt with base

$$R-\overset{\overset{O}{||}}{C}-OR' + KOH \xrightarrow[heat]{} R-\overset{\overset{O}{||}}{C}-\overset{-}{O}\overset{+}{K} + R'-OH$$

Ester — Potassium carboxylate — Alcohol

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Example

$$CH_3-\overset{\overset{O}{\|}}{C}-OCH_3 + KOH \xrightarrow{heat} CH_3-\overset{\overset{O}{\|}}{C}-OK + CH_3OH$$

Methyl acetate — Potassium acetate

3. Hydrolysis Alkyl halide with Sodium Hydroxide

- An alcohol is formed when an alkyl halide reacts with sodium hydroxide.

$$R-CH_2-X + NaOH \xrightarrow{heat} R-CH_2-OH + NaX$$

Alkll halide (X = Cl , Br, I) — Alcohol

Example

$$CH_3-CH_2-Br + NaOH \xrightarrow{heat} CH_3-CH_2-OH + NaBr$$

Note: Method 1 is addition reaction

- Methods 2 and 3 are substitution reactions
- Reduction of aldehyde, ketone, carboxylic acid and ester also yields alcohols.

Industrial Preparation of Ethanol (C_2H_5OH)

- **Ethanol** is the second alcohol next to methanol in the homologous series of monohydric alcohols.
- It is an alcohol found in all alcoholic beverages.
- It can be prepared industrially by

1. Catalytic Hydration of Ethene

$$CH_2=CH_2(g) + H_2O(g) \xrightarrow[373\ K/60atm]{heat} CH_3CH_2OH(g)$$

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2. Fermentation of Carbohydrate Such as Sucrose ($C_{12}H_{22}O_{11}$)

$$\underset{\text{sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{\text{Invertase}} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

$$\underset{\text{glucose}}{C_6H_{12}O_6} \xrightarrow{\text{Zymase}} \underset{\text{Ethanol}}{2C_2H_5OH} + 2CO_2$$

- The process that takes place in beer factory can be summarized as follows

$$\underset{\text{starch}}{2(C_6H_{12}O_6)n} + nH_2O \xrightarrow{\text{diastase}} \underset{\text{maltose}}{nC_{12}H_{22}O_{11}} \xrightarrow[\text{maltose}]{nH_2O} \underset{\text{glucose}}{2nC_6H_{12}O_6}$$

$$\underset{\text{glucose}}{C_6H_{12}O_6} \xrightarrow{\text{zymase}} \underset{\text{Ethanol}}{2C_2H_5OH} + 2CO_2$$

Properties of Ethanol

- It is a colorless liquid with pleasant smell and burning taste.
- It boils at 78.3^{0}C and freezes at - 117^{0}C.
- It is miscible with water in all proportions
- It is non - poisonous and has an intoxicating effect.
- It is a hypnotic (sleep producer) alcohol
- It is volatile alcohol.
- It burns with a blue flame.

Uses of Ethanol

- It is a constituent of all alcoholic beverages.
- It is a good solvent for many organic compounds that are insoluble in water such as paints, dyes, perfumes etc.
- It is used as fuel for cars and spirit lamps in laboratories.
- It is used to make denatured alcohol (95% ethanol and 5% methanol)

Illustrative Example

24. The major product "X" in the reaction

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$C_6H_{12}O_6 \xrightarrow{\text{Fermentation}} X + CO_2$, X is

A. CH_3CH_2OH C. $C_5H_{11}OH$

B. CH_3CH_3COOH D. H_2O

Solution

The major product of fermentation of glucose is ethyl alcohol.

$$C_6H_{12}O_6 \xrightarrow{\text{Fermentation}} 2C_2H_5OH + 2CO_2$$

Answer: A

25. All of the following reactions produce an alcohol. EXCEPT

A. $2CH_3CH_2COOH + 2Na \longrightarrow$

B. $CH_3-CH=CH_2 + H_2O \xrightarrow{H_2SO_4}$

C. $CH_3-CH_2-CH_2-Br + NaOH \longrightarrow$

D. $CH_3-\overset{\overset{O}{\|}}{C}-H + H_2 \xrightarrow{Ni}$

Solution

A. $2CH_3CH_2COOH + 2Na \longrightarrow 2CH_3CH_2COONa + H_2$

B. $CH_3-CH=CH_2 + H_2O \xrightarrow{H_2SO_4} CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

C. $CH_3-CH_2-CH_2-Br + NaOH \longrightarrow CH_3-CH_2-CH_2-OH + NaBr$

D. $CH_3-\overset{\overset{O}{\|}}{C}-H + H_2 \xrightarrow{Ni} CH_3-CH_2-OH$

Answer: A

26. The type of alcohol that is found in all common alcoholic beverages like "Tella", "Areqi", "Beer", etc.

A. Propanol B. Butanol C. Methanol D. Ethanol

Solution

An alcohol that is found in all alcoholic beverages is ethanol.

Answer: D

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27. Complete and balance the following chemical equations

a. $CH_3-\underset{\underset{CH_2}{\|}}{C}-CH_3 + H_2O \xrightarrow{H^+}$

b. $(CH_3)_2CHOOCCH_3 + NaOH \longrightarrow$

c. $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-Cl + NaOH \longrightarrow$

Solution

a. $CH_3-\underset{\underset{CH_2}{\|}}{C}-CH_3 + H_2O \xrightarrow{H^+} CH_3-\underset{\underset{OH}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3$

b. $(CH_3)_2CHOO\,C\,CH_3 + NaOH \longrightarrow (CH_3)_2CH\,OH + Na\,OOCCH_3$

c. $H_3C-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-Cl + NaOH \longrightarrow H_3C-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OH + NaCl$

28. What is methylated spirits or denatured alcohol?

Solution

Methylated spirits is mainly ethanol mixed with a little methanol, which is poisonous. Colouring is added and has a horrible taste to stop people drinking it.

29. Why wine goes 'sour' when left open in air?

Solution

Because the alcohol (ethanol) in it slowly oxidized to ethanoic acid, by bacteria in air.

$$C_2H_5OH\,(l) + 2O_2(g) \longrightarrow CH_2COOH\,(aq) + H_2O(l)$$

30. State the properties, preparation and uses of methanol

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Solution

- Methanol is also known as wood alcohol
- It is a deadly poison alcohol.
- It can cause blindness and eventually total loss of motor control and death.
- It can be prepared by reacting carbon monoxide with hydrogen.

$$CO + 2H_2 \longrightarrow CH_3OH$$

- It is used to make formaldehyde (methanal)

Structure and Nomenclature of Ethers

Ether has two hydrocarbon groups bonded to oxygen.

$$R—O—R'$$

- If R and R' are same group, the given ether is known as **a symmetric ether**

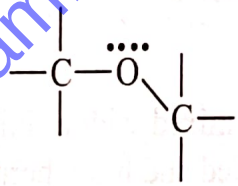
Example $CH_3—O—CH_3$

- If R and R' are different, the given ether is known as **unsymmetrical ether.**

Example

$$CH_3-O-CH_2\,CH_3$$

- The function group of an ether is



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Nomeclature of Ethers

- There are two ways of naming ethers. For simple ethers, the name is obtained by naming each of the organic groups on oxygen atom and adding the word ether.
- When the hydrocarbon groups are same add the prefix di before their names.

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Example

a. $CH_3O\,CH_3$ dimethyl ether

b. $C_2H_5O\,C_2H_5$ diethyl ether

c. $CH_3CH(CH_3)O\,CH(CH_3)CH_3$ di isopropyl ether

- If the hydrocarbon groups attached to oxygen are different, write them in alphabetical order.

Example

a. $CH_3O\,CH_2CH_3$ Ethyl methyl ether

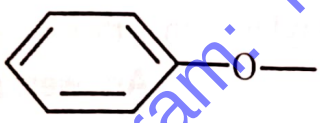
b. $CH_3O\,CH_2CH_2CH_2CH_3$ Butyl methyl ether

- For more complex molecules, the simplest organic group along with the oxygen atom is named as an **alkoxy or alryloxy group** and considered to be a substitute on a more complex chain.
- Some important alkoxy or aryloxy groups are the following.

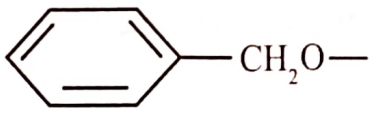
CH_3O- Methoxy

CH_3CH_2O- Ethoxy

$CH_3CH_2CH_2O-$ Propoxy



Phenoxy

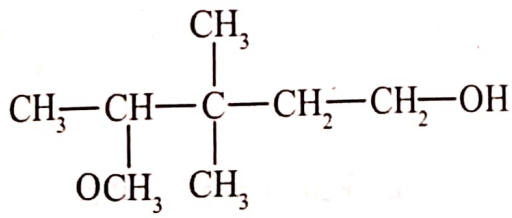


Benzyloxy

Example

Give the IUPAC name of the following ethers

a. $CH_3CH_2CH_2O\,CH_2CH_2CH_2CH_3$ 1 - propoxy butane

b.  4 - methoxy -3, 3 - dimethyl - 1 - pentanol

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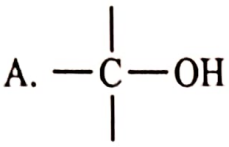
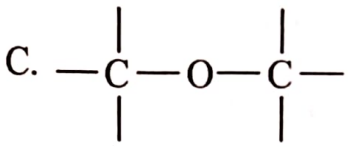
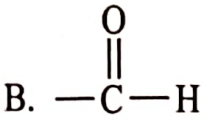
c. $CH_3O\,CH_2CH_2O\,CH_2CH_2CH_3$ 1 - methoxy - 2 - propoxyl ethane

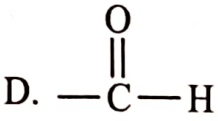
Illustrative Examples

31. Which of the following is the functional group of ether?

A. 

C. 

B. 

D. 

Solution

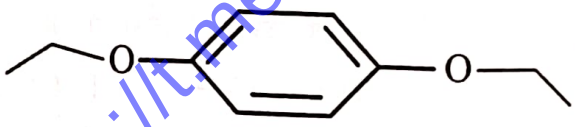
A. Alcohol B. Aldehyde C. Ether D. Ketone

Answer: C

32. Which of the following is commonly known alkoxy alkane?

A. Ketone B. Aldehyde C. Alcohol D. Ether

Answer: D

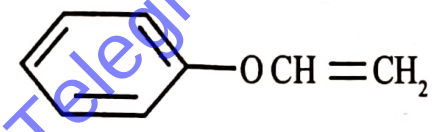
33. The IUPAC name of  is

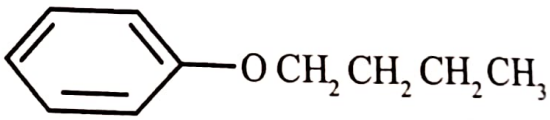
A. 1,4 - diethoxy benzene C. 1,4 - diethyl ether

B. 1,4 - dimethoxy benzene D. Diethyl benzene ether

Answer: A

34. Give common name for the following ethers

a. 

c. 

b. $CH_3\,CH(CH_3)\,CH_2O\,CH_2\,CH(CH_3)\,CH_3$

d. $CH_3\,CH_2O\,CH_2CH_2\,CH_3$

Solution

a. Phenyl vinyl ether

b. Diisobutyl ether

c. Pentyl phenyl ether

d. Ethyl propyl ether

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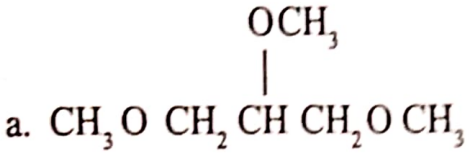
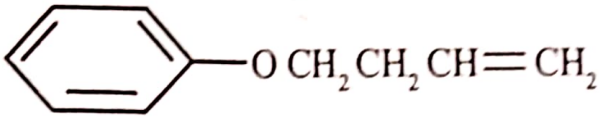
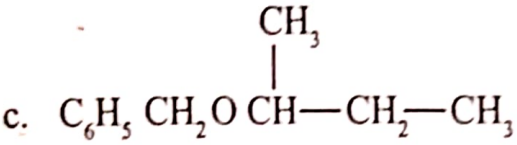
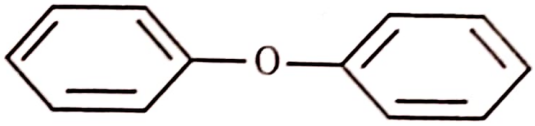
35. Give the IUPAC name of the following ethers

a. 

b. 

c. 

d. 

Solution

a. 1,2,3 - trimethoxy propane
b. 4 - phenoxy - 1 - butene
c. 2 - benzyloxy butane
d. Phenoxy benzene

Physical Properties of Ethers

Ethers have lower boiling point than alcohols of similar size because they do not form hydrogen bond.

Example

Boiling point of: $CH_3OCH_3 < CH_3CH_2OH$

- Like other organic compounds, as size increases, boiling point of ethers also increases.

Example

Boiling point: $CH_3OCH_3 < CH_3OCH_2CH_3 < CH_3OCH_2CH_2CH_3$

- Boiling point of ethers is comparable with hydrocarbons.
- An ether molecule has a dipole moment due to the carbon - oxygen bond.
- Lower member ethers with two and three carbon atoms are miscible in water.
- As carbon number increases, solubility of ether in water decreases.

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